

HCMC UNIVERSITY OF TECHNOLOGY AND EDUCATION
FACULTY OF INTERNATIONAL EDUCATION



HCMUTE

GRADUATION THESIS

**ENHANCING SPATIAL GEOMETRY
EDUCATION THROUGH INTERACTIVE
3D SOFTWARE, GAMIFICATION
AND AI - POWERED LEARNING**

INSTRUCTOR: M. Eng. NGUYỄN ĐĂNG QUANG

STUDENTS:

LÂM NGUYỄN HUY HOÀNG	-	21110028
DIỆP HOÀI AN	-	21110001
NGUYỄN ĐĂNG KHOA	-	21110045

ACADEMIC YEAR: 2021 - 2025

MAJOR: SOFTWARE ENGINEERING

Ho Chi Minh City, May 2025

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GRADUATION THESIS

Student Name: Lam Nguyen Huy Hoang ID: 21110028

Student Name: Diep Hoai An ID: 21110001

Student Name: Nguyen Dang Khoa ID: 21110045

1. **Major:** Software Engineering.
2. **Project title:** Enhancing Spatial Geometry Education Through Interactive 3D Software, Gamification, And AI – Powered Learning.
3. **Advisor:** M.Eng. Nguyen Dang Quang
4. **Original Data and Documents:**
 - Surveyed 92 high school students to identify cognitive barriers in spatial geometry.
 - Reviewed and applied pedagogical models: Van Hiele Theory, Cognitive Load Theory, Spatial Intelligence, Gamification, Systems Thinking.
 - Explored technologies including Unity 3D, Unity Netcode, GPT-4 API, Whisper (STT), ElevenLabs (TTS).
 - Researched interactive 3D drawing, AI-assisted geometry construction, and educational game design.

5. Project Implementation:

The team developed an integrated learning platform that supports geometry learning through 3D interaction, AI tutoring, and gamification. The core modules include:

- Drawing Tool: Allows users to construct geometric shapes manually or via text-based input.
- AI Drawing Assistant: Converts text-based geometry problems into visual 3D representations with data validation and error warnings.
- AI Tutor: Answers conceptual questions using voice interaction and a friendly animated character.

- Gamified Practice: Mini game “Collect Coin” delivers tagged geometry questions aligned with Van Hiele levels and logs user performance.
- Online Mode: Enables host-guest synchronization for real-time remote viewing of constructions.
- Data Logging System: Records user interactions and gameplay for future personalized analytics and profiling.

Ho Chi Minh City, May 2025

INSTRUCTOR

(Sign with full name)



ADVISOR'S EVALUATION SHEET

Student Name: Lam Nguyen Huy Hoang ID: 21110028

Student Name: Diep Hoai An ID: 21110001

Student Name: Nguyen Dang Khoa ID: 21110045

Major: Software Engineering.

Project title: Enhancing Spatial Geometry Education Through Interactive 3D Software, Gamification, And AI – Powered Learning.

Advisor: M.Eng. Nguyen Dang Quang

EVALUATION

1. Content of the project:

a. Theory: Unity 3D, Unity Netcode, GPT-4 API, Whisper, ElevenLabs, pedagogical frameworks (Van Hiele Model, Cognitive Load Theory, Spatial Intelligence, Gamification, Systems Thinking)

b. Experiments & Implementation:

- Conducted a survey with 92 high school students to investigate challenges in spatial geometry learning.
- Designed and built a 3D educational platform allowing interactive shape construction.
- Integrated AI Drawing Assistant to convert geometry problems into real-time 3D models.
- Implemented AI Tutor with text-to-speech and voice input for theory explanation.
- Developed the mini game “Collect Coin” to embed questions tagged with Van Hiele levels and misconception types.
- Built online host-guest synchronization for collaborative learning.
- Enabled logging of gameplay for future analytics.

c. Repository: <https://github.com/adairh/CapstoneProject>

2. Strengths:

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3. Weaknesses:

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.....

4. Approval for oral defense? (Approved or denied)

.....

5. Overall evaluation: (Excellent, Good, Fair, Poor)

.....

6. Mark:..... (in words:)

Ho Chi Minh City, May 2025

ADVISOR

(Sign with full name)



PRE-DEFENSE EVALUATION SHEET

Student Name: Lam Nguyen Huy Hoang ID: 21110028

Student Name: Diep Hoai An ID: 21110001

Student Name: Nguyen Dang Khoa ID: 21110045

Major: Software Engineering.

Project title: Enhancing Spatial Geometry Education Through Interactive 3D Software, Gamification, And AI - Powered Learning.

Reviewer: MSc. Nguyen Duc Khoan

EVALUATION

1. Content of the project:

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2. Strengths:

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3. Weaknesses:

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.....
4. Approval for oral defense? (Approved or denied)

.....
5. Overall evaluation: (Excellent, Good, Fair, Poor)

.....
6. Mark:..... (in words:)

Ho Chi Minh City, May 2025

REVIEWER

(Sign with full name)



**EVALUATION SHEET OF
DEFENSE COMMITTEE MEMBER**

Student Name: Lam Nguyen Huy Hoang ID: 21110028

Student Name: Diep Hoai An ID: 21110001

Student Name: Nguyen Dang Khoa ID: 21110045

Major: Software Engineering.

Project title: Enhancing Spatial Geometry Education Through Interactive 3D Software, Gamification, And AI - Powered Learning.

Name of Defense Committee Member:

Dr. Nguyen Thanh Tuan

Dr. Nguyen Thanh Son

MSc. Nguyen Tran Thi Van

EVALUATION

1. Content of the project:

a. Theory: Unity 3D, Unity Netcode, GPT-4 API, Whisper, ElevenLabs, pedagogical frameworks (Van Hiele Model, Cognitive Load Theory, Spatial Intelligence, Gamification, Systems Thinking)

b. Experiments & Implementation:

- Conducted a survey with 92 high school students to investigate challenges in spatial geometry learning.
- Designed and built a 3D educational platform allowing interactive shape construction.
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- Built online host-guest synchronization for collaborative learning.
- Enabled logging of gameplay for future analytics.

2. Strengths:

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3. Weaknesses:

.....

.....

.....

4. Approval for oral defense? (Approved or denied)

.....

5. Overall evaluation: (Excellent, Good, Fair, Poor)

.....

6. Mark:..... (in words:)

Ho Chi Minh City, June 2025

COMMITTEE MEMBER

(Sign with full name)

ACKNOWLEDGMENT

We would like to express our sincerest gratitude to our instructor, Mr. Nguyen Dang Quang, for his invaluable guidance, encouragement, and continuous support throughout the development of our graduation project, *“Enhancing Spatial Geometry Education Through Interactive 3D Software, Gamification, and AI-Powered Learning.”*

His practical insights and constructive feedback played a crucial role in shaping the direction of our work and helping us stay on track. He was always available to answer questions, offer timely suggestions, and provide thoughtful advice, enabling us to complete the project within the given timeframe.

We also extend our heartfelt appreciation to all the lecturers and staff of the Faculty of International Education, especially those in the Software Engineering program, whose dedication and expertise provided us with a solid foundation of knowledge. This project has given us the valuable opportunity to apply and deepen our understanding of both spatial geometry and interactive educational technologies.

Additionally, we are thankful to our classmates and friends for their encouragement, meaningful discussions, shared resources, and constructive critiques, all of which contributed greatly to the improvement of our work.

We recognize that, due to time limitations and our current level of technical experience, the project may still contain certain imperfections. Therefore, we humbly welcome any feedback and suggestions from our respected lecturers to further enhance our understanding and better prepare us for future endeavors.

Finally, we would like to wish all our teachers and mentors good health, happiness, and continued success in their noble mission to educate and inspire.

Once again, we sincerely thank you.

Ho Chi Minh City, May 2025

STUDENT

Lam Nguyen Huy Hoang

Diep Hoai An

Nguyen Dang Khoa

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CHAPTER 1. INTRODUCTION

This thesis introduces an innovative educational software solution designed to revolutionize spatial geometry education for high school students, addressing long-standing challenges in traditional learning methods and outlining the project's core vision and objectives.

1.1. Barriers in Spatial Geometry Learning and the Need for a New Approach

Spatial geometry, especially in three dimensions, is one of the most cognitively demanding areas of mathematics for high school students. *Spatial geometry* (also known as solid geometry) is the branch of mathematics that studies three - dimensional figures, including their shapes, dimensions, positions, and relationships in space. Unlike plane geometry, which deals with two - dimensional figures, spatial geometry involves depth and requires reasoning about volume, perspective, and spatial transformation. It requires abstract reasoning, mental visualization, and geometric transformation skills - abilities that many students struggle to develop using traditional instructional methods. In the Vietnamese high school curriculum (grades 10 to 12), spatial geometry is a core component not only for graduation exams but also for future pathways in architecture, engineering, and computer science.

Despite its importance, spatial geometry is often perceived by students as difficult and intimidating. This is reflected in cognitive learning models such as Van Hiele's theory, which describes students' progression through levels of geometric understanding. Many learners remain stuck at the early levels, where they rely on visual recognition but cannot yet analyze or reason about spatial relationships. This stagnation is frequently reinforced by static instructional tools such as chalkboard sketches or textbook diagrams, which fail to convey the dynamic and visual nature of 3D geometry.

To better understand these challenges, our team conducted a survey with 92 *Vietnamese high school students*. The results revealed key barriers:

- Only **23.9%** of students rated their understanding of 3D transformations (e.g., rotation, symmetry) as good or excellent. In contrast, over **51.1%** felt their

CHAPTER 1: INTRODUCTION

understanding was average, and **16.3%** admitted to poor or below - average comprehension.

Bạn nghĩ mình hiểu các mối quan hệ giữa các đối tượng 3D, chẳng hạn như xoay, đối xứng và phép biến hình đến mức nào?

92 responses

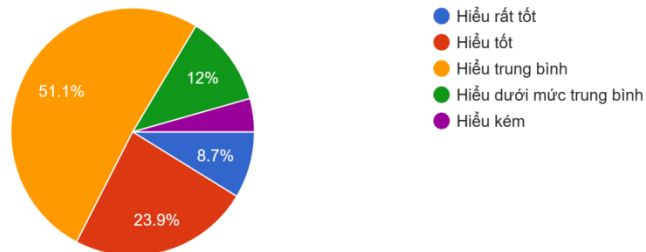


Figure 1.1 Students rated their understanding of 3D transformations

- Regarding the effectiveness of traditional teaching methods, such as chalkboard diagrams or physical models, only **39.6%** considered them effective. **20.7%** found them ineffective, and **40.2%** rated them as merely average.

Theo trải nghiệm của bạn, các phương pháp truyền thống (sách giáo khoa, mô hình vật lý, bảng vẽ) có hiệu quả như thế nào trong việc giúp bạn hình dung và hiểu hình học không gian 3D?

92 responses

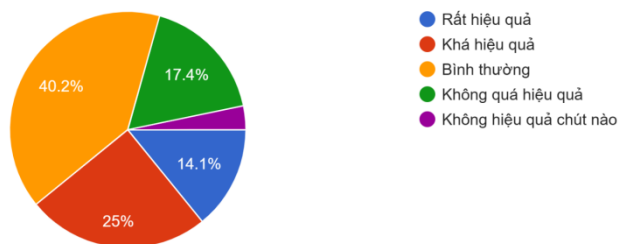


Figure 1.2 Effectiveness of traditional teaching methods

- In terms of engagement, only **16.3%** of students found traditional teaching approaches engaging, **16.3%** found them not engaging at all, while **67.4%** said they were only occasionally engaging or not consistently appealing.

Bạn có thấy các phương pháp truyền thống hấp dẫn khi học về hình học không gian không?
92 responses

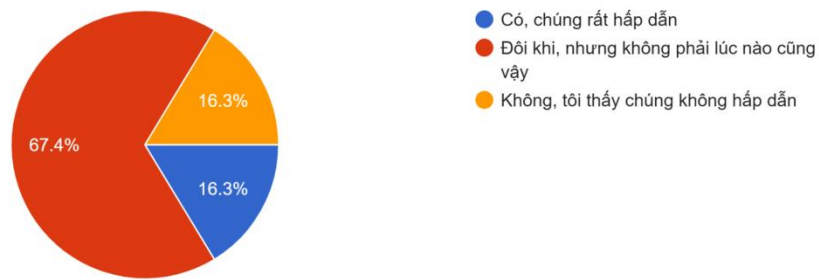


Figure 1.3 Rate of traditional teaching approaches engaging

These statistics highlight not only gaps in comprehension, but also emotional disengagement - a combination that leads to learning fatigue and avoidance.

To address these challenges holistically, this project proposes a comprehensive digital solution built on four foundational strategies:

1. *Interactive 3D Drawing Tools* – Enabling students to construct, manipulate, and explore geometric shapes in real time helps build intuition and bridges the gap between static diagrams and dynamic spatial reasoning.
2. *AI - Powered Tutoring* – By integrating an AI assistant capable of interpreting student input and providing immediate explanations (via ChatGPT), learners receive personalized support that adapts to their pace and confusion points.
3. *Gamification of Learning* – Educational mini-games transform practice into an active and rewarding experience, reducing anxiety and reinforcing core concepts through play.
4. *Multiplayer Collaboration* – Real-time drawing sessions (host/guest model) encourage peer learning, teacher-student interaction, and shared exploration of geometry problems.

Together, these components reflect a shift from *passive to active learning* - from isolated problems - solving to social, visual, and intelligent engagement. This project does not treat 3D drawing, AI tutoring, or gamification as isolated features, but as interconnected responses to a deeper educational need. In doing so, it introduces a new

approach to spatial geometry: “*Interact – Assess – Support*”. Students interact through 3D drawing and online collaboration; they are assessed via embedded mini - games that reveal cognitive patterns; and they are supported through an AI tutor and real - time in - tool guidance. These three elements form a continuous learning loop that responds to how students learn spatial geometry.

1.2. Objectives of the Project

The primary goal of this project is to create an educational software solution that enhances high school students’ learning experience in spatial geometry through technology.

Building on the multifaceted barriers identified in spatial geometry learning, this project proposes an integrated educational software platform with the following objectives:

- *Objective 1:* Develop an interactive 3D drawing environment that allows students to construct and explore geometric shapes intuitively, with snapping, labeling, and constraint features.
- *Objective 2:* Incorporate game-based learning features, including mini-games designed to reinforce geometric concepts while assessing specific learning difficulties via a mechanically designed analytical question bank to diagnose spatial geometry difficulties.
- *Objective 3:* Enable real-time multiplayer collaboration, allowing teachers to host drawing sessions and students to work on problems collaboratively in class or remotely.
- *Objective 4:* Integrate an AI-powered virtual tutor, using ChatGPT to interpret prompts, suggest geometric constructions, explain errors, and support self-guided learning.
- *Objective 5:* Provide data-information that indirectly assesses students’ spatial skills and identifies whether their learning obstacles are due to visualization, logic, or procedural misunderstanding.

By fulfilling these objectives, the project not only bridges the gap between pedagogical theory and technological advancement, but also directly addresses the motivational and cognitive barriers described in *Section 1.1*.

1.3. Scope of the Project

In response to both logistical and pedagogical challenges outlined earlier, this system is designed for high accessibility, cognitive support, and cross-device usability. The primary users of the application are Vietnamese high school students, along with mathematics teachers seeking to improve classroom engagement and conceptual clarity. The platform has been intentionally designed to be simple and easily accessible, requiring no user account or login - following the usability model of widely adopted platforms like *Kahoot* - to ensure seamless integration in both formal classroom settings and independent learning scenarios.

The application is developed primarily as a mobile app, allowing students to conveniently engage with geometry lessons anytime and anywhere, whether in the classroom, at home, or on the go. This mobility not only supports flexible learning schedules but also aligns with students' increasing preference for mobile-first educational tools.

Included in the system's functional scope are:

- A 3D shape drawing engine that supports geometric construction, transformation, and manipulation with snapping and constraint logic.
- Real-time multiplayer capabilities utilizing Unity Netcode to enable collaborative exploration and group-based learning.
- An AI assistant powered by the ChatGPT API, capable of interpreting geometry-related prompts and providing context-aware guidance.
- A set of educational mini-games and interactive challenges designed to be both engaging and diagnostic-reinforcing key concepts while evaluating individual learning difficulties.

- A visually optimized user interface based on principles of color psychology and cognitive ergonomics, using calming tones such as green and yellow to reduce anxiety and maintain attention.

This combination of mobility, interactivity, AI support, and game-based logic makes the application well-suited to address both logistical and pedagogical challenges in current spatial geometry education.

1.4. Scientific and Practical Significance

From a scientific perspective, this project draws upon educational theory, human-computer interaction, and cognitive psychology to reimagine how students learn geometry - especially considering the disengagement and cognitive overload issues revealed in our student survey. It integrates:

- *Educational Psychology*: Guided by the Van Hiele model to align features with students' cognitive development stages.
- *Cognitive Science*: Addressing spatial visualization, transformation, and encoding difficulties.
- *Human-Computer Interaction (HCI)*: Designing interfaces that reduce cognitive load and enhance usability.
- *Gamification Theory*: Embedding motivation and formative assessment through game mechanics.
- *AI in Education*: Integrating ChatGPT and Whisper to provide context-aware, natural-language tutoring.
- *Systems Thinking*: Structuring the app as an interconnected ecosystem where modules (drawing, AI, game) inform and support one another.
- *Diagnostic Assessment*: Introducing a geometry-specific cognitive profiling model inspired by MBTI-style categorization, using gameplay data to identify learning difficulties.

From a practical standpoint, the system offers an implementable solution for modern classrooms. It is low-friction, platform-agnostic (Android and PC), and designed for both individual and collaborative learning. With its no-login model and modular

features, the application can be easily adopted by schools without requiring systemic infrastructure changes.

Ultimately, this project reflects the next step in educational innovation: transforming abstract math into something students can see, touch, experiment with, and enjoy - all while being supported by AI and modern game-based logic. It does not merely modernize delivery; it directly counters the core difficulties students face in spatial reasoning, as evidenced in *Section 1.1*

1.5. Structure of Report

This thesis is structured to guide the reader through the comprehensive development of our educational software, from foundational concepts to advanced implementation. Each chapter builds upon the last, providing a clear roadmap of our project's journey.

Chapter 1. Introduction: This chapter sets the stage by outlining the project's core idea, scope, and the current challenges in spatial geometry education, providing the fundamental rationale for our work.

Chapter 2. Background and Design Foundations: This chapter delves into the research and theoretical underpinnings that shaped our design, detailing pedagogical insights, cognitive challenges, and the strategic choices behind our system's architecture and educational impact.

Chapter 3. System Requirements: Translating our design principles into actionable specifications, this chapter defines the essential functional and non-functional requirements, key actors, and use cases that delineate the system's intended capabilities.

Chapter 4. System Analysis and Design: This chapter presents detailed technical architecture and component-level design, illustrating how each module of our interactive learning platform is engineered from a technical perspective.

CHAPTER 1: INTRODUCTION

Chapter 5. System Implementation: Moving from design to execution, this chapter outlines the development and deployment environment, technological integrations, and rigorous testing methodologies that brought our system to life.

Chapter 6. System Evolution: This pivotal chapter highlights the transformative innovations that distinguish our current system from its predecessor, showcasing how it revolutionizes the initial concept and fulfills its core ideals.

Chapter 7. Conclusion: The final chapter summarizes the project's achievements, acknowledges its strengths and limitations, and outlines a strategic vision for future enhancements, encapsulating our commitment to ongoing innovation in spatial geometry education.

In conclusion, this chapter has laid the groundwork for our project by articulating the pressing challenges in traditional spatial geometry education, validating these difficulties through student survey data, and clearly defining our project's objectives and scope. It establishes the critical need for an interactive, AI-powered, and gamified learning solution, setting the stage for the detailed design and implementation that follows

CHAPTER 2. BACKGROUND AND DESIGN FOUNDATIONS

Building upon the critical need identified in Chapter 1 for a more interactive and effective spatial geometry learning environment, this chapter delves into the comprehensive research and foundational design principles that underpin our system. It explores the technical landscape of existing educational tools, analyzes the cognitive challenges students face, and details the pedagogical theories and empirical insights from our student survey that directly informed every design decision, ensuring our solution is not merely technological but deeply rooted in educational effectiveness.

2.1. Theoretical and Cognitive Foundations

2.1.1. Spatial Geometry and Cognitive Barriers

Spatial geometry is the branch of mathematics concerned with the representation and reasoning of figures in three-dimensional space. Unlike planar geometry, it requires learners to visualize shapes from multiple perspectives, perform mental transformations, and comprehend complex spatial relationships. These tasks pose significant cognitive challenges, especially for high school students who are still developing these abstract reasoning abilities.

The *Van Hiele Model of Geometric Thought* is foundational in diagnosing geometry learning levels. It categorizes students' understanding of geometry into five hierarchical levels:

1. Visualization (Level 0) – Students identify shapes purely based on visual appearance without understanding their properties.
2. Analysis (Level 1) – Students recognize shapes based on their properties (e.g., sides, angles) but do not understand relationships between different shapes.
3. Informal Deduction (Level 2) – Students understand relationships between properties and can make logical arguments about these relationships.
4. Deduction (Level 3) – Students use formal logic to derive geometric proofs and theorems.

5. Rigor (Level 4) – Students understand geometry through axiomatic systems and can compare different geometrical frameworks.

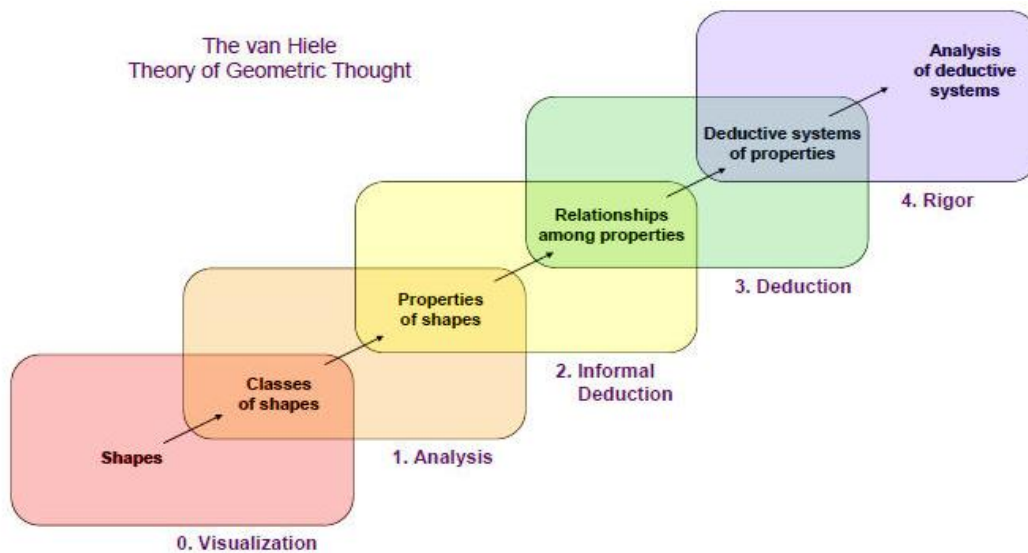


Figure 2.1 Van Hiele Model of Geometric Thought

While curricula and national assessments typically demand reasoning at Level 2 or 3, most students surveyed in Vietnamese high schools remain at Level 0 or 1. This gap leads to deep-rooted misunderstandings and low performance in spatial geometry.

From our survey of 92 Vietnamese highschool students:

- Only **23.9%** expressed confidence in performing 3D transformations.
- **39.6%** found traditional methods (e.g., blackboard diagrams) effective.
- Only **16.3%** felt regularly engaged by these methods.

These challenges manifest in three typical error types (*Rosnawati & Lukman, 2019*):

- Comprehension Errors: Misinterpreting diagrams or geometric terminology.
- Transformation Errors: Difficulty rotating, slicing, or transforming shapes mentally.
- Encoding Errors: Struggles with converting ideas into correct constructions.

2.1.2. Educational Theories Applied

Our system design draws from five foundational educational theories:

Theory	Role in Design
Van Hiele Model	Provides structure for student progression from Level 0 - 1 to Level 2 - 3 through scaffolded tasks and feedback.
Cognitive Load Theory	Informs interface design to reduce extraneous cognitive load through clear feedback, simplified controls, and visual cues.
Spatial Intelligence Theory	Emphasizes tasks that enhance mental rotation, object manipulation, and spatial reasoning.
Gamification Theory	Driving motivation uses challenges, progression, and feedback to sustain student engagement.
System Thinking	Encourages design of interconnected learning tools that reinforce each other in a loop of interaction, diagnosis, and support.

Table 2.1 Role of foundational education theories

Each theory plays a specific role in system architecture, ensuring alignment between pedagogy and digital affordances.

2.2. Design Philosophy and System Thinking Integration

2.2.1. From Problem to Intervention: A Systems View

The challenges in spatial geometry are not isolated. Difficulty in visualization leads to incorrect construction, which reduces confidence and causes disengagement. Viewing the student journey as a *dynamic system*, we structured our solution around the ***Interact*** → ***Diagnose*** → ***Support cycle***.

- Interact: The student constructs geometry with feedback from AI.
- Diagnose: Mistakes are logged, categorized, and analyzed.

CHAPTER 2: BACKGROUND AND DESIGN FOUNDATIONS

- Support: Personalized suggestions are generated through AI and integrated game mechanics.

This design ensures that each module does not function in isolation but contributes to the learning ecosystem.

2.2.2. Feature Integration via Morphological Analysis

We applied morphological analysis to recombine effective features from existing systems:

Feature	Source	Purpose in System
Constraint Snapping	GeoGebra	Ensures precision and immediate visual feedback during drawing.
Peer Collaboration	Miro	Facilitates synchronous learning and peer-to-peer instruction.
Voice-Based AI Interaction	ChatGPT + Whisper	Allows students to query conceptual content naturally without typing.
Retry + Star Rewards	Kahoot	Reduces fear of failure and encourages iterative learning.
Prompt Interpretation	Custom-built	Converts natural-language questions into visual construction support.

Table 2.2 Feature Intergration and purpose

This modular yet cohesive design enhances both usability and pedagogical alignment.

2.3. System Architecture and Modules

2.3.1. Overview of Components

CHAPTER 2: BACKGROUND AND DESIGN FOUNDATIONS

Module	Description	Technology
3D Drawing Tool	Enables students to build and manipulate geometric shapes in real time with snapping and constraints.	Unity, UGUI, C#
Drawing Assistant	Accepts natural-language problem input, interprets context, suggests construction, and highlights possible errors.	ChatGPT API, ShapeData Interpreter
AI Tutor (Siri-style)	Offers conceptual support via voice and text, such as definitions, distinctions, and geometric facts - not direct problem solving.	ChatGPT, Whisper, Unity TTS
Gamification Game	Embeds quiz questions into gameplay; logs responses to identify student errors across Van Hiele levels.	Unity, Logger
Online Session	Allow teachers or peers to join collaborative geometry sessions in real-time.	Unity Netcode, Relay & Lobby

Table 2.3 Overview of Components

2.3.2. Learning Outcomes per Module

Module	Expected Learning Outcome
3D Drawing Tool	Develop procedural fluency and accurate construction ability.
Drawing Assistant	Help students interpret word problems into precise shapes, reducing cognitive load.
Gamification Game	Identify and categorize student misconceptions (comprehension, transformation, encoding).
AI Tutor	Reinforce understanding of definitions and core concepts in an interactive, non-threatening way.
Online Session	Increase motivation and collaboration through social presence and co-construction.

Table 2.4 Learning Outcomes per Module

2.4. Diagnostic Game Design: The Collect Coin Module

The *Collect Coin* mini game serves as a gamified diagnostic tool integrated into the learning loop of *Interact* → *Diagnose* → *Support*. It combines path-based coin collection with tagged geometry questions to evaluate spatial reasoning skills.

Each question is categorized by Van Hiele level and cognitive error domain (comprehension, transformation, encoding), allowing the game to function as a lightweight assessment. A forgiving scoring system, retry mechanics, and progression through levels enhance engagement while reducing math anxiety.

Player responses and interaction data are logged for further analysis, forming the basis of future AI-driven recommendations and personalized learning paths.

To understand how a seemingly simple coin-collection mechanic is transformed into a structured cognitive diagnostic tool - complete with tagged geometry questions, dual-layered difficulty, and gameplay-driven insight - readers are invited to explore *Appendix D: Game Design Document (Collect Coin Module)*, which details the full design rationale, question architecture, and educational integration strategies.

2.5 Conclusion

This chapter has provided a comprehensive foundation for understanding the rationale behind our system. Grounded in cognitive theory, learning psychology, and system thinking, the design builds an integrated solution tailored to elevate students from Van Hiele Level 0 - 1 to Level 2 - 3. Through modular tools that reinforce each other - drawing, AI support, gameplay, and collaboration - the system is designed not only to teach geometry, but to reshape how students experience and understand it.

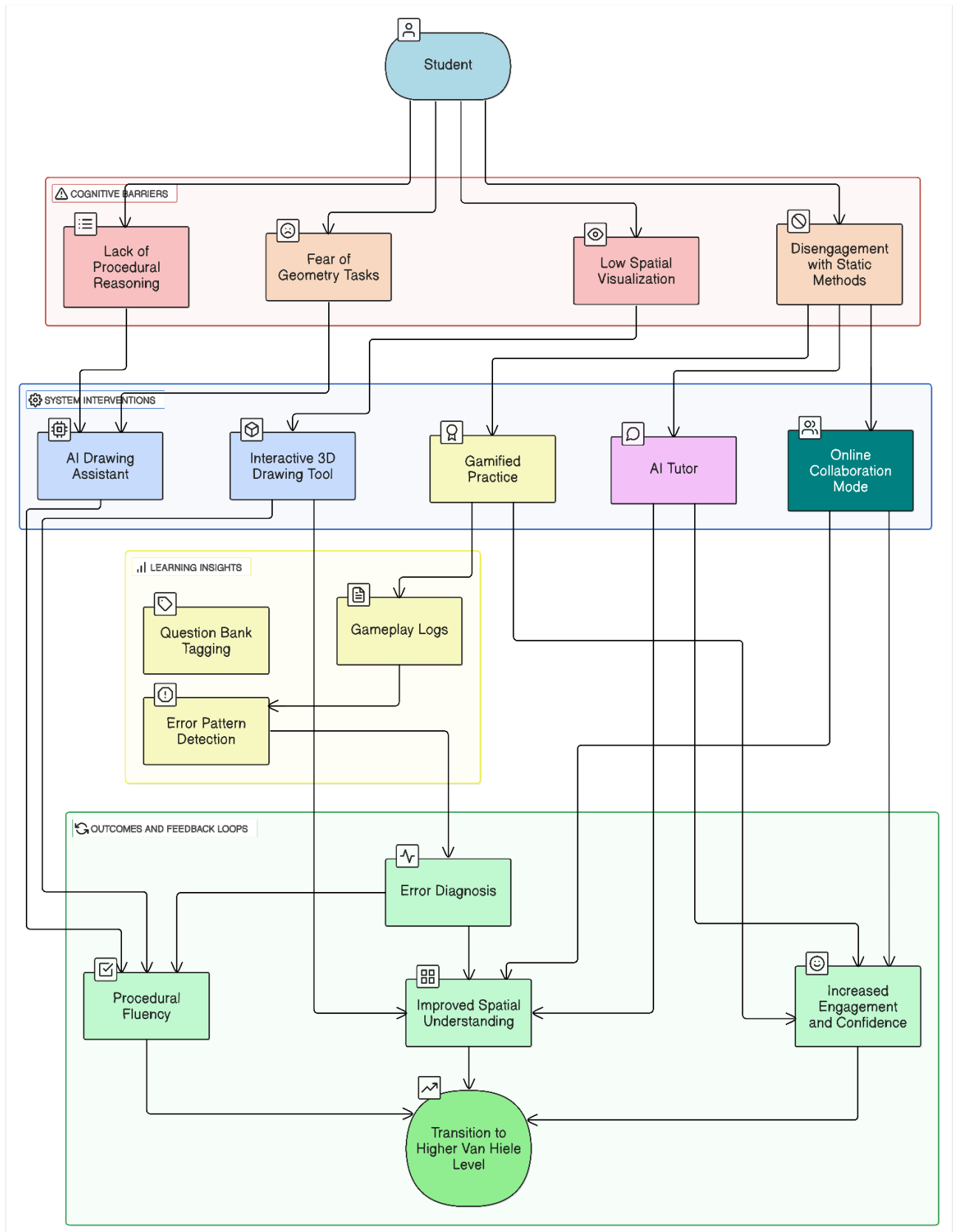


Figure 2.2 Conceptual Framework of System: From Barriers to Understanding

CHAPTER 2: BACKGROUND AND DESIGN FOUNDATIONS

This chapter has comprehensively explored the theoretical and empirical underpinnings of our system's design. By detailing the cognitive difficulties in spatial geometry, integrating insights from our student survey, and justifying our selection of technologies and pedagogical strategies - including gamification and UI/UX considerations - we have established a robust, research - informed foundation for developing an effective and engaging learning environment.

CHAPTER 3. SYSTEM REQUIREMENTS

With the pedagogical and technical foundations established in Chapter 2, this chapter translates our design philosophy into a precise set of system requirements. It outlines the essential functionalities and non-functional criteria that our software must satisfy to effectively address the identified learning challenges and achieve the project's objectives, setting the stage for the detailed system analysis and design.

3.1 User Requirements

Based on survey results, pedagogical analysis, and target user interviews, the following user requirements were identified for the application with:

- Learner: Any user engaging with the app to learn or practice spatial geometry.
- Host / Guest: Roles within online session.
- Analyst: Anyone using gameplay data to assess or improve understanding (e.g., AI, teacher, self-learner).
- All Users: Universal needs applicable to all interaction roles.

No.	Requirement	Role
UR1	Users should be able to draw and explore 3D geometric shapes interactively without needing prior training.	Learner
UR2	The system should allow users to collaboratively draw geometry in real time.	Host, Guest
UR3	Users should receive immediate feedback and guidance during gameplay and drawing.	Learner
UR4	The application should support voice and text interaction with an AI tutor for geometry-related questions.	Learner
UR5	The UI must be intuitive and suitable for both mobile and PC platforms.	All Users

CHAPTER 3: SYSTEM REQUIREMENTS

UR6	No login or account should be required to use core functionalities.	All Users
UR7	Game mechanics should be simple but informative, helping diagnose weaknesses in spatial understanding.	Learner, Analyst

Table 3.1 User requirement

3.2 System Requirements

This section defines the system - level requirements that translate user needs into technical terms. The system must enable seamless interaction, ensure learning efficacy, and maintain usability across diverse learning environments.

3.2.1. Functional Business Requirements

No.	Requirement
BR1	Provide an interactive drawing module for spatial geometry, supporting point placement, segment creation, and constraints.
BR2	Support multiplayer collaboration between users via online host/guest sessions.
BR3	Include a game - based module (“Collect Coin”) for engaging practice and low - stakes assessment.
BR4	Integrate an AI tutor capable of understanding geometry - related natural language queries and responding via text or speech.
BR5	Store game and interaction logs locally for potential assessment analytics.
BR6	Ensure accessibility on both Android and Windows platforms.

Table 3.2 Functional business requirements

3.2.2 Functional Requirements

CHAPTER 3: SYSTEM REQUIREMENTS

ID	Functional Requirement
FR1	The system allow users to draw and manipulate geometric objects in 3D space.
FR2	The system shall support snapping and constraint features during construction (e.g., perpendicular, equal length).
FR3	The system shall enable undo/redo operations for drawing actions.
FR4	The system shall provide a Collect Coin game where players answer geometry questions after slicing actions.
FR5	The system shall trigger multiple - choice questions based on level progression.
FR6	The system shall log correct/incorrect answers and attempt times.
FR7	The system shall allow a user to host or join a multiplayer drawing session using room codes.
FR8	The system shall synchronize drawing and constraint states in real - time between host and guest.
FR9	The system should provide an AI Tutor panel for students to input questions via text or speech.
FR10	The system shall return contextual explanations using the ChatGPT API.
FR11	The system should use text-to-speech to vocalize AI responses when required.
FR12	The system shall allow switching between public and private room visibility in drawing sessions.

Table 3.3. Functional Requirements

3.2.3 Non - Functional Requirements

ID	Non - Functional Requirement
NFR1	The application shall run smoothly on mid-range Android devices and standard Windows PCs.
NFR2	The system shall have a responsive interface, with loading times for any screen below 3 seconds.
NFR3	The UI shall use a pedagogically informed color palette (e.g., green, yellow) to support user focus and reduce anxiety.
NFR4	The app shall not require user login or authentication for normal operation.
NFR5	Multiplayer synchronization latency shall be under 200ms in standard network conditions.
NFR6	The AI Tutor shall respond within 2-4 seconds for standard prompts under normal internet conditions.
NFR7	All game scores and performance logs shall be stored locally without requiring cloud services.
NFR8	The interface shall support both touch and mouse input without requiring platform - specific training.

Table 3.4 Non - Functional Requirements

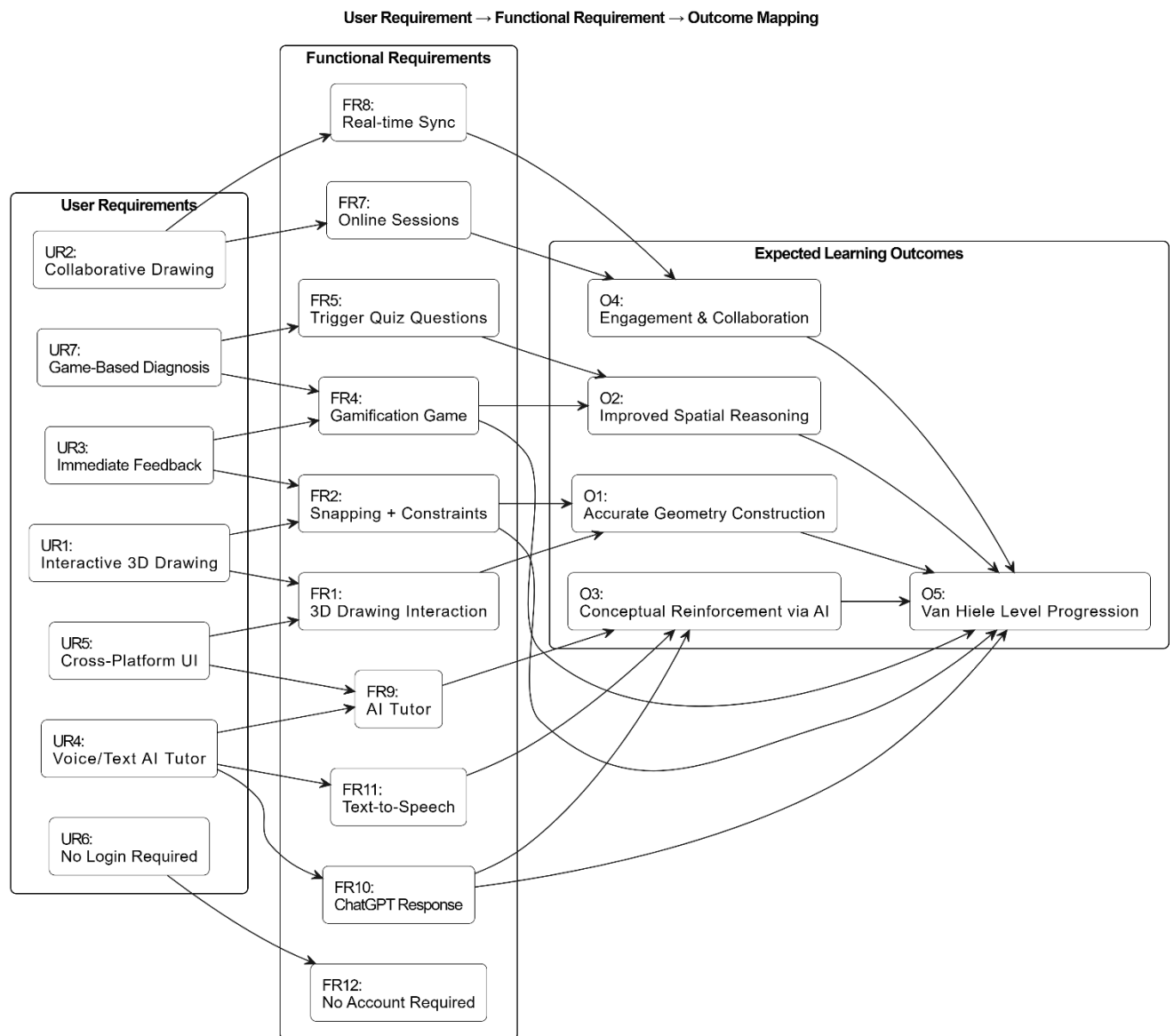


Figure 3.1 User Requirement to Functional Requirement to Outcome Mapping

3.3 Identify Actors and Use Cases

The system defines several distinct actors, each representing a specific role in how users interact with the application's functionalities:

- **Learner:** This is the primary actor of the system, representing any individual who accesses the app - including both students and teachers. Since the application does not require login or account creation, all general users are treated equally. They can draw geometric shapes, interact with the AI tutor, and participate in educational games.

CHAPTER 3: SYSTEM REQUIREMENTS

- Host (in Online session): A user who creates a online session for collaborative geometry drawing. The host can initiate the session, share access codes, and lead real-time interactions in the shared workspace.
- Guest (in Online session): A user who joins an existing online session. Guests collaborate in the drawing session and can contribute shapes and ideas but rely on the host to manage the session context.

These actors are not defined by user accounts or permissions, but by *contextual roles* within the application’s learning and gameplay flow. The following table outlines the specific actions available to each actor, based on their responsibilities and interactions within the system.

Actor	Action / Role
Learner (User)	– Launch application – Navigate modes (Draw, Game, AI, Online) – Open Geometry Drawing mode – Place a new point – Connect two points with a segment – Apply constraints (e.g., equal, perpendicular) – Use snapping to align shapes – Undo drawing action – Redo drawing action – Delete a shape or clear the board – Select tools (point, segment, polygon...) – Draw prebuilt shapes – Create shapes by dynamic attributes panel – View Built-in Drawing Assistant – Enter text-based question – Open AI Tutor panel – Use voice input to ask question – View AI’s response in text

	– Listen to AI’s speech response – Play ‘Collect Coin’ game – Slice character to correct coins – Avoid hitting obstacles – Answer quiz triggered after slicing – View quiz result/feedback – Restart drawing session – Replay a game
Host	– Create online session – Share room code – Draw collaboratively with other users – Broadcast drawing updates to guests – Control drawing flow (host - determined)
Guest	– Enter join code to access room – Draw collaboratively in host session – View shapes drawn by host in real-time – Receive host updates (geometry, constraints)

Table 3.5 Actor Roles and Corresponding Use Cases

3.4. Detail Description

3.4.1 Actor Description

Actor	Action / Role
Learner	– Launch the application (no login required) – Navigate through Home menu to select functions – Enter Geometry Drawing Mode – Place points in 3D space – Connect points to form line segments – Apply snapping between geometric elements

	– Use constraint logic (e.g., equal length, perpendicular) – Perform Undo actions – Perform Redo actions – Input question via text – Delete or reset shapes during drawing – Switch between drawing tools (point, segment, polygon, etc.) – Switch to public room mode – View room information – Copy room information – Access AI Tutor panel – Input question via voice (speech - to - text) – Receive AI explanation as text – Receive AI explanation via speech (text - to - speech) – Play ‘Collect coin’ game – Slice character to correct coins – Avoid hitting obstacles – Answer multiple - choice questions triggered – View game results (correct/wrong, score) – Restart or replay a game session – Reset drawing canvas to begin a new construction
Host	– Create a multiplayer room – Share room code with guest – Draw shapes in shared canvas
Guest	– Join a multiplayer room via code – Draw collaboratively in shared canvas – View updates in real-time from host actions

Table 3.6 Detailed Action Mapping by Actor Type

CHAPTER 3: SYSTEM REQUIREMENTS

3.4.2 Functional Description

The following table outlines *a representative set of core functionalities* in the application. While the system includes a broader range of interface-level features (e.g., sub-tools, prebuilt shape shortcuts, mode transitions, etc.), only *key user-driven actions* are documented here to capture the learning flow from launch to collaborative usage. Full component-level implementation can be found in *Chapter 5*.

No	Function	Description
1	Launch Application	Opens the app without requiring login or registration.
2	Navigate Modes	Switch between Drawing Mode, Game Mode, and AI Tutor Mode from the Home screen.
3	Draw Geometry	Place points, draw segments, and construct shapes in 3D.
4	Undo/Redo	Reverse or re-apply recent actions while drawing.
5	Apply Constraints	Apply geometric rules (equal length, perpendicular, etc.) and use snapping logic.
6	Use AI Tutor	Ask geometry questions via voice or text and receive AI - generated answers.
7	Play ‘Collect Coin’ Game	Slice designated 3D shapes while avoiding incorrect ones in a reaction - based mini - game.
8	Answer Quiz Questions	Respond to multiple - choice questions that appear within games.

9	Replay or Reset	Restart the drawing session or game to try again or start fresh.
10	Create Multiplayer/Online Room	Host creates a collaborative drawing room and shares access code.
11	Join Multiplayer/Online Room	Guests join a shared room by entering the host's code.

Figure 3.2 Functional Description

3.5 Use Case Diagram

Due to the modular nature of the platform and its numerous UI elements, this section provides a *high-level abstraction* of user interaction logic through a set of Use Case diagrams. These diagrams are designed to illustrate *conceptual user goals and flows* across key modules (Drawing, Game, AI, Online), not to depict every UI control or state transition.

3.5.1 All Use Case Diagrams

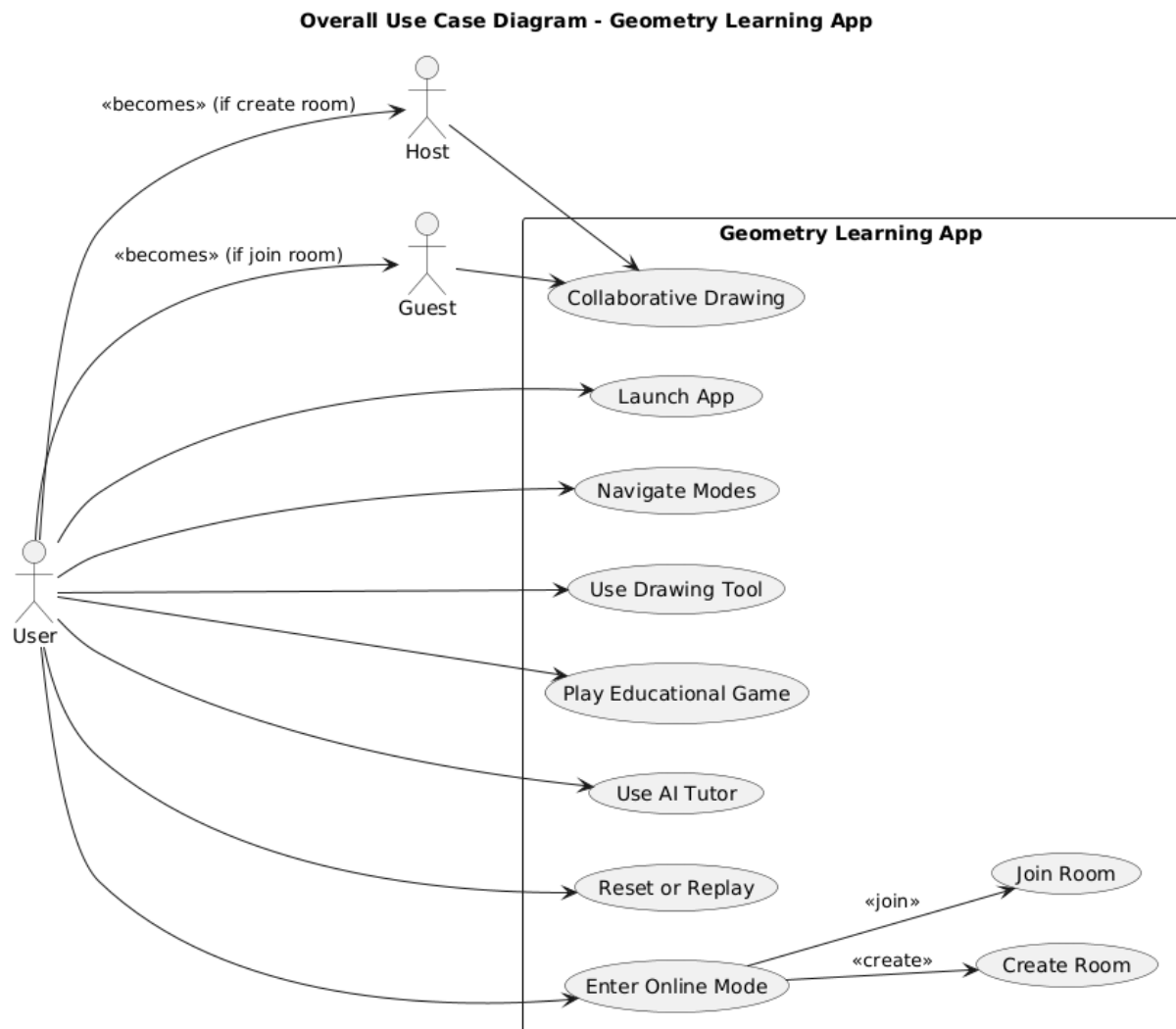


Figure 3.3.3 Use case diagram for overall application

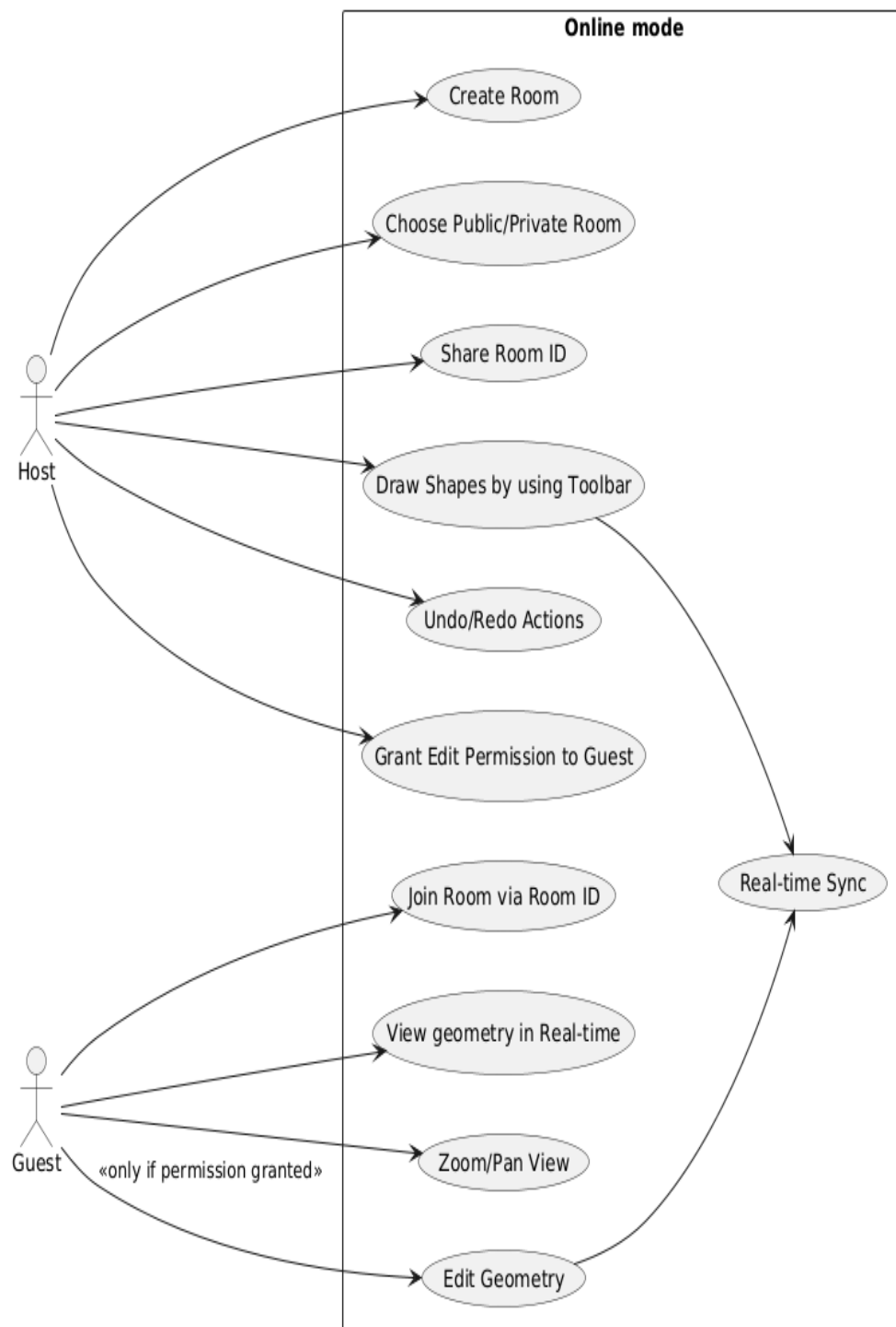


Figure 3.4 Usecase diagram for Online mode

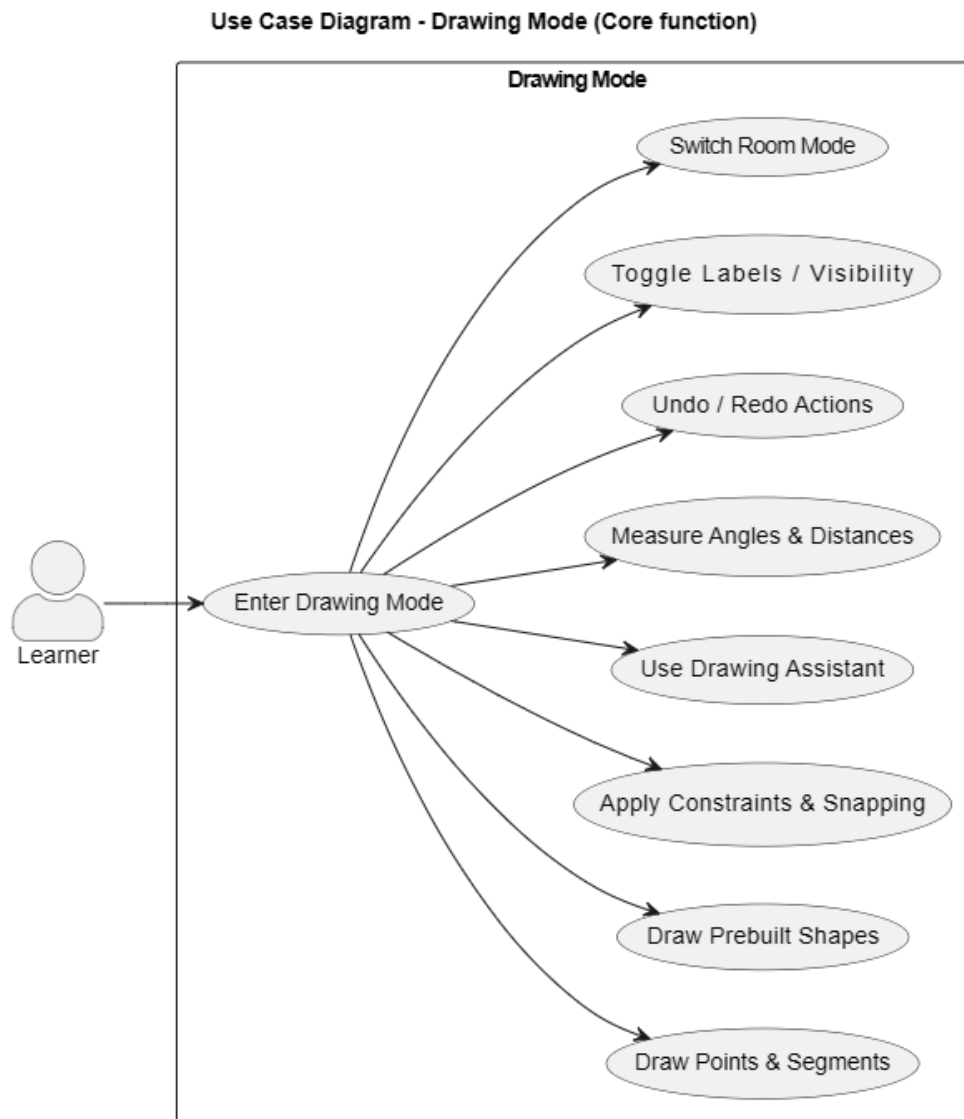


Figure 3.5 Usecase diagram for drawing tool

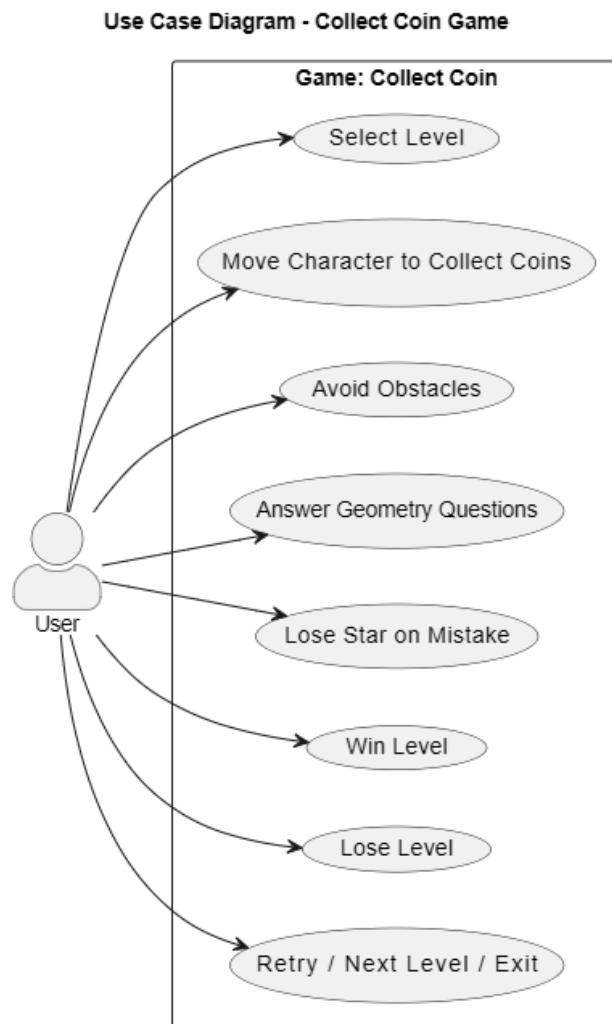


Figure 3.6 Usecase diagram for Collect Coin game

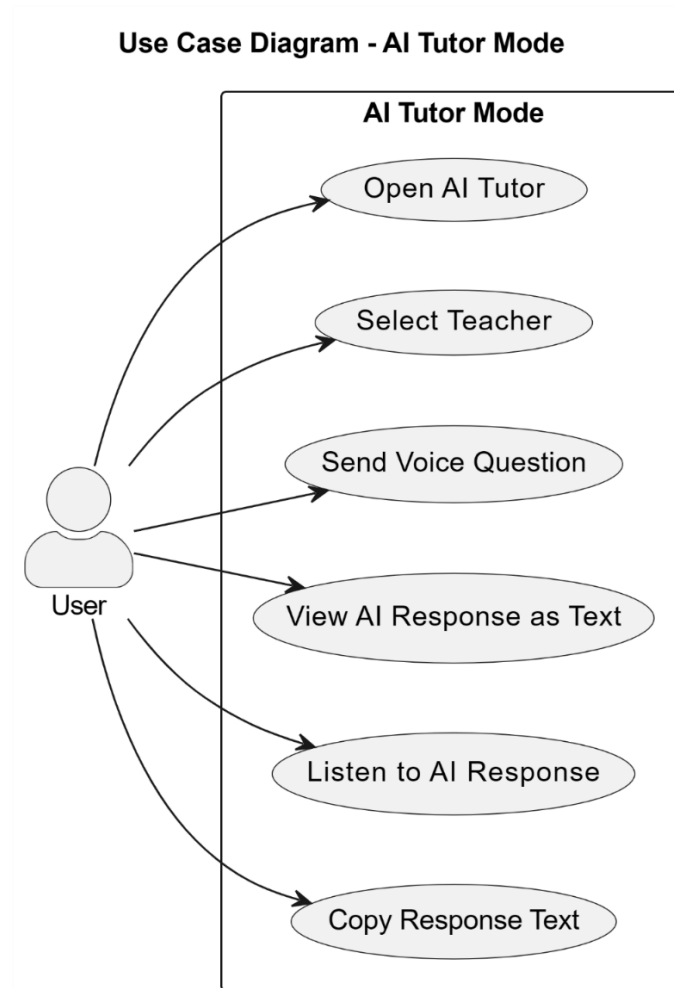


Figure 3.7 Usecase diagram for AI Tutor mode

3.5.2 Use Case Specification

Use Case: Launch Application	
Description	User starts the app to begin interacting with the system.
Actor	User
Pre - condition	Device is on and app is installed.
Steps	1. Tap app icon 2. App loads and displays onboarding screen 3. Tap the 'Start' button to navigate to the Home screen
Post - condition	User reaches the home menu

Table 3.7 Table Use Case: Launch Application

Use Case: Navigate Modes	
Description	User switches between Drawing, AI, and Game modes.
Actor	User
Pre - condition	App is launched
Steps	1. Select mode from Home menu 2. App loads selected mode
Post - condition	User is in the desired mode.

Table 3.8 Table Use Case: Navigate Modes

Use Case: Draw Geometry	
Description	User creates shapes using points and segments in 3D space.
Actor	User
Pre - condition	User is in Drawing Mode.
Steps	1.Select drawing tool (point, segment, polygon...). 2.Tap to place points in space. 3.Connect points with segments. 4.Form larger shapes like polygons or pyramids.
Post - condition	Geometric shapes are built.

Table 3.9 Table Use Case: Draw Geometry

Use Case: Undo/Redo Drawing	
Description	User reverses or re - applies actions while drawing.
Actor	User
Pre - condition	Drawing actions have occurred.
Steps	Tap Undo (or press Z) to revert an action. Tap Redo (or press Y) to reapply an undone action.
Post - condition	Drawing is updated accordingly.

Table 3.10 Table Use Case: Undo/Redo Drawing

Use Case: Apply Constraints and Snapping	
Description	User applies geometric constraints or enables snapping between elements.

Use Case: Apply Constraints and Snapping	
Actor	User
Pre - condition	At least two elements are selectable.
Steps	1.Select elements to constrain. 2.Choose a constraint type (e.g., equal length, perpendicular). 3. Enable snapping toggle (if needed).
Post - condition	Geometric logic is applied and visually updated.

Table 3.11 Table Use Case: Apply Constraints and Snapping

Use Case: Use AI Tutor	
Description	User asks a geometry question and receives an AI - generated response.
Actor	User
Pre - condition	Internet connection is available.
Steps	1. Access AI Tutor panel. 2. Enter questions by press ‘Talk’ button 3. Submit the question by press ‘Talk’ button again 4. View or listen to the AI’s response.
Post - condition	User receives an explanation relevant to their query.

Table 3.12 Table Use Case: Use AI Tutor

Use Case: Play ‘Collect Coin’ Game	
Description	User slices to the target coin while avoiding incorrect ones.
Actor	User
Pre - condition	Game is running.
Steps	1. Observe the target coins. 2. Slice game character only to the specified target coin. 3. Trigger a quiz if the coin is correctly sliced. 4. Collect all the coins shown while avoiding incorrect ones.
Post - condition	Score and feedback are shown.

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Table 3.13 Table Use Case: Play 'Collect Coin' Game

Use Case: Answer Quiz Questions	
Description	User responds to multiple choice questions.
Actor	User
Pre - condition	Question is triggered in - game.
Steps	1. Read the question and options. 2. Select an answer. 3. Receive the response.
Post - condition	System records the result and gives feedback.

Table 3.14 Table Use Case: Answer Quiz Questions

Use Case: Replay or Reset	
Description	User restarts a drawing or gameplay session from scratch.
Actor	User
Pre - condition	User is within an active session.
Steps	1. Tap "Reset" or "Replay" on screen 2. Confirm restart if prompted.
Post - condition	Session restarts with a blank canvas or fresh game state.

Table 3.15 Table Use Case: Replay or Reset

Use Case: Create Online Room	
Description	Host starts a collaborative drawing session.
Actor	Host
Pre - condition	The Internet is available.
Steps	1. Select Online mode. 2. Tap "Create Room" button. 3. Input RoomID and password 4. Tap 'Create' button 5. Enter online room 6. Share join code to join room.
Post - condition:	Room is active and waiting for participants.

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Table 3.16 Table Use Case: Create Online Room

Use Case: Join Multiplayer Room	
Description	Guests enter a join code to participate in a collaborative session.
Actor	Guest
Pre - condition	Valid room code is received.
Steps	1. Tap “Join Room”. 2. Enter join code (RoomID and password) 3. Connect to session.
Post - condition	User joins session.

Table 3.17 Table Use Case: Join Multiplayer Room

Use Case: Draw Collaboratively	
Description	Host and Guest view and draw together in a synchronized shared canvas.
Actor	Host, Guest
Pre - condition	Both host and guests are connected to the same room.
Steps	1. Host draws or modifies a shape. 2. System syncs updates across devices in real time. 3. Guest can view, rotation and scale shape and draw (if has permission from host)
Post - condition:	Shared canvas updates for both.

Table 3.18 Table Use Case: Draw Collaboratively

To summarize, this chapter has meticulously defined the essential user and system requirements, translating our project’s vision into concrete functional and non - functional specifications. By identifying key actors and detailing their interactions through use cases, we have established a clear framework that guides the subsequent architectural design and ensures the developed system meets its intended purpose and user needs.

CHAPTER 4. SYSTEM ANALYSIS AND DESIGN

Having defined the comprehensive system requirements in Chapter 3, this chapter presents the architectural blueprint and detailed design of our interactive learning platform. It dissects the system into its core components, illustrating how each module—from the Geometry Engine and Collect Coin game to the AI Tutor System—is meticulously designed to fulfill functional demands, integrate seamlessly, and provide a robust foundation for implementation.

4.1 System Architecture

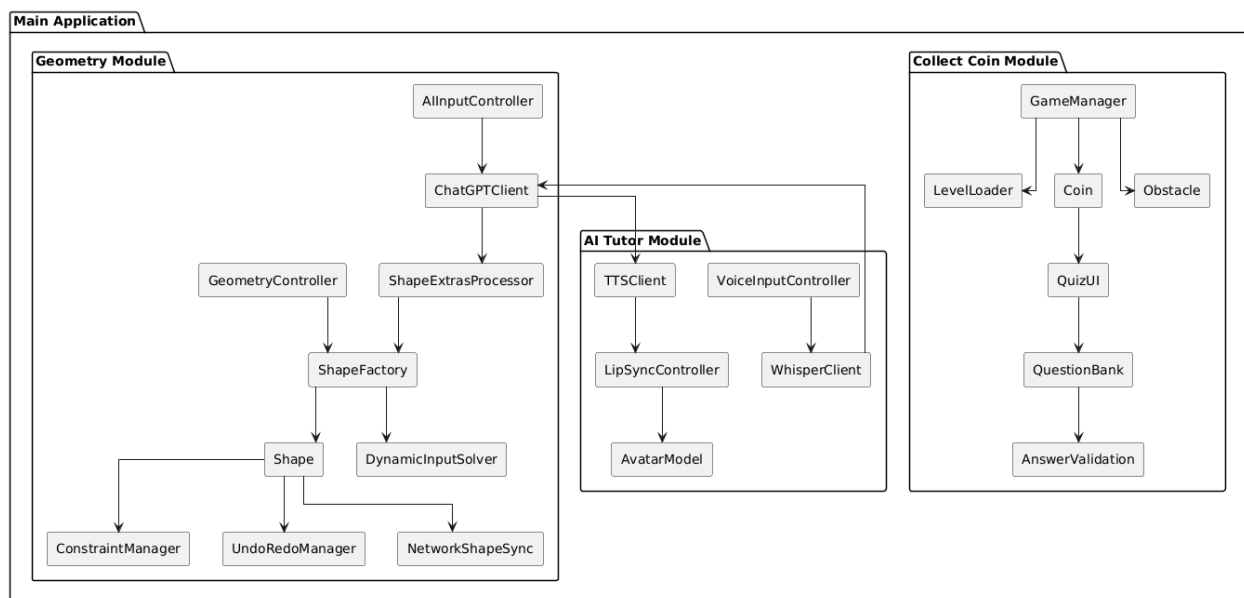


Figure 4.1 Main Application Diagram

4.1.1 Geometry Drawing Module: Interactive Geometry Construction System

The Geometry drawing module is a sophisticated interactive geometry construction tool. Its architecture is highly modular, breaking down the complex task of creating and manipulating geometric shapes into clear components. It supports drawing points, lines, triangles, and more advanced shapes, with features like measurement tools, constraints (parallel, perpendicular, etc.), an AI - assisted drawing helper, and even multiplayer collaboration. The design emphasizes extensibility - new shape types or tools can be added with minimal changes to existing code - and maintainability through well-defined responsibilities for each class.

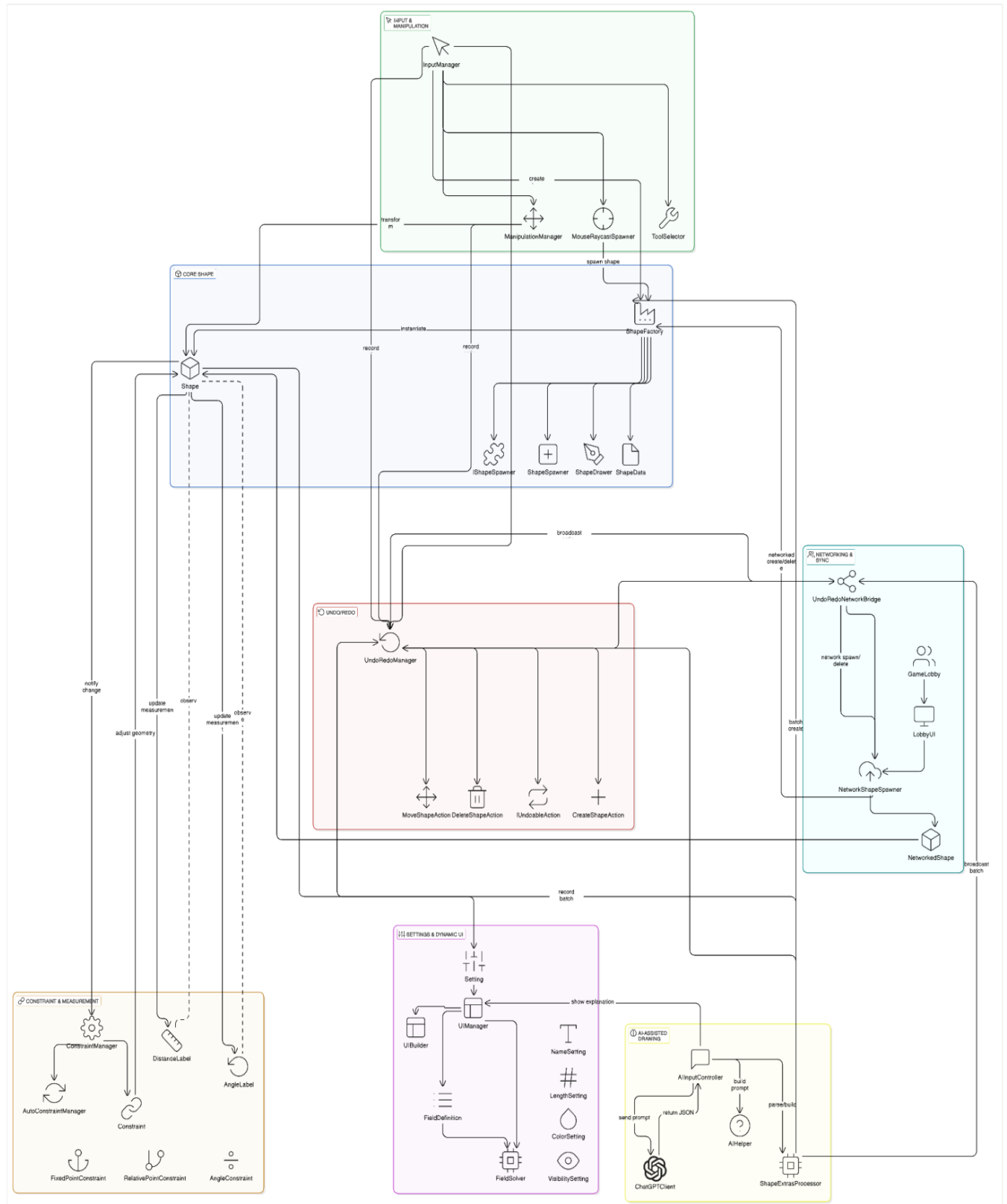


Figure 4.2 Architecture Diagram of Networking Geometric Drawing Modules

4.1.1.1. Key Components and Managers in Geometry

Component	Role & Design	Type / Classification
Shape Manager	Orchestrates all shape operations, connects UI/undo/network and maintains scene state	Manager / Singleton
Shape Factory	Provides a single interface to instantiate shapes by type using Factory Pattern	Factory / Creator
Shape Classes	Each shape (Point, Segment, Polygon, etc.) is a class holding its geometric data and basic logic	Data Class / Entity
Shape Interaction Behaviors	Modular, attachable components that add interactivity (select, drag, highlight) to shapes	Behavior / Component
Tool & Constraint Controllers	Separate classes for drawing tools and geometric constraints, using Command Pattern	Tool / Controller / Command
Undo/Redo System	Stack of action classes for reversible operations, using Command Pattern	Action System / Command Pattern
UI Managers & Dynamic UI Builder	Controls UI panels and generates input fields dynamically for different shapes/tools	UI Manager / Builder
Shape Buttons & Toolbars	UI buttons/scripts for tool/shape activation; ensures only one active tool at a time	UI Control / Toolbar

*Table 4.1 Geometry Module Key Components***4.1.1.2. Communication Flow and Design Patterns**

The Geometry drawing subsystem uses multiple *design patterns* to achieve a clean separation of concerns:

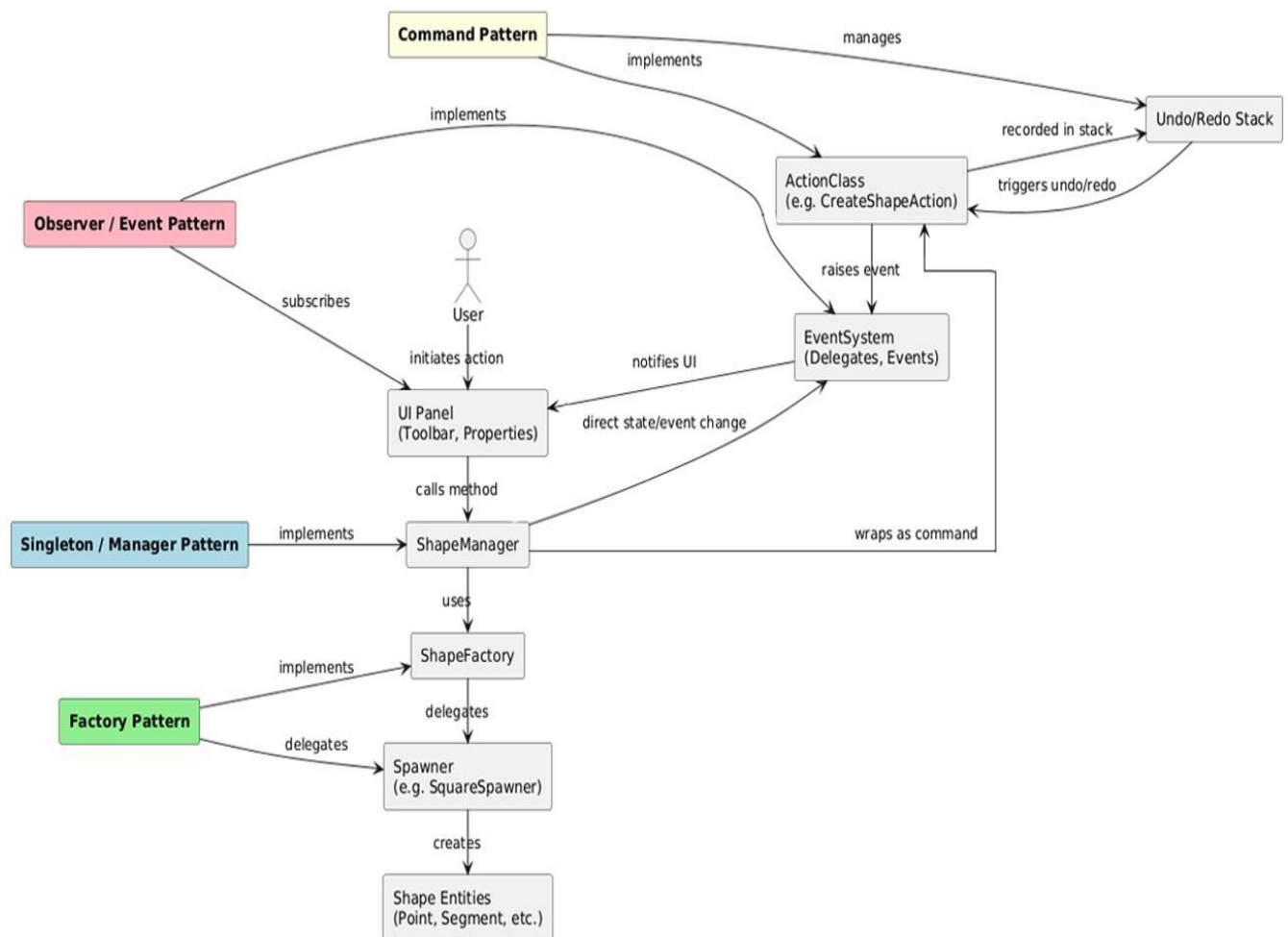


Figure 4.3 Communication Flow and Design Patterns

4.1.1.3. Advanced Features: AI Integration and Multiplayer Infrastructure

One standout aspect of the Geometry drawing module is how it *integrates AI assistance* and *collaborative multiplayer*, all within the same coherent architecture.

- AI Assistance:

The geometry system integrates an AI-driven module to support natural language-based shape construction. This subsystem utilizes a dedicated service class. Security best practices are observed by sourcing API keys from configuration files rather than hard-coding. When a user provides a textual description of a geometric figure, the service returns a structured result containing points, segments, or other elements, formatted as JSON. This output is then translated into standardized creation commands and

processed via the existing shape factory and action architecture within the core geometry system.

```
"CustomPoints": {  
  "A": { "type": "absolute", "position": [0, 0, 0] },  
  "B": { "type": "absolute", "position": [1, 0, 0] },  
  "C": { "type": "absolute", "position": [0.5, 0.866, 0] },  
  "D": { "type": "absolute", "position": [0.5, 0.2887, 0.8165] },  
  },  
  "M": { "type": "midpoint", "from": ["A", "B"] },  
  "N": { "type": "midpoint", "from": ["C", "D"] },  
  "I": { "type": "equidistant", "from": ["A", "B", "C", "D"] }  
}
```

Figure 4.4 Json template for ChatGPT to response

Thus, AI-generated shapes are fully compatible with core features such as *undo/redo*, *shape management*, and *state synchronization*. This design ensures modularity and extensibility: AI logic remains decoupled from core systems and can be enabled or replaced with minimal impact. Auxiliary classes assist with prompt preparation and post-processing, maintaining robustness and flexibility in the AI workflow.

- Multiplayer Collaboration:

The system also supports real-time collaborative editing through a multiplayer module that leverages Unity's Relay and Lobby services. A dedicated manager handles room creation, matchmaking, and session joining, integrating closely with the Unity Netcode for message routing and object synchronization. Once connected, all shape-related actions - such as creation, modification, or deletion - are funneled through the same centralized core logic as local actions.

High-level commands are broadcast to all clients via Remote Procedure Call (RPC) or network synchronization mechanisms, ensuring that each participant's state remains consistent. The architecture ensures that, regardless of origin, all geometric actions are handled uniformly by the core subsystems.

4.1.2. Collect coin Module: Arcade Gameplay with Quiz Integration

The Collect Coin module is a modular arcade-style game that seamlessly integrates real-time action mechanics with an educational Question & Answer (Q&A) system. The codebase is architecturally divided into two principal subsystems: the core gameplay (managing player interactions, obstacles, collectibles, and level progression) and a dedicated Q&A module that delivers quiz content at key points in the game flow.

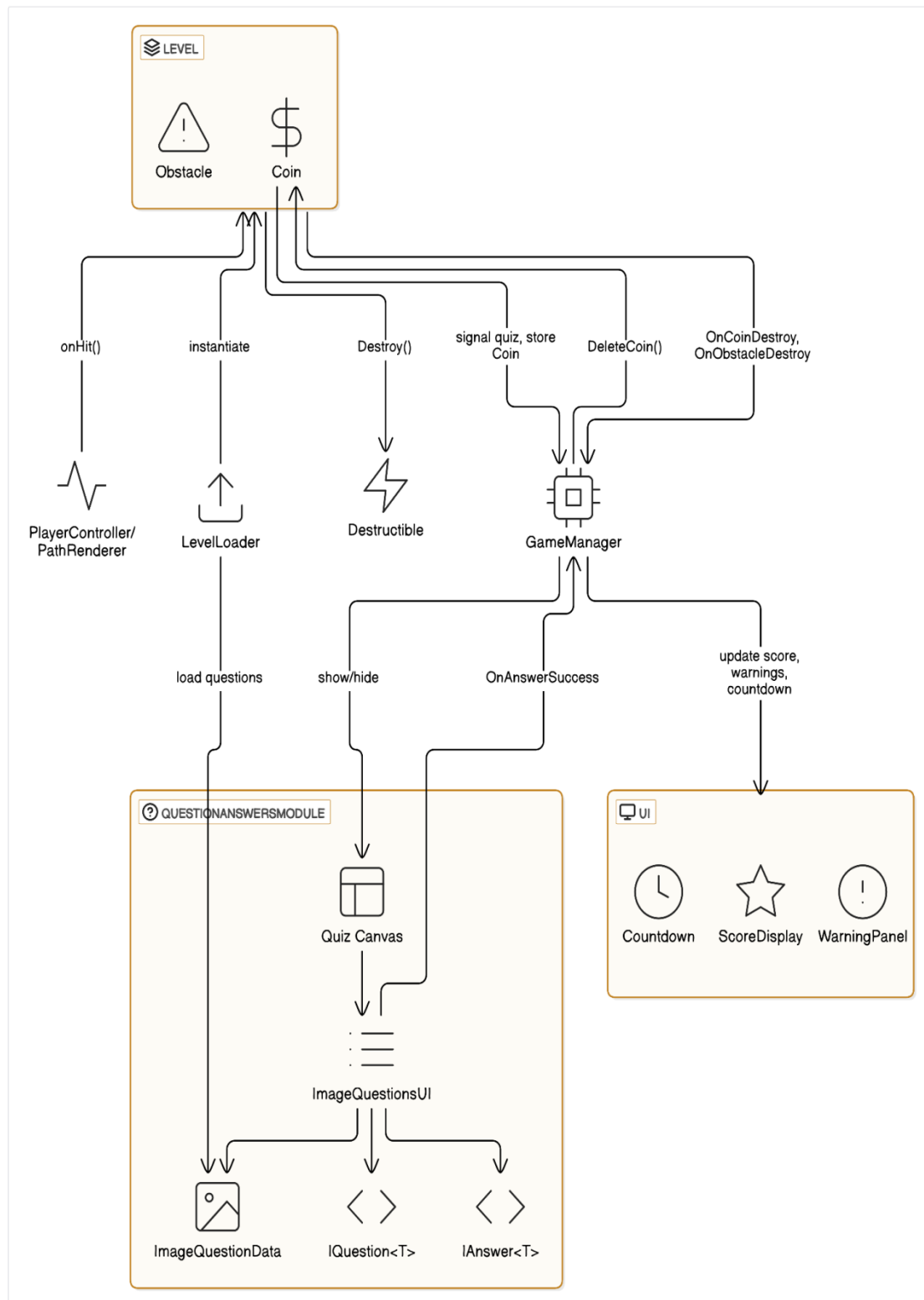


Figure 4.5 Architecture Diagram of Collect Coin Module

Key Components of the Collect Coin Game

Component	Role & Design	Type / Classification
GameManager	Orchestrates game state, flow, and win/lose logic	Manager / Singleton
LevelLoader	Loads levels and configures coins and obstacles	Loader / Factory
PlayerController	Handles player input and movement logic	Controller
Coin	Collectible object, increases score	Entity (inherits Destructible)
Obstacle	Hazard, penalizes player on collision	Entity (inherits Destructible)

Table 4.2 Collect Coin Module Key Component

4.1.2.1. Question - Answers Module Integration

One unique aspect of Collect Coin is the **QuestionAnswersModule**, which adds an educational quiz component to the game. This subsystem was designed to be **modular and reusable**, almost like a mini - framework for Q&A that could be embedded in any game.

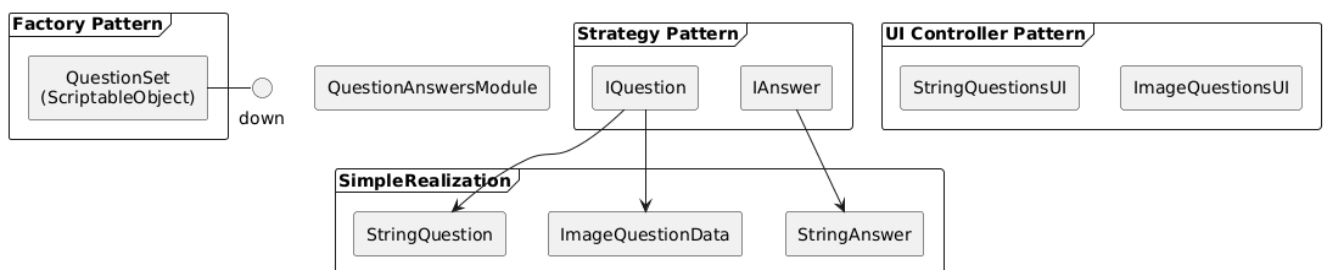


Figure 4.6 Question - Answers Modules structure

Component	Role & Design
QuestionAnswersModule	Modular Q&A system for embedding educational quizzes; loosely coupled, reusable, supports varied question/answer types.

IQuestion, IAnswer	Abstractions for questions and answers; support polymorphism and new formats (text, image, future extensions).
StringQuestion, StringAnswer, ImageQuestionData	Concrete implementations for basic Q&A types (text, image); provide out-of-the-box formats for quiz content.
QuestionSet (ScriptableObject)	Stores a set of questions and references to acceptable answers; enables data-driven content via Unity assets.
ImageQuestionsUI, StringQuestionsUI	UI handlers that display and process player responses for each Q&A format; interface with core module and update feedback.
GameManager Integration	Triggers quiz UI at relevant game moments, links Q&A session to gameplay (reward, gating, tracking used questions).

Table 4.3 Question - Answers components table

4.1.2.2. Communication Between Game and Quiz

Collect Coin structure effectively means it has two models: the real - time action model and the quiz model.

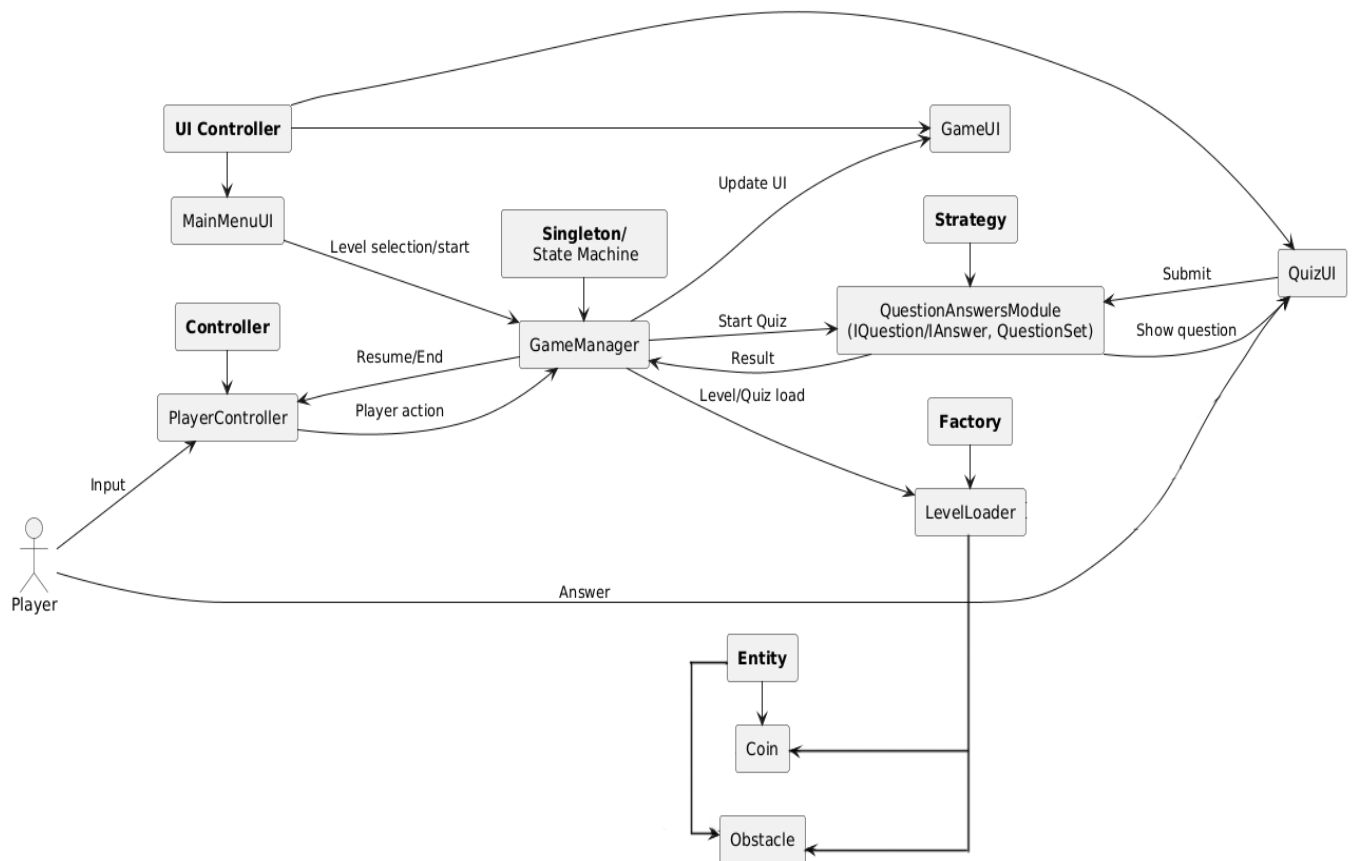


Figure 4.7 Flow between Collect Coin game and Question - Answers module

4.1.3. Tutor AI Module: Intelligent NPC Conversational Tutor

The Tutor AI system (referenced by the NPC-GPT scene and described in the project scope) is an ambitious feature that brings an interactive conversational character into the project. While tightly integrated with external AI services rather than being heavy on in-engine mechanics, its architecture is worth exploring. The goal of Tutor AI is to allow natural conversation between the user and a virtual character (an NPC), using *OpenAI's GPT - 4 (or similar) for dialogue*, *Whisper for speech - to - text*, and *ElevenLabs for text - to - speech*. Essentially, it's a pipeline connecting the user's spoken input to an intelligent response and back to audio output. Achieving this in a game context requires careful orchestration of asynchronous tasks and a design that keeps the user's experience smooth despite network calls.

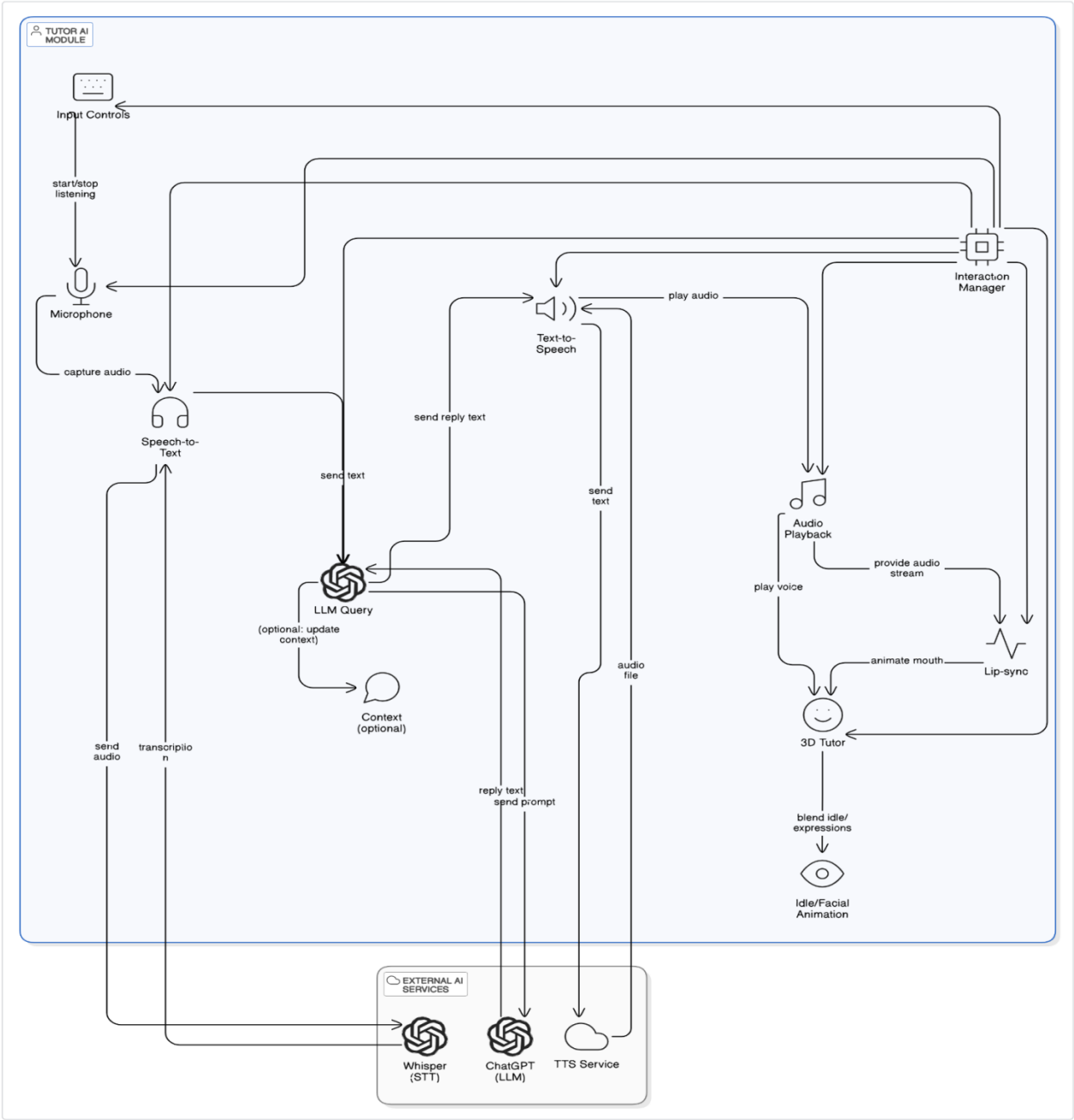


Figure 4.8 Architecture Diagram of AI Tutor Modules

4.1.3.1. Components

Component	Role / Responsibility	Pattern / Principle
SpeechInputController	Captures microphone input, manages audio recording, input UI	Single Responsibility, Controller

WhisperClient	Sends audio to speech-to-text (STT) API and returns transcribed text	Single Responsibility, Asynchronous
DialogueManager	Orchestrates conversation flow, tracks state, manages events/callbacks	State Machine, Facade, Observer
ChatGPTClient	Sends user input/context to GPT API, receives AI-generated reply	Single Responsibility, Asynchronous
TextToSpeechClient	Converts text reply to speech using TTS API, returns audio clip	Single Responsibility, Asynchronous
Controller	Plays reply audio, animates NPC, displays visual feedback	Single Responsibility, State Machine
UI	Displays transcript, prompt/feedback, error states	UI View, Observer/Event

Table 4.4 AI Tutor modules components table

4.1.3.2. Design Patterns and Workflow

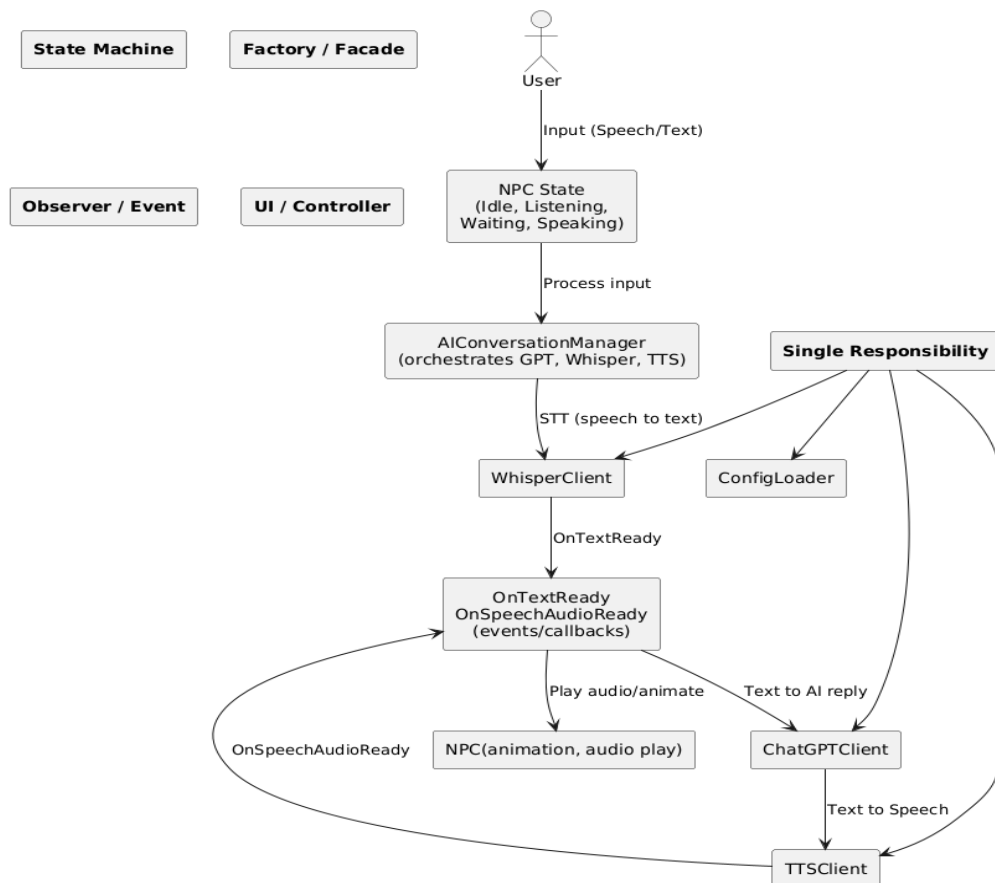


Figure 4.9 Design Pattern and Work Flow diagram of AI Tutor modules

4.2 Component Design

4.2.1. Component Diagrams

4.2.1.1 Main Architecture

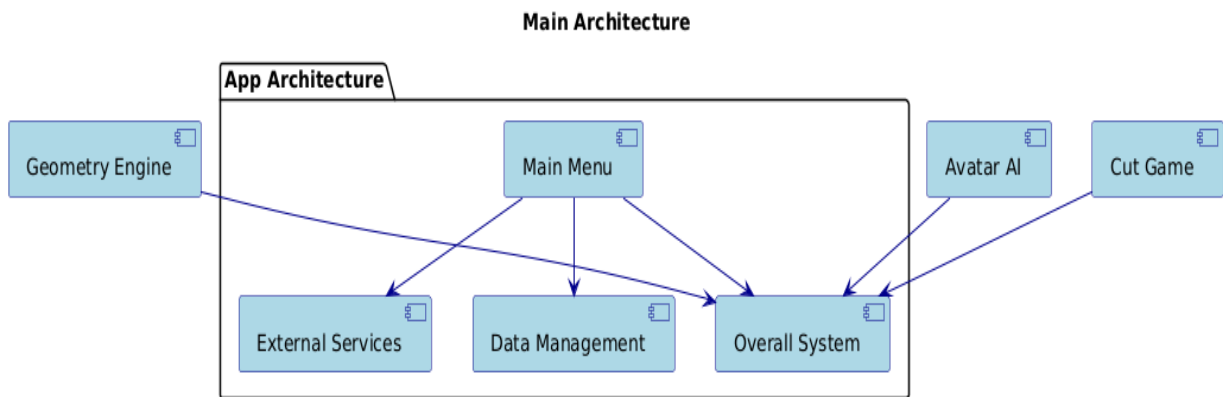


Figure 4.10 Main architecture component diagram

4.2.1.2 Geometry Engine

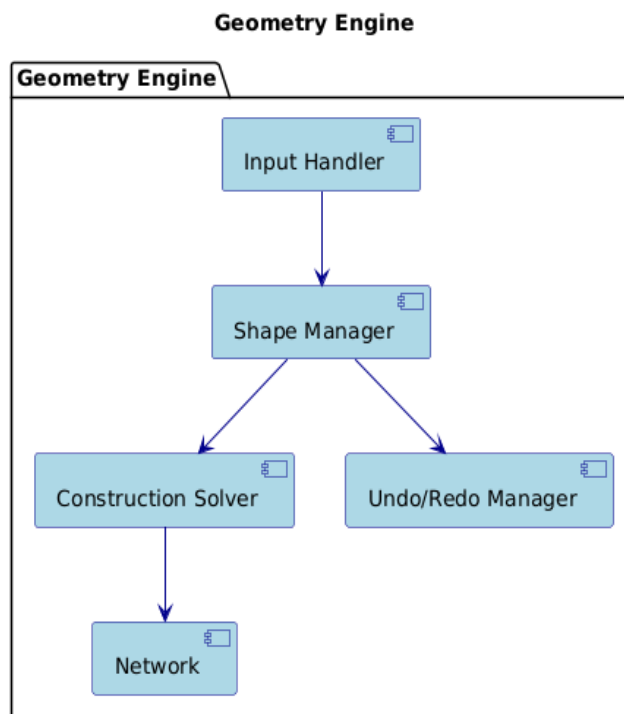


Figure 4.11 Geometry engine component diagram

4.2.1.3 Coin Collecting Game

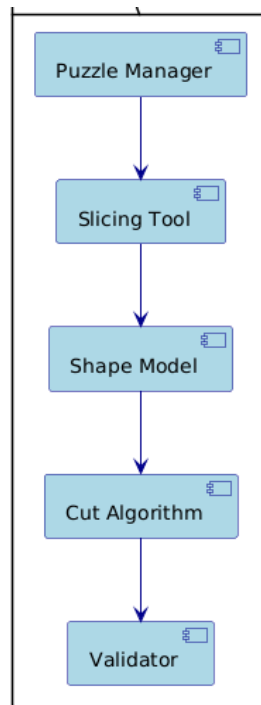


Figure 4.12 Coin collecting component diagram

4.2.1.4 Tutor AI Assistant

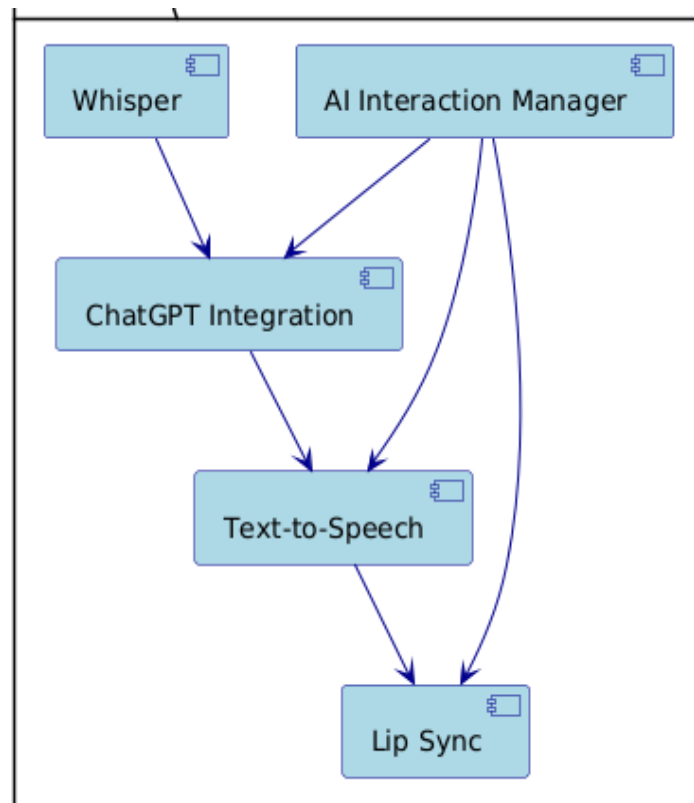
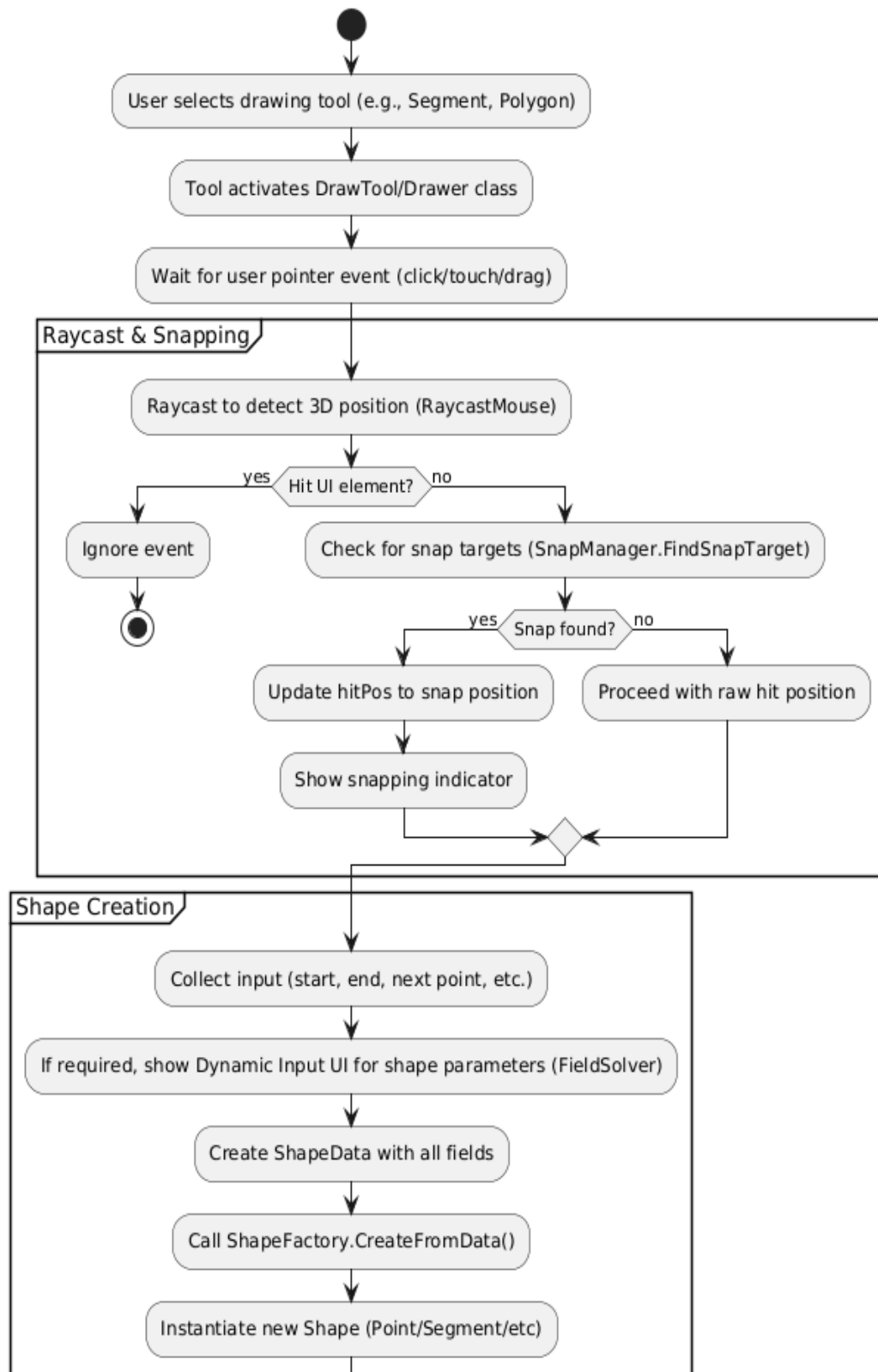


Figure 4.13 GPT AI assistant component diagram

4.2.2 Activity Diagrams

4.2.2.1. Geometry Drawing Activity Diagram



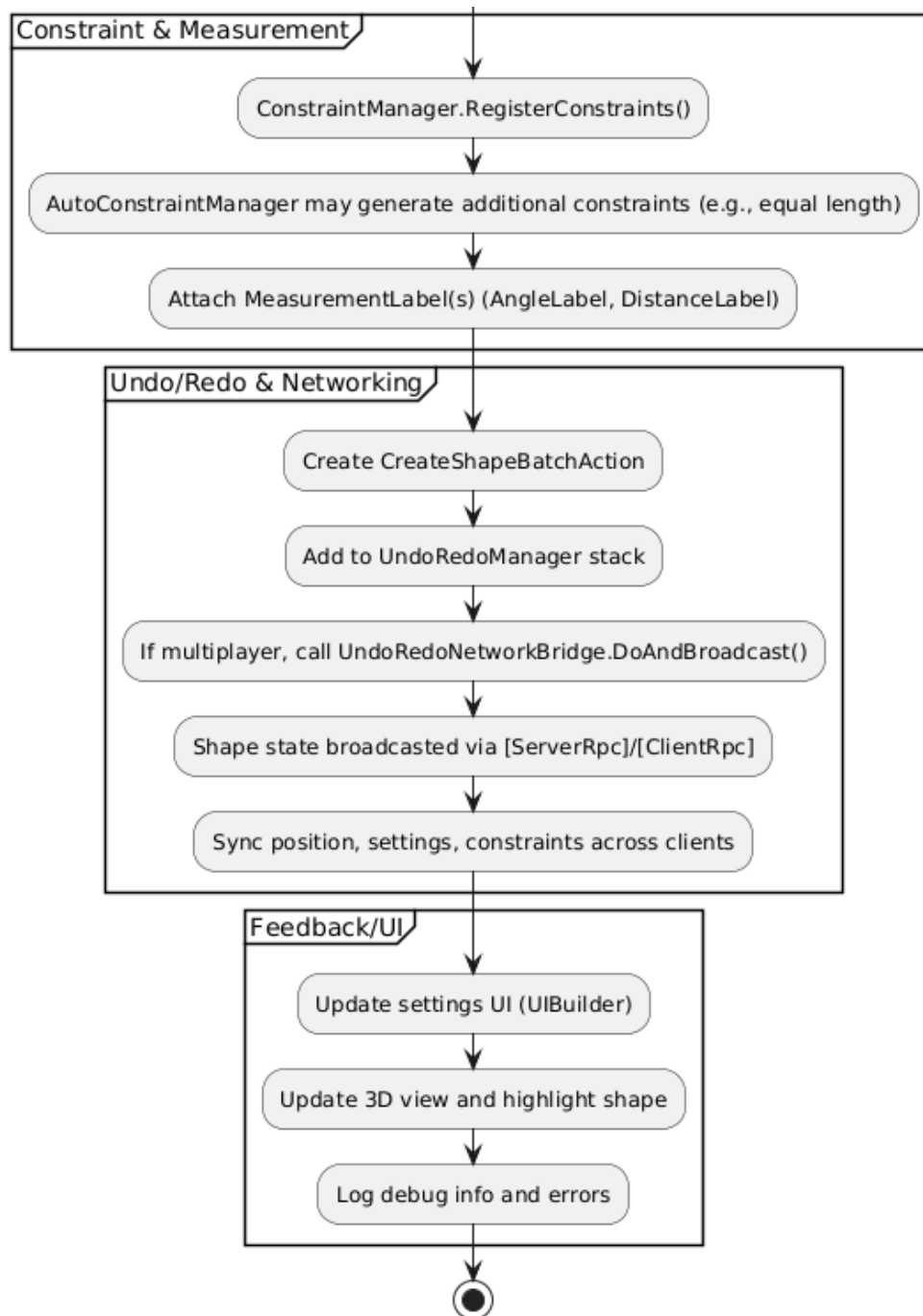


Figure 4.14 Geometry Drawing Activity Diagram

4.2.2.2. Geometry Constraint Solving Activity Diagram

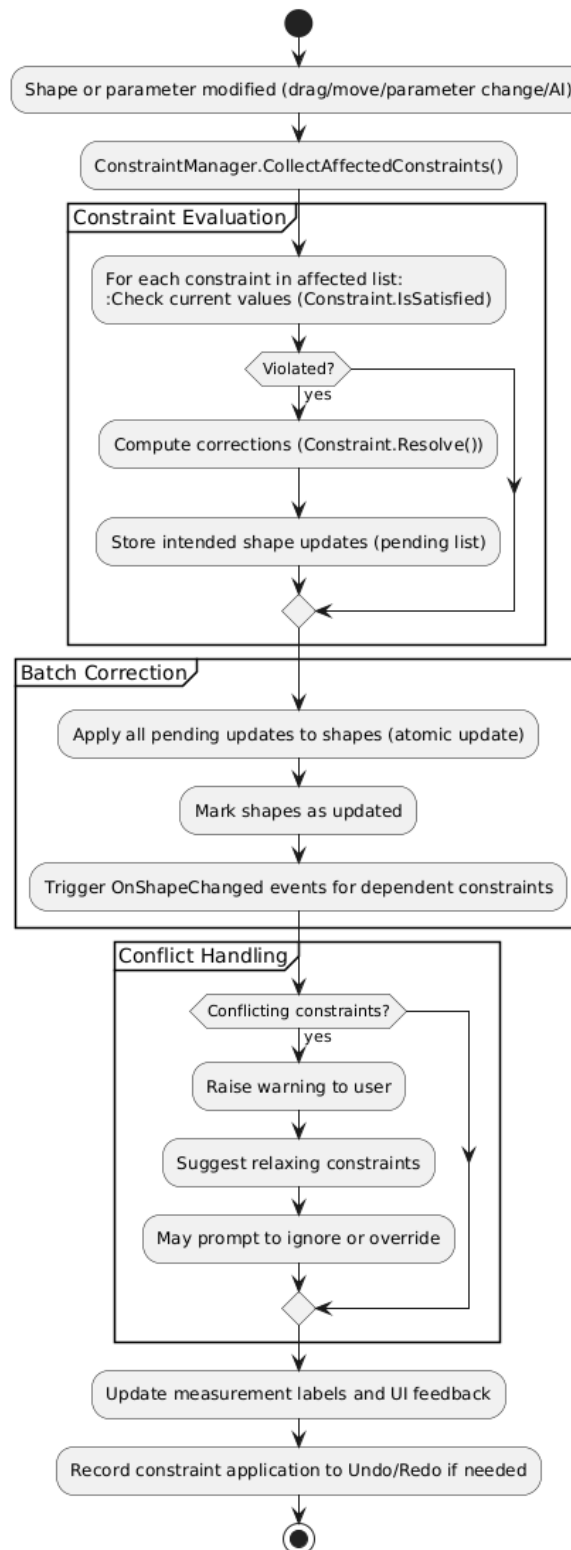


Figure 4.15 Geometry Constraint Solving Activity Diagram

4.2.2.3. Geometry Snapping Activity Diagram

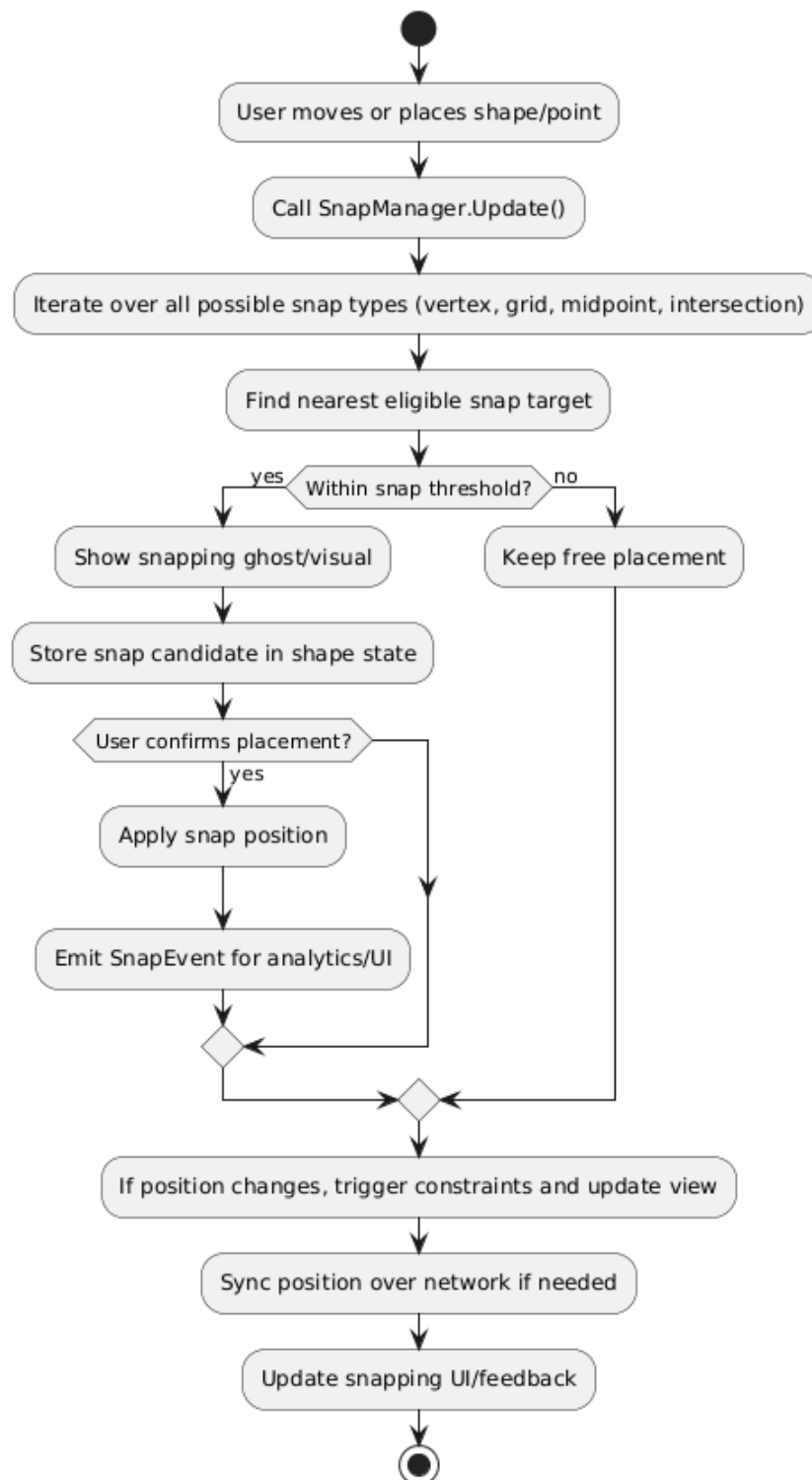
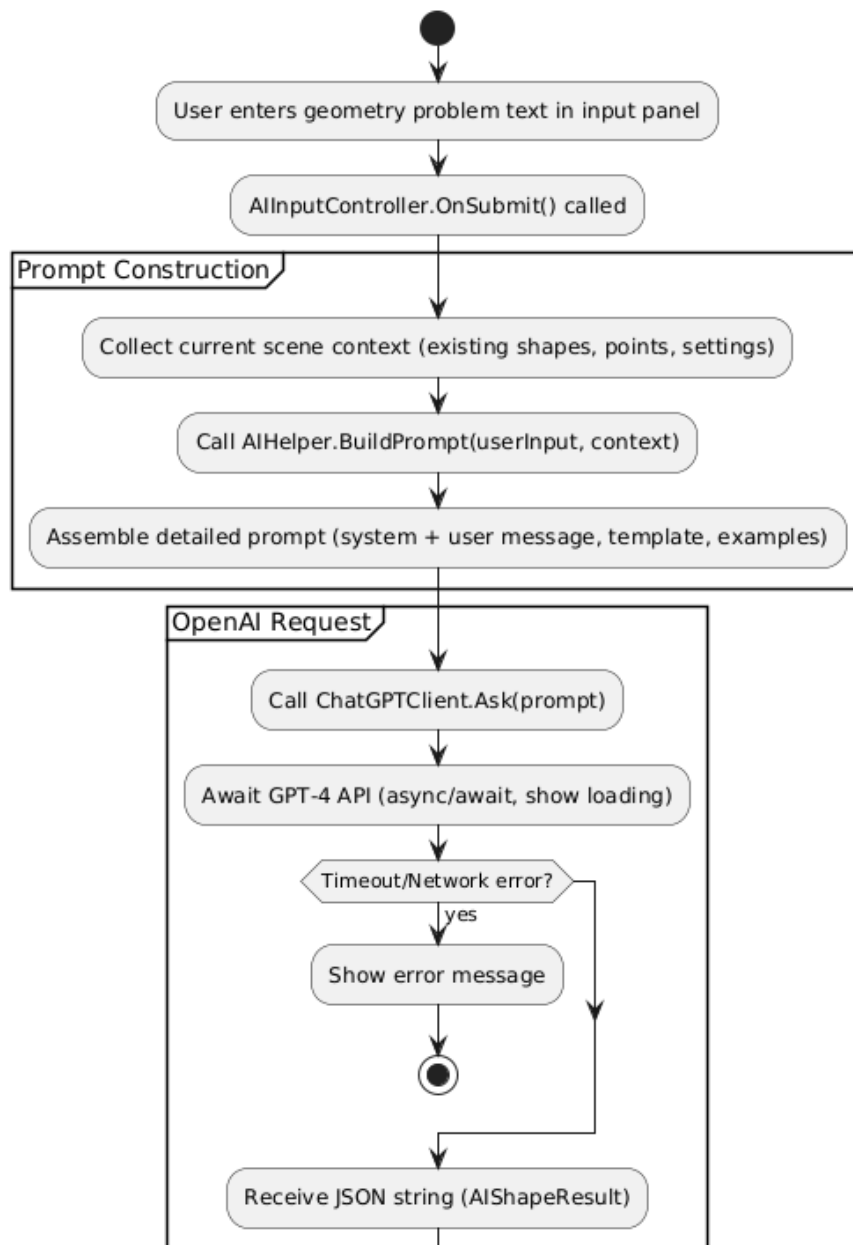


Figure 4.16 Geometry Snapping

4.2.2.4. AI - Assisted Input Activity Diagram



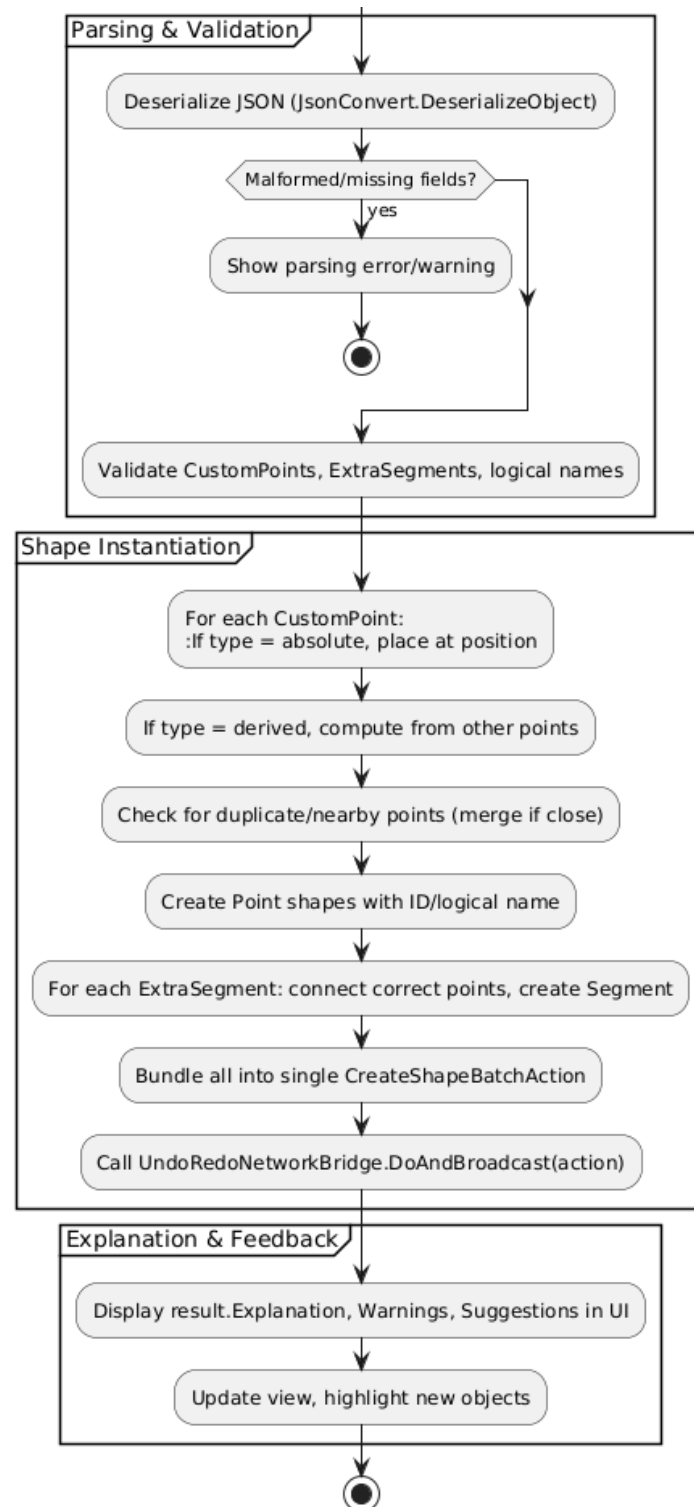
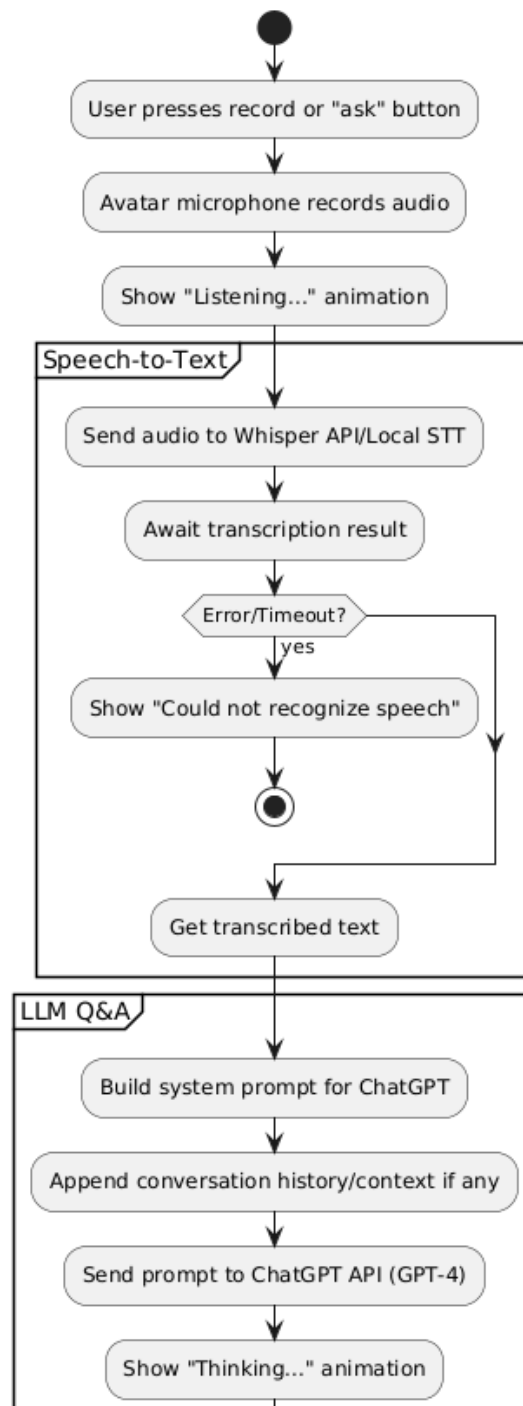


Figure 4.17 AI - Assisted Input Activity Diagram

4.2.2.5. AI - Assisted Input Activity Diagram



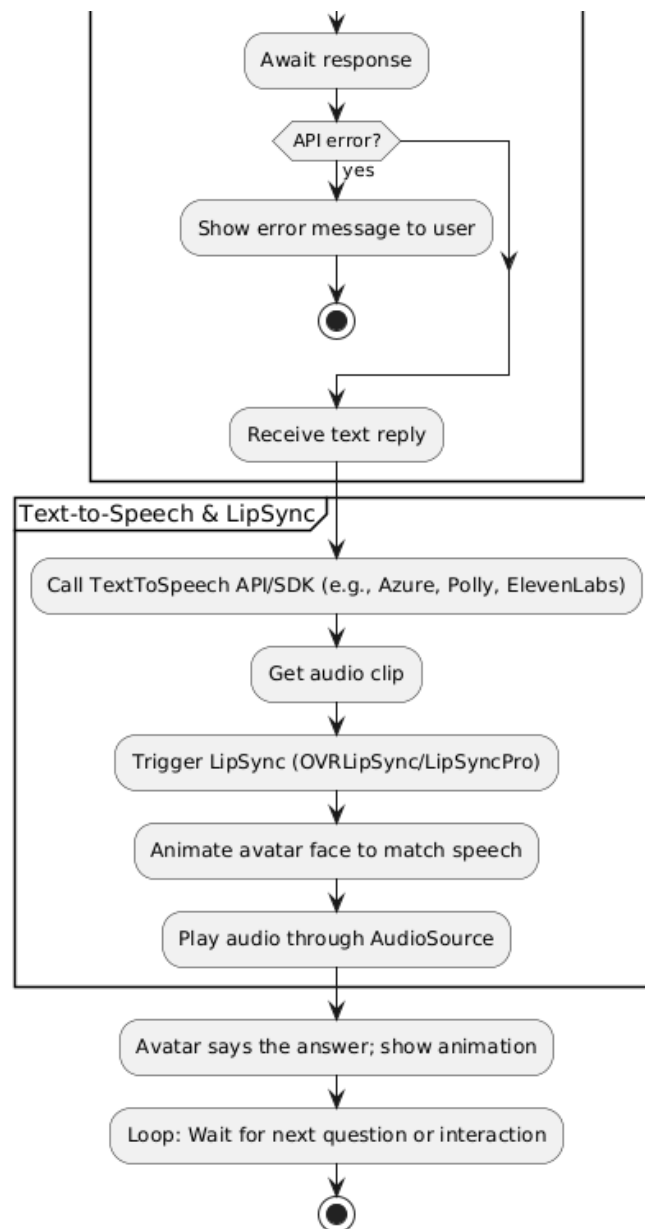


Figure 4.18 Tutoring Conversation Activity Diagram

4.2.2.6. Speech Recognition Activity Diagram

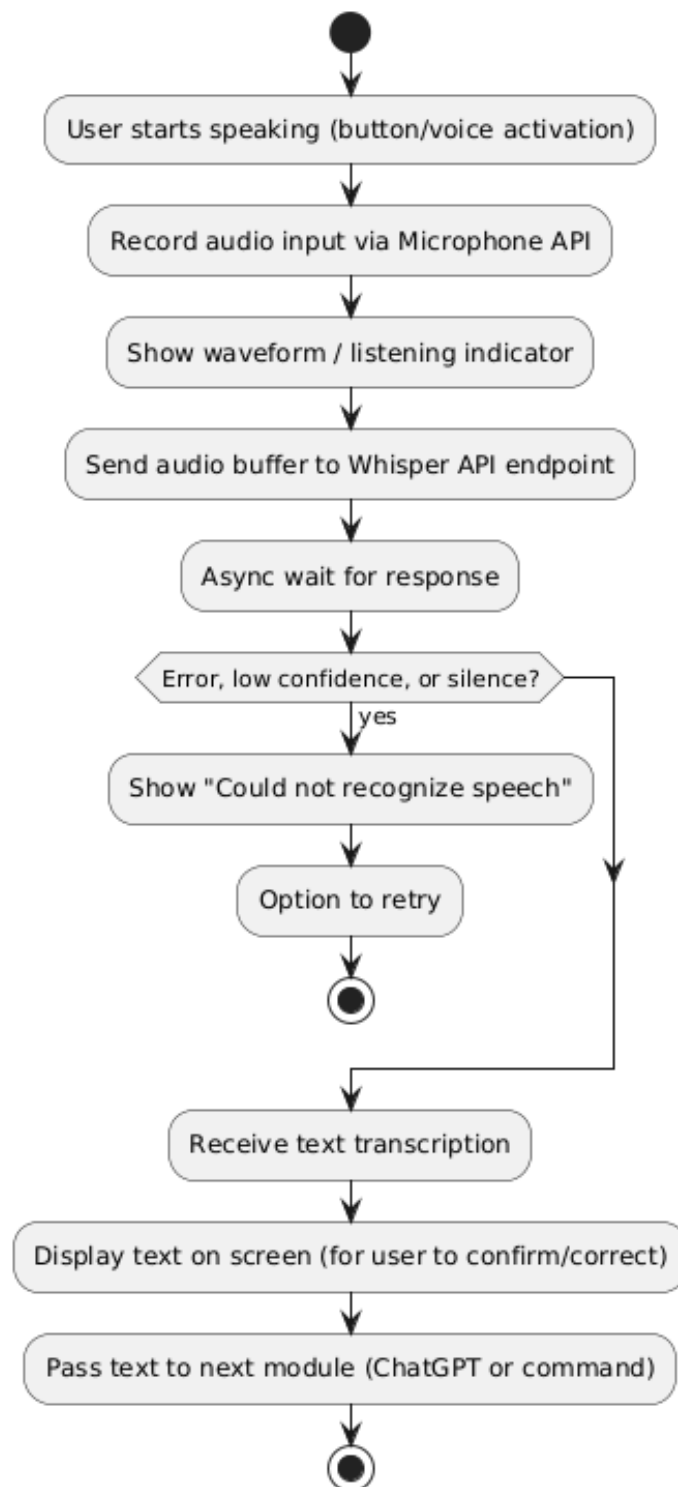


Figure 4.19 Speech Recognition Activity Diagram

4.2.2.7. Collect Coin Game Activity Diagram Activity Diagram

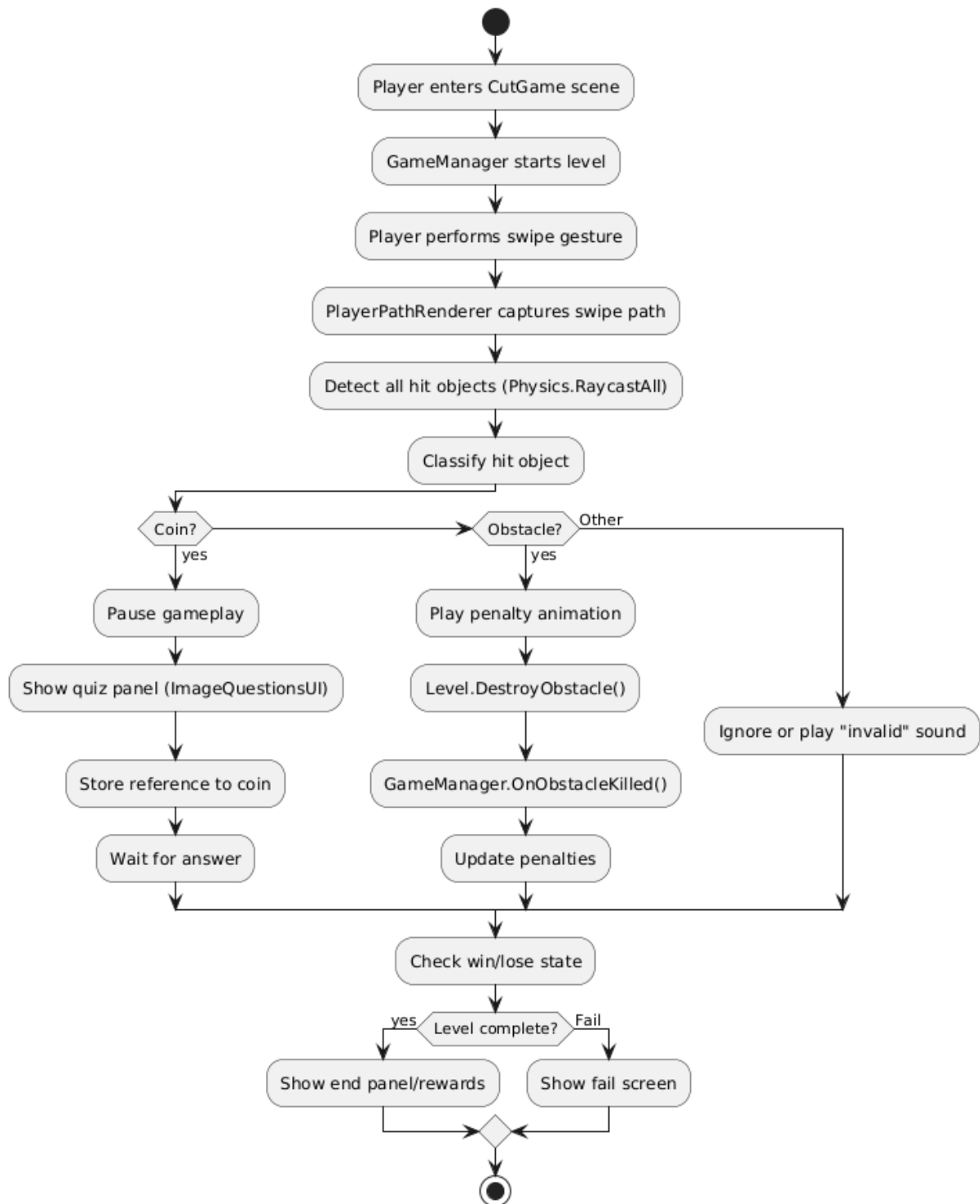


Figure 4.20 Collect Coin Game Activity Diagram

4.2.2.8. Collect Coin Answer Validation Activity Diagram

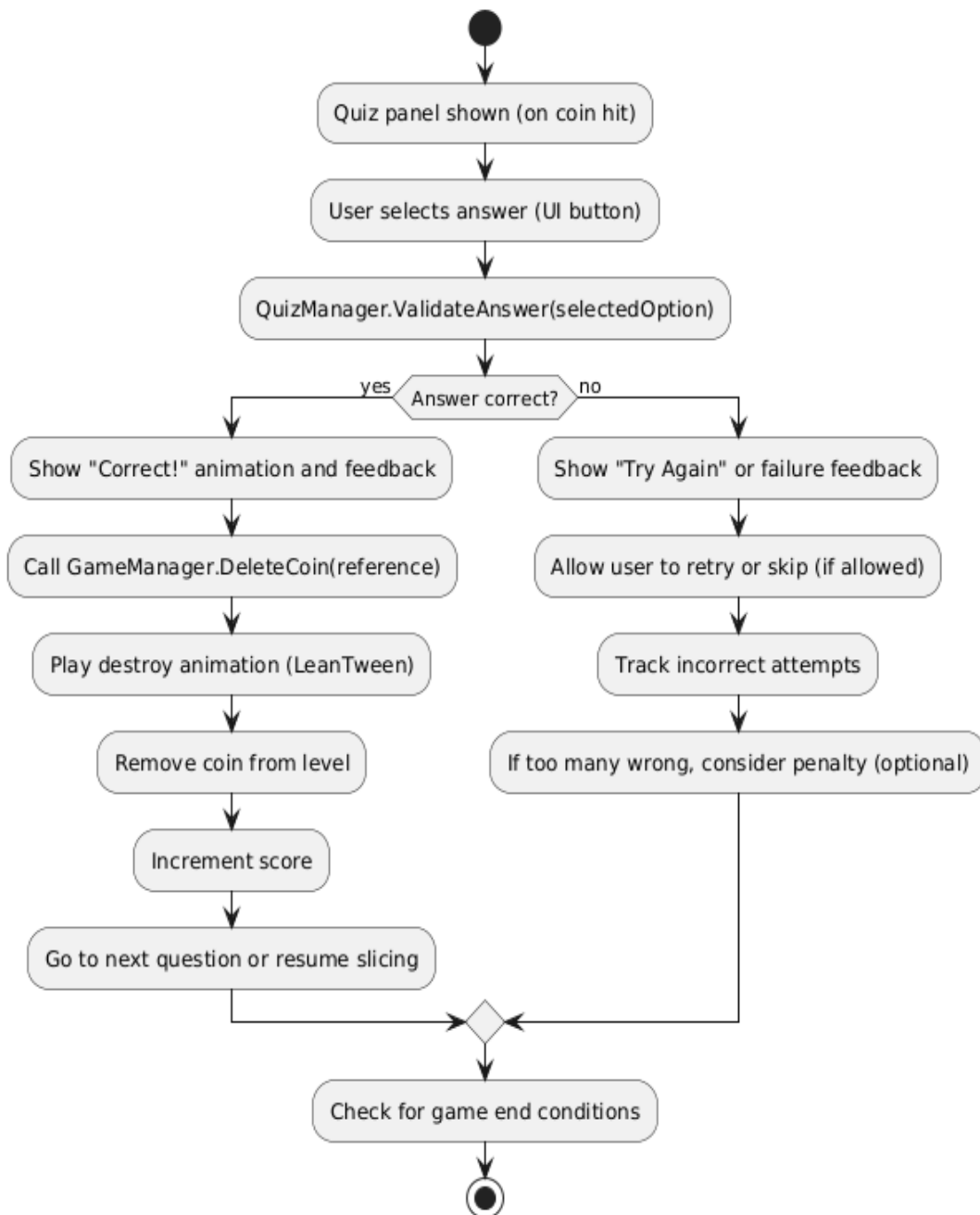


Figure 4.21 Collect Coin Answer Validation Activity Diagram

4.3 Data Storage

The project employs a unified, yet modularized data storage architecture designed to support diverse functional requirements across its three primary modules: Geometry, Collect Coin, and Tutor AI. Each module utilizes tailored data structures and storage

strategies that collectively ensure seamless operation, synchronization, and extensibility across the application.

4.3.1. Geometry Drawing Module

The Geometry drawing module handles the most complex data operations, owing to its real - time collaborative drawing and shape manipulation capabilities. It utilizes a combination of in-memory runtime data structures, JSON serialization, and synchronized network variables.

In - Memory Shape Storage

- All runtime shapes (e.g., Point, Segment, Polygon) are tracked by a centralized ShapeStorage system.
- Each shape contains properties such as ID, Type, Position, ConnectedPoints, Settings, and Constraints.
- These are stored in Dictionary<string, Shape> or List<Shape> collections that enable fast lookup, modification, and removal.

4.3.2. Collect Coin Module

The Collect Coin module handles gameplay - specific data tied to levels, objects (e.g., coins, obstacles), and quiz content.

Level and Object Storage

- Each level prefab includes serialized references to destructible objects.
- On runtime, LevelManager maintains dynamic collections of active Coin and Obstacle objects, along with their state (e.g., hit, destroyed, in quiz).

Quiz Data Handling

- Questions and answers are loaded from ScriptableObject assets (e.g., ImageQuestionData) or external files (e.g., JSON or XML for bulk import).

- Active quizzes are managed by ImageQuestionsUI, which tracks current questions, selected answers, and correctness state.

Game State Persistence

- Current progress (e.g., score, time left, number of correct/incorrect answers) is stored in GameManager's runtime variables.
- End - of - level data can optionally be exported for logging or analytics expansion.

4.3.3. Tutor AI Module

The Tutor AI system handles AI conversations, speech data, and response playback.

Prompt and Chat History

- Conversations are stored temporarily in memory using List<Message> (system/user/assistant roles).
- This supports context - aware prompts when needed.

AI Input/Output Storage

- Prompts and AI responses are processed through ChatGPTClient, which builds structured JSON requests and parses structured responses.
- The data received is also stored (optionally) in local logs or transcripts for analysis.

Audio and Lipsync Storage

- TTS responses from ElevenLabs or other providers are cached as AudioClip objects.
- LipSync uses a viseme buffer to store frame-by-frame mouth shape data for animation playback, often handled by third-party systems like OVR_LipSync.

4.3.4. Shared and Infrastructure - Level Storage

Network Communication

- All host/client synchronization is powered by Unity's Netcode, using NetworkVariable, ServerRpc, and ClientRpc to transmit ShapeData, ActionData, and collaboration events.
- The server maintains the source of truth and broadcasts diffs or full state as needed.

Undo/Redo Log

- Each user action is logged in an UndoRedoManager, which stores a stack of IUndoableAction objects.
- These are serializable and synchronized across the network via UndoRedoNetworkBridge.

4.4. Object-Oriented Modeling

4.4.1. Class Diagram

4.4.1.1. Geometry Drawing Module



Figure 4.22 Core Shape System

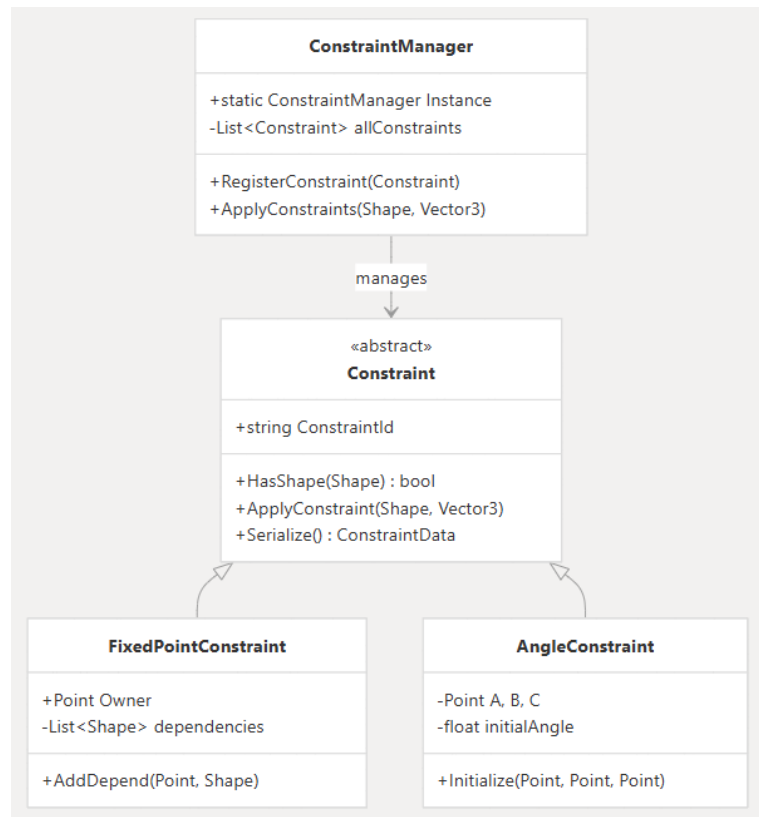


Figure 4.23 Constraint System

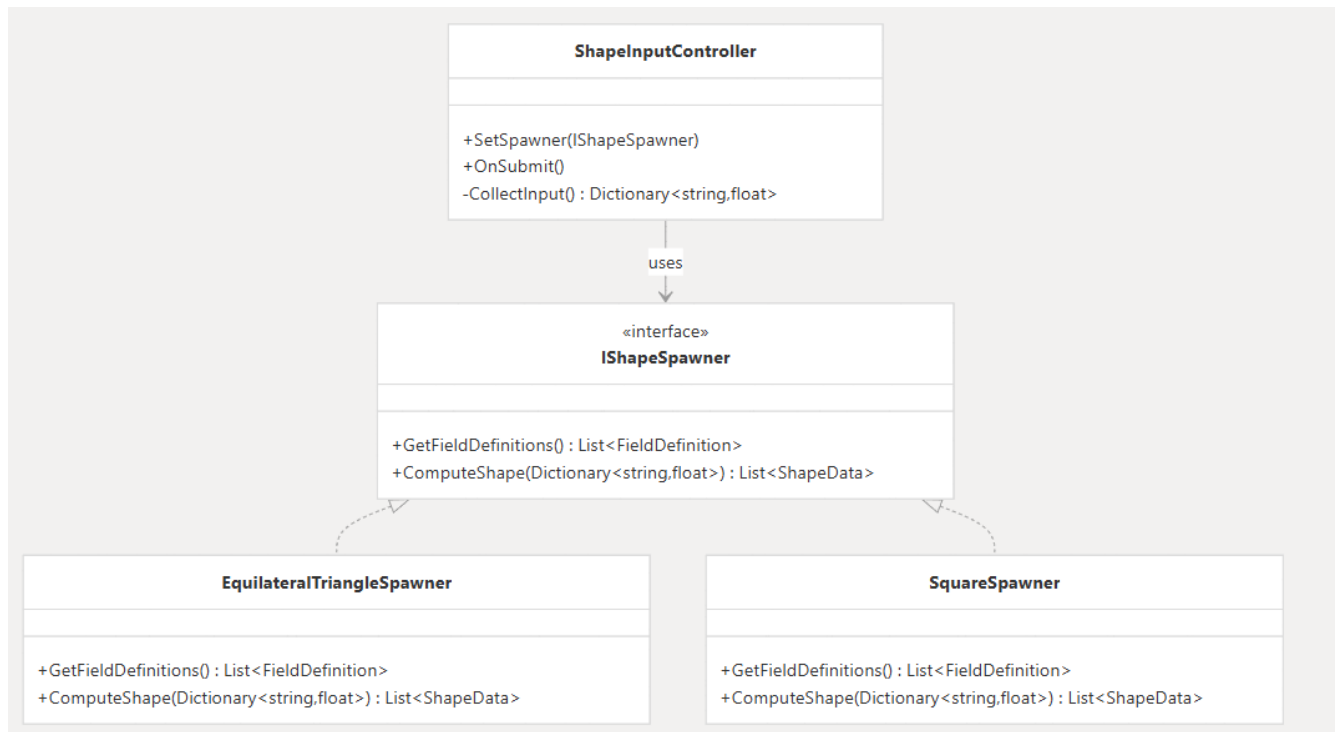


Figure 4.24 Prebuilt Shape System

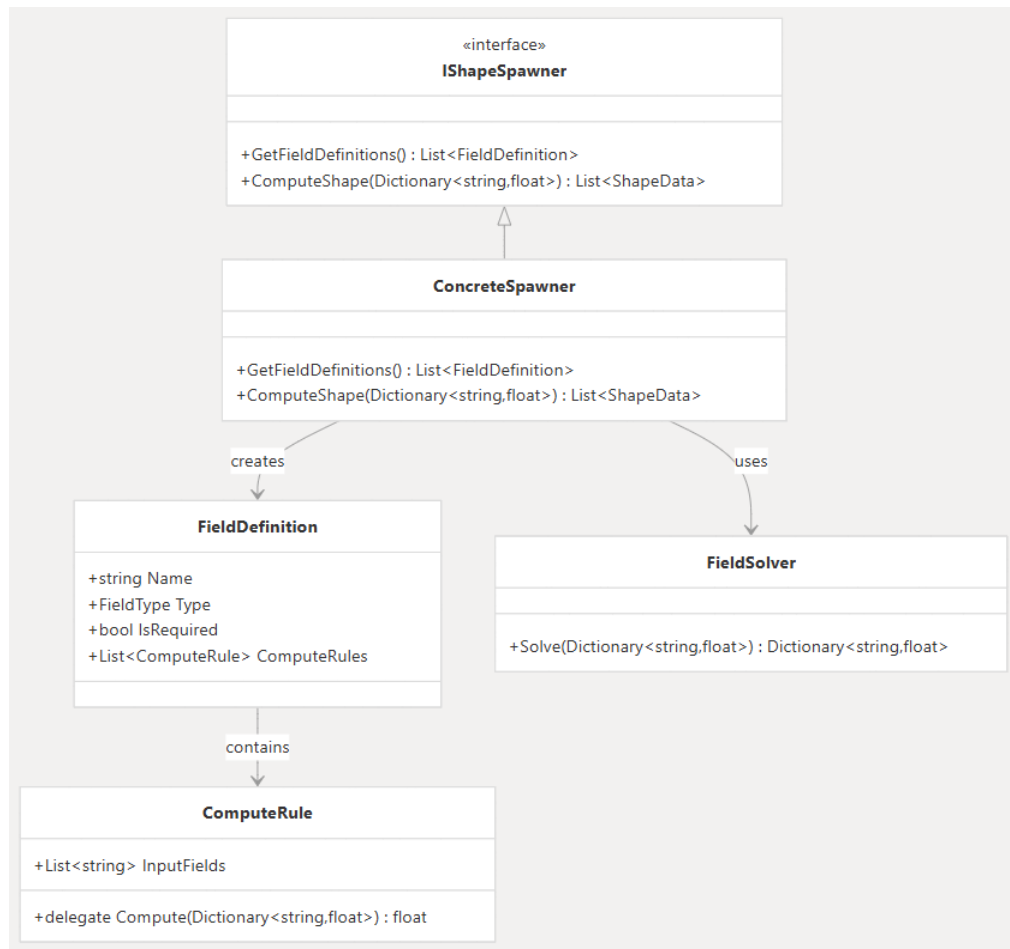


Figure 4.25 Shape Spawner System

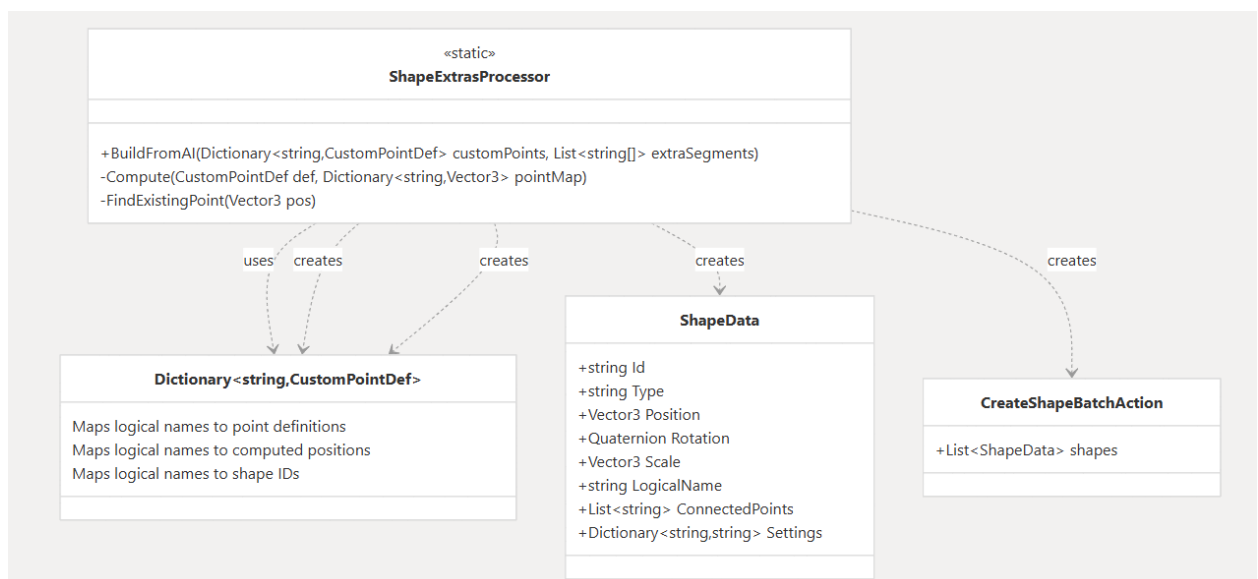


Figure 4.26 Extras Shape Processors

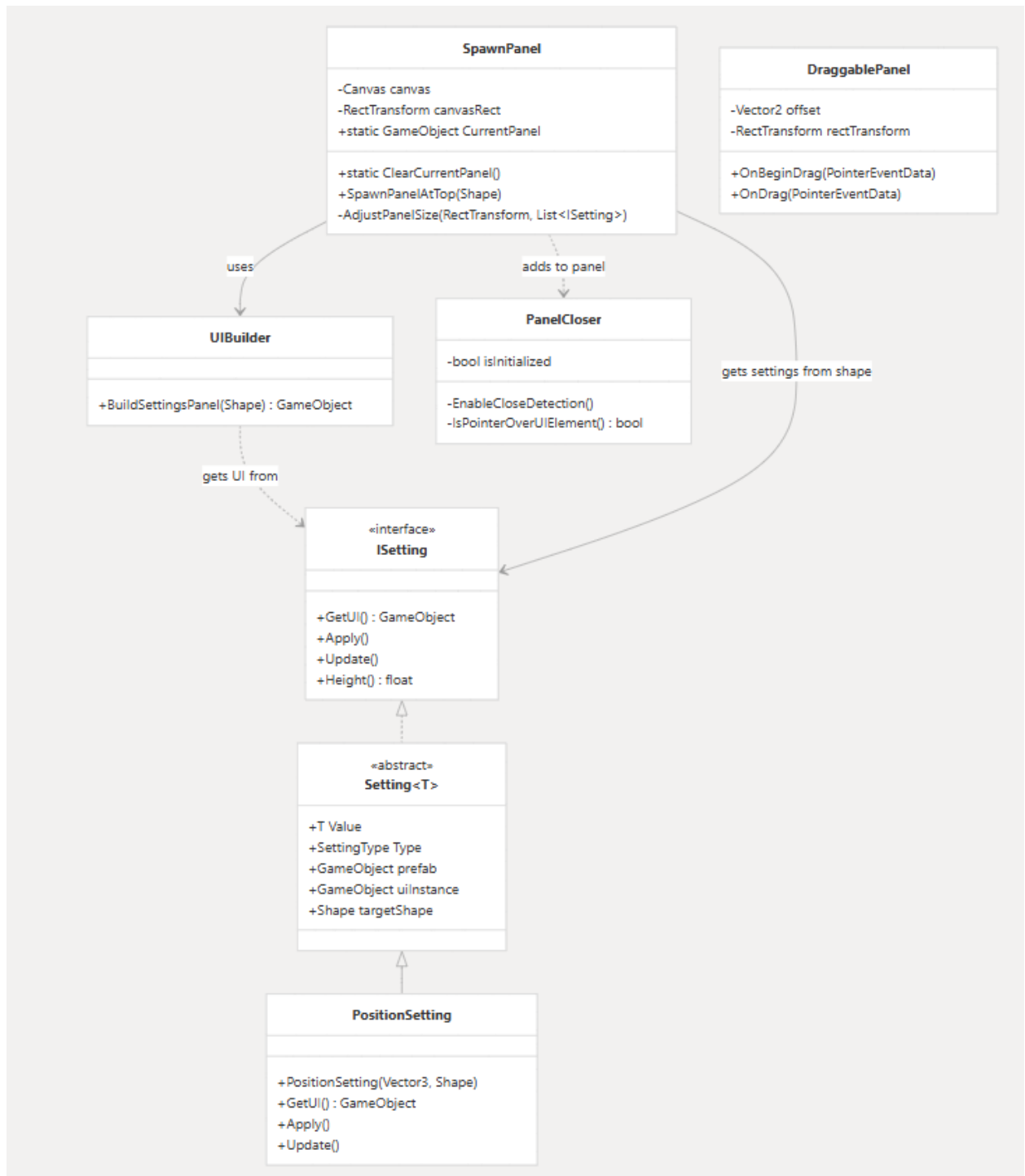


Figure 4.27 Dynamic Shape Settings Panel

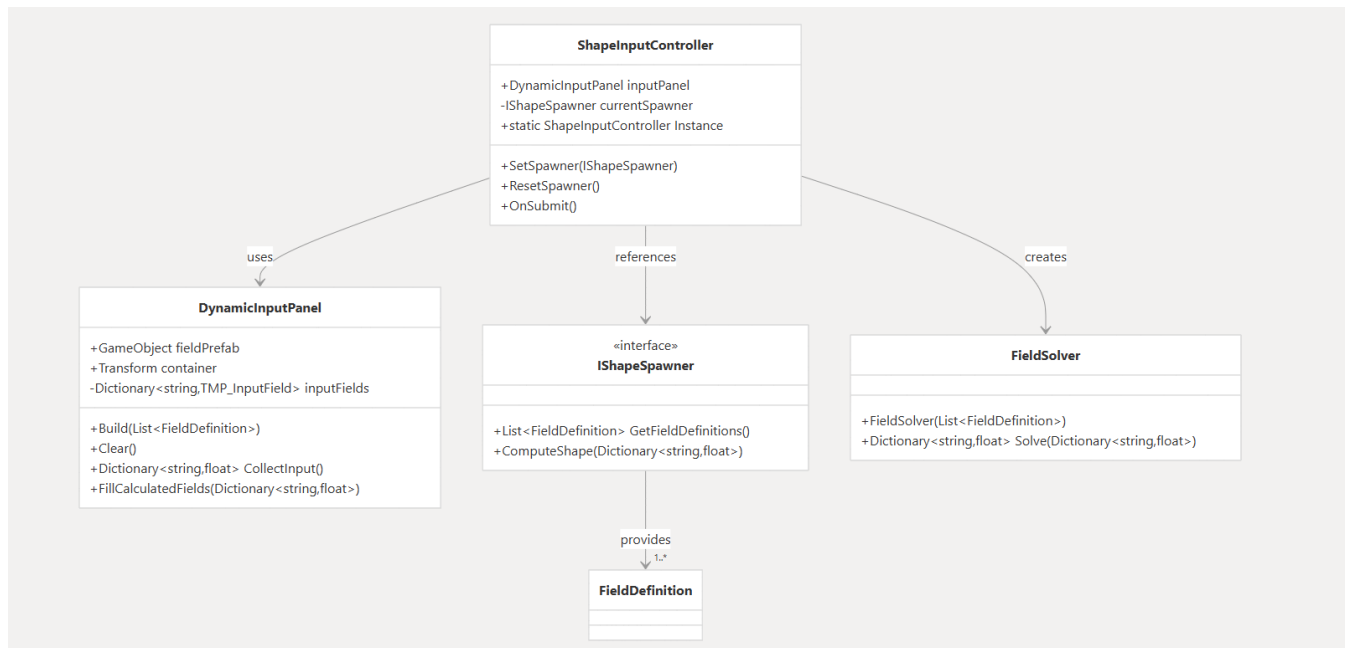


Figure 4.28 Dynamic Shape Spawner Panel

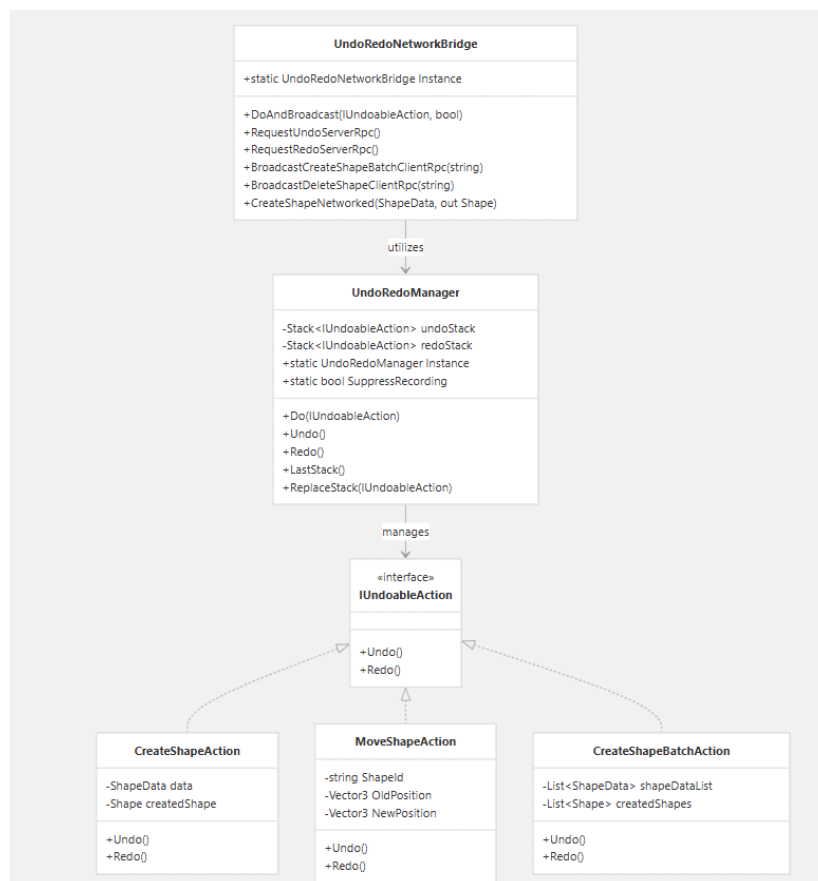


Figure 4.29 Undo Redo System

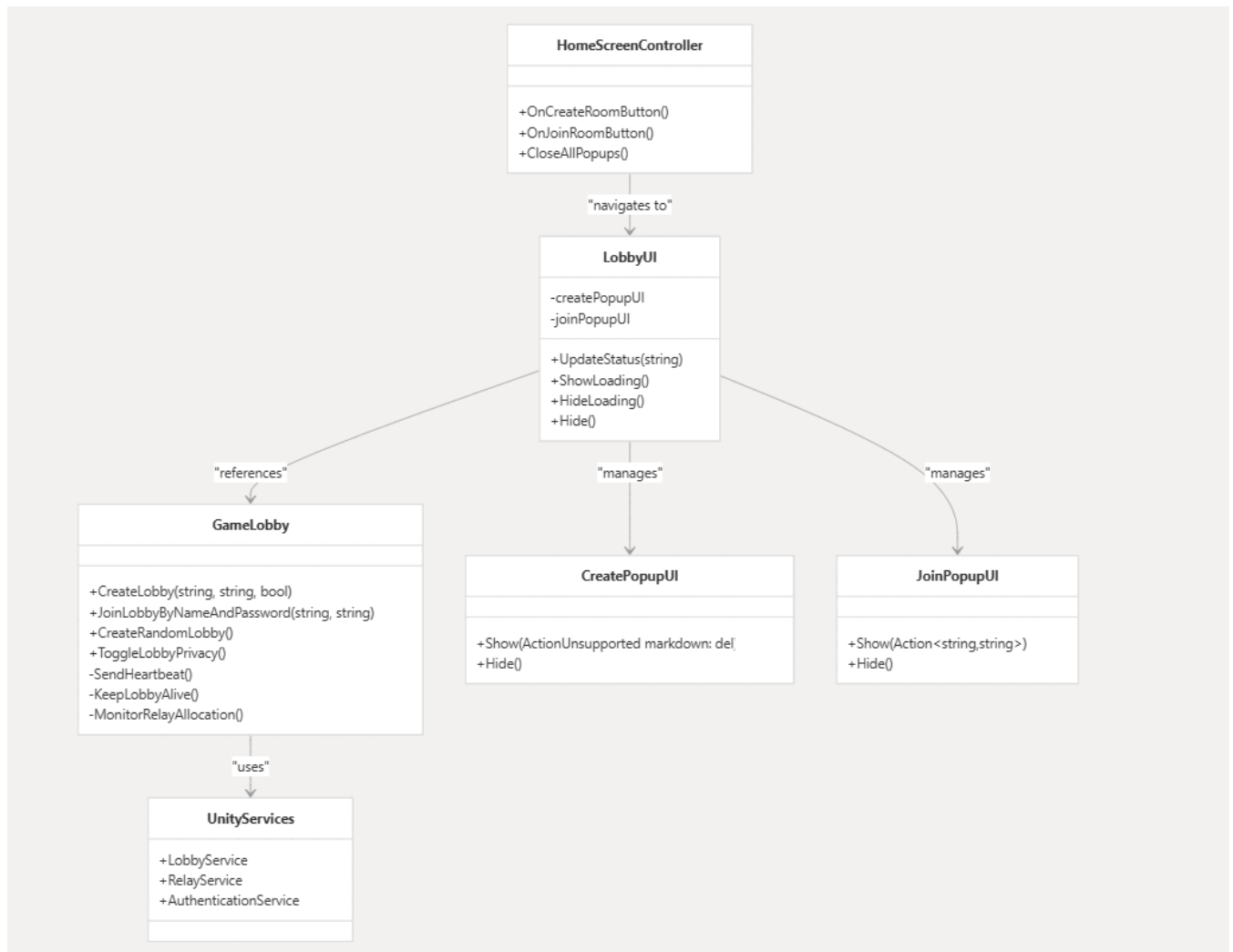


Figure 4.30 Lobby system

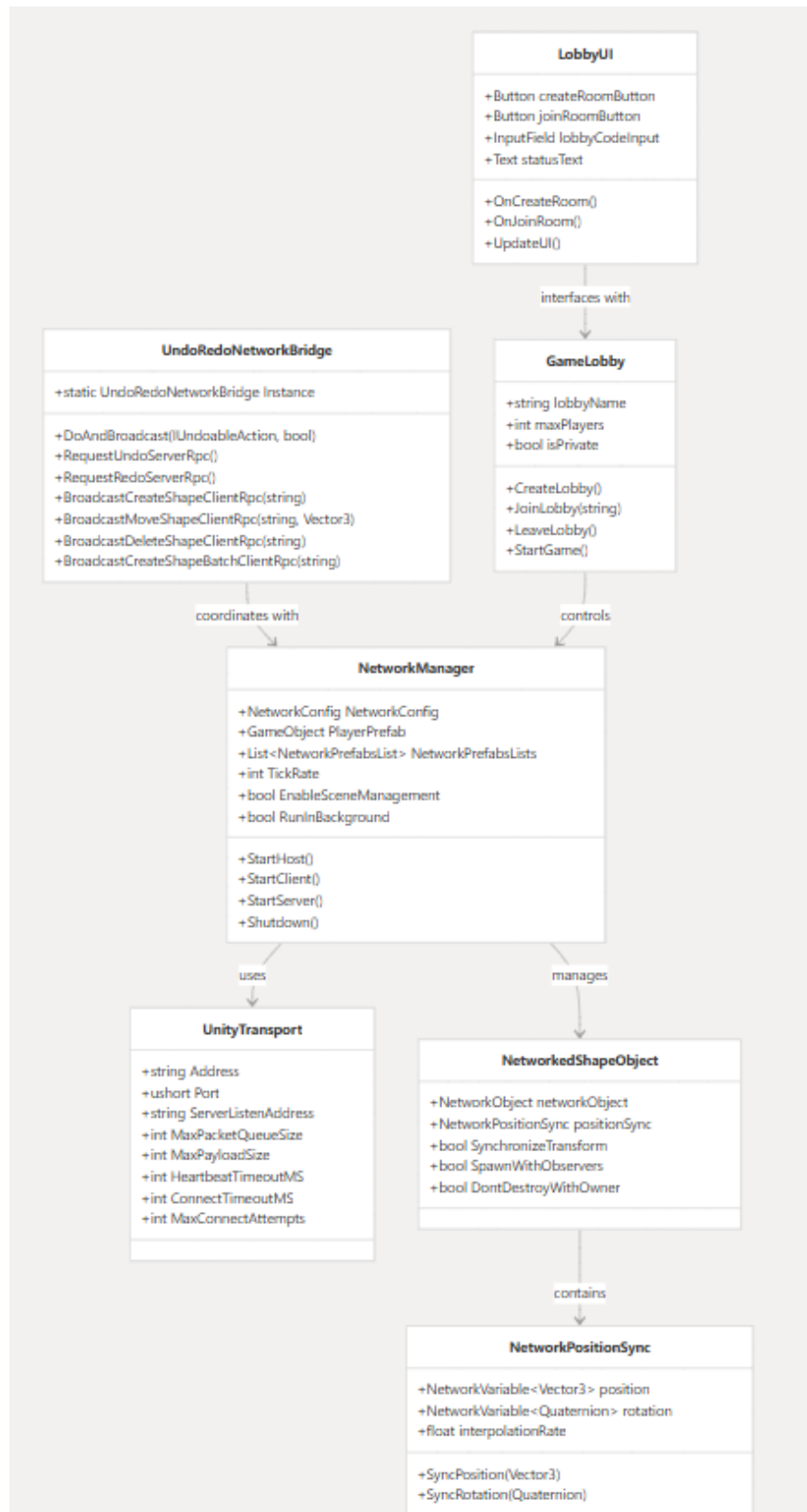


Figure 4.31 Networking System

4.4.1.2. Collect Coin Game Module

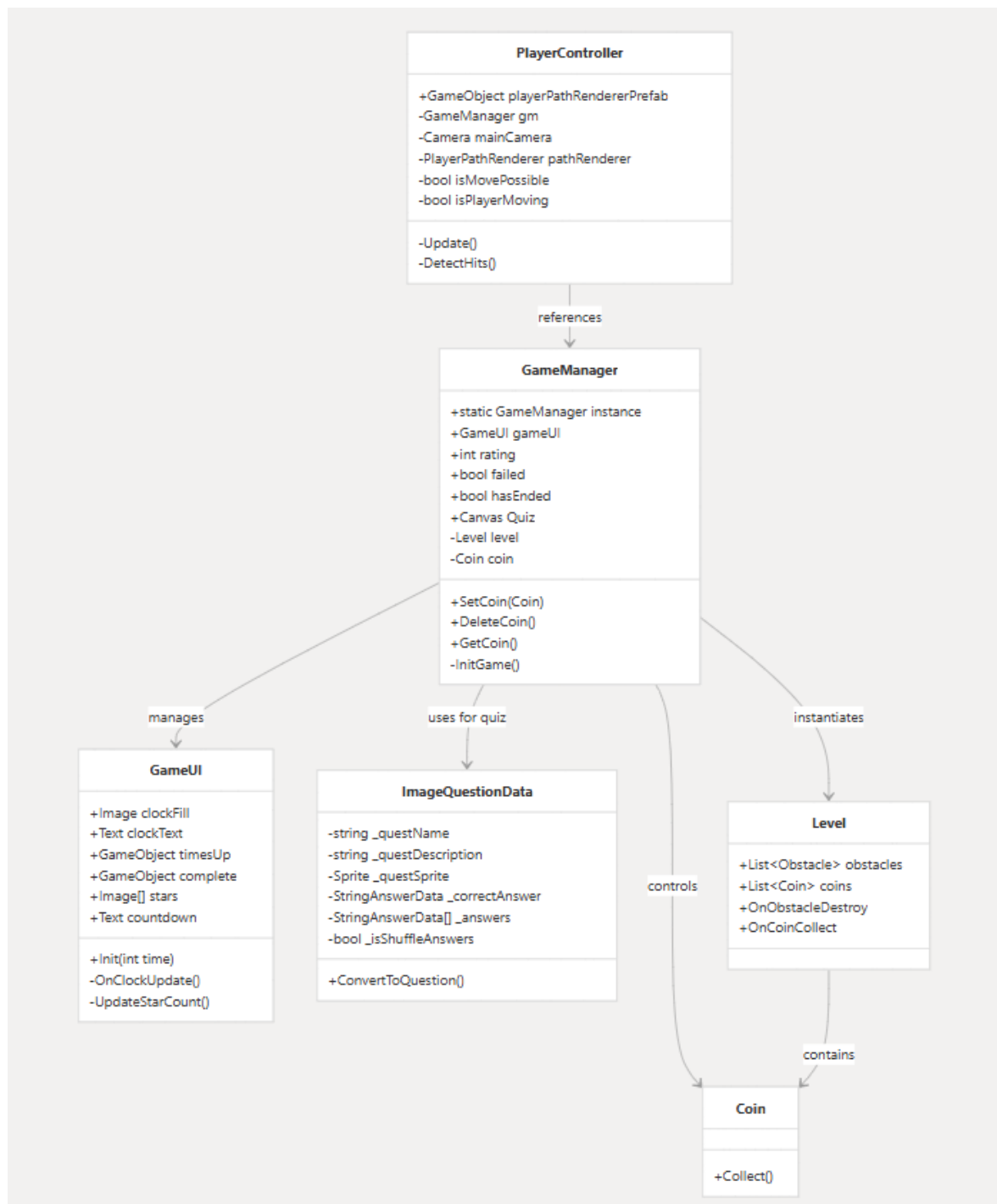


Figure 4.32 Collect Coin Game System

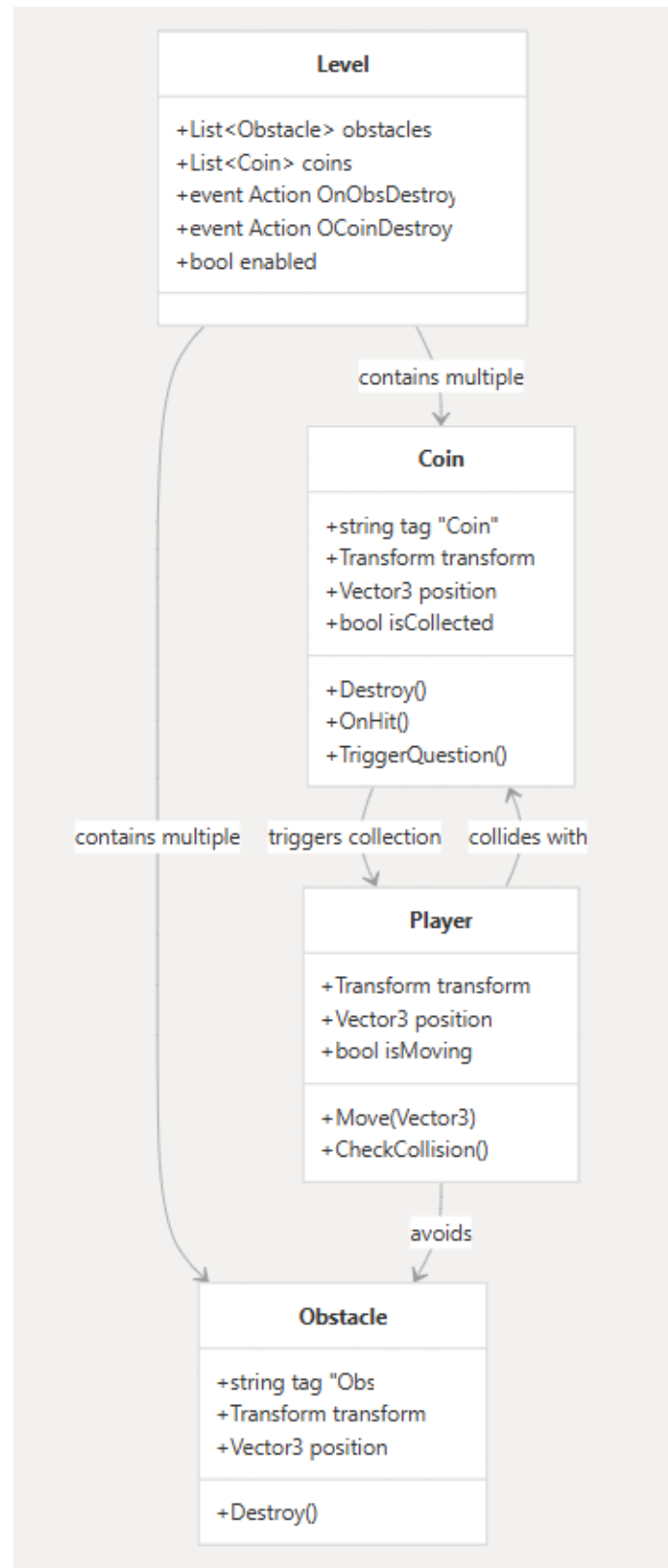


Figure 4.33 Entity System

4.4.1.3. Tutor AI Modules

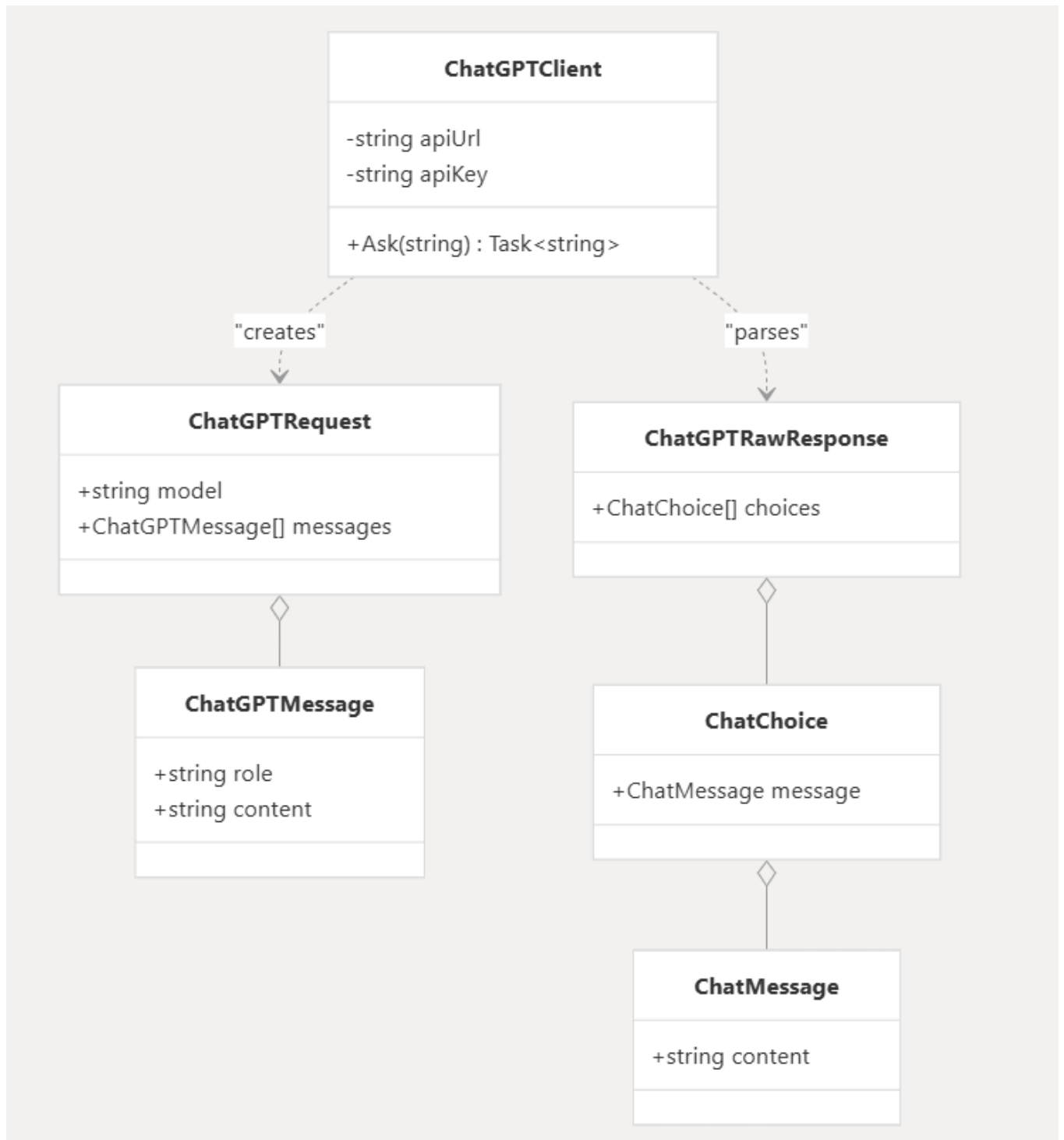


Figure 4.34 Chat Client System

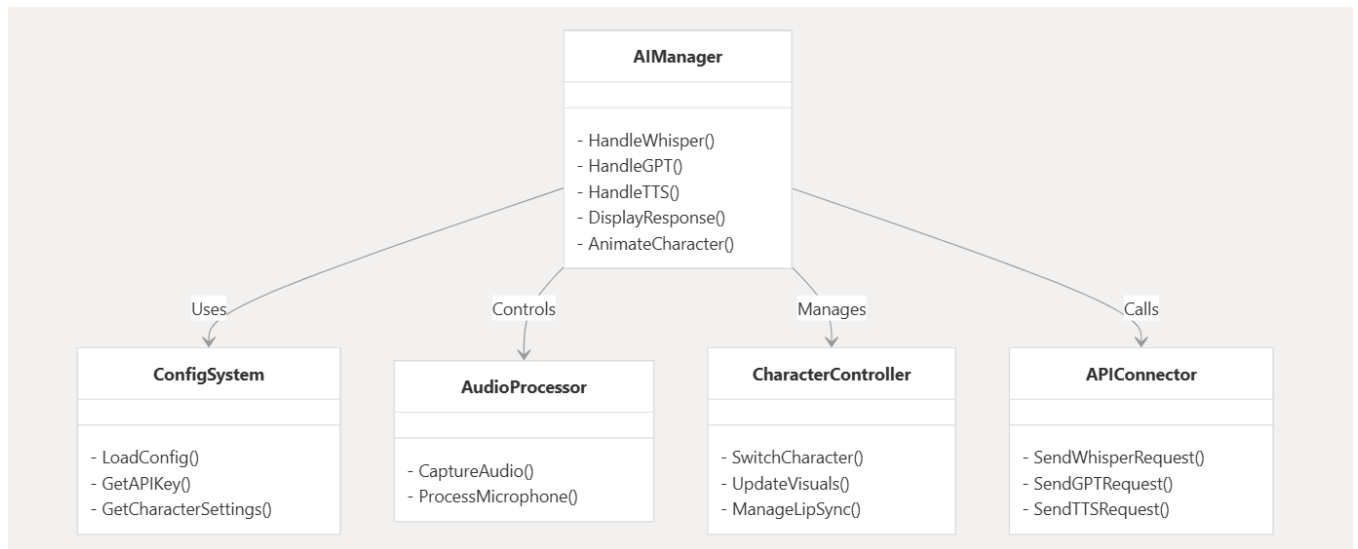


Figure 4.35 AI Core System

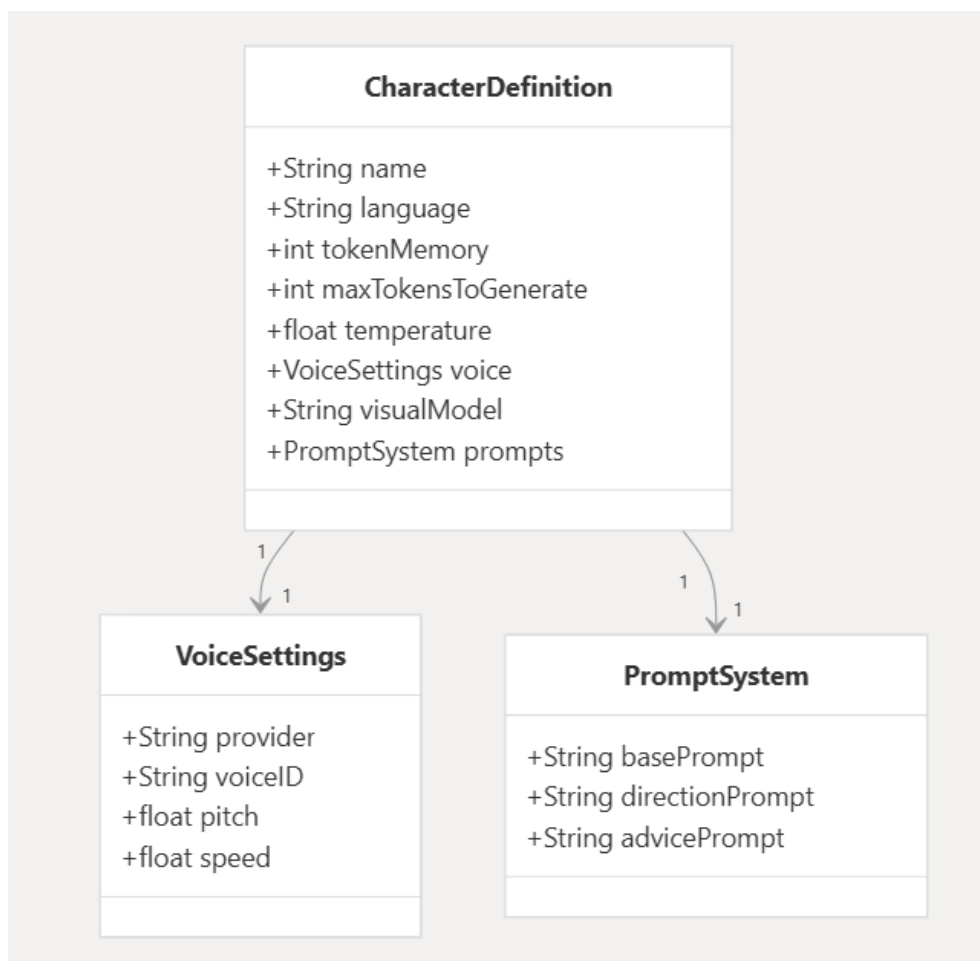


Figure 4.36 NPC settings System

CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

4.4.2. Sequence Diagram

4.4.2.1. System Overview of Unity Multiplayer Services Flow

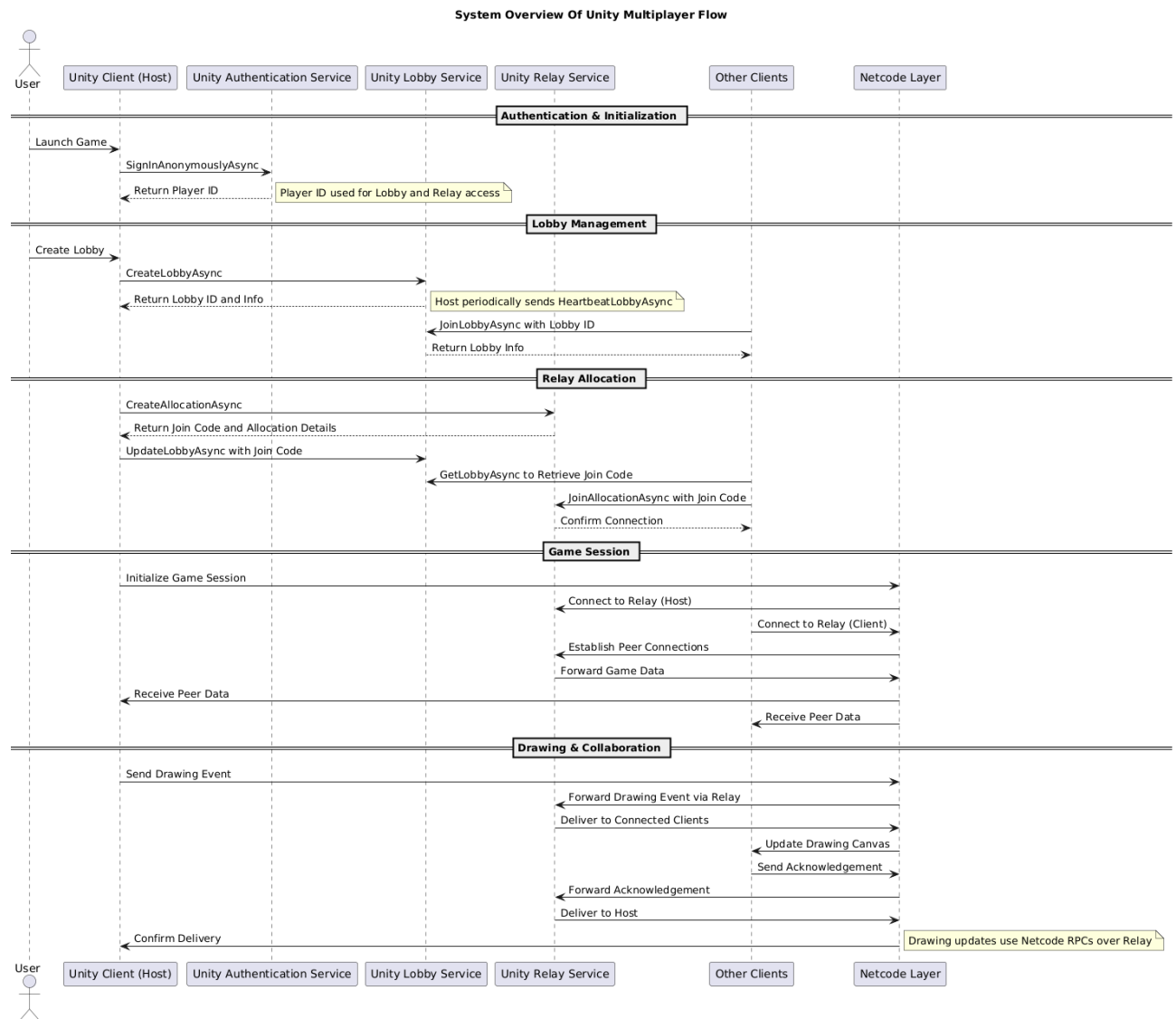


Figure 4.37 System Overview of Unity Multiplayer Services Flow

4.4.2.2. Online Lobby Creation

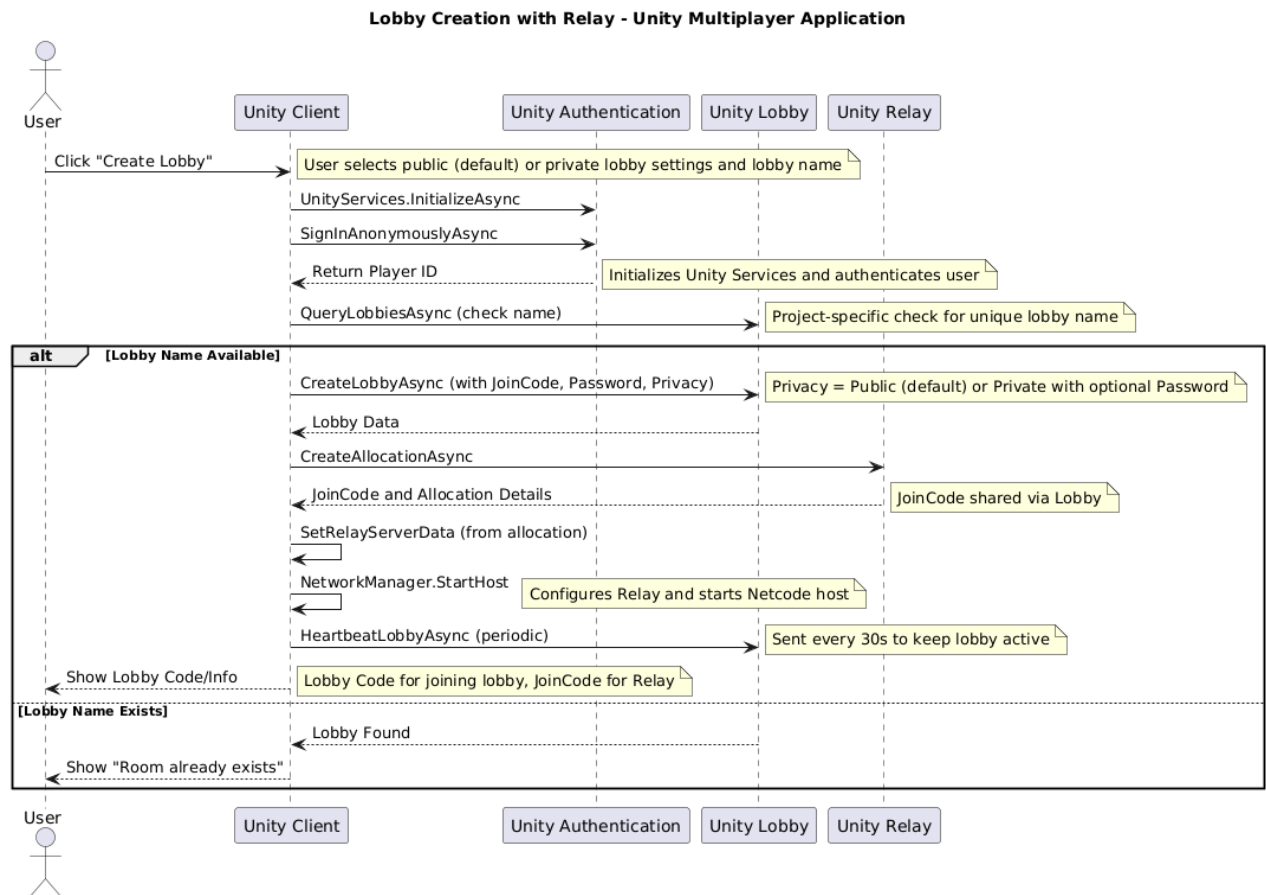


Figure 4.38 Lobby creation sequence diagram

4.4.2.3. Client Join Room

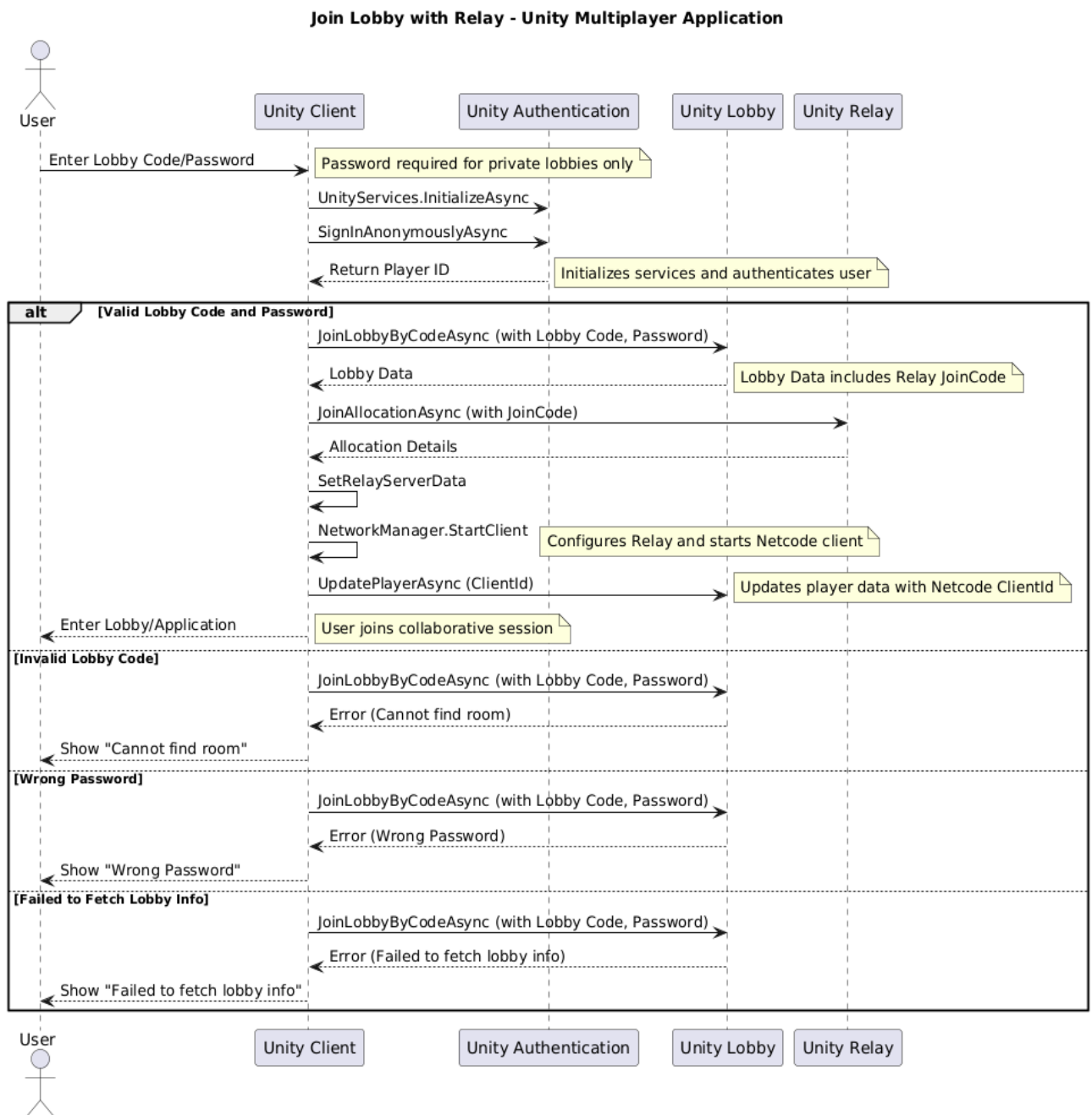


Figure 4.39 Client join lobby sequence diagram

CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

4.4.2.4. Share Lobby

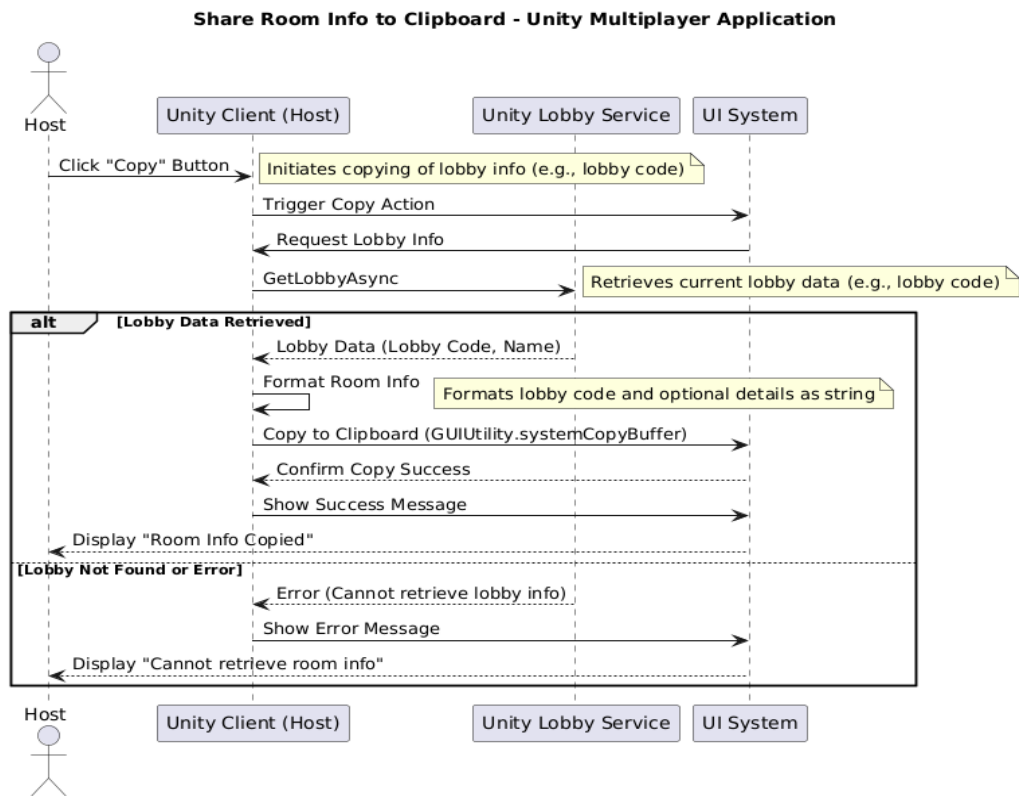


Figure 4.40 Share lobby information sequence diagram

4.4.2.5. Client Join Notification

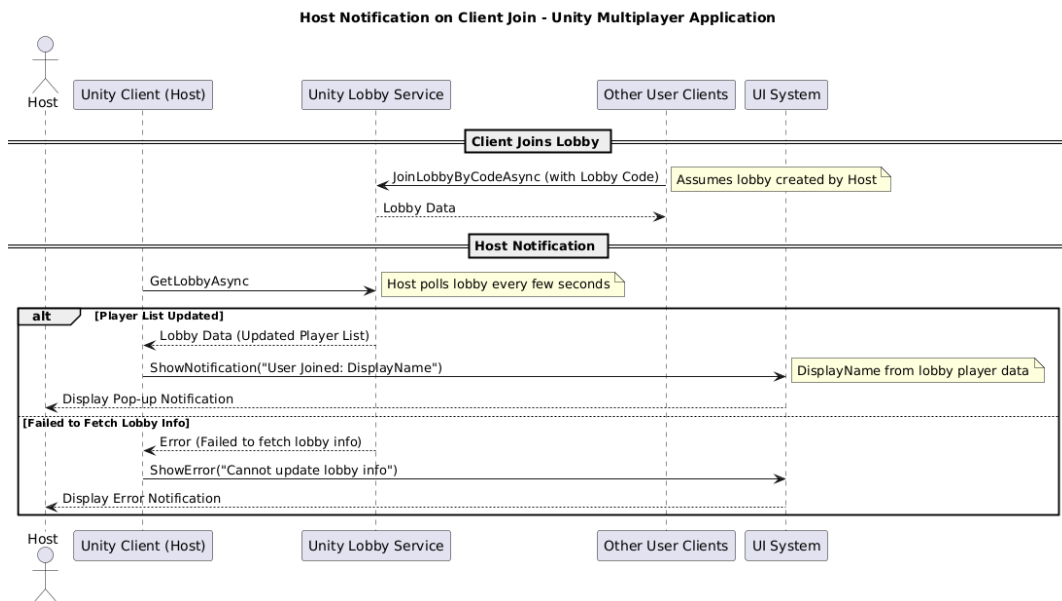


Figure 4.41 Client join notification sequences diagram

CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

4.4.2.6. Application Navigation Flow

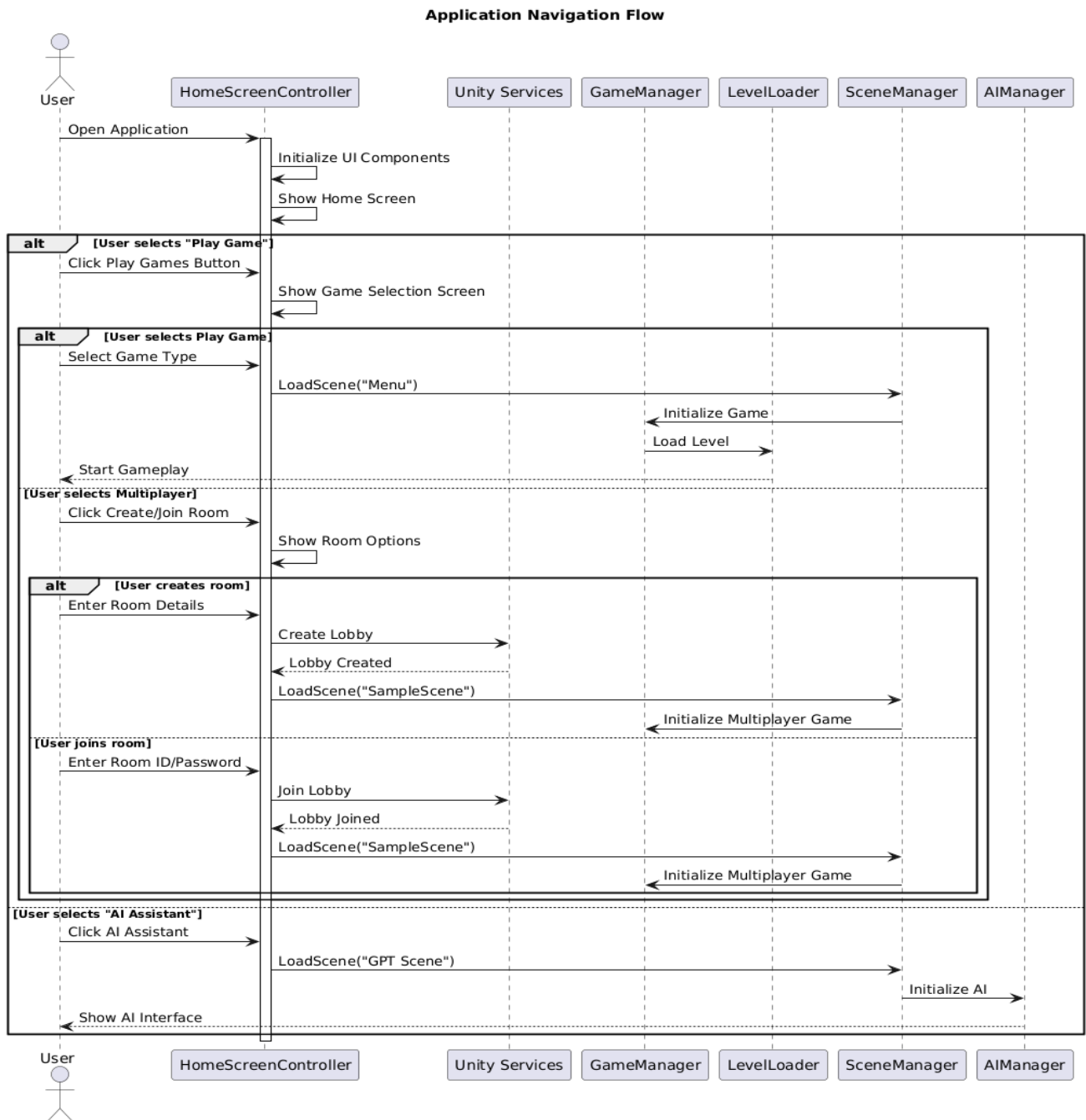


Figure 4.42 Application navigation sequence diagram

4.4.2.7. System Navigation Module

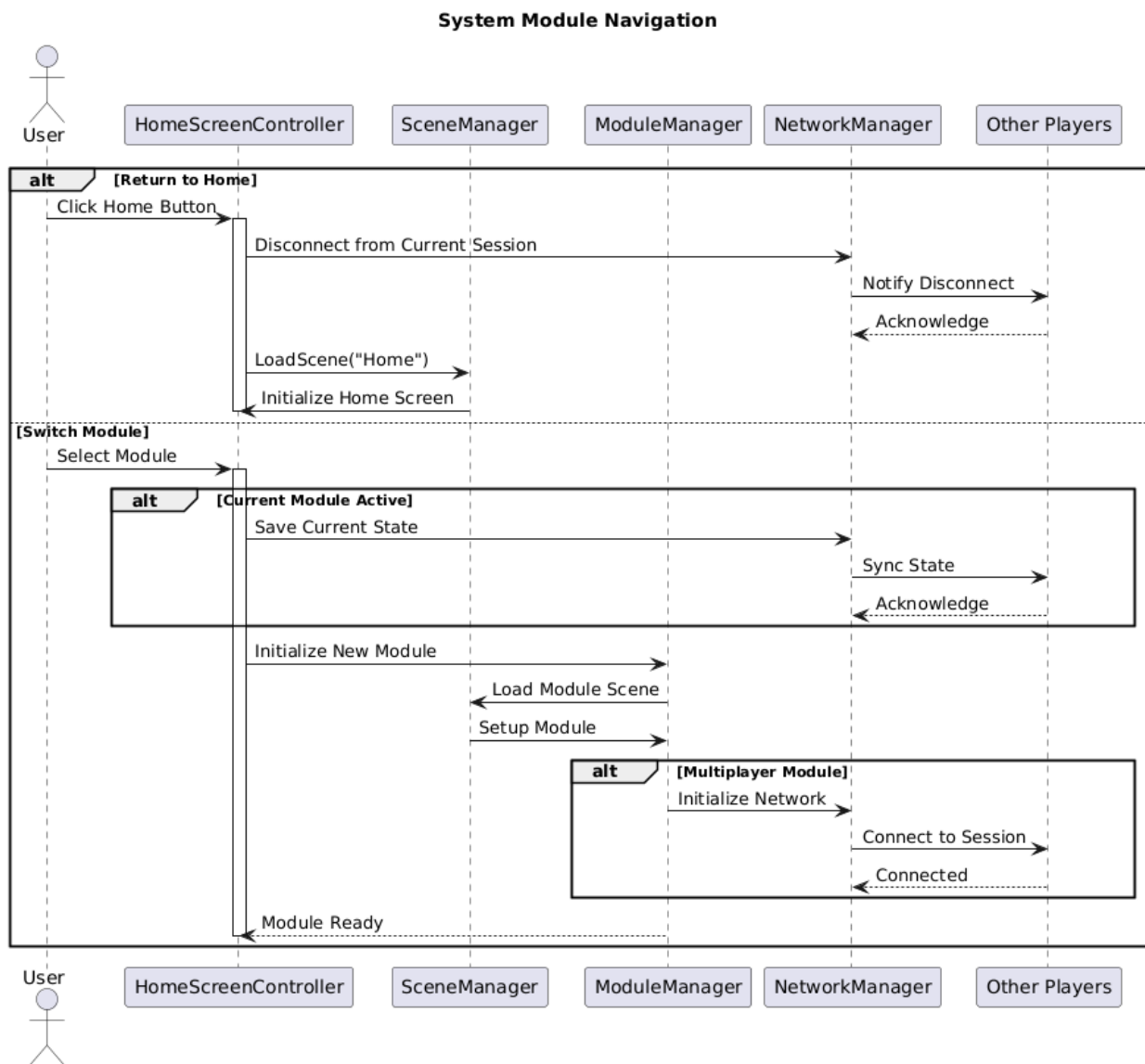


Figure 4.43 System navigation module sequence diagram

4.4.2.8. Geometry Multiplayer Sync

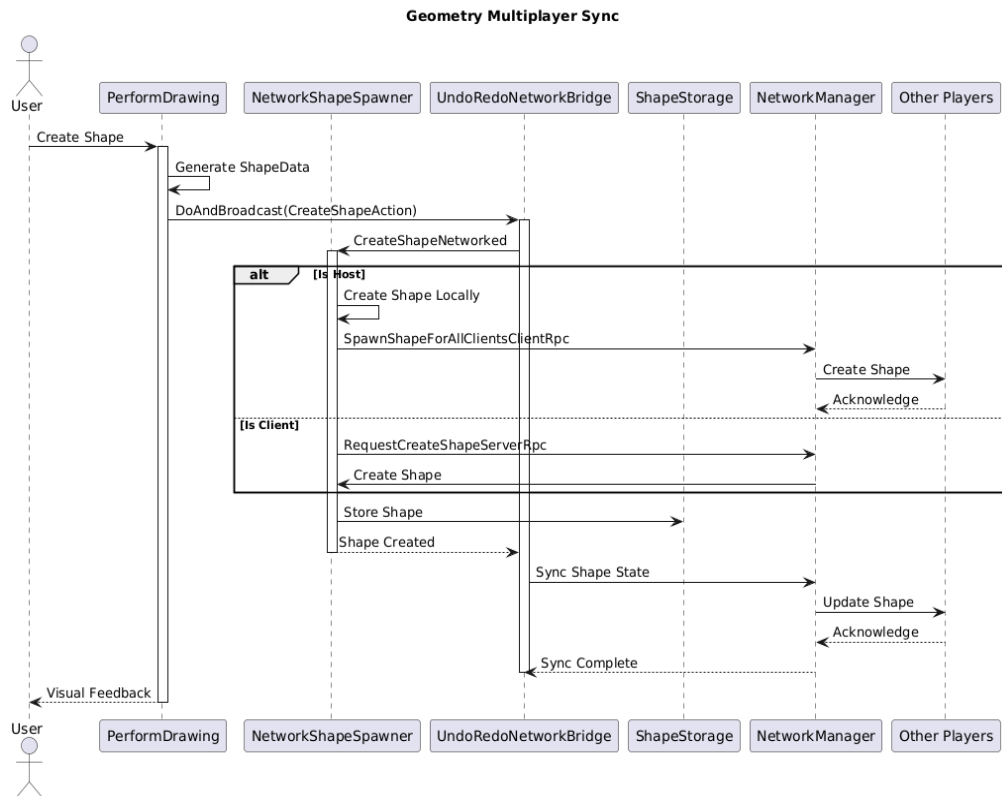


Figure 4.44 Geometry multiplayer sync sequence diagram

4.4.2.9. Shape Selection and Manipulation

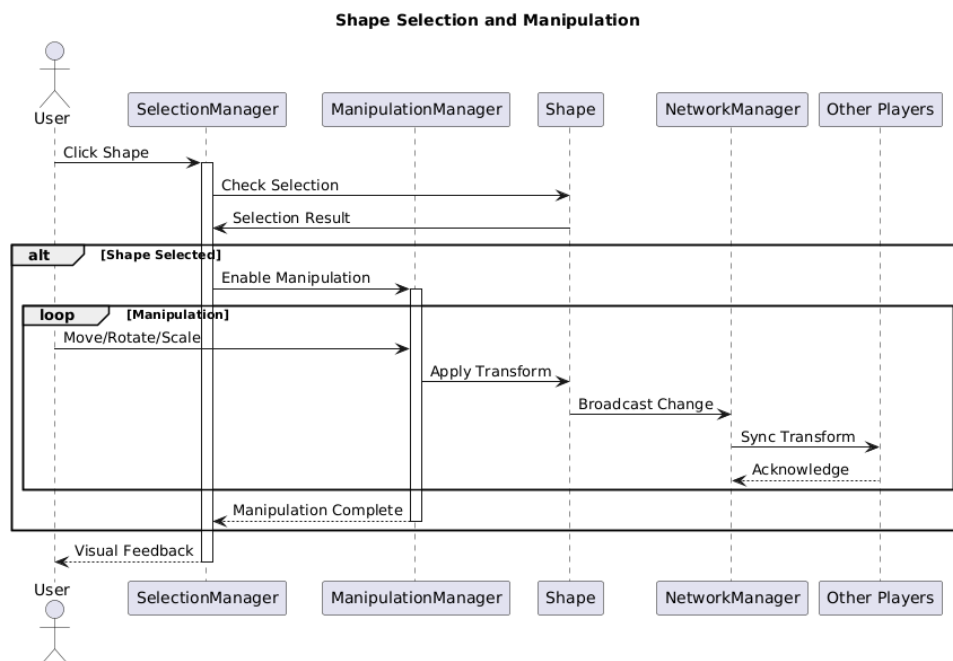


Figure 4.45 Shape selection and manipulation sequence diagram

CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

4.4.2.10. Geometric Shape Creation

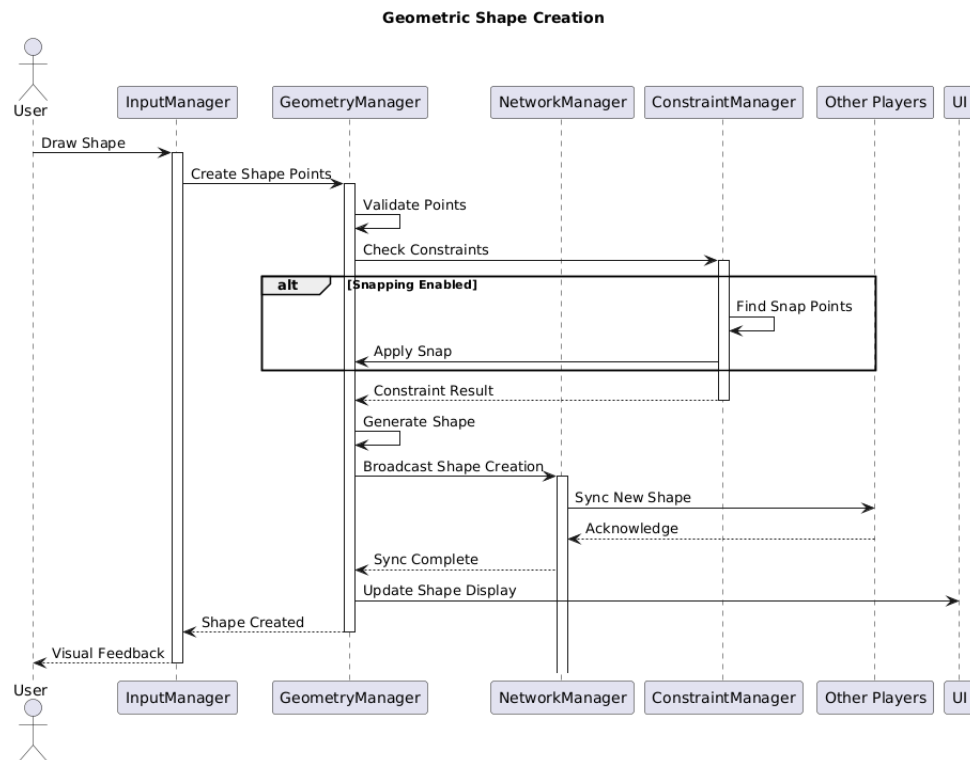


Figure 4.46 Geometric shape creation sequence diagram

4.4.2.11. Shape Measurement System

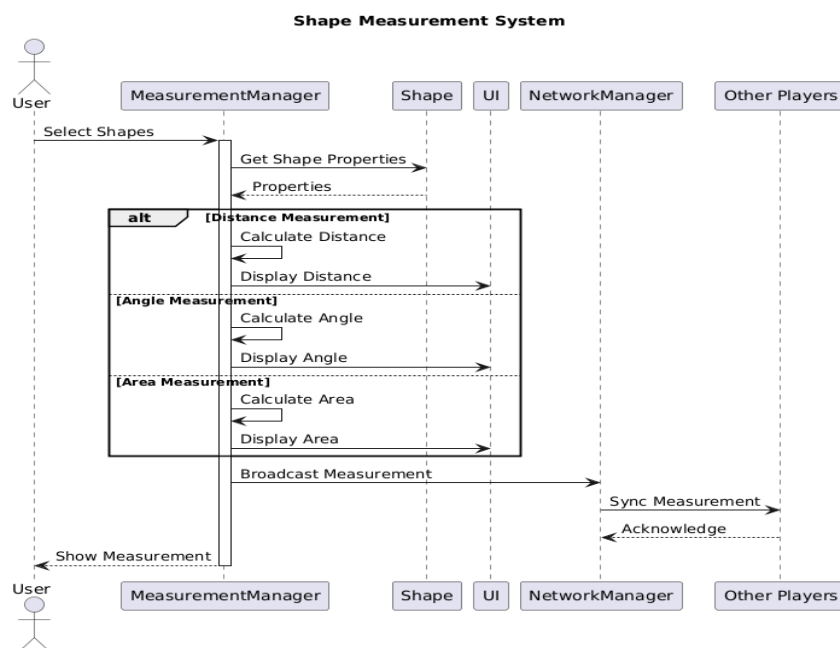


Figure 4.47 Shape measurement

4.4.2.12. Shape Snapping System

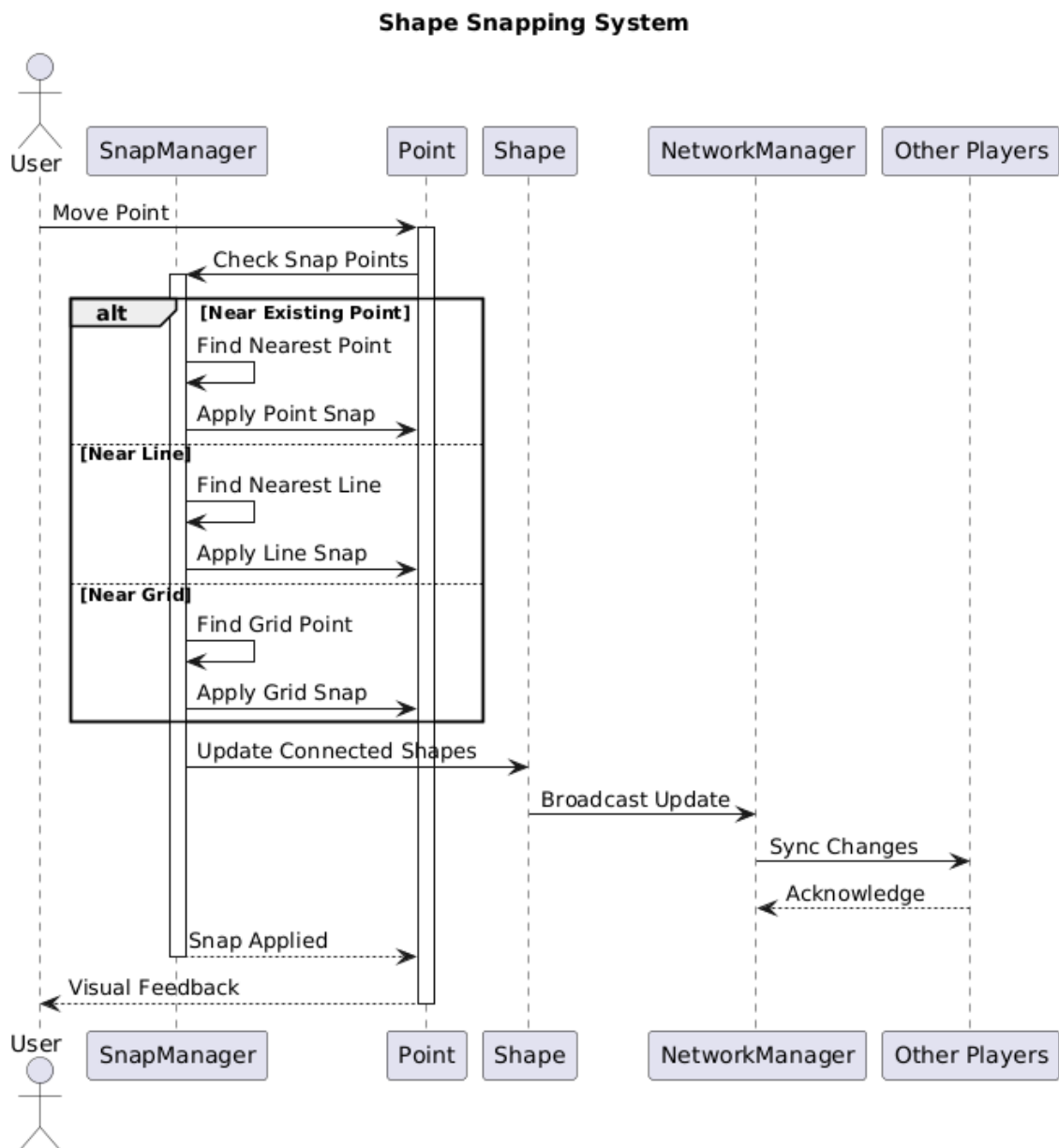


Figure 4.48 Shape snapping sequence diagram

4.4.2.13. Shape Constraint System

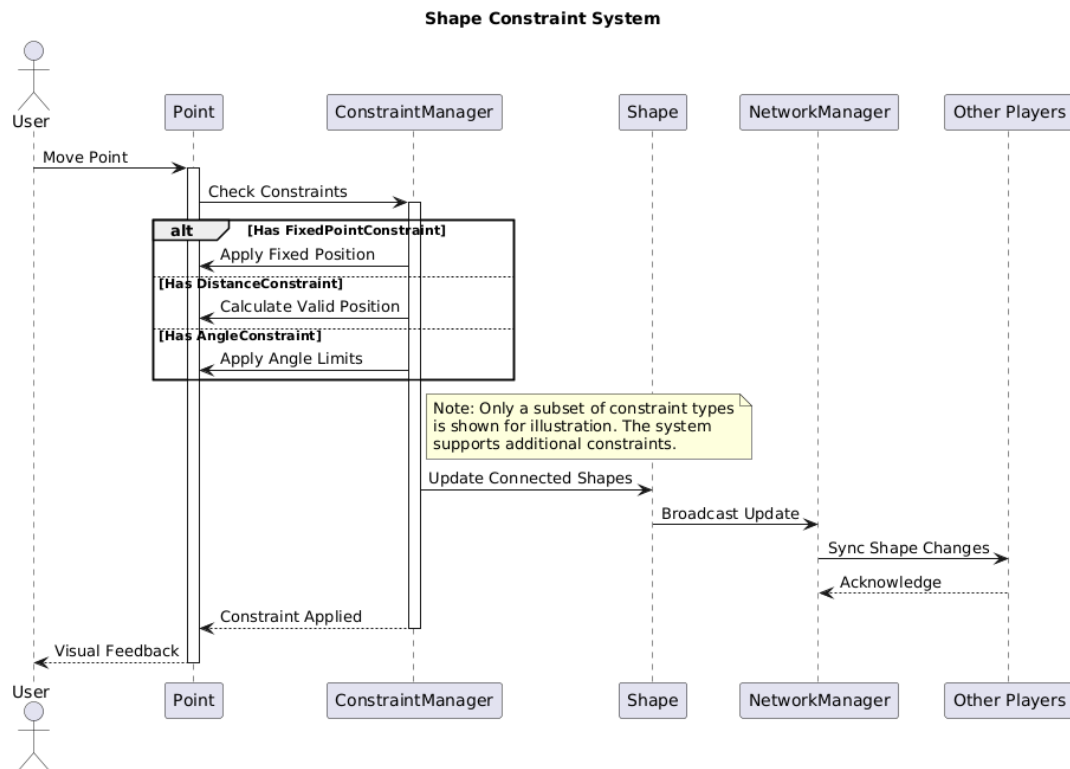


Figure 4.49 Shape constraint sequence diagram

4.4.2.14. Shape History System

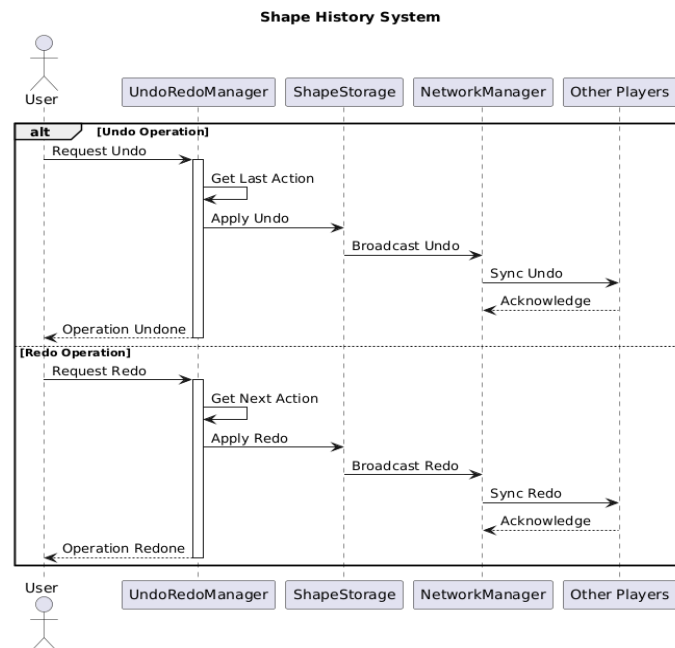


Figure 4.50 Shape history system sequence diagram

4.4.2.15. Geometry Shape Undo/Redo

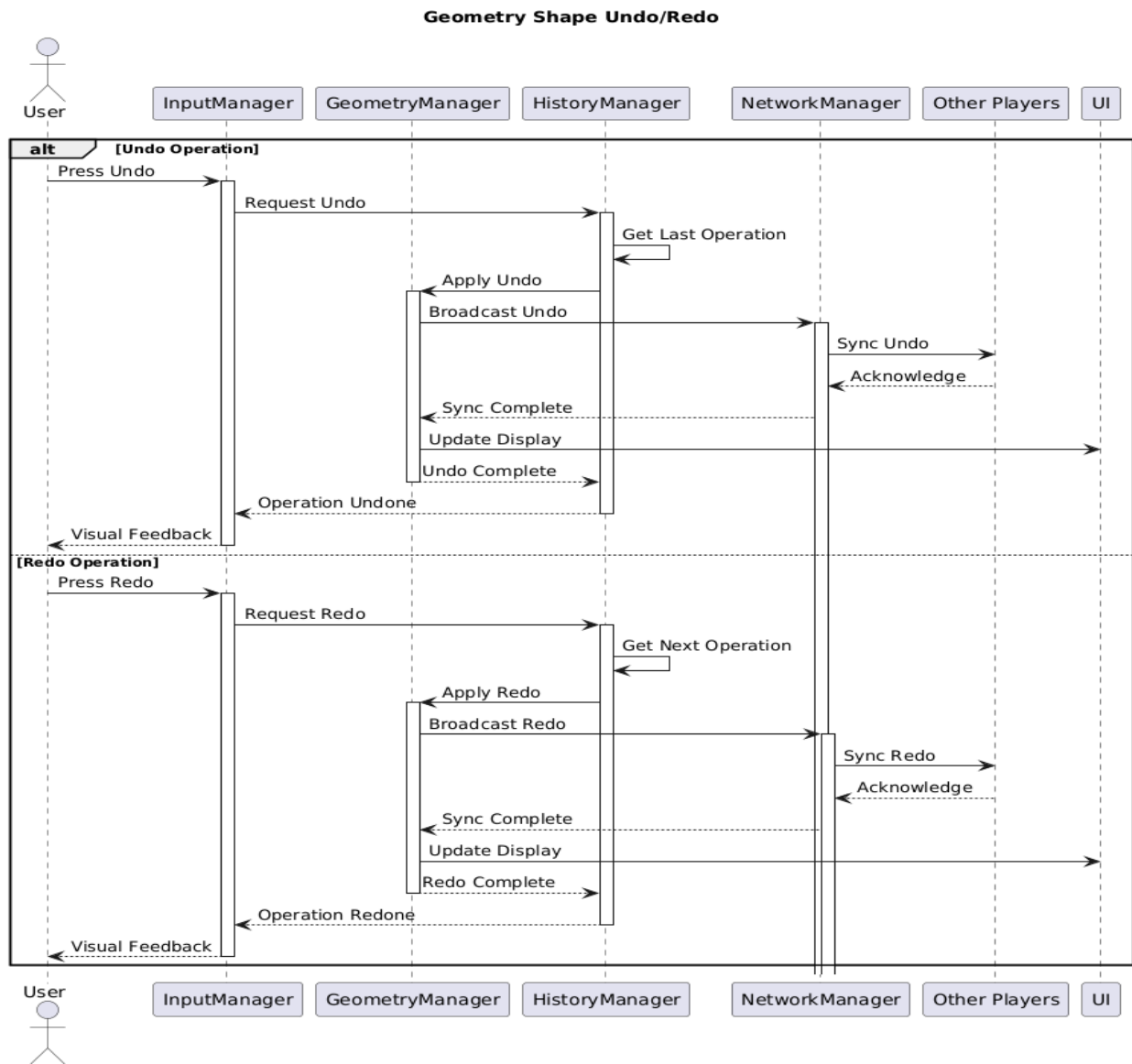


Figure 4.51 Undo/redo sequence diagram

4.4.2.16. Geometry Multiplayer Shape Sync

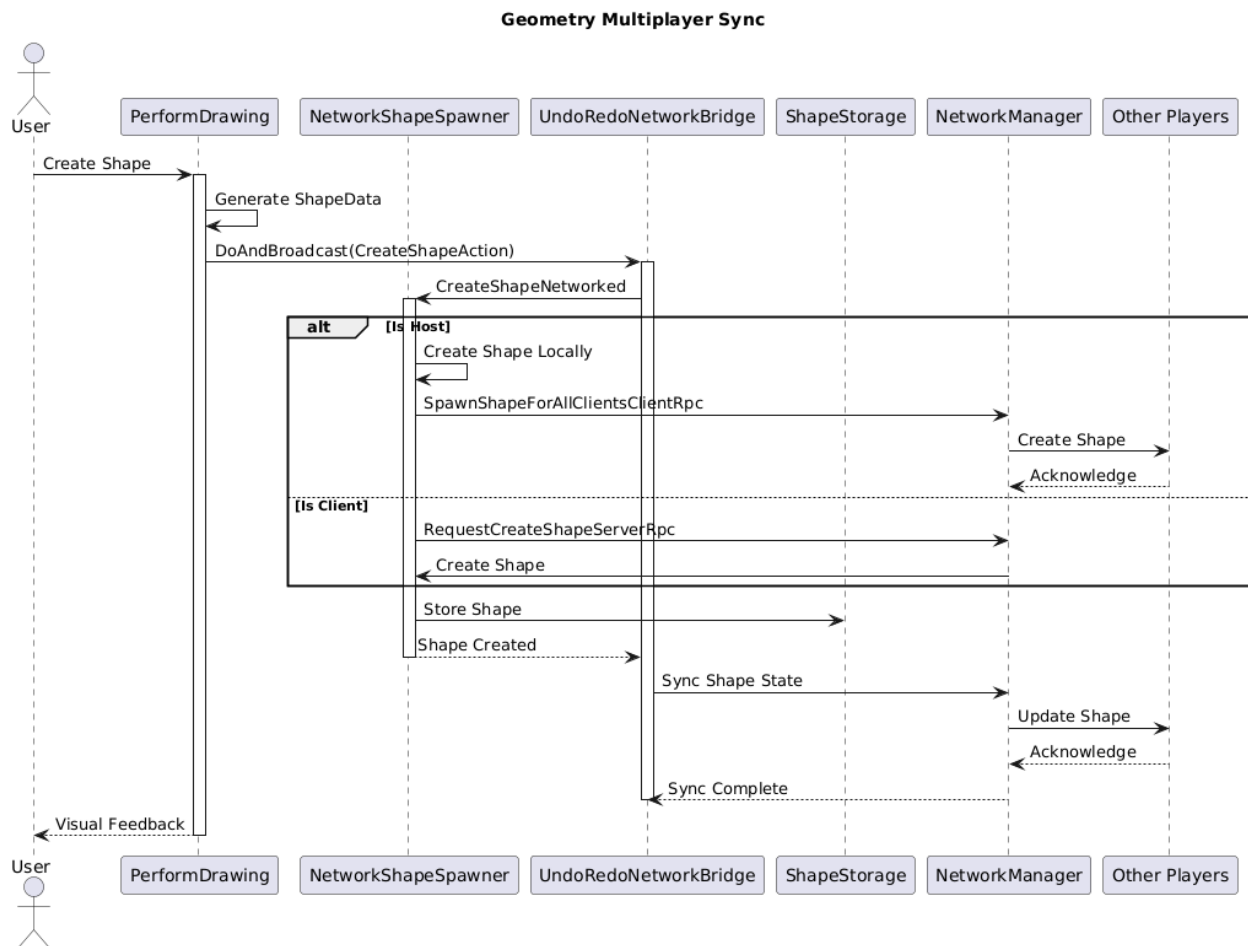


Figure 4.52 Multiplayer synchronization sequence diagram

CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

4.4.2.17. Coin Collecting Game System

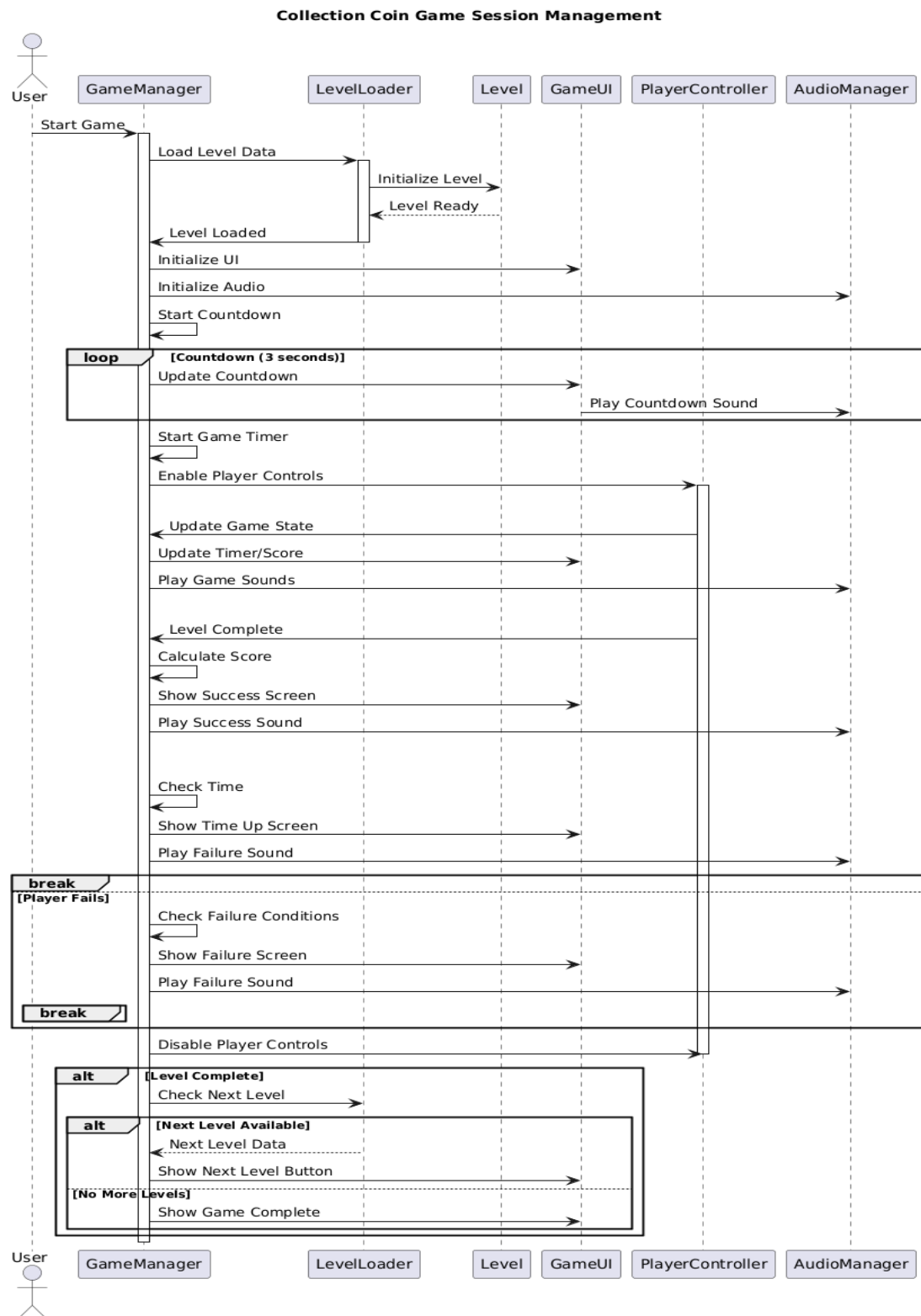


Figure 4.53 Coin collecting game sequence diagram

CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

4.4.2.18. Tutor AI Assistant

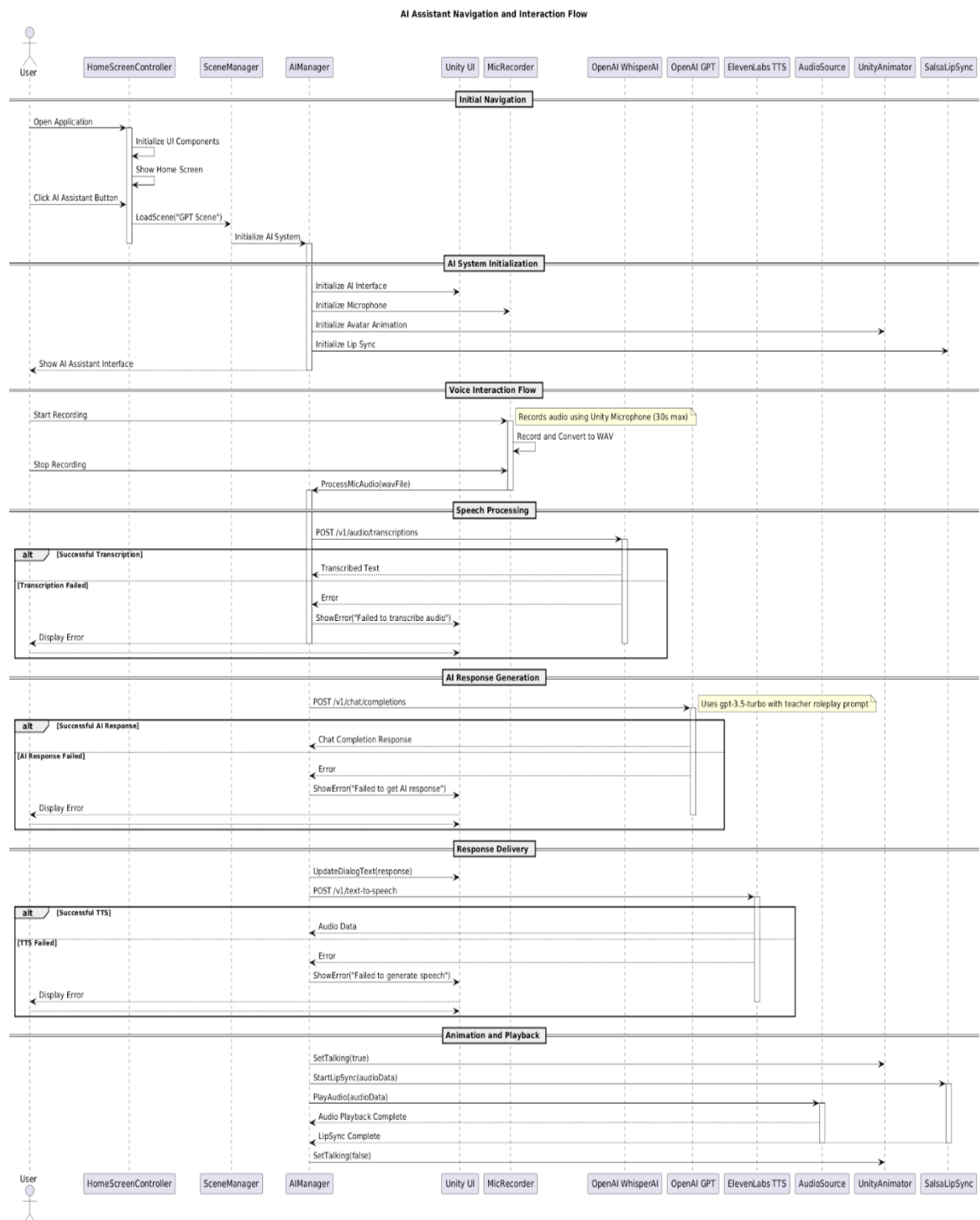


Figure 4.54 Tutor AI Assistant Sequence diagram

CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

4.4.2.19. GPT Assistant Voice Query System

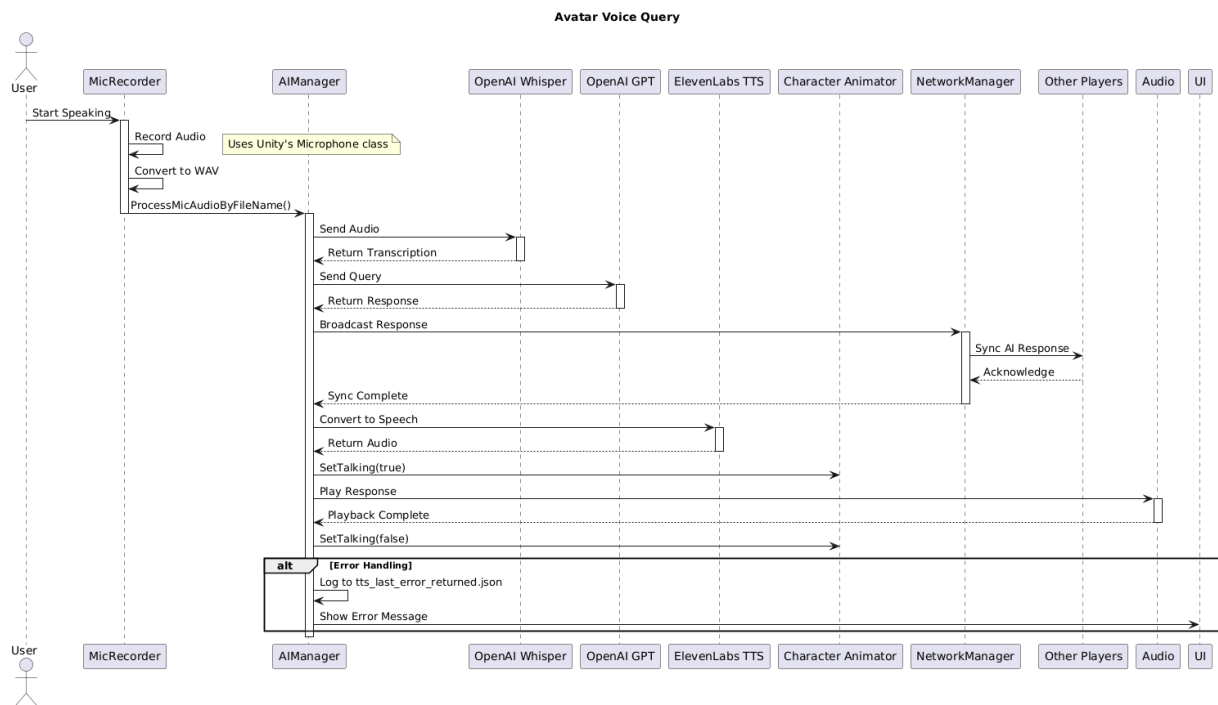


Figure 4.55 GPT assistant voice query sequence diagram

4.4.2.20. GPT Assistant Text Query System

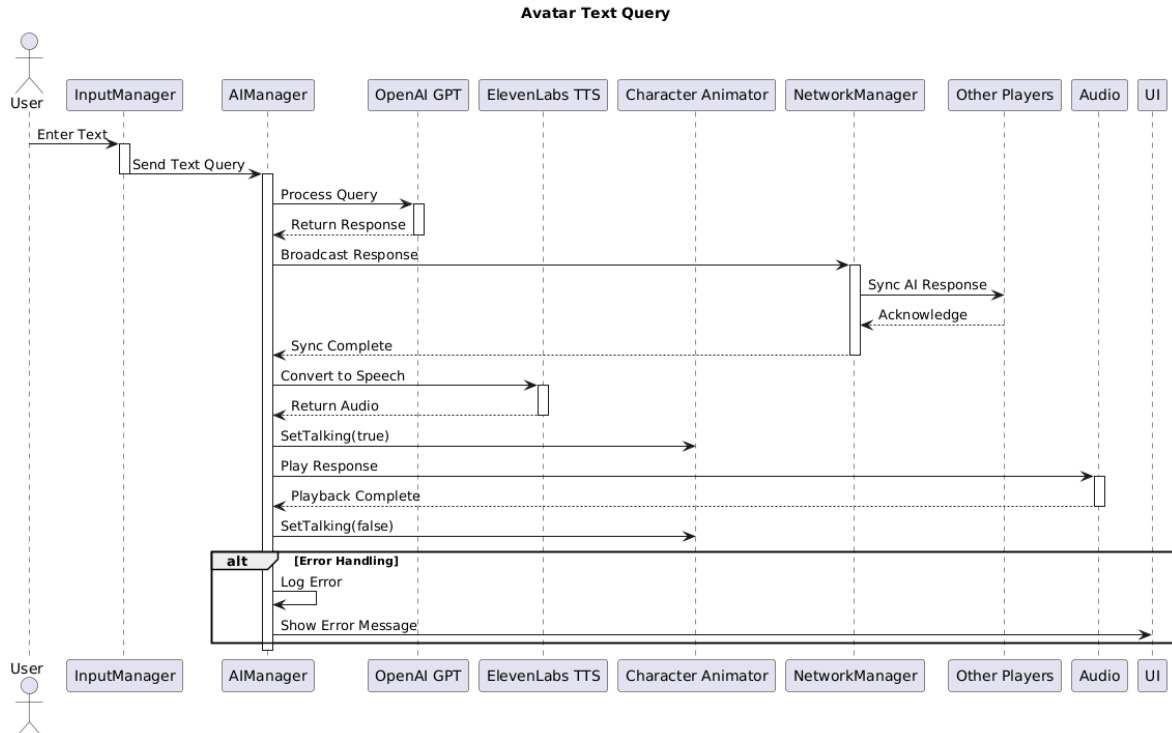


Figure 4.56 GPT assistant text query sequence diagram

4.4.2.21. GPT Drawing Assistant System

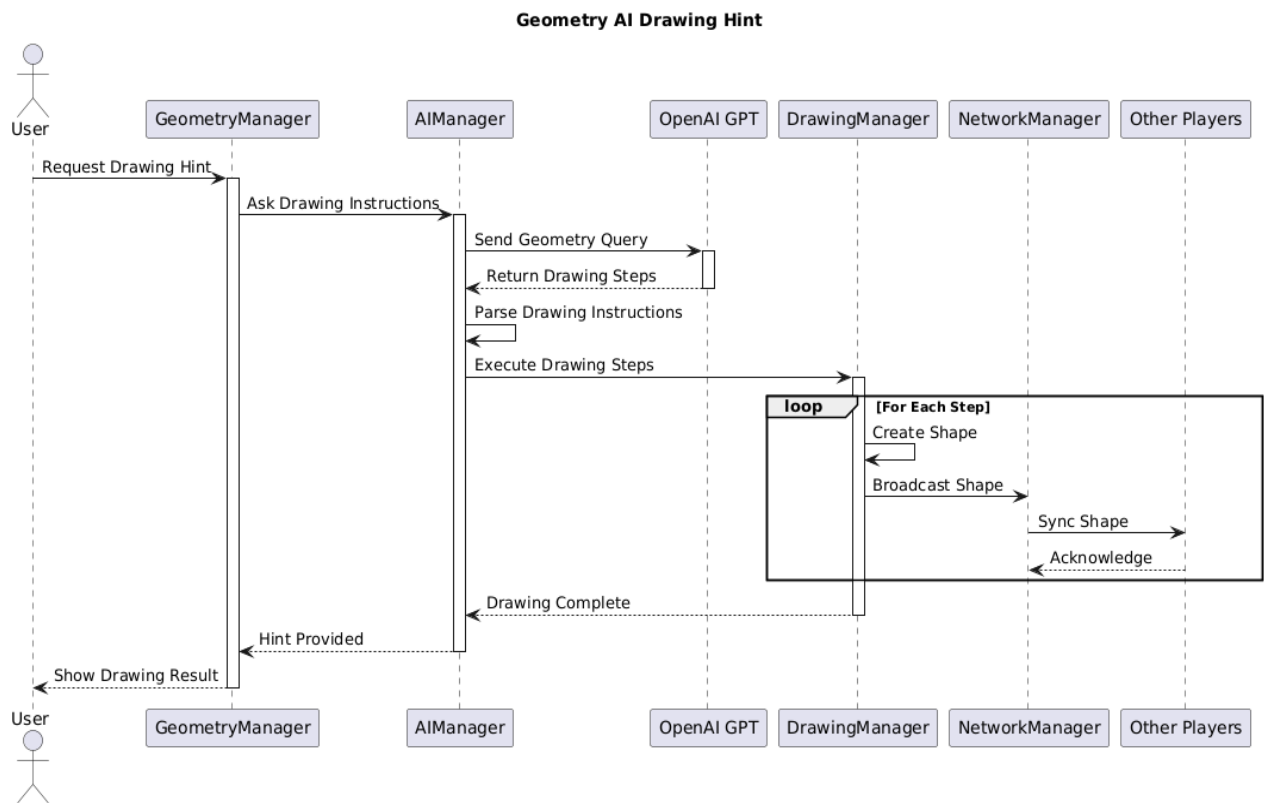


Figure 4.57 GPT drawing assistant sequence diagram

4.5 Algorithms

4.5.1 Geometry Drawing Raycasting

4.5.1.1 Definition and Motivation

Raycasting is a foundational technique in interactive 3D applications, used to translate 2D user input (such as mouse position) into meaningful 3D world coordinates.

4.5.1.2 Algorithmic Process

Screen-to-World Mapping:

The method `Camera.main.ScreenPointToRay(Input.mousePosition)` projects a ray from the camera through the mouse cursor position into the 3D scene.

Collision Detection:

The `Physics.Raycast(ray, out hit)` call checks whether the ray intersects any physical colliders in the scene. If a hit is detected, the hit location (`hit.point`) is extracted as the precise 3D coordinate.

```
Frequently called 13 usages adairh +1
public static bool RaycastMouse(out Vector3 hitPos)
{
    var ray = Camera.main.ScreenPointToRay(Input.mousePosition);

    if (Physics.Raycast(ray, out var hit))
    {
        hitPos = hit.point;
        return true;
    }

    hitPos = Vector3.zero;
    return false;
}
```

Figure 4.58 Raycasting determination

4.5.2 Multiplayer Synchronization

4.5.2.1 Operation - Based Network Model

Definition and Rationale

Real-time collaboration in educational geometry environments demands high responsiveness and absolute consistency of the geometric state across all participants. In this system, synchronization is achieved through an ***operation-based model***: rather than transmitting entire geometric states, only discrete “operations” (state transitions) are broadcast. This approach draws from principles in distributed systems, notably operational transformation and event sourcing, but is tailored for pedagogically sequenced geometry construction.

Algorithmic Workflow

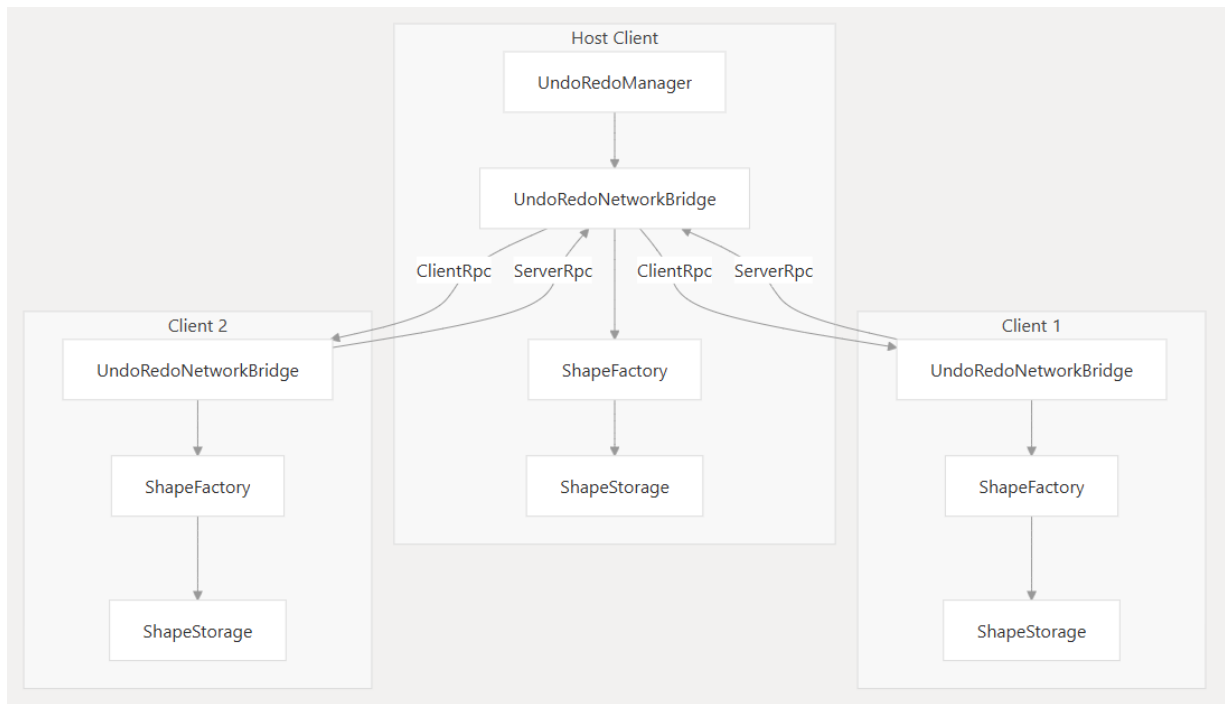


Figure 4.59 RPC core Algorithmic Workflow

4.5.2.2 Shape Data Serialization

Definition

Serialization transforms the in-memory geometric model into a platform-agnostic format for transmission and storage. This system implements a *flat, versioned schema* optimized for minimalism and human readability.

Schema Example

```
{ "points": [ {"id":1, "pos":[0,0,0]}, ... ],  
  "edges": [ {"id":101, "p1":1, "p2":2}, ... ],  
  "faces": [ {"id":201, "vertices":[1,2,3]}, ... ],  
  "labels": {"1":"A", "2":"B"}}
```

Technical Details:

Flat Structure: All primitives are serialized in separate lists, referenced by unique IDs. Dependencies (e.g., edge endpoints) are represented by ID, ensuring referential integrity.

```
public class Shape : NetworkBehaviour
{
    [SerializeField] public string ShapeId;  ⚙️ Unchanged

    [SerializeField] private string shapeType;  ⚙️ Unchanged

    protected readonly List<Point> pivotPoints = new();

    protected readonly List<ISetting> settings = new();
    👤 adairh
    public string ShapeType => shapeType;

    ⚡ Frequently called  📄 15 usages
    public ShapeData Data { get; protected set; }
    📄 1 usage  👤 adairh
    public IReadOnlyList<Point> PivotPoints => pivotPoints;
    👤 adairh
    public IReadOnlyList<ISetting> Settings => settings;
```

Figure 4.60 ShapeData class to store serializable object

4.5.3 AI Assistant

4.5.3.1 Prompt Construction Strategy

Prompt construction in this project is the process of assembling a structured, context-aware input for the GPT-based assistant. The goal is to maximize the accuracy, relevance, and clarity of AI-generated responses within the geometry learning context. Diving into creating the friendliest and tutor - like AI.

4.5.3.2 Algorithmic Steps

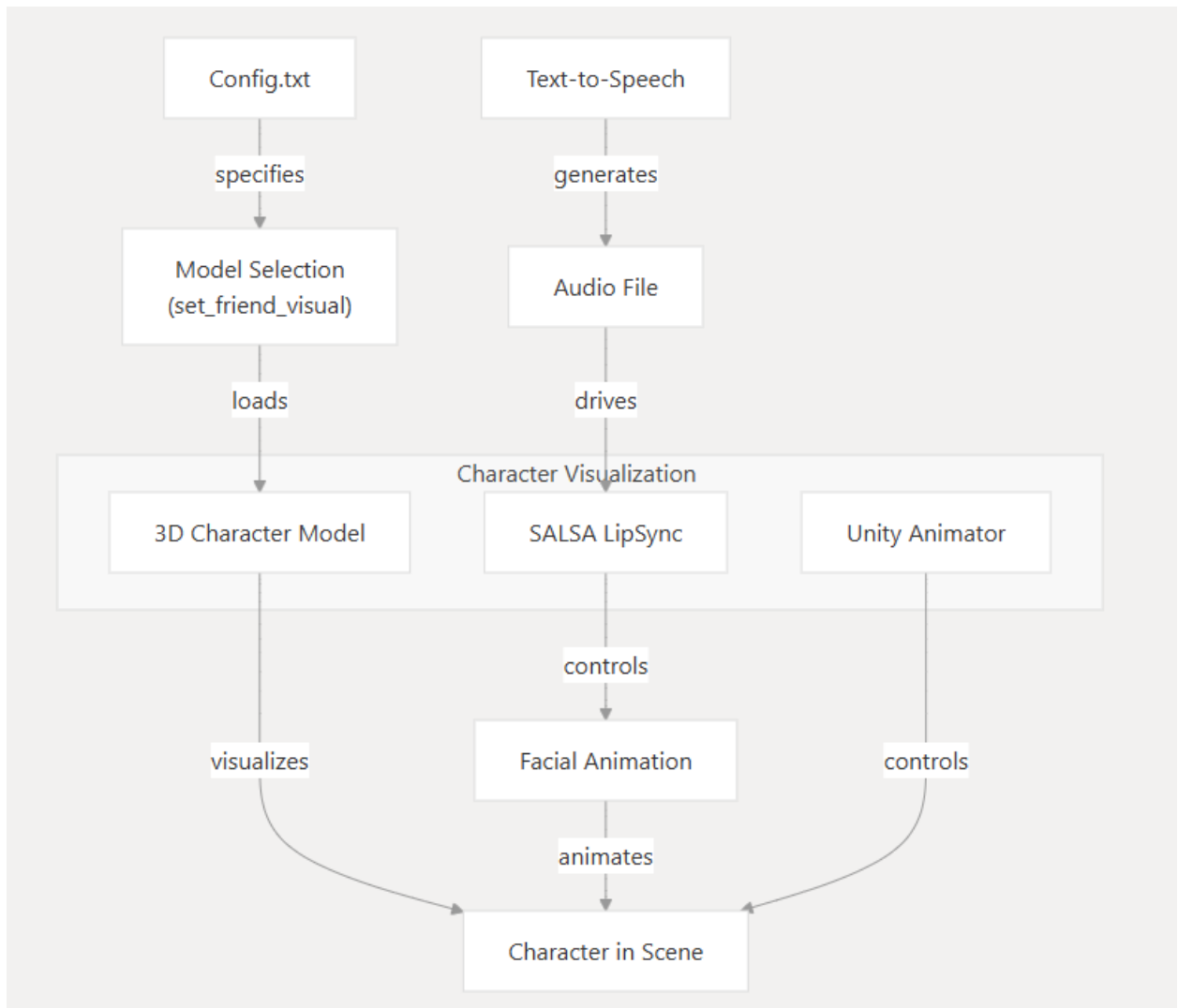


Figure 4.61 AI Character builder Algorithmic Workflow

4.5.4. AI - Driven Shape Generation Pipeline

4.5.4.1. Definition and Motivation

The AI-Driven Shape Generation Pipeline is the algorithmic process by which the system interprets structured outputs from an LLM (Large Language Model, e.g., ChatGPT) and instantiates corresponding geometric constructs directly within the collaborative 3D environment. This pipeline operationalizes “AI-to-shape” translation, enabling students or teachers to describe complex geometric situations in natural language (or select AI-generated suggestions), and have the software deterministically realize those constructions as editable objects - preserving all the system’s features for constraint enforcement, multiplayer sync, and undo/redo.

4.5.4.2. Algorithmic Workflow

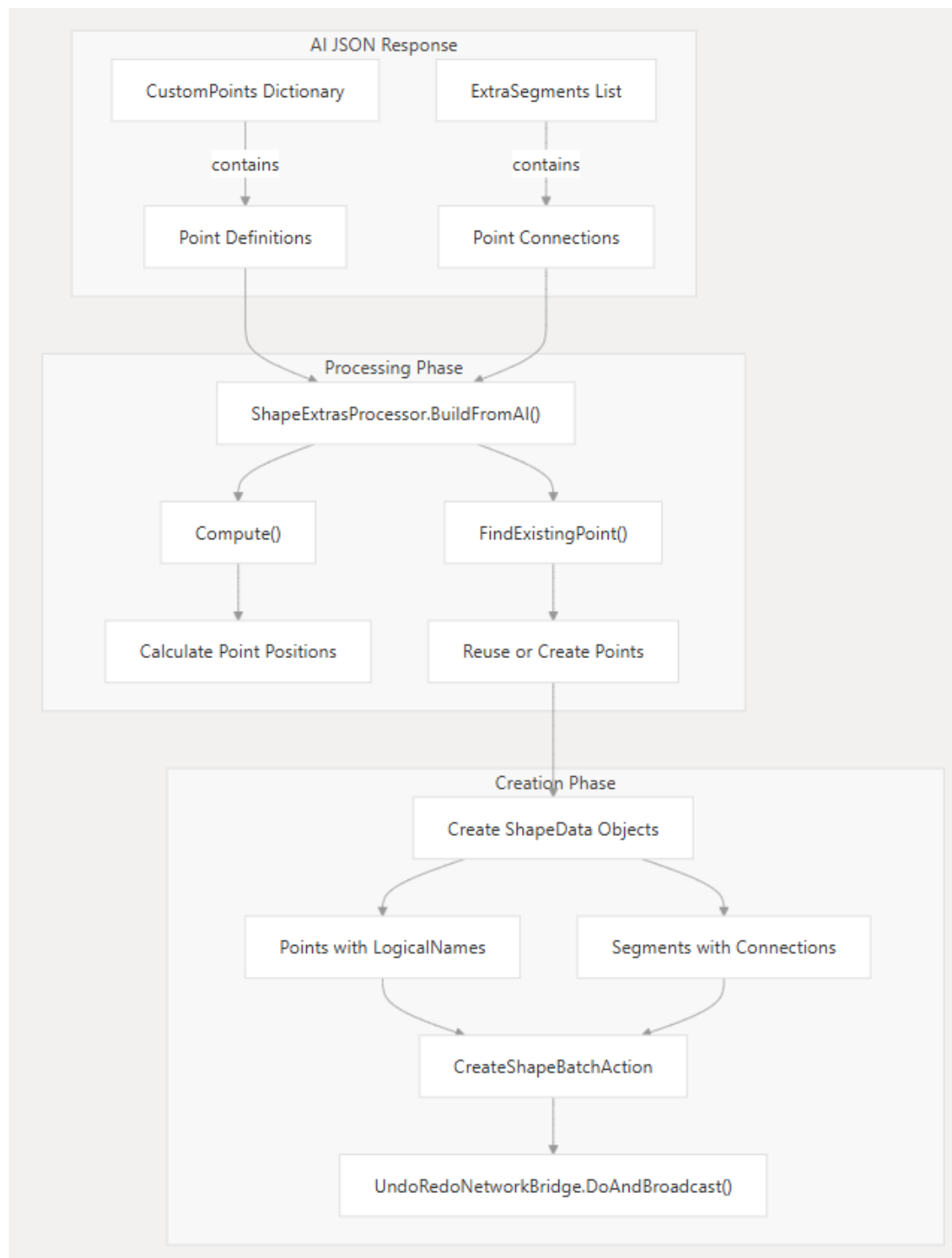


Figure 4.62 AI Create shape Algorithmic Workflow

4.5.5 Manual Shape Construction via Partial Input Fields

4.5.5.1 Educational Motivation and User Flow

This feature enables users to build geometric shapes by entering *partial information*, without requiring them to provide every field upfront. The system automatically infers the missing data using built - in geometric formulas and relationships.

For example, to construct a rectangular prism (box), users might input just the width and volume. The system will compute the height and length using internal logic and complete the shape.

This mirrors *real - world learning scenarios* in spatial geometry, where students often receive *incomplete data* and must deduce the rest. The system provides feedback, error checking, and interactive construction, making it ideal for both exploration and verification in classroom or self - study settings.

4.5.5.2 System Architecture and Inference Engine

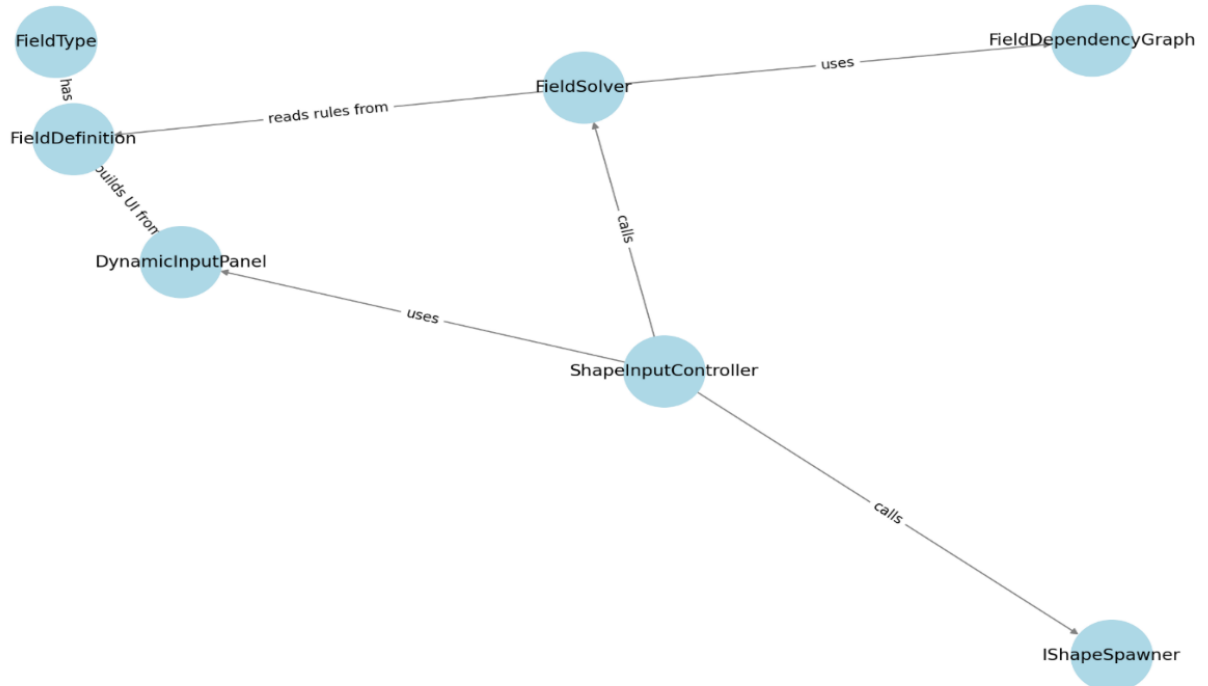


Figure 4.63 Working flow of Manual Shape Construction via Partial Input Fields

4.6 User Interface Design

4.6.1 UI Design Goals

The User Interface (UI) of the application is designed with *three primary goals*:

- *Pedagogical Support*: Facilitate spatial geometry learning through clear visuals, intuitive interaction patterns, and minimal cognitive load. The UI allows learners to focus on geometric reasoning rather than interface complexity.
- *Functional Integration*: Seamlessly support core technical modules - 3D drawing tools, AI tutoring, and gamified learning - with minimal transitions and consistent design language.
- *User Experience (UX)*: Ensure smooth user flow across different functionalities (drawing, gaming, collaboration), regardless of the user's technical proficiency. This includes quick access, real-time feedback, and accessibility on both PC and mobile devices.

4.6.2 Application Layout

The application follows a module-based navigation system, accessible from a unified Home Screen. Each mode presents its own specialized interface:

Drawing Mode

- Toolbar with icons for Points, Segments, Polygons, Vectors, Constraints, and Snapping
- Real-time 3D viewport with interactive manipulation
- Undo/Redo and Delete buttons always visible
- AI helper for generating shapes and explaining assignments

Online Mode

- Online selection menu: Create room and Join room
- Popup with input information: Room ID and Password to create or join room
- Simple enter new session and exit flow

Game Mode

- Game selection menu: Collect Coin and an upcoming game

- Each game has its own progression screen, feedback panel, and scoring elements
- Simple retry and exit flow

AI Tutor Mode

- Panel with voice input and some other button
- Animated teacher and classroom environment
- Response displayed as text and audio

4.6.3 Color Palette & Fonts

To minimize visual fatigue and align with color psychology for education:

- Main colors: Calm green and bright yellow, evoking clarity and motivation.
- Text and Highlight: Black-white-grey for contrast, yellow-orange gradient for important action buttons and grey for normal buttons.
- Fonts: A modern font (Montserrat SemiBold) chosen for readability across screen sizes.

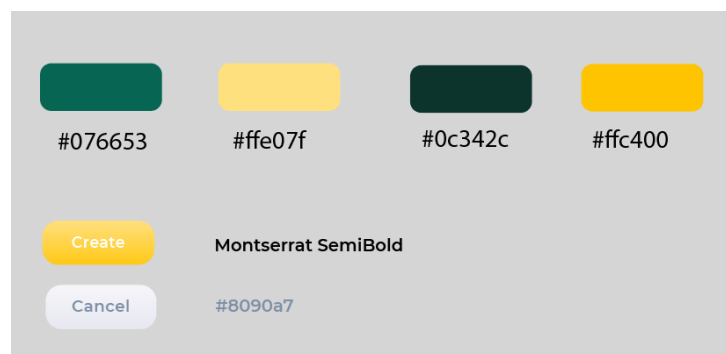


Figure 4.64 Color Palette

4.6.4 User Flow & Routing Logic

The interface was designed according to a *structured user flow*, supporting smooth navigation between screens

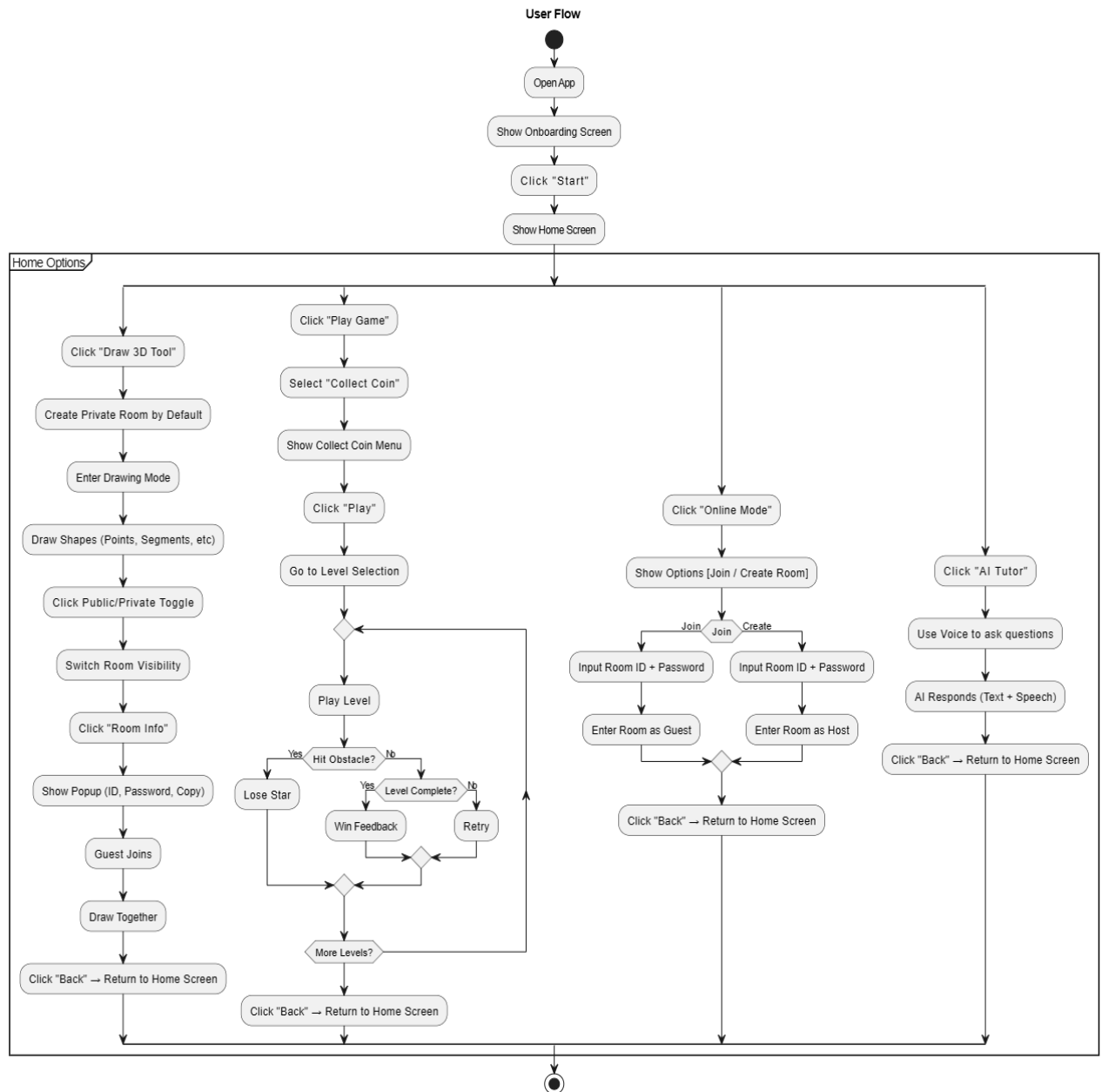


Figure 4.65 User flow Diagram

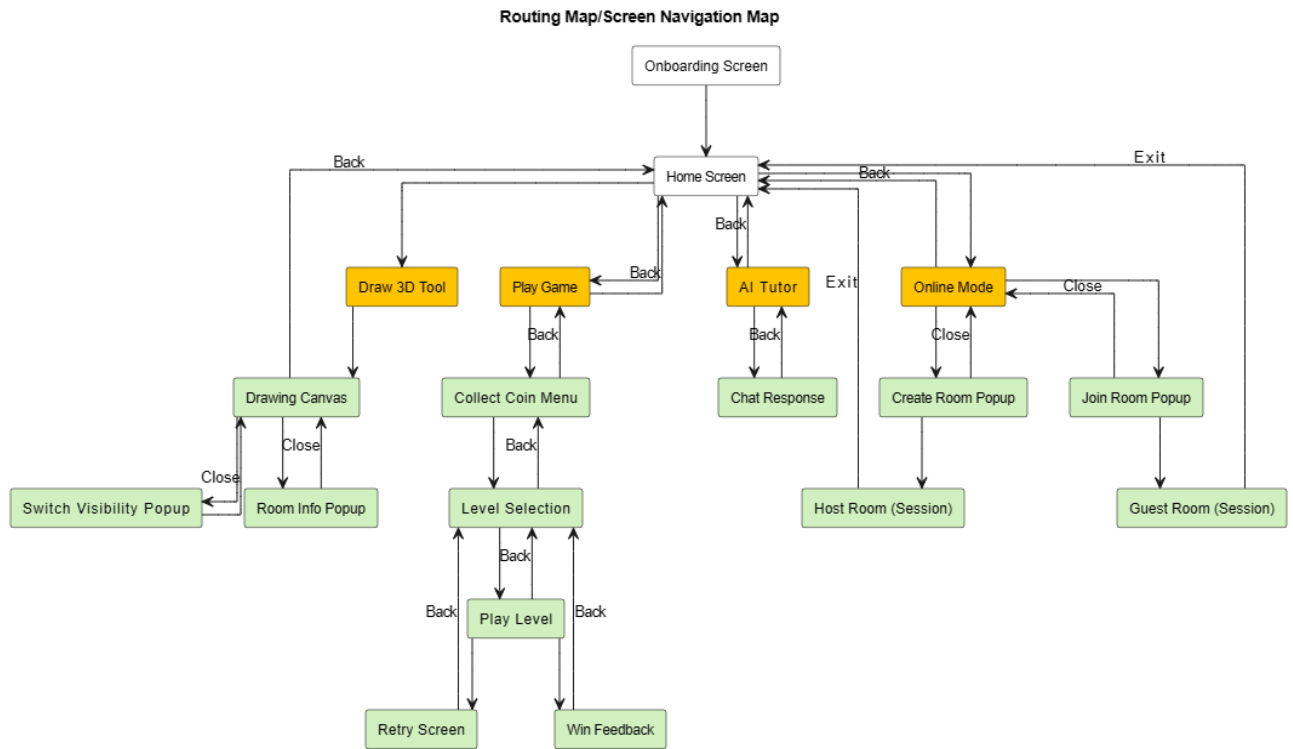


Figure 4.66 Routing Map

This structure helps reduce friction and enables users to resume learning quickly at any point in the application.

4.6.5 Detail Screens Design

Screen 1 - Onboarding Screen



Figure 4.67 Onboarding screen

Item	Description
Screen Name	Onboarding Screen
Purpose	Welcome the user and introduce the app's learning slogan
Main Elements	App logo, mascot character, slogan ("Tap to learn - Turn to explore - Build to remember"), Start button
User Actions	Tap "Start" to enter the Home Screen

Table 4.5 Onboarding Screen Specification

Screen 2 - Home Screen

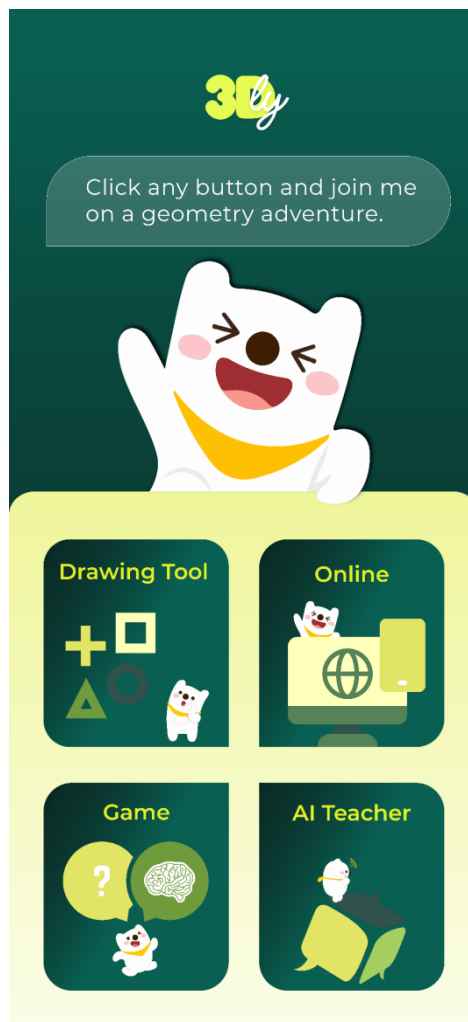


Figure 4.68 Home screen

Item	Description
Screen Name	Home Screen
Purpose	Central hub to access Drawing Tool, Game, Online mode and AI Tutor
Main Elements	4 big function cards, mascot waving, motivational speech bubble
User Actions	Tap any module card to enter its section

Table 4.6 Home screen Specification

Screen 3 - Online Mode Menu Screen

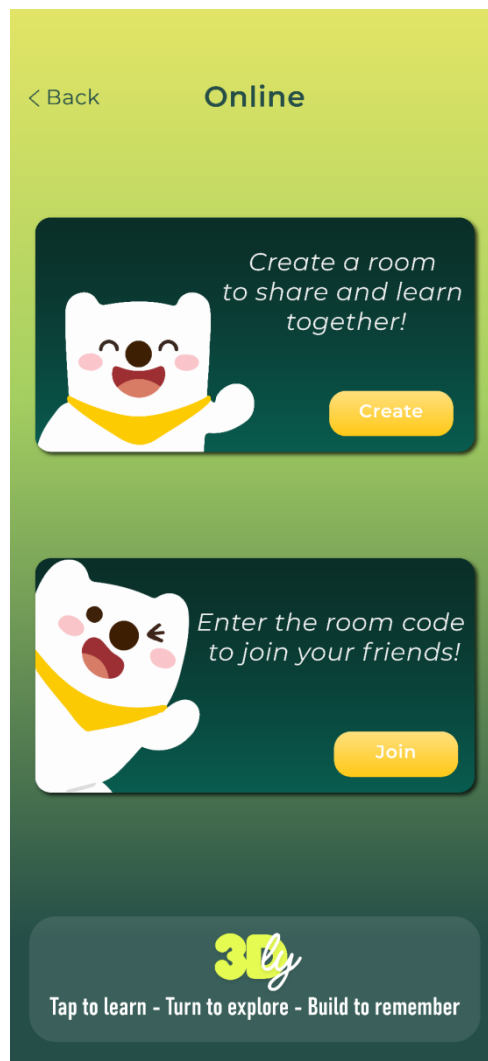


Figure 4.69 Online mode screen

Item	Description
Screen Name	Online Mode Selection Screen
Purpose	Lets user choose between joining or creating a collaborative session
Main Elements	Create Room card, Join Room card, mascot illustrations
User Actions	Tap “Create” or “Join” to enter multiplayer popup

Table 4.7 Online mode screen Specification

Screen 4 – Game Menu Screen

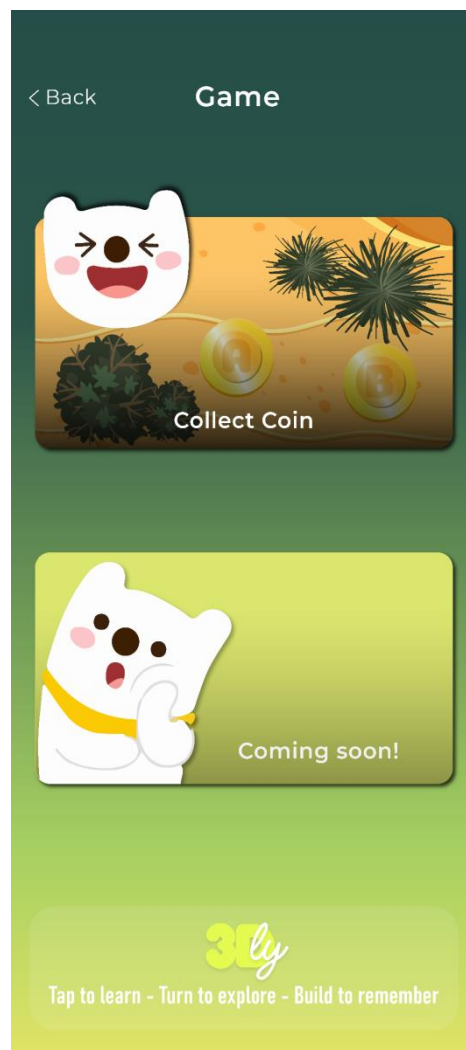


Figure 4.70 Game mode screen

Item	Description
Screen Name	Game Selection Menu
Purpose	Lets user pick a game to play (currently: Collect Coin game)
Main Elements	Game cards (Collect Coin - active, Coming Soon - disabled), mascot expressions

User Actions	Tap “Collect Coin” to view level selection
---------------------	--

Table 4.8 Game mode screen Specification

Screen 5 – AI Tutor Screen

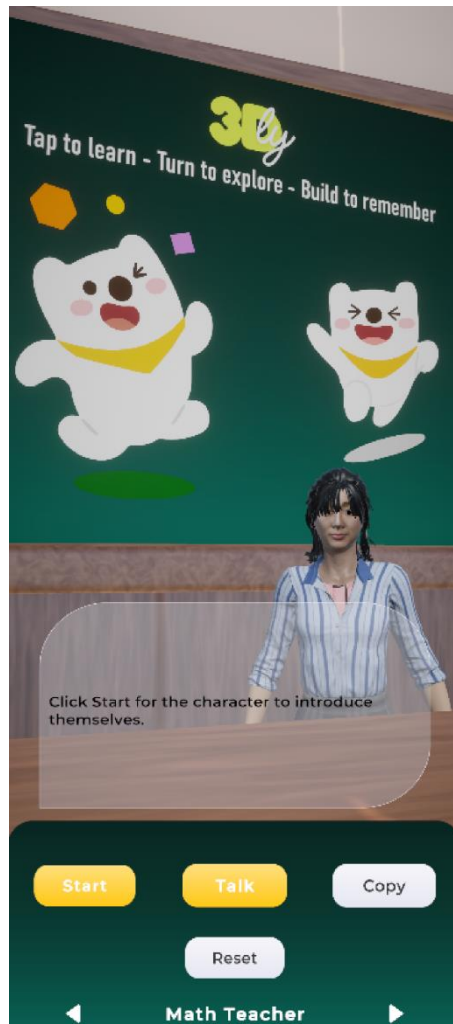


Figure 4.71 AI Tutor screen

Item	Description
Screen Name	AI Tutor Interaction Panel
Purpose	Allows user to interact with the AI teacher via voice input and view responses.

Main Elements	3D teacher model, chalkboard background, mascot visuals, voice input buttons, response area
User Actions	Tap “Start” for intro, use “Talk” to ask questions, “Copy” to copy response, “Reset” to clear chat.

Table 4.9 AI Tutor screen Specification

Screen 6 – Join Room (Popup)

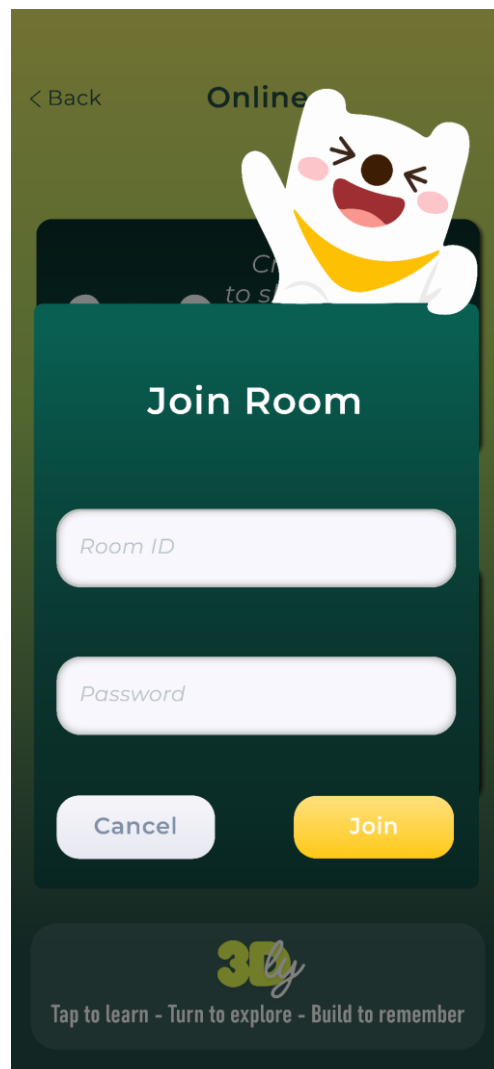


Figure 4.72 Join room screen

Item	Description
Screen Name	Join Room Popup

Purpose	Allows a user to join an existing multiplayer room as Guest.
Main Elements	Room ID field, Password field, mascot overlay, Cancel and Join buttons
User Actions	Input valid room info and tap “Join” to enter session or tap “Cancel” to return.

Table 4.10 Join room pop-up screen Specification

Screen 7 – Create Room (Popup)

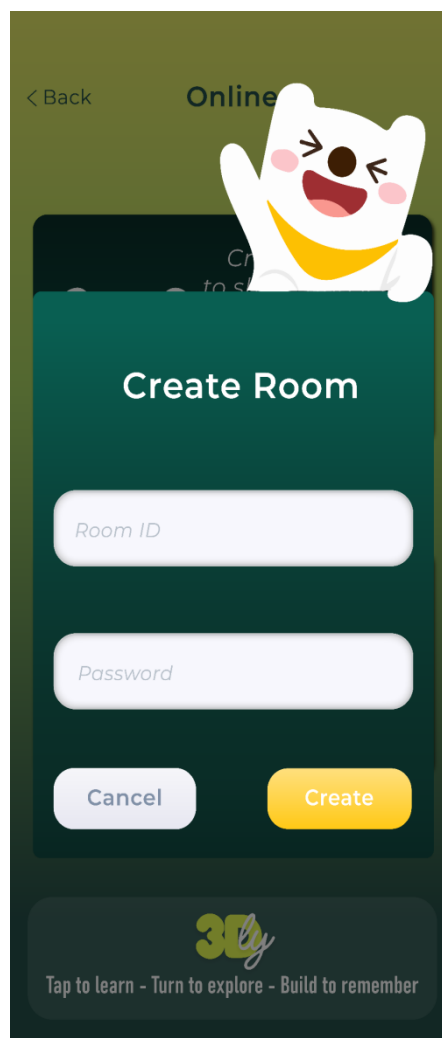


Figure 4.73 Create room screen

Item	Description
Screen Name	Create Room Popup

Purpose	Enables users to create a new multiplayer (online) room as Host.
Main Elements	Input fields for Room ID and Password, mascot artwork, Cancel and Create buttons
User Actions	Enter room credentials and tap “Create” to begin hosting, or “Cancel” to go back.

Table 4.11 Creation room pop-up screen Specification

Screen 8 – Drawing Mode

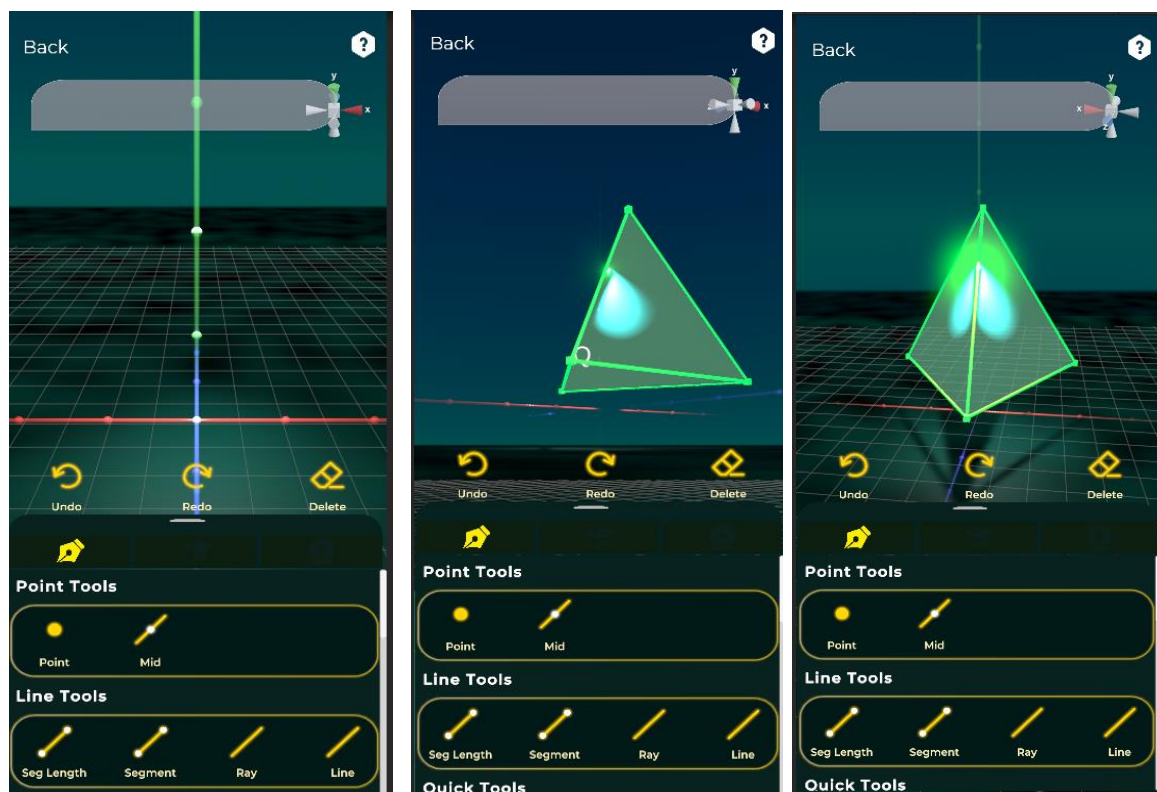


Figure 4.74 In - app drawing UI

Item	Description
Screen Name	Drawing Canvas
Purpose	Allows users to interactively draw and manipulate 3D geometric shapes

Main Elements	– 3D Viewport with X, Y, Z axes – Basic Tool Panel (Move, Point, Segment...) – Shape Spawning Section (Equilateral Triangle, Tetrahedron, etc.) – Undo/Redo/Help Icons
User Actions	1. Tap on a tool to activate drawing mode 2. Tap on shapes to spawn with custom parameters 3. Long press or drag to manipulate 4. Tap AI helper to get help

Table 4.12 Geometry Drawing mode Specification

Screen 9 – Spawn Shape Input

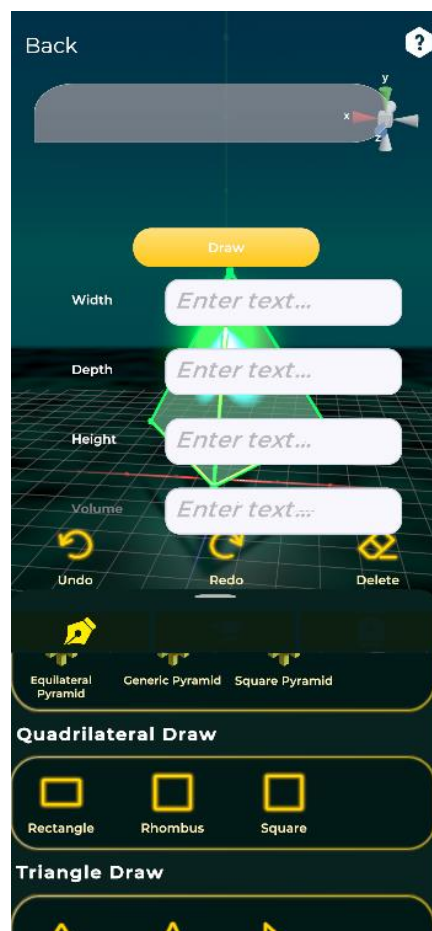


Figure 4.75 Spawn Shape Input

Item	Description
Screen Name	Shape Input Popup

Purpose	Allow users to specify dimensions before spawning complex shapes
Main Elements	– Form fields (e.g., Width, Height, Length) – Apply and Cancel buttons
User Actions	1. Fill in values 2. Press “Apply” to spawn the shape into the 3D space

Table 4.13 Spawn shape input pop-up screen Specification

Screen 10 – AI Helper (in Drawing Mode)

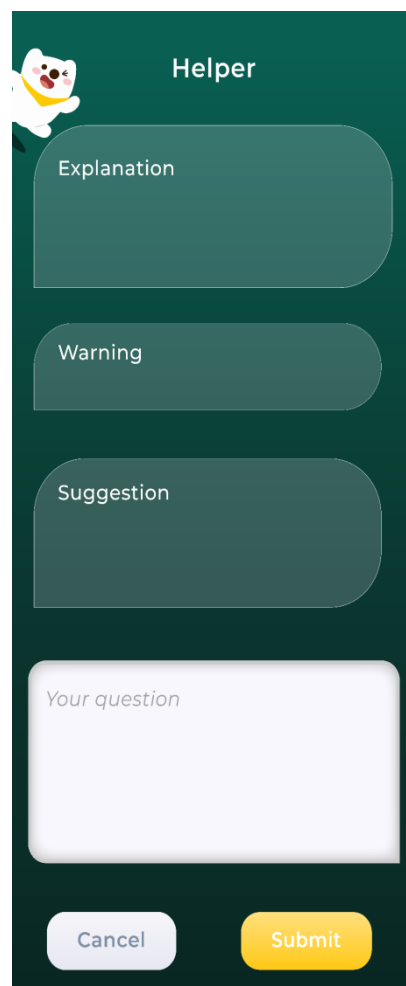


Figure 4.76 Drawing Assistant screen

Item	Description
Screen Name	Drawing Assistant
Purpose	Provide feedback, suggestions, and warnings based on user input

Main Elements	– Response boxes (Explanation, Warning, Suggestion) – Input textbox – Submit and Cancel buttons
User Actions	1. Type a question about current drawing 2. Tap “Submit” to receive AI-generated shape and get analysis 3. View categorized responses for better understanding

Table 4.14 Drawing Assistant screen Specification

Screen 11 – Collect coin Game Menu

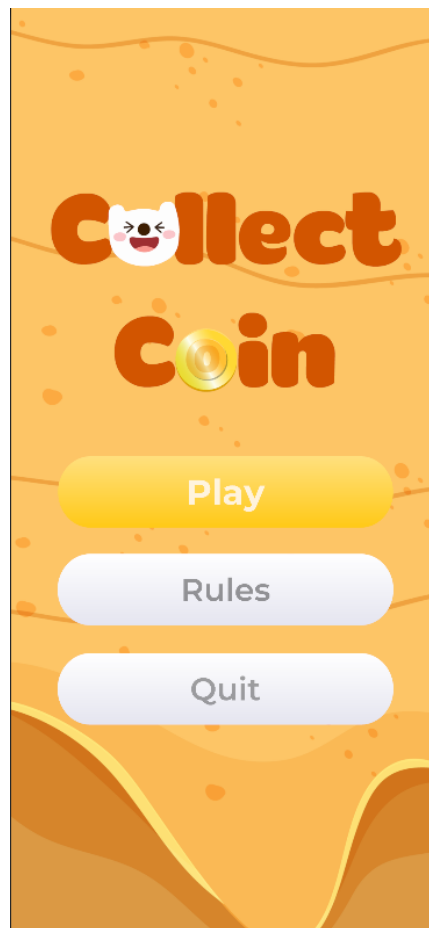


Figure 4.77 Game UI

Item	Description
Screen Name	Collect coin Game Selection Menu

Purpose	Let users choose a mini game to play
Main Elements	Game name tag and 3 buttons with Play, Rules and Quit
User Actions	1. Tap a game card to proceed to its specific menu 2. Select the menu: – Play to go level selection screen, – Rules to read How to play game and reset game, – Quit to exit game

Table 4.15 Collect coin Selection menu Specification

Screen 12 – Level Selection



Figure 4.78 Level UI

Item	Description
Screen Name	Level Selection
Purpose	Allow users to pick a level in “Collect Coin” mini - game
Main Elements	10 level buttons with star ratings (locked/unlocked status can be added)
User Actions	Tap a level number to enter that level, use back button to return to game menu

Table 4.16 Level selection Specification

Screen 13 – Level Gameplay

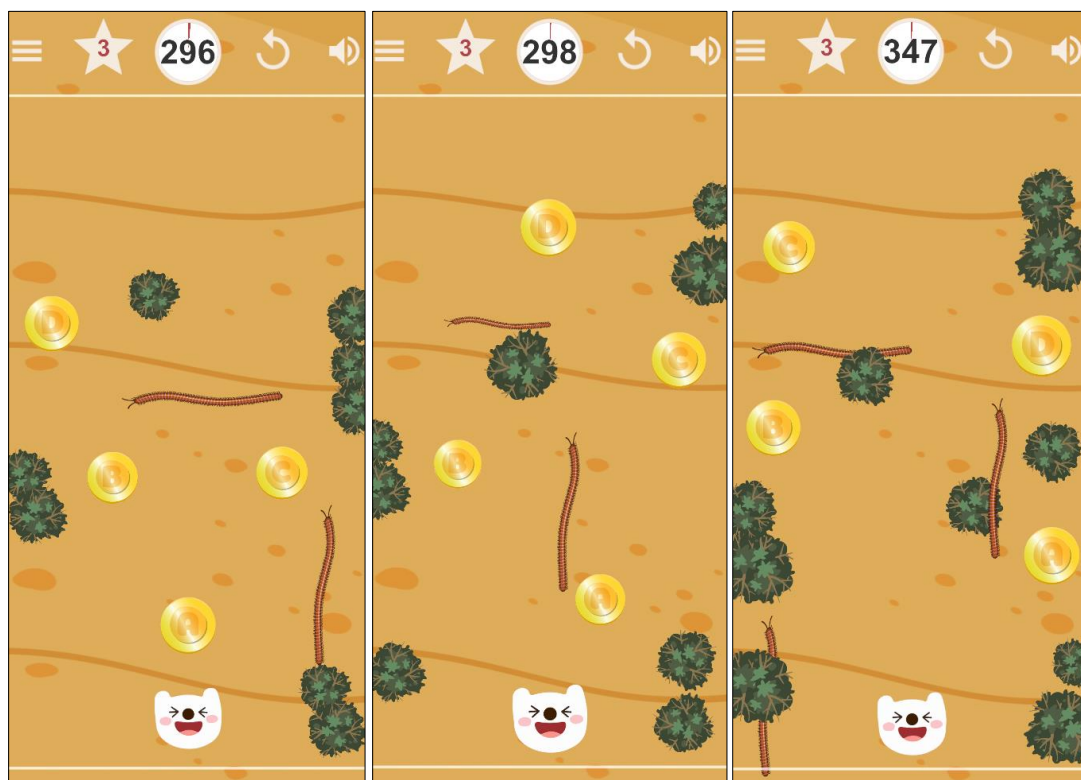


Figure 4.79 Game scene UI

Item	Description
------	-------------

Screen Name	Collect Coin – Gameplay
Purpose	Navigate character to collect coins
Main Elements	Timer, Star counter, Restart and Mute buttons, controllable mascot character, coins (A, B, C...)
User Actions	Drag character to collect coins and avoid obstacles

Table 4.17 Gameplay Specification

Screen – Quiz Popup

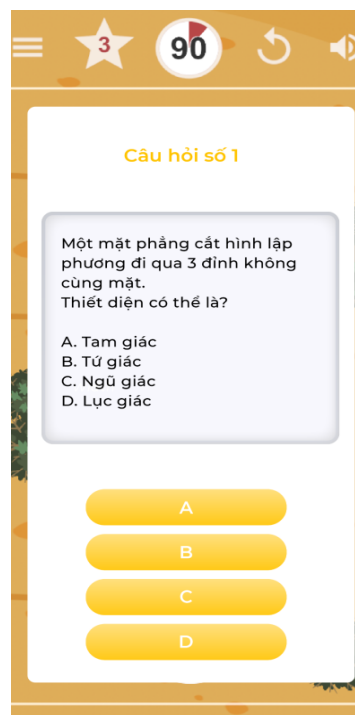


Figure 4.80 Quiz Popup

Item	Description
------	-------------

Screen Name	Coin - triggered Quiz
Purpose	Assess geometry understanding through MCQs after each coin collected
Main Elements	Geometry question, multiple choice answers (A, B, C, D), styled container
User Actions	Tap on the answer to proceed; correct answers retain stars, incorrect lose one

Table 4.18 Quiz pop-up Specification

In conclusion, this chapter has presented the comprehensive architectural and component - level design of our interactive learning platform. From the modular Geometry Engine and Collect Coin game to the intelligent AI Tutor System, we have detailed how each part is structured, specific User Interface design and how key algorithms enable features, providing a robust blueprint for implementation.

CHAPTER 5. SYSTEM IMPLEMENTATION

Following the detailed system analysis and design presented in Chapter 4, this chapter outlines the practical implementation of our educational software solution. It elaborates on the development environment, the specific technologies employed, and the methodologies used to translate the architectural designs into a functional and interactive learning platform, culminating in the testing and validation of the final product.

5.1 Development Environment

5.1.1 Unity 2022 LTS – Real-Time 3D Engine

To build a highly interactive learning environment focused on spatial geometry, this project uses Unity 2022 LTS, a real-time 3D development platform typically used in game and simulation industries. Unlike conventional tools for building educational apps (e.g., Android Studio, React Native), Unity offers a physics-enabled 3D space, advanced rendering capabilities, and intuitive interaction models - making it especially powerful for spatial learning tasks.

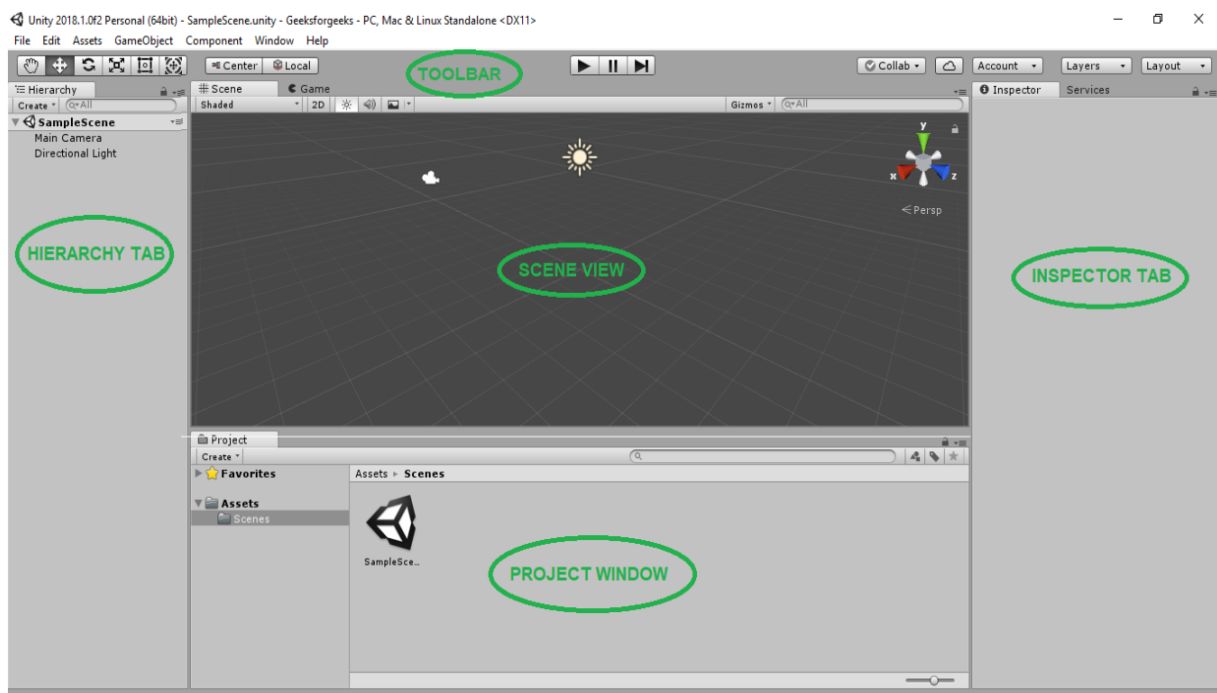


Figure 5.1 Unity default layout

Why Unity?

- Real-time 3D Visualization: Unlike static diagrams or 2D drawing libraries, Unity allows users to interact with geometry in three-dimensions - manipulating shapes dynamically, which is crucial for developing spatial reasoning skills.
- Cross-Platform Output: Unity enables the same project to be exported for both Windows and Android, ensuring consistent experience across PC and mobile devices.
- Visual + Code Hybrid Workflow: Unity supports both drag-and-drop UI creation (via UGUI) and detailed logic scripting (in C#), which made it possible to build features such as snapping, constraint logic, and game-based interaction within one ecosystem.

In this project, Unity serves as more than a game engine - it acts as an educational visualization engine. For example:

- Drawing Tools allow students to construct shapes with real-time feedback.
- TextMeshPro ensures clear, scalable in-app text (e.g., labels, AI responses).
- UGUI is used to build interactive menus, floating panels, and heads-up displays (HUD).

By choosing Unity, the development team could go beyond traditional mobile interfaces and create a spatially immersive, hands-on learning tool - directly addressing the cognitive barriers in geometry education identified in earlier chapters.

5.1.2 C# – Core Programming Language

The application logic was fully implemented in C#, Unity's primary scripting language.

Usage areas:

- Geometry creation and manipulation (e.g., Point, Segment, ShapeData classes).
- Constraint enforcement logic.
- AI communication (via HTTP APIs and JSON).
- Game logic (state control, scoring, star system, level progression).
- Undo/Redo and multiplayer sync.

The language's object-oriented nature allowed for clean separation between gameplay, geometry, and networking systems, improving scalability and maintainability.

5.1.3 Visual Studio 2022 – Integrated Development Environment

Development was performed using Visual Studio 2022 Community Edition, which integrated seamlessly with Unity.

Benefits:

- IntelliSense and auto-completion for faster development.
- Breakpoint debugging directly in Unity's Play Mode.
- Integration with NuGet for third-party libraries (e.g., Newtonsoft.Json for parsing API responses).

This IDE was essential for maintaining productivity during iterative development cycles.

5.1.4 Git & GitHub Desktop – Version Control and Collaboration

The project used Git (via GitHub Desktop) as the version control system, enabling distributed and organized collaboration.

Workflow:

- Feature branches were used for each module (Drawing, AI, Multiplayer, UI).
- Pull requests ensured code review and functional verification before merging.
- GitHub Issues were used for tracking bugs and planning tasks.

To simplify use across all team members, GitHub Desktop was adopted as the GUI client, enabling:

- Visual diff and commit history.
- Conflict-free branching and merging.
- Clear collaborative process even for less experienced developers.

5.1.5 Unity Netcode for GameObjects - Multiplayer Networking

For the collaborative drawing feature, the system used Unity Netcode for GameObjects - a high-level networking solution built for Unity.

Features:

- Real-time sync of geometric shapes across multiple devices.
- Authority management (Host/Guest).
- Automatic state propagation with minimal boilerplate code.

This allowed creation of a shared interactive board, supporting real-time classroom interaction or peer collaboration.

5.1.6 Unity Relay & Lobby - Multiplayer Connectivity

Instead of managing custom servers, the system leveraged Unity Relay and Unity Lobby services to:

- Enable serverless room creation and discovery using room codes.
- Ensure secure peer-to-peer communication behind NAT firewalls.
- Support 2–4 user sessions without infrastructure setup.

This minimized development complexity while maintaining scalability for classroom or small group learning environments.

5.1.7 OpenAI APIs – GPT & Whisper Integration

To power the AI tutor, the system integrated:

- OpenAI GPT (ChatGPT) for geometry explanations and feedback.
- OpenAI Whisper for speech-to-text processing of voice input.
- Integration was handled through Unity’s UnityWebRequest system. Responses were parsed via JSON deserialization and rendered via TextMeshPro, while voice feedback was delivered via Unity’s built-in Text-to-Speech and avatar animation.

This enabled natural language interaction, improving accessibility and learning flexibility.

5.1.8 Adobe Illustrator – UI/UX Asset Design

For UI design and custom visuals, the team used:

- Adobe Illustrator for iconography, color testing, and branding elements.
- Wireframing on paper for interaction planning and user flow diagrams on draw.io
- Design principles followed cognitive psychology recommendations: low-saturation colors (green, yellow), minimalist layouts, and large touch targets for mobile usability.

5.2 Deployment Environment

The deployment environment defines the hardware and software contexts in which the 3D geometry learning platform operates. The system was designed for use on local end-user devices rather than cloud-based servers, focusing *on offline accessibility, cross-platform consistency, and low-setup deployment* for students and educators in Vietnamese high schools.

5.2.1 Target Platforms and Hardware Specifications

The application supports deployment on the following platforms:

Platform	Build Type	Minimum Hardware Requirements	Target Users
Android	APK (v5.0+)	2GB RAM, ARM64 CPU, HD display (1280x720)	Students (mobile learning)
Windows	Portable .exe (64-bit)	Dual-core CPU, 4GB RAM, DirectX 11 GPU, 1366x768 screen	Teachers, students (lab/class)

Table 5.1 Platform deployment Specification

This dual-platform deployment ensures flexibility for both individual study (on mobile) and classroom instruction (on desktop).

iOS and macOS are not supported due to the lack of Apple hardware, developer provisioning, and testing infrastructure within the academic scope.

5.2.2 Software Environment and Runtime Requirements

Component	Specification/Version
Game Engine	Unity 2022 LTS
Programming Language	C# (Unity .NET scripting runtime)
IDE	Visual Studio 2022 Community
APIs	OpenAI (GPT-3.5, Whisper), Unity Relay & Lobby
UI Framework	Unity UGUI + TextMeshPro
Build Tools	Android SDK, Unity Build Pipeline
OS Compatibility	Android 5.0+ / Windows 10+ (64-bit)

Table 5.2 Software environment version

Each build is compiled and exported via Unity’s native build system. Conditional preprocessor directives (e.g., if UNITY_ANDROID) ensure input systems and UI layouts adapt automatically to platform-specific environments.

5.2.3 Hosting and Deployment Model

This application follows an *on-device deployment model* rather than centralized hosting. Each device runs the application natively without relying on cloud services, enabling offline usage and minimizing technical barriers for schools with limited infrastructure.

Deployment Method	Description
Android	APK file is sideloaded or distributed via USB; no Google Play dependency.
Windows	Standalone portable executable (.exe) run locally; no installation required.

Table 5.3 Deployment Method description

All multiplayer connections (e.g., shared geometry boards) are peer-to-peer using Unity Relay and Lobby Services, which eliminates the need for dedicated backend servers.

5.2.4 Accessibility and Usability Considerations

- No login required: Students can use the app immediately - mirroring lightweight tools like *Kahoot*.
- Touch and mouse support: Designed for both mobile gestures and desktop precision.
- Resolutions: UI scales to a fixed resolution (1284x2778 on Android/iOs), it is a limitation of the application.
- Color scheme: Green/yellow palette chosen to reduce visual fatigue and math anxiety.
- Multiplatform consistency: All features (drawing, AI tutor, games) are available on both platforms.

5.2.5 Performance and Network Optimization

To ensure smooth performance across device types:

Optimization Target	Techniques Used
Mobile	Lightweight shaders, compressed textures, object pooling, capped resolution
Desktop	High-fidelity rendering, real-time multiplayer with <200ms sync latency
Networking	Operation-based synchronization via Unity Netcode & Relay

Table 5.4 Optimization target and techniques used

Test results show consistent frame rates (30-60 FPS on Android, >60 FPS on Windows) under normal classroom use.

Summary

The system is designed to operate directly on user devices, leveraging Unity’s cross-platform capabilities to ensure scalable deployment without requiring cloud infrastructure. This ensures ease of access, low setup time, and performance reliability in both mobile and classroom learning contexts.

5.3 System Testing

This section presents the testing strategy, documented test cases, and test results used to evaluate the stability, correctness, and cross-platform functionality of the developed geometry learning platform. The testing process ensured that all major modules (Drawing Mode, AI Tutor, and Collect Coin Game) performed as intended under realistic usage conditions.

5.3.1 Testing Approach

A multi-layered testing methodology was employed throughout development:

- Unit Testing: Applied to geometry logic (e.g., snapping, constraint enforcement), ensuring that shape calculations, constraint application, and undo/redo actions performed as expected in isolation.
- Integration Testing: Focused on subsystems such as AI interaction (voice-to-text to response) and multiplayer synchronization. This ensured coherent behavior between different components such as UI input, geometry logic, and networking.

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- System Testing: Conducted using both Unity Editor simulations and deployment on physical devices (Android smartphones and Windows laptops) to simulate real-world learning scenarios.
- Platform-Specific Testing: Special attention was paid to:
 - Performance on low-end Android devices (e.g., 2GB RAM, 720p screen).
 - UI responsiveness on small screen sizes.
 - Behavior in offline mode and multiplayer reconnection.

Testing objectives

- Validate core learning workflows (drawing, AI support, gameplay).
- Ensure multiplatform stability (Windows, Android).
- Assess UI usability and scalability.
- Confirm multiplayer session consistency.

5.3.2 Test Cases

No.	Test Case Description	Expected Result	Actual Result	Status
TC01	Draw a segment between two points	Segment appears correctly with snapping if enabled	Segment drawn with correct snapping	Pass
TC02	AI Tutor answers a voice question	Converts voice to text and provides correct response	Answer generated; minor phrasing issue	Pass
TC03	Collect Coin: correct slicing unlocks quiz	Coin collected; quiz appears with four options	Quiz loaded correctly with four choices	Pass
TC04	Incorrect quiz answer in game	Deduct one star	Star deducted properly; stable behavior	Pass

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TC05	Online join session (guest)	Guests see host's drawing actions in real time	Updates visible with ~1 second delay	Pass
TC06	Online disconnection recovery	App attempts to reconnect or shows an error	No reconnect; returns silently to Home	Fail
TC07	Use AI Tutor offline	Detects offline mode; shows fallback message	No response shown; silent failure	Fail
TC08	Game performance on 2GB Android	Loads in <10s; maintains >30 FPS	Loads in 9s; FPS drops to ~20 when too many objects	Partial
TC09	Undo/Redo drawing action	Undo or redo updates geometry and constraints correctly	Undo/Redo worked with accurate constraint handling	Pass
TC10	Resize UI on a small 5-inch screen	UI remains visible and clickable	Some overlap in quiz answer buttons	Partial

Table 5.5 Key Test Cases and Outcomes

5.3.3 Test Results

Total Test Cases: 10

Passed: 6

Partial: 2

Failed: 2

Summary of issues identified

Issue Area	Description	Planned Action
Offline AI Fallback	No response displayed when offline	Add user feedback message for offline mode
Multiplayer Reconnection	Silent failure after disconnection	Implement reconnect prompt or auto-retry

UI Scaling on Small Devices	UI elements overlap on small screens	Improve responsive UI layout
Mobile Game FPS Drop	Visual performance degrades with too many shapes	Optimize mesh rendering and pooling

Table 5.6 Summary of issues identified

The system was deemed functionally stable for core educational use cases, including drawing, AI tutoring, and gameplay. While some user experience limitations were observed under constrained conditions, these did not block primary learning tasks.

This chapter has outlined the development environment, the specific technologies employed, and the rigorous testing processes undertaken. By leveraging Unity, C#, and integrated AI and networking APIs, we have successfully translated our design into a functional and responsive educational tool, validating its performance and adherence to pedagogical goals

CHAPTER 6. SYSTEM EVOLUTION

The current iteration of our interactive learning platform represents a significant evolution from its foundational predecessor. This chapter details the transformative advancements that have revolutionized the system, showcasing how our new architecture and integrated features - including Modularized codebase, Pedagogical Enhancements, User Experience and Interface Design, Project Vision profoundly deepen its ability to satisfy the project's scope and embody its core educational ideology.

6.1. Technical Improvements

6.1.1. Modularized codebase

In the previous version, our development relied extensively on specific functionalities created through extensive hard coding. This approach initially appeared effective; however, due to limited experience with real-world scenarios, the application demonstrated technical complexity without providing opportunities for future enhancements or adaptability. The system became narrowly experimental, restricting its broader applicability.

The new project iteration involved a comprehensive refactoring of the entire system into a modularized, flexible architecture specifically tailored to educational purposes. This restructuring introduced intentional constraints to simplify user interactions, making the application accessible to non-technical users. Additionally, new integrated modules such as AI-driven tutoring and gamified learning components significantly broadened the scope and utility of the application.

This transformation effectively resolved the previous limitations by aligning the application's functionalities directly with the pedagogical needs of students. The modularized codebase now supports continuous enhancement and scalability, ensuring that the platform remains competitive and relevant within the educational technology market. Furthermore, it lays a solid foundation for future developments, promoting an adaptable and expandable ecosystem suited to evolving educational requirements.

6.1.2. Networked multiplayer

In the previous system, network communication was implemented using peer-to-peer web sockets, which presented significant limitations. This architecture restricted scalability, responsiveness, and practical usability, particularly impacting the user experience due to limited real-time collaborative capabilities.

The current iteration replaced the web socket-based approach with Unity's NetCode system, fundamentally redesigning the networking logic to support a robust client-server architecture. Now, users can effortlessly create and join rooms with clearly defined roles, with hosts managing interactions and clients participating actively in real-time.

This innovation significantly enhanced the collaborative potential of the platform, allowing virtually unlimited user connections and collaborative interactions. It simplified the process for educators and students alike, facilitating dynamic and scalable educational sessions and paving the way for future enhancements involving detailed permission controls and sophisticated collaboration mechanics akin to leading collaborative platforms like *Canva*.

6.1.3. Integrated AI services

Previously, the AI integration was limited to a basic chatbox utilizing OpenAI's API, functioning essentially as a standalone, generalized chatbot with minimal practical impact on user experience. The application offered little differentiation from directly using standard ChatGPT services, leading to suboptimal usage and engagement.

The enhanced system introduces specialized AI modules with distinct educational functionalities. One AI module is specifically designed for spatial geometry problem - solving, automatically parsing natural language queries and generating precise geometric constructions directly within the app. Another AI component is an interactive NPC Tutor employing Whisper for speech recognition and ElevenLabs for text- to - speech capabilities, significantly enhancing the interactive and intuitive experience.

These improvements dramatically elevated user interaction by removing dependency on manual input, creating a more intuitive and human-like tutoring environment. Consequently, the AI features are now integral to the learning process, significantly enhancing the app's educational effectiveness and practical applicability in real - world teaching scenarios.

6.1.4. Dynamic input field solver

In the former system, input fields for geometry exercises required manual specification of each parameter, making complex spatial geometry problems cumbersome and error prone. Users often face difficulty managing intricate dependencies between geometric elements, resulting in frustration and reduced educational effectiveness.

The updated version implements a sophisticated Dynamic Input Field Solver that automatically resolves interdependencies between input fields. By intelligently inferring necessary geometric constraints and values from minimal user inputs, the system significantly simplifies the process of solving complex geometry problems.

This innovation substantially improves user experience by minimizing manual calculations and reducing the likelihood of errors. It supports learners by automatically guiding them through problem- solving processes, enhancing clarity, educational value, and overall engagement with geometry exercises.

6.2. Pedagogical Enhancements

6.2.1. Natural Language Problem Input

*The previous system **lacked*** any capability to process natural language, requiring users to manually interpret and input structured geometry data. This presented a barrier, especially for students unfamiliar with geometric syntax or terminology. The learning experience was hindered by the rigid input expectations, resulting in increased cognitive load and diminished accessibility.

In the enhanced system, users can now input entire problem statements in natural language, mimicking real exam or textbook questions. This is achieved through the

AIBuilder module, which integrates OpenAI's GPT model to analyze the prompt, extract geometric intent, and generate JSON instructions to render the correct shape. Furthermore, the system also explains the problem in simpler terms and suggests problem-solving steps.

This innovation significantly lowers the entry barrier for learners, supports diverse learning styles, and enables contextual understanding. It positions the application as a real educational assistant capable of bridging the gap between abstract language and concrete geometry visualization.

6.2.2. Gamification - Collect Coin and Quiz Mechanics

*The previous approach to engagement **lacked** motivational design elements, providing a purely functional geometry workspace. This led to reduced long-term user retention and less enthusiasm among learners who benefit from game-like interactions.*

*In the new version, the Collect Coin game introduces *Fruit Ninja* - style slicing mechanics combined with embedded educational content. When users slice a target, they are presented with multiple - choice geometry questions, blending kinetic interaction with cognitive recall. Answering correctly grant progress; incorrect answers prompt retries and feedback.*

This gamified approach turns learning into an active and enjoyable experience. It reinforces memory through repetition and introduces challenge-driven motivation. By merging physical actions with mental processing, the platform aligns with research-backed principles of gamified learning.

6.2.3. Multimodal Learning

*Traditional geometry tools often **limit themselves** to visual or textual inputs, excluding students with different learning needs or preferences. This exclusion reduces inclusivity and effectiveness in diverse classroom settings.*

The new system supports multimodal learning through voice input (Whisper), audio explanations (ElevenLabs TTS), text annotations, visual geometry, and

interactive manipulation. Learners can draw, speak, hear, read, and engage kinesthetically in a unified environment.

This variety empowers students to engage through their preferred modes, ensures accessibility for learners with disabilities, and aligns with modern pedagogical models that recognize learning diversity. The result is a more inclusive, engaging, and effective educational experience.

6.3. User Experience and Interface Design

6.3.1. Unified Theme and Visual Identity

In the earlier iteration, the application ***lacked*** consistent aesthetic and design language, which fragmented the user experience and made onboarding confusing. Visual inconsistency also reduced the application's perceived professionalism and polish.

The current design features a cohesive theme across all modules. UI components follow a consistent color scheme, typography, spacing, and iconography. Whether in Geometry Drawing, AI Tutor, or Collect Coin, the visual cues and interactions maintain a unified identity.

This consistency creates a seamless, intuitive experience that reinforces brand identity, reduces cognitive load, and improves user confidence when switching between tools or modes.

6.3.2. In - App Mascot/Tutor with Voice, Lipsync, and Dialogue

Prior versions had no in - app guide or human-like interface, leaving users to explore on their own. This absence of interactivity limited engagement, particularly for younger or non-technical users who benefit from guided experiences.

The upgraded version features an animated AI tutor that speaks, listens, and provides support via real - time lipsync and expressive gestures. This NPC is not just a helper but a companion that users can interact with using natural language.

This mascot humanizes the learning process, builds emotional engagement, and offers a personalized touch to digital tutoring. It reduces user intimidation and fosters a mentor

- student dynamic. Although the implementation is not developed completely, it is the base for future improvements.

6.4. Project Vision

The core philosophy of this project evolved from mere digitalization - replicating traditional exercises in digital form - to transformation: *reimagining how spatial geometry is taught and learned.*

- *AI as a co-educator is foundational.* AI is no longer a feature or a chat tool, but an active agent in the learning process, interpreting input, explaining logic, guiding thought, and even providing voice interaction. The integration of AI is systemic, not superficial.
- *Learning as play is another tenet.* Rather than treat engagement and evaluation as separate layers, games and interactive quizzes are built into the learning loop. The student is encouraged to explore, make mistakes, and grow.
- *The application is built to be open, extensible, and community driven.* Modular code, well-structured APIs, and dynamic UIs make it possible to add new shapes, games, learning modes, or AI agents.
- *Scalability is considered at every step:* from a single student to a classroom of peers learning together, to nationwide deployment in educational institutions.
- *The design respects accessibility:* text, voice, visual, and social interaction are all supported.

Finally, this project is *a future - forward platform*. It is more than a geometry tool; it is an experimental sandbox for educational innovation - where new ideas in AI, UX, gamification, and pedagogy can be implemented, tested, and evolved.

In summary, this chapter has highlighted the profound evolution of our project from its foundational Project on Software Engineering, predecessor to the advanced Capstone Project system. Through significant technical improvements in modularity, our current system truly embodies a revolutionary approach to spatial geometry education, aligning with its core ideology.

CHAPTER 7. CONCLUSION

Bringing together the concepts, design, and implementation detailed in the preceding chapters, this conclusion provides a comprehensive overview of our achievements in developing interactive 3D software for spatial geometry education. It reflects on the project's successes, acknowledges its current limitations, and outlines a strategic future development plan, reaffirming our commitment to continuously enhancing the learning experience through technological innovation.

7.1. Summary of Achievements

7.1.1. Project Accomplishment Summary

This project successfully delivers a complete, extensible, and pedagogically grounded 3D geometry learning system, combining state-of-the-art software engineering, artificial intelligence, and learning science. It responds directly to cognitive challenges faced by students in spatial geometry - particularly low spatial visualization, procedural misunderstanding, and disengagement with static instruction methods.

1. Core System Architecture

The platform is built with modular, extensible design principles using industry-standard patterns. The core drawing engine supports:

- Dynamic 3D construction of geometric primitives (points, segments, polygons, etc.)
- Prebuilt complex shapes (prisms, pyramids, special rectangles) with parametric generation
- Auxiliary constructions (e.g., perpendiculars, parallels)
- Live measurement tools for angle, length, and area
- Visual customization (colors, labels, visibility toggling)

Its attribute-based construction system allows learners to input known values (e.g., area, height) and infer the rest, supporting a more conceptual and explorative approach to problem-solving.

2. AI-Powered Tutoring Experience

A standout accomplishment is the development of an *AI conversational tutor* integrated with:

- Speech recognition (Whisper) for hands-free interaction
- GPT-4 language model for natural language understanding and feedback
- Text-to-speech with facially animated avatar, creating a high-engagement, human-like experience
- Lip-synced 3D animation, blinking, and idle behavior to enhance emotional connection and trust

The tutor acts as a responsive guide, capable of interpreting both voice and text input and dynamically generating explanations, examples, or geometry constructions tailored to student queries.

3. Gamified Assessment and Error Diagnosis

The Collect Coin mini-game integrates curriculum-aligned geometry questions with gamified mechanics. Questions are tagged with:

- Van Hiele levels of geometric thinking
- Cognitive error categories (e.g., transformation, encoding, comprehension)

This system not only reinforces learning but also enables *implicit diagnostic tracking*. It logs user behavior (attempts, retry patterns, reaction time) to help teachers understand conceptual bottlenecks.

4. System-Level Insights and Feedback Loops

A logging backend continuously captures gameplay and drawing data for:

- Error pattern detection
- Progress monitoring
- Insightful feedback generation

This supports a *feedback-driven learning loop* where insights from student actions inform both AI responses and instructional decisions.

5. Collaborative and Inclusive Design

The platform supports:

- Real-time collaborative drawing, enabling teachers to demonstrate while students follow
- Touch-friendly mobile and desktop UI, usable in both classrooms and at home
- No-login, low-friction design optimized for accessibility in Vietnamese school settings
- Anxiety-reducing visuals and playful mascot design to engage young learners

Conclusion

This project combines AI, game mechanics, and cognitive diagnostics to address real challenges in spatial geometry education. It offers a future-ready educational tool that is technically robust, pedagogically aligned, and highly accessible. By integrating drawing, tutoring, and assessment into one unified system, it lays a strong foundation for scalable, adaptive learning in Vietnamese STEM education.

7.1.2 Alignment with Chapter 1 Objectives

Objective	Achievement Summary
Objective 1 Develop an interactive 3D drawing environment with snapping, labeling, and constraint features.	Fully Achieved – The application features a 3D drawing system supporting point/segment/polygon creation, real-time snapping, dynamic labeling (A, B, C...), constraint logic (parallelism, right angles), and visibility toggles. – All geometry updates are visualized interactively with minimal latency.
Objective 2 Incorporate game-based learning with diagnostic question banks.	Fully Achieved – The Collect Coin game presents spatial geometry questions tied to movement-based coin collection. – Each question is tagged by error domain and Van Hiele level.

	– Player responses are logged for future analysis and feedback.
Objective 3 Enable real-time online collaboration.	Partially Achieved – The system supports real-time synchronized drawing via Unity Netcode with Host-Guest roles. – While full co-editing is limited to the Host, Guests observe all updates live, supporting classroom or tutoring use. – Future iterations may extend editing rights to Guests.
Objective 4 Integrate an AI-powered tutor using ChatGPT.	Fully Achieved – The AI Tutor accepts natural language via voice input, processes geometry questions through GPT-4, and returns vocalized explanations using Text-to-Speech, all presented via a lip-synced 3D avatar. – This supports self-guided, immersive learning.
Objective 5 Provide indirect assessment of spatial skills and error types.	Achieved. – The game and drawing modules log detailed interaction data (errors, retry behavior, timings), which can help infer whether a student struggles with visual rotation, conceptual understanding, or procedural logic. – These patterns serve as diagnostic signals for personalized support for future improvement

Table 7.1 Achievements with defined objectives

7.2. Strengths and Limitations

7.2.1. Strengths

The project demonstrates robust strengths across architecture, pedagogy, and user experience:

- ***Pedagogical Rigor:*** The system is firmly grounded in established educational theories such as the Van Hiele model and cognitive error taxonomy. These

frameworks not only shape the game mechanics and AI feedback but also ensure alignment with real classroom challenges.

- ***Modular and Scalable Architecture***: The codebase follows clean separation of concerns—drawing logic, AI interaction, and game mechanics operate in distinct, reusable modules. This architecture allows the system to scale easily with new features (e.g., adding new shapes, AI upgrades, or analytics dashboards).
- ***Extensibility-First Design***: Whether it's supporting new shape types, AI modalities, or data logging extensions, the system is designed to evolve. Key interactions like drawing, measuring, and tutoring are built using customizable pipelines with clear extension points.
- ***User-Centric Accessibility***: The app requires no account creation, supports touch-based interfaces, and is optimized for both classroom and home environments. This ensures low entry barriers, especially in under-resourced or rural schools.
- ***Emotional and Cognitive Engagement***: Using a 3D animated mascot and immersive conversational AI, the system creates a friendly and motivating environment that reduces math anxiety and sustains attention.

7.2.2. Limitations

Despite its strengths, several constraints and unfinished integrations remain:

- ***Incomplete Language Localization***: While most UI components are in English for technical consistency, question content remains in Vietnamese to match local curricula. This partial localization may pose scalability challenges when expanding to other regions or learners.
- ***UI Responsiveness Gaps***: The interface currently performs best at a fixed resolution. On small or large screens, certain elements may appear misaligned or clipped due to the lack of full responsive layout adaptation.
- ***Lack of Onboarding and Guidance for Drawing Tools***: Although the drawing tools are functional, they currently lack in-app tutorials or contextual hints. For

novice users, this may create a steep learning curve when interacting with constraints or prebuilt shape logic.

- ***Gameplay Data Not Fully Utilized:*** While the game engine logs valuable learning data (e.g., retry count, error categories), these insights are not yet visualized or used to personalize feedback within the app.
- ***Limited User Retention Mechanics:*** The app currently lacks motivational elements such as progress tracking, badges, or notifications. This limits its ability to sustain regular use beyond the first few sessions.
- ***Module Interoperability Gaps:*** While each module functions well independently, deeper integration is still pending. For instance, the AI tutor does not yet adapt its responses based on gameplay performance or drawing errors.
- ***Suboptimal App Size for Mobile Devices:*** Due to the inclusion of 3D assets, AI integration, and animated avatars, the current build version has a relatively high file size, which may affect downloadability and performance on lower-end mobile devices.
- ***Unoptimized Mobile Interaction Flow:*** Certain UI interactions, such as selecting, rotating, or zooming shapes, have not been fully optimized for touch-based gestures, leading to usability friction on smartphones and tablets.

7.3. Future Development Plan

To address the identified limitations and expand the system’s pedagogical and technical depth, the following development directions are proposed:

7.3.1. Full Localization and Multilingual Support

Introduce a localization manager system to support dynamic language switching. This would allow the entire experience - from UI to question content and AI tutor prompts - to be delivered in multiple languages. It will also enable expansion to international markets or bilingual environments.

7.3.2. Responsive UI and Device Optimization

Redesign the UI layout using flexible components (e.g., Unity UI Toolkit or grid-based layouts) to support screen size adaptation. Focus will be placed on ensuring visual clarity and usability across tablets, smartphones, and desktops.

7.3.3. Integrated User Feedback and Smart Tutorial System

Develop contextual tooltips, onboarding flows, and real-time guidance in the drawing environment. These systems will monitor user behavior (e.g., frequent undo, tool misuse) and provide corrective suggestions through the mascot or overlay hints.

7.3.4. Behavioral Analytics Dashboard

Expose learning logs as actionable visual reports. Teachers and students will be able to review performance over time (e.g., error heatmaps, retry frequency), enabling diagnosis of conceptual gaps. This will create a loop between gameplay, analysis, and AI tutoring.

7.3.5. Enhanced AI-Gameplay Integration

Enable the AI tutor to access gameplay logs and drawing history. For example, after a failed game round, the tutor could provide personalized scaffolding based on the user's specific mistake (e.g., misjudged angle, incorrect visualization).

7.3.6. Retention and Motivation Features

Implement a light gamification layer: badges, milestone rewards, daily practice streaks, and smart notifications. These features aim to promote sustained engagement and foster habit-building around geometry practice.

7.3.7. Curriculum-Wide Expansion

Scale the question bank and shape library to cover a full-year geometry curriculum. Include complex 3D concepts like sphere intersections, coordinate geometry, and

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applied math problems, all supported by interactive drawing and AI explanation modules.

This concluding chapter has summarized the significant achievements of our project, demonstrating how our interactive 3D software, gamification, and AI - powered learning solution successfully addresses the challenges in spatial geometry education. While acknowledging current limitations, it also outlines a clear future development plan, reaffirming our commitment to continuous innovation and enhancing the learning experience for students worldwide.

CHAPTER 8. REFERENCES

1. A Review of Van Hiele's Theory. (n.d.). *ResearchGate*.
https://www.researchgate.net/publication/370946482_A_Review_of_Van_Hiele's_Theory
2. Senior high school student's visual - spatial intelligence according to van Hiele geometric thinking theory. (n.d.). *ResearchGate*.
https://www.researchgate.net/publication/332302737_Senior_high_school_student's_visual_spatial_intelligence_according_to_van_hiele_geometric_thinking_theory
3. Ganley, C. M., & Vasilyeva, M. (2011). Gender differences in mathematical problem - solving and spatial visualization among secondary school students. *Frontiers in Psychology*.
4. Wintari, & Patahuddin. (2020). Students' mental rotation ability as a function of gender in a disadvantaged community in Indonesia. *ERIC*.
5. Rodán, A., et al. (2022). Gender differences in mental rotation tests: A geometry - teaching perspective. *Redalyc*.
6. Reinking, D., & Martín, L. A. (2018). Gender differences in STEM education: A review of the literature on the role of motivation and self - efficacy. *Educational Psychology Review*.
7. Uttal, D. H., Meadow, N. G., Tipton, E., Hand, L., & Newcombe, N. S. (2013). The malleability of spatial skills: A meta - analysis of training studies. *Psychological Bulletin*, 139(2), 352–402.
8. Pajares, F., & Kranzler, J. H. (1995). Self - efficacy beliefs and mathematical problem - solving of gifted students. *Contemporary Educational Psychology*, 20(4), 426–443.
9. Battista, M. T. (1990). Spatial visualization and gender differences in mathematics education: A review of the literature. *Springer*.
10. Hyde, J. S., Lindberg, S. M., Linn, M. C., Ellis, A. B., & Williams, C. C. (2008). Gender similarities characterize math performance. *Science*, 321(5888), 494–495.

11. Jansen, P., & van der Lubbe, R. H. J. (2013). The effect of stimulus complexity on mental rotation performance: Evidence from behavioral data and ERPs. *MDPI*. [https://www.mdpi.com/2227 - 7390/9/23/3067](https://www.mdpi.com/2227-7390/9/23/3067)
12. Chapter: The van Hiele theory in mathematics education. (n.d.). *SpringerLink*. [https://link.springer.com/chapter/10.1007/978 - 94 - 6300 - 561 - 6_4](https://link.springer.com/chapter/10.1007/978-94-6300-561-6_4)
13. Cognitive research on spatial abilities. (n.d.). *European Proceedings*. <https://www.europeanproceedings.com/article/10.15405/epes.23056.9>
14. Mental rotation in cognitive research. (n.d.). *Cognitive Research Journal*. [https://cognitiveresearchjournal.springeropen.com/articles/10.1186/s41235 - 021 - 00274 - 5](https://cognitiveresearchjournal.springeropen.com/articles/10.1186/s41235-021-00274-5)
15. Constructivism theory in learning. (n.d.). *Simply Psychology*. <https://www.simplypsychology.org/constructivism.html>
16. Multimedia learning theory. (n.d.). *WSU Open Textbooks*. [https://opentext.wsu.edu/theoreticalmodelsforteachingandresearch/chapter/multimedia - learning - theory/](https://opentext.wsu.edu/theoreticalmodelsforteachingandresearch/chapter/multimedia-learning-theory/)
17. Mayer, R. E. (2005). *The Cambridge Handbook of Multimedia Learning*. Cambridge University Press.
18. Moreno, R., & Mayer, R. E. (2000). Engaging students in active learning: The case for personalized multimedia messages. *Educational Psychologist*, 35(2), 85–94.
19. Article on spatial visualization. (n.d.). *ScienceDirect*. <https://www.sciencedirect.com/science/article/abs/pii/B978008045046900423X>
20. Medical publication on spatial skills. (n.d.). *PubMed*. <https://pubmed.ncbi.nlm.nih.gov/18400925/>
21. Moè, A. (2016). Gender differences in mathematics: A review of research on the role of motivation and self - efficacy. *Educational Psychology Review*, 28(4), 689–707.
22. Titze, M., Wirth, J., & Höffler, T. N. (2010). Gender differences in spatial ability: A meta - analysis. *Learning and Individual Differences*, 20(5), 507–512.

23. Heil, M., & Jansen, P. (2008). Gender differences in spatial ability: Evidence from a large sample study using different measures. *Learning and Individual Differences*, 18(1), 85–92.
24. Voyer, D., Voyer, S. D., & Bryden, M. P. (1995). Magnitude of sex differences in spatial abilities: A meta - analysis and consideration of critical variables. *Psychological Bulletin*, 117(2), 250–270.
25. Gesture during math instruction. (n.d.). *ResearchGate*.
[https://www.researchgate.net/publication/342059930_Gesture_during_math_in
struction_specifically_benefits_learners_with_high_visuospatial_working_me
mory_capacity](https://www.researchgate.net/publication/342059930_Gesture_during_math_instruction_specifically_benefits_learners_with_high_visuospatial_working_memory_capacity)
26. Cresswell's research methods. (n.d.). *UCG*.
[https://www.ucg.ac.me/skladiste/blog_609332/objava_105202/fajlovi/Creswell
.pdf](https://www.ucg.ac.me/skladiste/blog_609332/objava_105202/fajlovi/Creswell.pdf)
27. Mental rotation studies. (n.d.). *PMC*.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8915228/>
28. Sampling methods in research. (n.d.). *ResearchGate*.
[https://www.researchgate.net/publication/371985656_Sampling_Methods_in_
Research_A_Review](https://www.researchgate.net/publication/371985656_Sampling_Methods_in_Research_A_Review)
29. Simply Psychology on Vygotsky. (n.d.).
<https://www.simplypsychology.org/vygotsky.html>
30. Educational strategies using AR. (n.d.). *ScienceDirect*.
<https://www.sciencedirect.com/science/article/abs/pii/S1747938X18301805>
31. Augmented reality in education. (n.d.). *Interaction Design Foundation*.
[https://www.interaction-design.org/literature/article/augmented-reality-the-
past-the-present-and-the-future](https://www.interaction-design.org/literature/article/augmented-reality-the-past-the-present-and-the-future)
32. Azuma, R. (1997). A survey of augmented reality. *Presence: Teleoperators and Virtual Environments*, 6(4), 355–385.
33. Azuma, R., Bailiot, Y., Behringer, R., Feiner, S., Julier, S., & MacIntyre, B. (2001). Recent advances in augmented reality. *IEEE Computer Graphics and Applications*, 21(6), 34–47.

34. Bornik, A., Beichel, R., Reitingner, B., Gotschuli, G., Sorantin, E., Leberl, F., & Sonka, M. (2003). Computer - aided liver surgery planning: An augmented reality approach. *Medical Imaging, SPIE Proceedings*, 5029, 235–245.
35. Caudell, T. P., & Mizell, D. W. (1992). Augmented reality: Heads - up display technology in manufacturing processes. *Hawaii International Conference on System Sciences*, 659–669.
36. Durlach, N. I., & Mavor, A. S. (1995). *Virtual reality: Scientific and technological challenges*. National Academy Press.
37. Fitzmaurice, G., & Buxton, W. (1993). 3D spatial design and interaction. *INTERCHI '93*.
38. Winyu, C., Burstein, R., Seibel, E., & Furness, T. (2001). Head tracking using a virtual retinal display. *IEEE Symposium on Augmented Reality*, October 29 - 30, New York, NY.
39. Video - augmented trade fair. (n.d.). *ResearchGate*.
[https://www.researchgate.net/publication/302919746_Enhancing_Project - Based_Learning_Through_Student_and_Industry_Engagement_in_a_Video - Augmented_3 - D_Virtual_Trade_Fair](https://www.researchgate.net/publication/302919746_Enhancing_Project_-_Based_Learning_Through_Student_and_Industry_Engagement_in_a_Video_-_Augmented_3_-_D_Virtual_Trade_Fair)
40. Challenges in AR education. (n.d.). *ResearchGate*.
https://www.researchgate.net/publication/235703112_Current_status_opportunities_and_challenges_of_augmented_reality_in_education
41. Technology and equity in digital education. (n.d.). *ResearchGate*.
https://www.researchgate.net/publication/250185477_New_Technology_and_Digital_Worlds_Analyzing_Evidence_of_Equity_in_Access_Use_and_Outcomes
42. Spatial visualization in engineering education. (n.d.). *ResearchGate*.
[https://www.researchgate.net/publication/248974874_Educational_Research_in_Developing_3 - D_Spatial_Skills_for_Engineering_Students](https://www.researchgate.net/publication/248974874_Educational_Research_in_Developing_3_-_D_Spatial_Skills_for_Engineering_Students)
43. Influence of visualization training. (n.d.). *ResearchGate*.
https://www.researchgate.net/publication/335276238_The_Influence_of_Spatial_Visualization_Training_on_Students'_Spatial_Reasoning_and_Mathematics_Performance

CHAPTER 9. APPENDICES

APPENDIX A. SURVEY

A.1. Survey Questions

The survey aims to gather data on how high school students perceive and interact with Augmented Reality (AR) tools in learning spatial geometry. The responses will provide insights into how AR impacts spatial reasoning skills, engagement, and overall learning experience. The information collected will be used for research purposes and remain confidential.

A.2. Questions list for the survey

A.2.1. Demographic Information

1. Gender:
 - Male
 - Female
 - Prefer not to say
2. Grade Level:
 - 10th Grade
 - 11th Grade
 - 12th Grade
3. Do you generally enjoy learning mathematics and geometry?
 - Yes
 - No
 - Neutral

A.2.2. Pre-existing Knowledge and Spatial Abilities

5. How comfortable do you feel with visualizing 3D geometric shapes based on 2D representations (e.g., in textbooks or board drawings)?

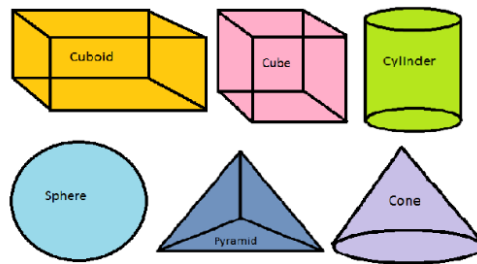


Figure A.1. Image in Survey form

- Comfortable
 - Somewhat comfortable
 - Neutral
 - Somewhat uncomfortable
 - Very uncomfortable
6. How well do you think you understand the relationships between 3D objects, such as rotation, symmetry, and transformations?

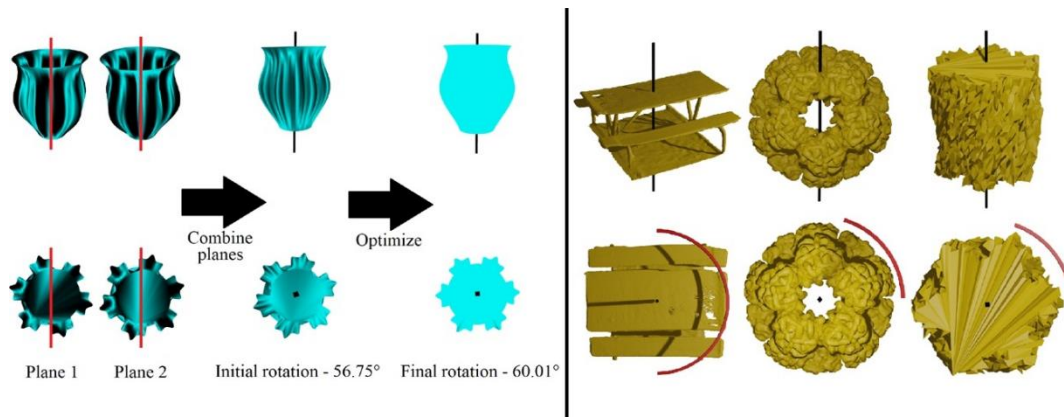


Figure A.2. Image in Survey form

- Excellent understanding
- Good understanding
- Average understanding
- Below average understanding
- Poor understanding

A.2.3. Experience with Traditional Geometry Learning

7. In your experience, how effective are traditional methods (textbooks, physical models, chalkboard drawings) in helping you visualize and understand 3D geometry?
- Very effective
 - Somewhat effective
 - Neutral
 - Not very effective
 - Not effective at all
8. Do you find traditional methods engaging when learning about spatial geometry?
- Yes, they are engaging.
 - Sometimes, but not always.
 - No, I find them unengaging.
9. Have you used Augmented Reality (AR) tools to learn geometry before?

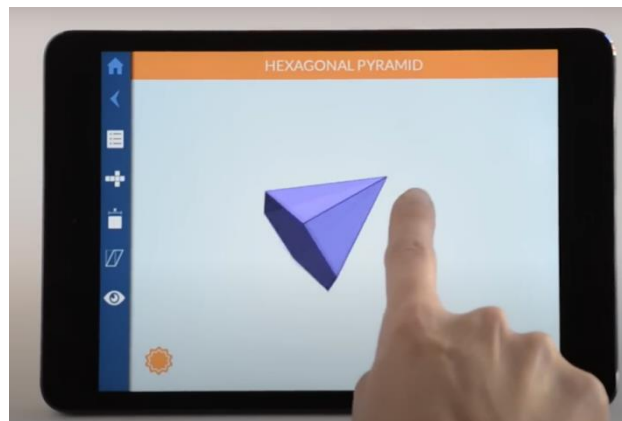


Figure A.3. AR in learning geometry (captured from a Youtube video¹)

- Yes
- No
- I am not sure

¹

[https://pmc.ncbi.nlm.nih.gov/articles/PMC8915228/#:~:text=Mental%20rotation%20performance%20is%20mainly,non%2Dmirror%20reversed\)%20or%20%E2%80%9C](https://pmc.ncbi.nlm.nih.gov/articles/PMC8915228/#:~:text=Mental%20rotation%20performance%20is%20mainly,non%2Dmirror%20reversed)%20or%20%E2%80%9C)

A.2.4. Experience with Augmented Reality (AR)

10. If yes, how did AR help you in understanding 3D shapes? (If you haven't used AR, skip to section 8)

- Significantly improved understanding
- Moderately improved understanding
- Neutral
- Slightly improved understanding
- No improvement at all

11. Which of the following features of AR do you find most helpful in understanding spatial geometry? (Select all that apply)

- Ability to rotate and manipulate 3D models
- Visualizing 3D shapes in real-world environments
- Gesture recognition to control shapes
- Interacting with models through zooming in/out and exploring details

12. Did using AR make learning geometry more engaging and enjoyable?

- Yes, it was much more engaging
- Slightly more engaging
- Neutral
- Less engaging than traditional methods
- Not engaging at all

13. How likely are you to recommend using AR tools for learning geometry to other students?

- Very likely
- Likely
- Neutral
- Unlikely
- Very unlikely

A.2.5. Impact on Spatial Reasoning and Problem Solving

14. Do you feel that using AR has improved your ability to solve geometry problems involving 3D shapes?

- Yes, significantly
- Yes, to some extent
- Neutral
- No, not really
- No, not at all

15. After using AR tools, how much easier is it for you to visualize transformations like rotations, reflections, and translations of 3D objects?

- Much easier
- Somewhat easier
- Neutral
- Somewhat harder
- Much harder

A.2.6. User Experience and Interaction with AR

16. How easy or difficult was it to use the AR tools (e.g., gesture recognition, manipulating 3D models)?

- Very easy
- Somewhat easy
- Neutral
- Somewhat difficult
- Very difficult

17. Which aspect of AR did you find the most difficult to use?

- Gesture recognition
- Manipulating 3D models
- Understanding the interface
- Other (please specify): _____

18. Would you prefer learning spatial geometry with AR over traditional methods?

- Yes
- No
- I prefer a combination of both

A.2.7. Open-Ended Questions

19. In your opinion, how could AR tools be improved to enhance the learning of spatial geometry?

20. What suggestions do you have for teachers who want to use AR tools in their geometry lessons?

Justification for Questions and Form Structure:

- **Quantitative Questions:** These questions focus on collecting measurable data regarding students' spatial abilities, experience with AR, and preferences for learning tools. The pre- and post-test style questions (questions 5, 14, and 15) are essential for understanding how AR improves (or fails to improve) spatial visualization and problem-solving skills. This is aligned with studies by Sorby (2009), which emphasize the value of interactive technologies in improving spatial intelligence.
- **User Experience Metrics:** Questions 10, 11, and 16 are aimed at understanding how students interact with AR tools, which is vital for evaluating usability and engagement. According to Lowrie and Logan (2023), the ability to interact with 3D models can significantly improve engagement, but user interface challenges can hinder learning.
- **Qualitative Feedback:** Open-ended questions (19, 20) allow students to provide subjective insights on how AR can be improved. This data is crucial for making iterative changes to the technology, based on real-world classroom use.

A.2.8. Visual and Interactive Spatial Geometry Assessment

The questions in this section will incorporate images of 3D geometric objects and ask students to manipulate these objects mentally or visually, helping to directly assess their spatial reasoning.

1. Recognizing 3D shapes from 2D projections

APPENDIX A: SURVEY

Instruction: Look at the image below. Which of the following 3D shapes matches this 2D projection?

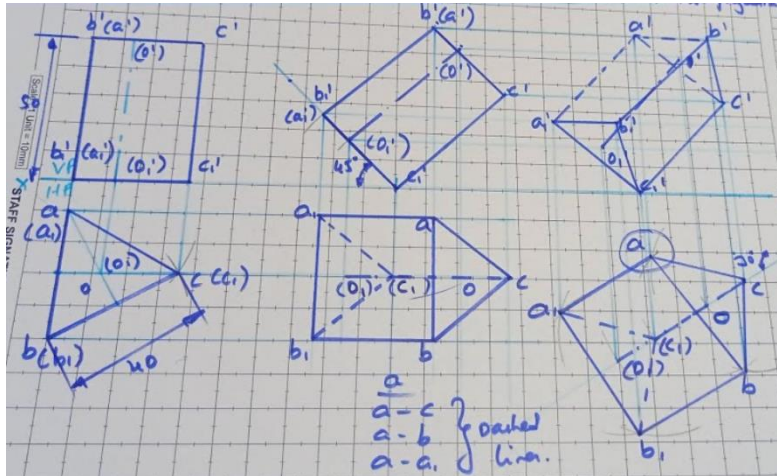


Figure A.4. Image in Survey form

Answer Options:

- A: Cube
- B: Square Pyramid
- C: Rectangular Prism
- D: Triangular Prism

Justification: According to research by Sorby (2009), one of the key challenges in spatial reasoning is the ability to recognize 3D shapes from their 2D projections. This question assesses the student's ability to mentally reconstruct a 3D object from a 2D view, a critical skill in spatial geometry.

2. Mental Rotation of a 3D Object

Instruction: The image below shows a 3D object. Imagine rotating it 90 degrees clockwise. What will the shape look like after the rotation? Select the correct image from the options below.

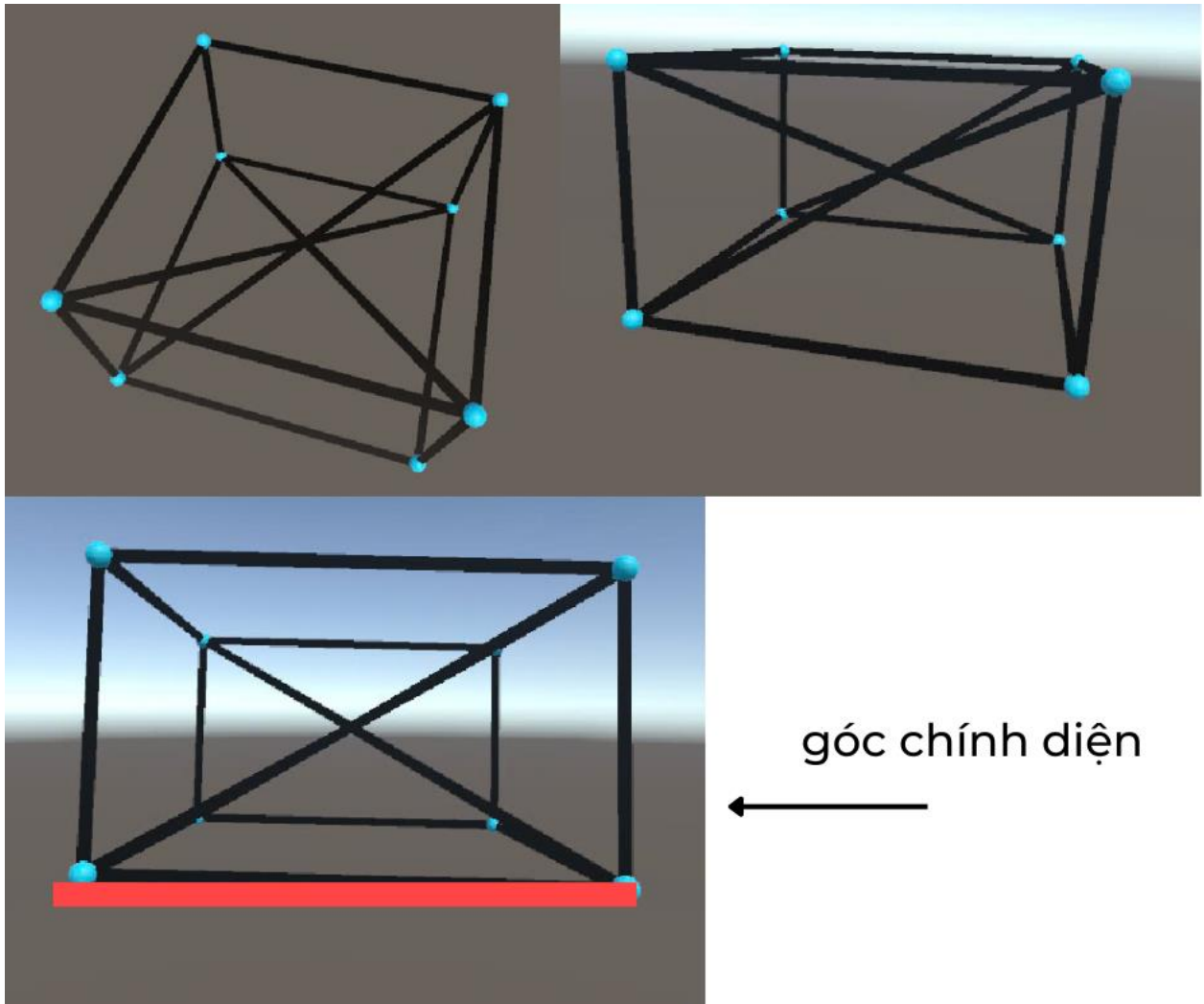
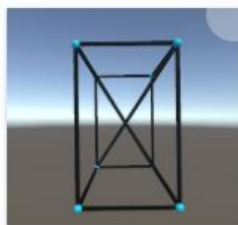
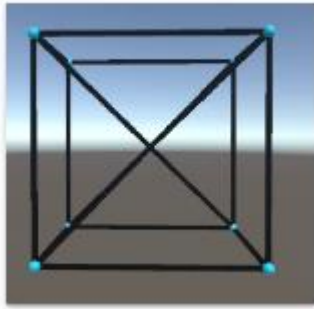


Figure A.5. Image in Survey form

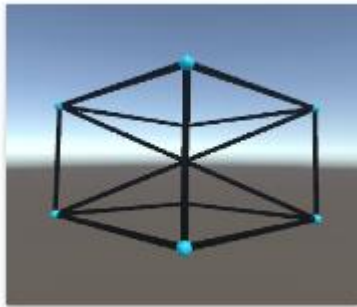
Answer Options:



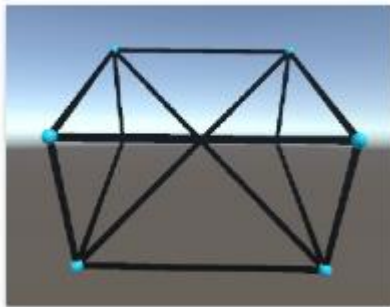
- A



- B:



- C:



- D:

Justification: This question is based on **mental rotation tasks**, as used in spatial reasoning tests by Shepard and Metzler (1971). Mental rotation is a strong indicator of spatial visualization abilities, and students' responses will reveal their proficiency in manipulating 3D objects in their minds².

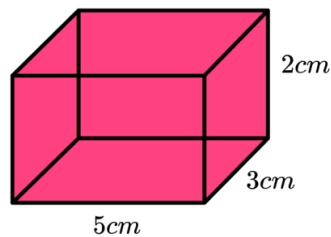
3. Surface Area and Volume of 3D Objects

Instruction: Observe the 3D shape below. Calculate its surface area and volume.

²

[https://pmc.ncbi.nlm.nih.gov/articles/PMC8915228/#:~:text=Mental%20rotation%20performance%20is%20mainly,non%2Dmirror%20reversed\)%20or%20%E2%80%9C](https://pmc.ncbi.nlm.nih.gov/articles/PMC8915228/#:~:text=Mental%20rotation%20performance%20is%20mainly,non%2Dmirror%20reversed)%20or%20%E2%80%9C)

APPENDIX A: SURVEY



Face	Area
Bottom	$5 \times 3 = 15$
Top	15
Front	$5 \times 2 = 10$
Back	10
Right side	$2 \times 3 = 6$
Left side	6

$$\begin{aligned}\text{Total surface area} &= 15 + 15 + 10 + 10 + 6 + 6 \\ &= 62\text{cm}^2\end{aligned}$$

Figure A.6. Image in Survey form

Answer:

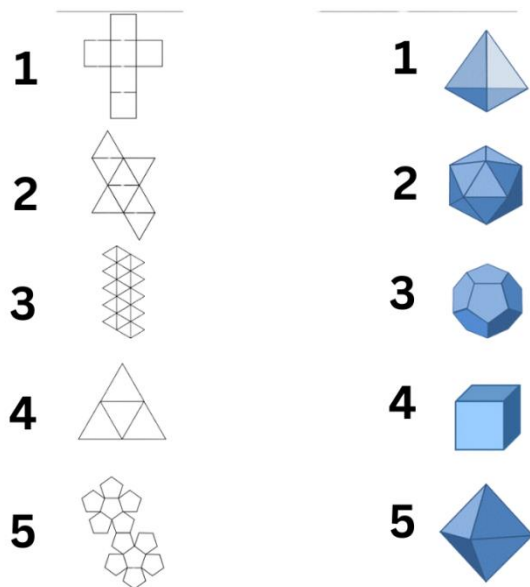
- Surface Area: ____ cm^2

Justification: This type of question tests students' ability to integrate geometric formulas with their understanding of the shape's spatial properties. According to **Cognitive Load Theory** (Sweller, 2011), combining visual and mathematical skills into a single task places cognitive demands on learners that reveal their depth of spatial understanding.

A.2.9. Advanced Spatial Problem-Solving

4. Symmetry and Reflection in 3D Shapes

Instruction: The image below shows a 3D shape. Connect the corresponding grids with the corresponding 3d shapes



APPENDIX A: SURVEY

Figure A.7. Image in Survey form

- Answer: 1 - 4, 2 - 2, 3 - 2, 4 - 1, 5 - 3

Justification: Identifying symmetry is a fundamental concept in spatial geometry, and this question tests the student's ability to find symmetry planes, which are crucial for understanding 3D transformations (Hoffer, 2006).

5. 2D Cross-Sections of 3D Shapes

Instruction: The image below shows a 3D shape. If you cut through the object along the given plane, what will the resulting 2D cross-section look like? Choose the correct shape from the options below.

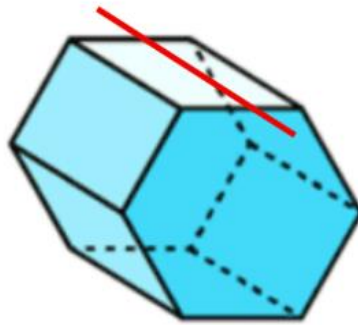


Figure A.8. Image in Survey form

Answer Options:

- A: Hexagon
- B: Ellipse
- C: Rectangle
- D: Square

Justification: The ability to visualize cross-sections is vital for comprehending spatial geometry. According to **Van Hiele's Levels of Geometric Thinking**, students who can correctly visualize 2D cross-sections from 3D objects are operating at a higher cognitive level, as they can mentally manipulate objects to assess different views.

A.3. The Final Survey form

After the discussion steps, the questions were finally compiled and put into the following form.

Survey form link: <https://forms.gle/4XvZ12nZdm7b7Xoz9>

Currently, the form has 92 responses.

APPENDIX B: MBTI-INSPIRED LEARNER ANALYSIS AND ADAPTIVE RESPONSE MECHANIC FOR GEOMETRY EDUCATION

APPENDIX B. MBTI-INSPIRED LEARNER ANALYSIS AND ADAPTIVE RESPONSE MECHANIC FOR GEOMETRY EDUCATION

B.1. The Imperative for Nuanced Learner Assessment

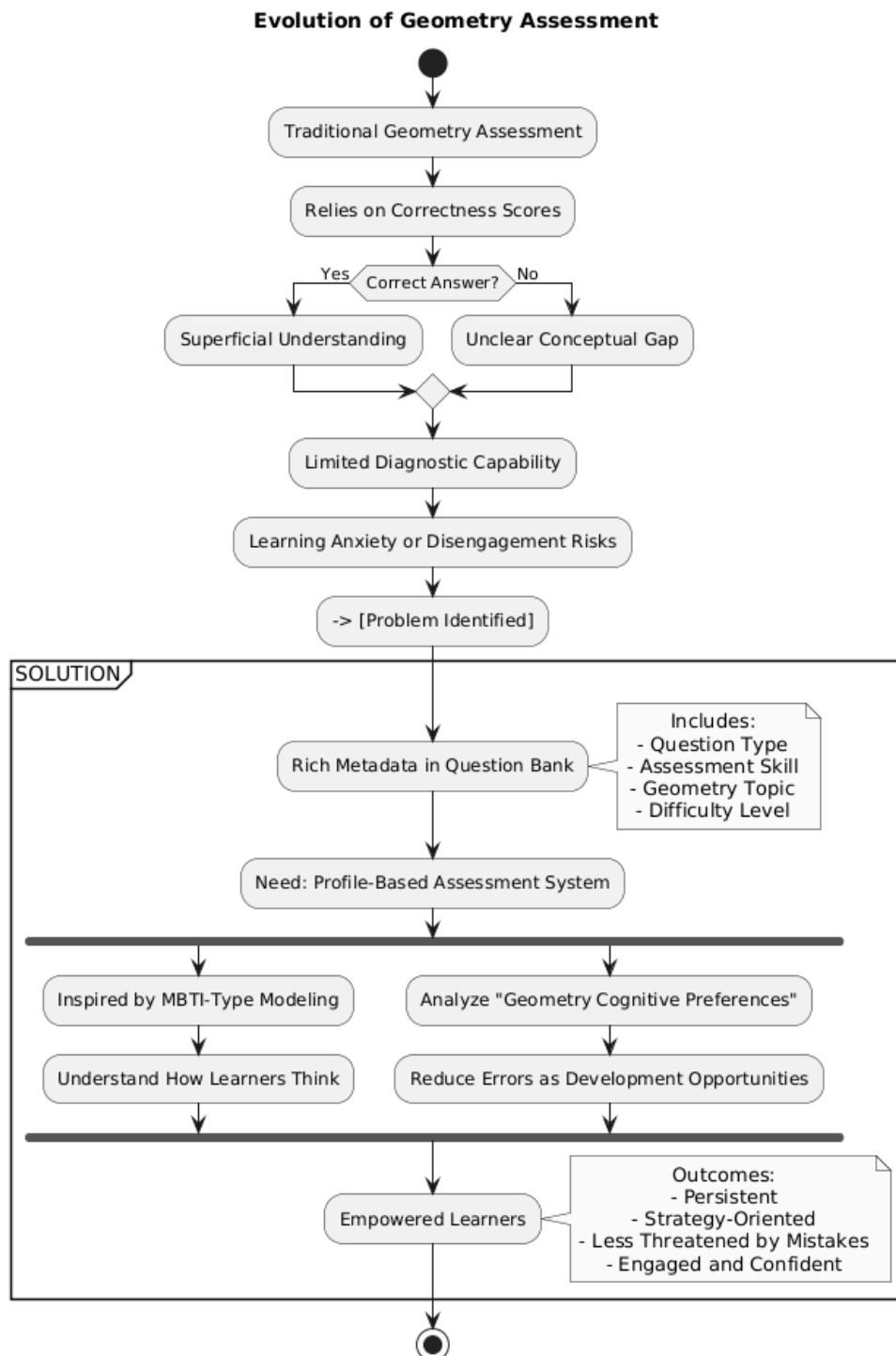


Figure B.1 The Imperative for Nuanced Learner Assessment

Traditional assessment methods in geometry education, which predominantly rely on correctness-based scoring, often fail to capture the underlying cognitive strategies of

APPENDIX B: MBTI-INSPIRED LEARNER ANALYSIS AND ADAPTIVE RESPONSE MECHANIC FOR GEOMETRY EDUCATION

learners. Correct responses do not necessarily reflect conceptual understanding, while incorrect answers rarely illuminate specific misconceptions. This diagnostic limitation leads to generalized feedback that inadequately supports individualized learning pathways.

By contrast, question banks enriched with metadata (e.g., question type, assessed skill, topic, and difficulty) offer untapped potential for deeper learner analysis. To move beyond performance-based evaluation, assessment must evolve toward identifying how students learn and what cognitive tendencies they exhibit.

Aligned with growth mindset theory, this approach reframes errors as indicators of developmental opportunity rather than fixed deficiency. Drawing inspiration from the Myers-Briggs Type Indicator (MBTI), this study proposes a profiling system for geometry learners' cognitive preferences, enabling adaptive feedback and personalized progression models.

B.2. Foundational Principles: Adapting Psychometric Insights from MBTI

The proposed mechanic is grounded in the psychometric principles of the MBTI framework, which classifies individuals across four dichotomies – Extraversion - Introversion, Sensing - Intuition, Thinking - Feeling, and Judging – Perceiving - yielding 16 personality types. MBTI Step II extends this model by identifying 20 facets, offering finer-grained insight into preference expression, while the Preference Clarity Index quantifies the strength of these inclinations.

Despite criticisms regarding its binary structure, test-retest reliability, and susceptibility to bias, MBTI remains a valuable tool for fostering self-awareness. When adapted for education, its strength lies not in labeling but in revealing cognitive tendencies - default strategies learners use when engaging with geometry tasks.

In this context, the framework supports the identification of learner profiles based on habitual approaches (e.g., visual-first vs. formula-driven), allowing educators to tailor feedback and recommend developmental pathways. Crucially, these preferences are viewed as flexible and trainable, transforming the assessment from a static classification into a dynamic learning scaffold.

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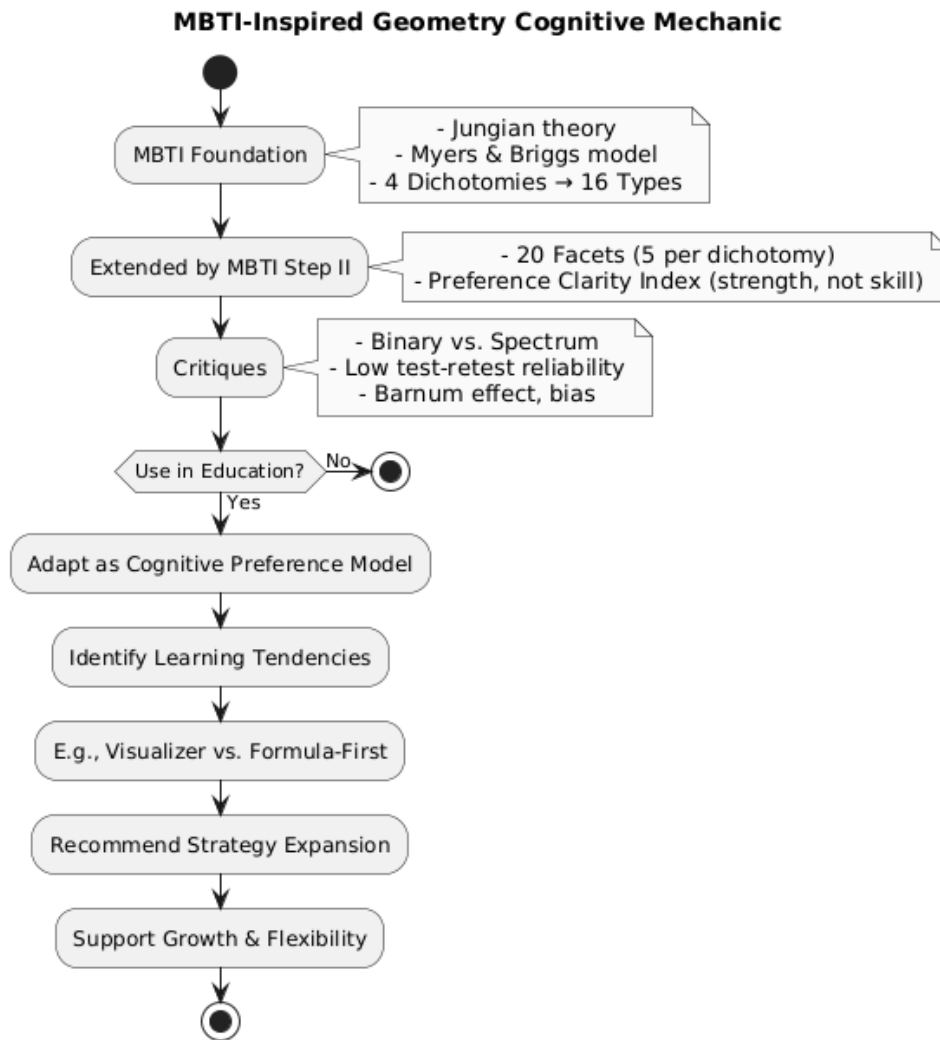


Figure B.2 MBTI-Inspired Geometry Cognitive Mechanic

B.3. The Geometry Question Bank: Data for Diagnostic Profiling

The proposed learner analysis mechanic draws its analytical power from the metadata-rich geometry question banks. Each question is annotated with attributes such as ID, type, correct/incorrect options, difficulty, and game-specific hints, alongside pedagogically significant tags like *Assessment Skill* (e.g., VIS, TRF, CMP) and *Geometry Topic* (e.g., cross-section, surface area).

Of relevance are the *Assessment Skills*, which serve as cognitive dimensions directly analogous to MBTI's facets. This predefined set enables targeted analysis of learner tendencies without the need to develop new constructs. For instance, repeated success on high-difficulty VIS-tagged items reveals a strong visualization tendency—akin to a high clarity score in MBTI.

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Difficulty level plays a critical role by modulating the weight of each correct response. Just as MBTI's Preference Clarity Index reflects how strongly a preference is expressed, this system uses item difficulty to refine cognitive profiling. Success on harder tasks suggests deep-seated strengths rather than superficial familiarity, enabling precise and meaningful identification of cognitive preferences in geometry learning.

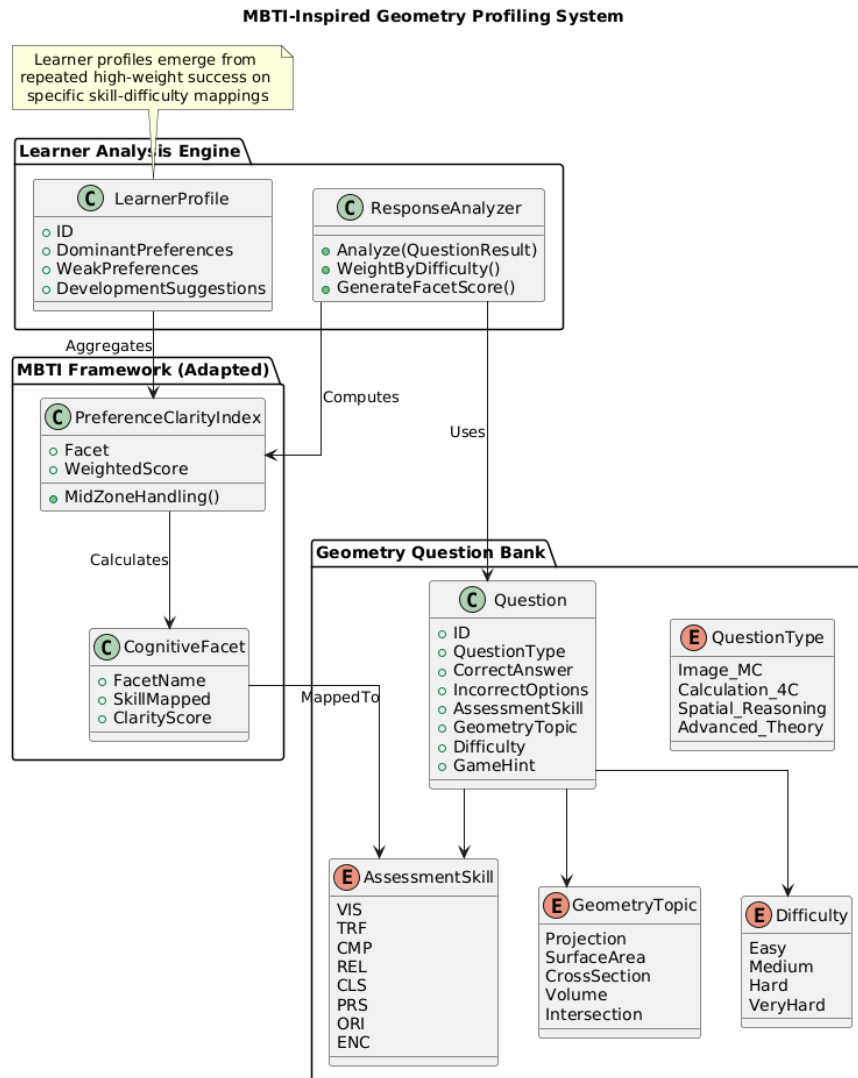


Figure B.3 MBTI-Inspired Geometry Profiling System

The following table summarizes the metadata available in the question banks, illustrating the richness and diversity of the data that underpins the proposed mechanic:

Category	Sub-Category / Example	Count (Set 1)	Count (Set 2)	Total Questions	Example Question IDs
Question Type	Image – MC	2	1	3	Q001, HQ001
	Calculation – 4 choices	2	0	2	Q002

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	Spatial Reasoning	1	1	2	Q008, HQ003
	Advanced Theory	0	3	3	HQ002, HQ010, HQ014
	Theoretical MC	1	0	1	Q005
	Multiple Choice	3	2	5	Q015, Q020, HQ001, HQ005
	Calculation	2	0	2	Q014, Q016
	Spatial Geometry Reasoning	0	1	1	HQ004
	Feature Counting	0	1	1	HQ006
	Coordinates – Center Selection	0	1	1	HQ007
	3D Calculation	0	1	1	HQ008
	Space – Cut Shape	0	1	1	HQ009
	Spatial Calculation	0	1	1	HQ011
	Cross-section Analysis	0	1	1	HQ012
	Model Geometry	0	1	1	HQ013
	Symmetry Analysis	0	1	1	HQ015
	Vector Calculation	0	1	1	HQ016
	Spatial Angle	0	1	1	HQ017
	Counting Question	0	1	1	HQ018
	Distance Calculation in Space	0	1	1	HQ019
	Geometric Position	0	1	1	HQ020
Assessment Skill	VIS (Visualization)	4	8	12	Q001, Q003, HQ001, HQ003
	TRF (Transformation)	3	4	7	Q002, Q016, HQ002, HQ007
	CMP (Comprehension)	5	5	10	Q005, Q009, HQ002, HQ005

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	REL (Relationship)	2	4	6	Q004, Q022, HQ003, HQ008
	CLS (Classification)	5	3	8	Q007, Q010, HQ005, HQ006
	PRS (Problem Solving)	2	3	5	Q002, Q014, HQ008, HQ011
	ORI (Orientation)	2	0	2	Q001, Q018
	ENC (Encoding)	1	0	1	Q006
Difficulty	Easy	9	0	9	Q002, Q005, Q007
	Medium	10	3	13	Q001, Q004, HQ010, HQ014
	Hard	5	13	18	Q003, Q008, HQ001, HQ003
	Very Hard	0	4	4	HQ002, HQ005, HQ007

Table B.1 Question Bank Metadata Summary

Some questions may be tagged with multiple skills, leading to skill counts that sum to more than the total number of questions. The example question IDs are illustrative and not exhaustive.

B.4. The MBTI-Inspired Learner Analysis Mechanic: Design and Functionality

This section details the core design of the proposed mechanic, systematically translating MBTI principles into a functional system for geometry education.

B.4.1. Defining Core Geometry Dimensions (Dichotomies)

To structure geometry-specific cognitive profiling, two overarching dichotomous dimensions are proposed, drawing conceptual inspiration from the MBTI framework. These dimensions reflect distinct yet complementary modes of geometric thinking:

Spatial - Perceptual (SP) vs. Analytical - Computational (AC)

This dimension captures how learners perceive and process geometric data.

SP (Spatial - Perceptual)

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Visually driven learners who excel at mental rotation, spatial awareness, and diagrammatic reasoning.

Facets: VIS, ORI, CLS (visual)

Sample Questions: Q001 (ORI/VIS), HQ004 (VIS)

AC (Analytical - Computational):

Learners who favor logical sequences, symbolic manipulation, and formula application.

Facets: PRS, TRF, CMP (deductive)

Sample Questions: Q002 (PRS/TRF), HQ007 (TRF/CMP)

Conceptual - Relational (CR) vs. Applied - Procedural (AP)

This dimension reflects the learner's orientation toward abstract reasoning versus practical execution.

CR (Conceptual–Relational):

Theory-oriented learners who focus on underlying principles, definitions, and interrelationships.

Facets: CMP, CLS (theoretical), REL

Sample Questions: Q009 (CMP), HQ010 (CMP)

AP (Applied–Procedural):

Methodical learners who excel at algorithmic application and step-by-step execution.

Facets: PRS, TRF, ENC

Sample Questions: Q016 (PRS/TRF), Q006 (ENC)

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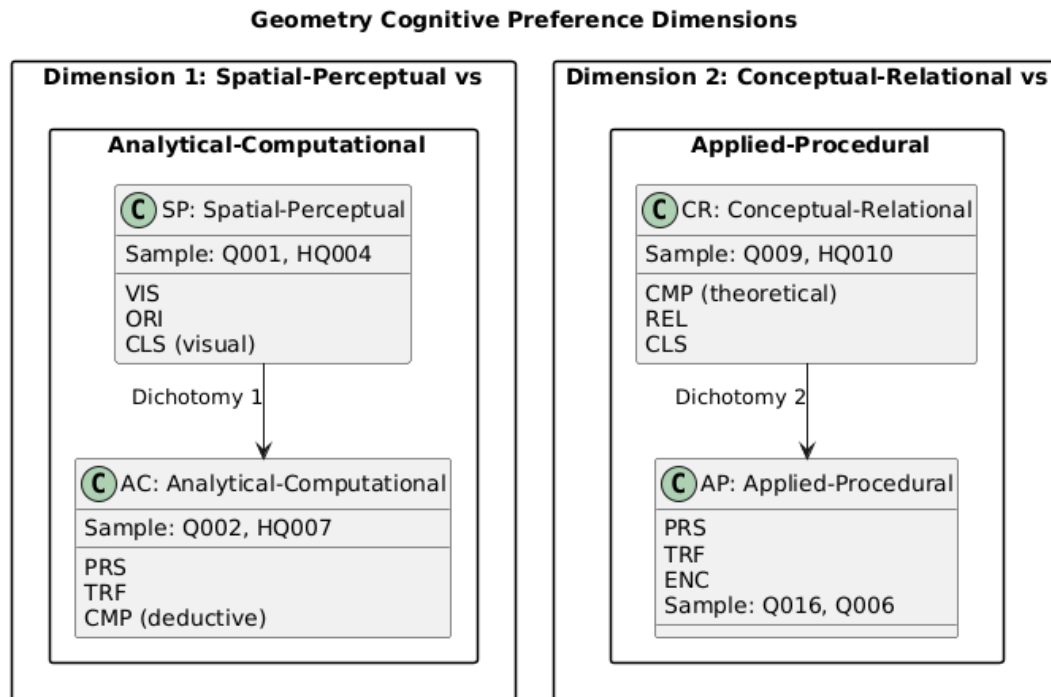


Figure B.4 Geometry Cognitive Preference Dimensions

B.4.2. Facet-Level Performance Analysis

Each Assessment Skill identified in the question bank metadata will function as a distinct facet within the geometry learning profile. There are 8 identified facets: VIS, TRF, CMP, REL, CLS, PRS, ORI, ENC. For each learner, their performance on questions tagged with these specific skills will be aggregated. This aggregation will involve tracking correctness, the number of attempts made, and the time taken to answer (if such data is available). For instance, a learner's performance across all questions tagged with "VIS" will contribute directly to their "Visualization" facet score.

The following table provides a clear, structured overview of how the granular assessment skills map to the broader cognitive dimensions, making the theoretical framework concrete by linking each facet to its definition and an actual question from the dataset. This transparency is crucial for understanding the mechanic's logic and for communicating the profile to learners.

Core geometry dimension	Corresponding pole	Geometry skill facet	Brief description of facet in geometry context	Example question ID

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Spatial-Perceptual	SP	VIS (Visualization)	Ability to mentally picture, manipulate, and interpret geometric shapes and their properties in 2D/3D space.	Q003, HQ004
	SP	ORI (Orientation)	Understanding of spatial positions, projections, and viewpoints of geometric figures.	Q001, Q018
	SP	CLS (Classification)	Identifying, categorizing, and distinguishing geometric figures based on their visual characteristics or defining properties.	Q007, Q010
Analytical-Computational	AC	PRS (Problem Solving)	Applying mathematical operations and logical steps to solve quantitative geometric problems.	Q002, Q014
	AC	TRF (Transformation)	Understanding and applying rules for geometric transformations (e.g., rotation, translation) or algebraic formulas in a spatial context.	Q016, HQ007
Conceptual-Relational	CR	CMP (Comprehension)	Grasping abstract geometric definitions, theorems, and the underlying theoretical principles.	Q005, Q009
	CR	REL (Relationship)	Understanding the connections and interdependencies between	Q004, Q022

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			different geometric elements (points, lines, planes, solids).	
Applied-Procedural	AP	ENC (Encoding)	Ability to formalize, present, or structure geometric solutions and conclusions accurately.	Q006

Table B.2 Proposed Geometry Skill Facets and Their Mapping to Core Dimensions

B.4.3. Scoring and Preference Clarity Index for Geometry

Each cognitive facet is scored on an **11-point clarity scale**, adapted from the MBTI Step II model. Correct responses to face-aligned questions incrementally increase the score toward the corresponding cognitive pole. Conversely, consistent errors may reflect midzone ambiguity or an opposite tendency.

A critical variable is **difficulty weighting**:

- Easy → low contribution
- Medium → moderate
- Hard → high
- Very Hard → maximum clarity gain

Facet scores are aggregated by dimension (e.g., SP vs. AC), accounting not only for polarity but also for **clarity strength**, ensuring the learner profile reflects robust, consistent cognitive patterns rather than isolated success.

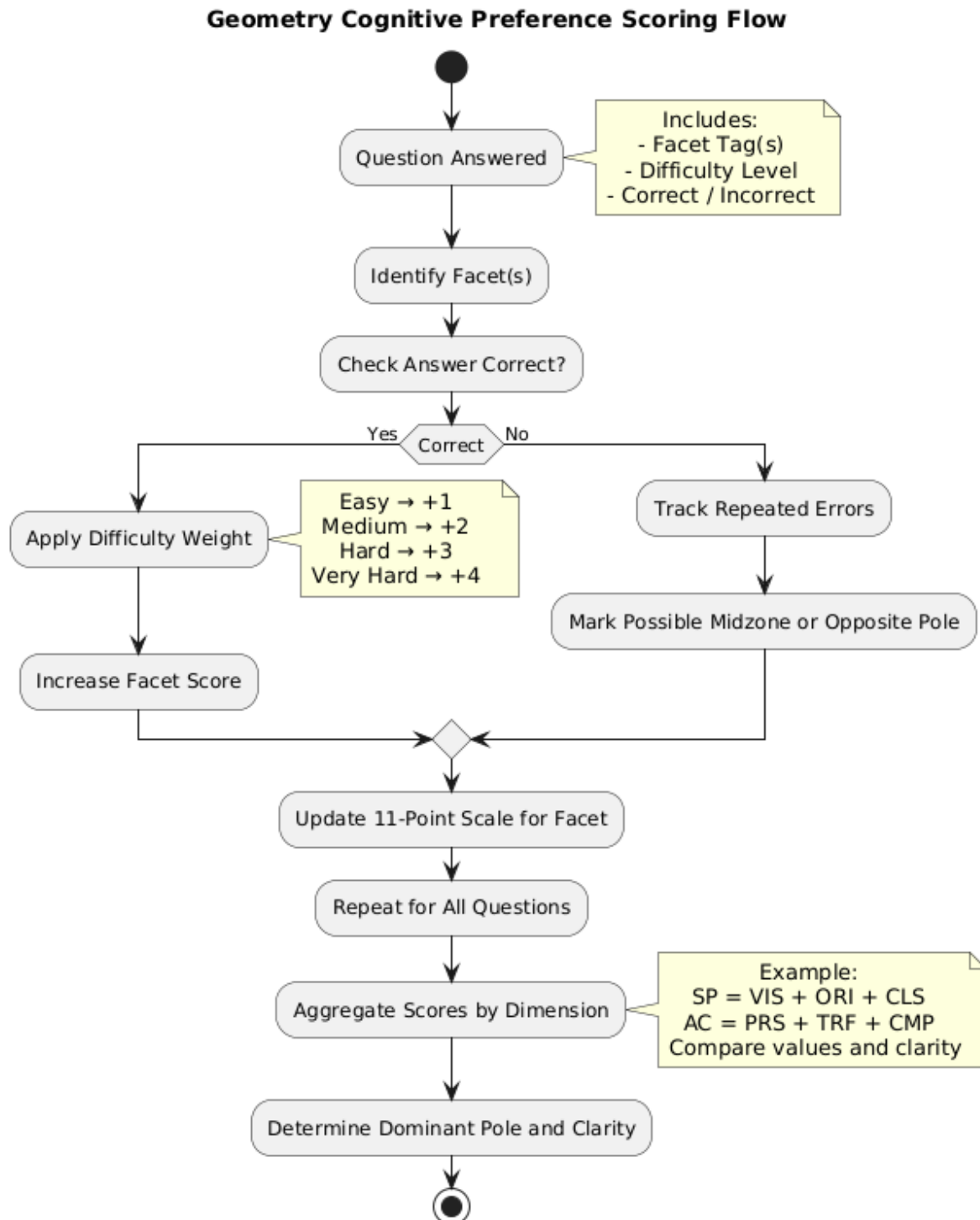


Figure B.5 Geometry Cognitive Preference Scoring Flow

The following table operationalizes the scoring logic, illustrating the quantitative basis for determining a learner's preference clarity. It explicitly shows how difficulty influences the strength of a preference and how both correct and incorrect responses contribute to the overall profile. This transparency is essential for understanding the mechanic's logic and for any future algorithmic implementation.

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Assessment Skill Facet	Difficulty Level	Points for Correct Answer	Points for Incorrect Answer (Consistent Error)	Contribution to 11-point Scale (Illustrative)	Thresholds for Interpretation (11-point scale)
VIS, ORI, CLS, PRS, TRF, CMP, REL, ENC	Easy	+1	-0.5	Low	0-1: Midzone
	Medium	+2	-1	Moderate	2-5: In-Preference / Out-of-Preference
	Hard	+3	-1.5	High	6-10: Strong In-Preference / Strong Out-of-Preference
	Very Hard	+4	-2	Very High	

Table B.3 Geometry Preference Clarity Scoring Matrix (Illustrative)

The 11-point scale would range from -5 to +5, where positive values indicate leaning towards one pole (e.g., SP, CR) and negative values towards the other (e.g., AC, AP). A score of 0 or +/-1 would typically fall into the “midzone”. The exact mapping of accumulated points to the 11-point scale would require calibration based on empirical data.

B.4.4. Generating the Geometry Learning Profile

Based on aggregated facet scores across cognitive dimensions, the system generates a **Geometry Learning Profile** that highlights tendencies rather than assigning fixed types. Each profile describes the learner’s *dominant, balanced, and developing preferences*, using constructive, growth-oriented language.

Examples include:

- **SP–AC (Spatial-Perceptual Dominant, Analytical-Computational Developing):** Strong visualization and spatial reasoning skills; developing proficiency in structured computation and formula-based reasoning.

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- **CR-AP (Conceptual-Relational Strong, Applied-Procedural Balanced):**
Robust theoretical understanding and abstract reasoning; procedural execution is consistent and poised for higher-level application.

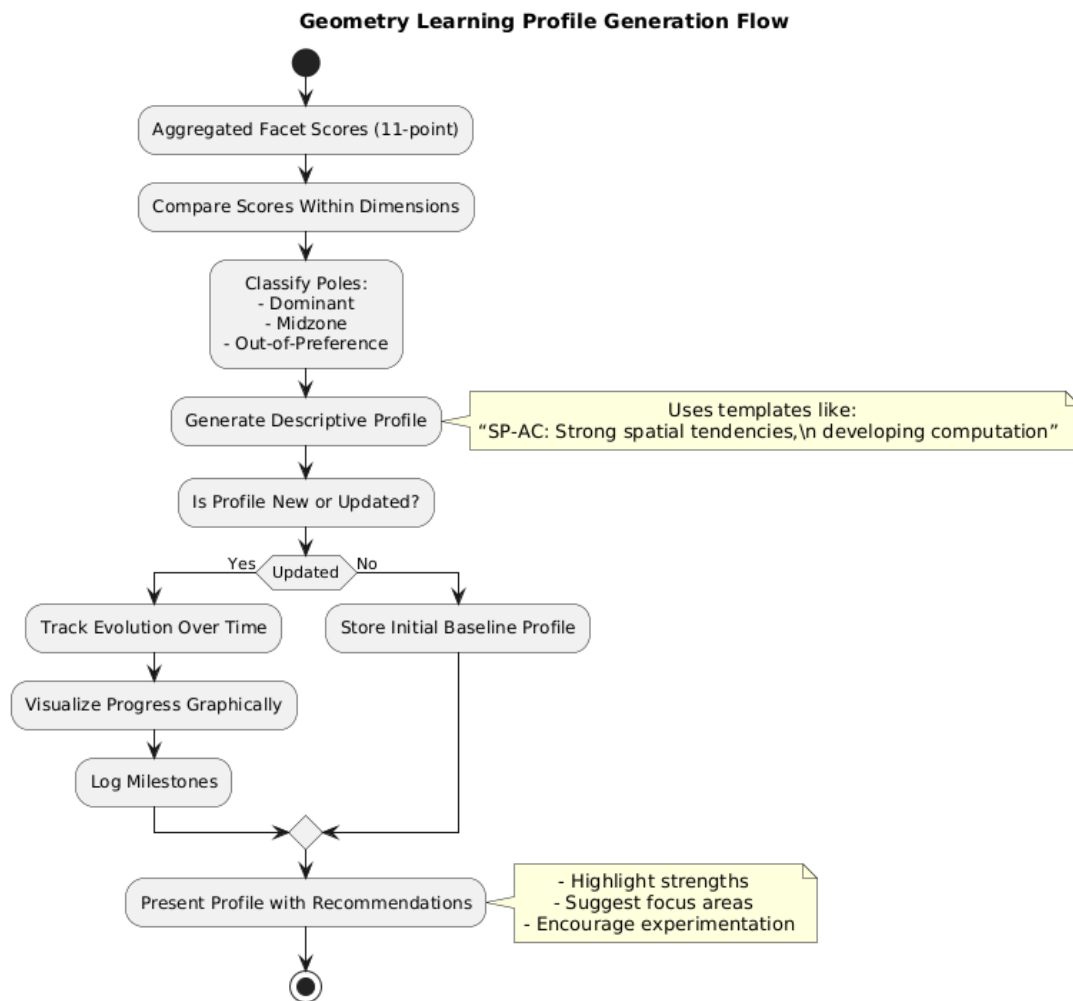


Figure B.6 Geometry Learning Profile Generation Flow

Each profile indicates:

- **In-Preference:** Confirmed strengths
- **Midzone:** Flexible or emerging areas
- **Out-of-Preference:** Growth opportunities

Importantly, the system is *dynamic*. Unlike static personality models, this profile evolves as learners interact with the system, allowing them to track progress in weaker areas and visualize the expansion of their cognitive repertoire. This adaptive feedback

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shifts emphasis from performance to growth, reinforcing a mindset of continuous learning and cognitive diversification.

B.5. Adaptive Response and Personalized Feedback System

The generated Geometry Learning Profile serves as the foundation for providing actionable, personalized support to learners.

B.5.1. Tailored Learner Insights and Communication

The comprehensive “Geometry Learning Profile” will be presented to learners via an intuitive dashboard or a narrative report. Communication will consistently employ positive, growth-oriented language, elucidating their unique cognitive strengths and areas ripe for development. Visualizations, such as radar charts illustrating facet strengths, can enhance the digestibility and impact of the profile. For instance, a report might state: “Your profile suggests a strong preference for Visualization (VIS), indicating a natural aptitude for understanding problems by picturing them in your mind. We also observe that your Problem Solving (PRS) skills are developing, which represents an excellent area for focused growth.”

B.5.2. Actionable Recommendations and Targeted Resources

Based on the identified “out-of-preference” or “developing” facets, the system will provide specific, personal recommendations:

- **Targeted Practice:** The system will suggest specific questions or mini modules from the question bank that are designed to target the identified weaker skills. For example, a learner developing their Analytical-Computational preference might be directed to more calculation-heavy problems.
- **Alternative Learning Strategies:** For a learner strong in Spatial-Perceptual skills but developing in Analytical-Computational approaches, the system might suggest attempting calculation problems by first sketching the geometric figures. Conversely, for a learner with strong Conceptual-Relational understanding but developing Applied-Procedural skills, the system could encourage breaking down complex problems into smaller, manageable procedural steps.

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- ***Curated Resources:*** The system will recommend external educational resources, such as videos, tutorials, or interactive simulations, that cater to different learning styles. This could include highly visual explanations for Spatial-Perceptual dominant learners or step-by-step procedural guides for those developing their Applied-Procedural skills.
- ***Error Analysis Prompts:*** When a learner consistently provides incorrect answers in an “out-of-preference” facet, the system could trigger self-reflection prompts. For an Analytical-Computational dominant learner struggling with Visualization, the prompt might be: “Did you try drawing this problem?” For a Conceptual-Relational dominant learner struggling with Problem Solving, the prompt could be: “Can you break this problem down into smaller, executable steps?”

B.5.3. Dynamic Learning Path Adjustments

The system is designed to dynamically adjust the sequence and type of questions presented to the learner in subsequent sessions. This adaptive approach ensures continuous engagement and targeted skill development.

- ***Reinforcing Strengths:*** Periodically, challenging questions aligned with a learner’s “in-preference” areas will be introduced to deepen their mastery and maintain engagement.
- ***Addressing Growth Areas:*** Questions that specifically require the use of “out-of-preference” skills will be prioritized, with a gradual increase in difficulty to systematically build proficiency in those areas.
- ***Gamified Integration:*** The Hint displayed in game column in the question banks provides a direct mechanism for integrating adaptive feedback within a gamified learning environment. Instead of generic hints, the system can trigger profile-specific hints.

B.5.4. Ethical Considerations in Interpretation

The implementation of this mechanic necessitates careful consideration of ethical implications, particularly regarding interpretation and communication of results.

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- ***Avoiding Rigid Labeling:*** It is paramount that the system avoids rigid categorization or stereotyping of learners. The generated profile must be presented as a dynamic snapshot of current cognitive tendencies, not a permanent label. Learners should be encouraged to view their profile as a starting point for exploration and growth, rather than a fixed identity.
- ***Emphasis on Development:*** The narrative accompanying the profile must consistently reinforce the principle that all cognitive preferences can be developed and that a well-rounded geometry learner effectively utilizes a variety of problem-solving approaches.
- ***Transparency and Agency:*** Learners should be fully informed about how their profile is generated and be granted agency to explore different learning paths, even those outside their “preferred” areas. This aligns with the MBTI’s original goal of fostering self-awareness for personal growth.
- ***Not for High-Stakes Decisions:*** Consistent with the limitations of the MBTI, this mechanic is designed purely as a learning tool. It should not be employed as an evaluative measure for grading, academic placement, or any other high-stakes decision.

B.6. Implementation Considerations and Future Directions

The successful deployment of this MBTI-inspired learner analysis mechanic requires careful consideration of several implementation aspects and opens avenues for future research.

B.6.1. Data Collection and Processing

Robust data collection mechanisms are essential to accurately track learner responses, the time taken for each question, and potentially more granular interaction patterns such as eye-tracking or mouse movements, if the learning environment supports such telemetry. Sophisticated algorithms will need to be developed and continuously refined for calculating facet scores and the overall preference clarity indices.

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B.6.2. Algorithm Development and Validation

Statistical methods will be critical for weighing questions by difficulty and for aggregating scores across various facets and dimensions. Ongoing validation studies will be indispensable to ensure that the mechanic accurately reflects learners' cognitive tendencies and that its recommendations are effective in improving learning outcomes. This includes rigorous A/B testing of different feedback mechanisms and adaptive pathways to optimize their impact.

B.6.3. User Interface (UI) and User Experience (UX) Design

The presentation of the "Geometry Learning Profile" must be designed to be intuitive, engaging, and, crucially, non-judgmental. The adaptive recommendations need to be seamlessly integrated into the existing learning platform or game, ensuring that the personalized support feels natural and helpful rather than intrusive.

B.6.4. Longitudinal Studies

Future research could involve conducting longitudinal studies to track how learners' profiles evolve over extended periods of engagement with the system. Such studies would provide invaluable insights into the malleability of cognitive preferences in an educational context and the long-term effectiveness of adaptive interventions.

B.6.5. Expansion to Other Domains

The underlying framework and principles of this mechanic possess the potential for adaptation to other STEM subjects or even broader academic domains. This expansion would be contingent upon the availability of similarly rich metadata pertaining to assessment skills, question types, and topics within those respective fields.

B.7. Empowering Geometry Learners Through Insight

This report has detailed the design of a novel MBTI-inspired mechanic for analyzing and responding to learner performance in geometry. By moving beyond conventional correctness scores, this system aims to profile learners' unique cognitive preferences and strengths within the domain of geometry.

Leveraging the rich metadata inherent in the provided question banks, particularly the Assessment Skills, the mechanic can generate a nuanced "Geometry Learning Profile"

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for each learner, drawing parallels to the personality types and facets identified by the MBTI. This profile empowers the system to deliver highly personalized feedback, targeted practice recommendations, and dynamically adjusted learning paths, thereby fostering a more engaging and ultimately more effective learning experience.

By emphasizing self-awareness, promoting a growth mindset, and encouraging the development of diverse problem-solving approaches, this mechanic holds significant potential to transform geometric education. It aims to empower learners not only to master geometric concepts but also to gain a deeper understanding of their own cognitive toolkit and how to expand it. The ultimate objective is to cultivate adaptable, resilient, and self-aware learners who are well-equipped to tackle complex challenges in mathematics and beyond.

APPENDIX C. CONCEPT AND THEME

C.1 Design Principles

The user interface (UI) of the application was designed with a learner-first mindset, following Human-Computer Interaction (HCI) principles that prioritize usability, visual clarity, and accessibility. The design philosophy was grounded in the understanding that spatial geometry already imposes a significant cognitive load on students; thus, the UI must serve as a cognitive aid, not a burden.

Three primary principles guided the design:

- Minimalism to reduce cognitive load: We intentionally removed non-essential elements from the screen to minimize distractions. This helps students focus on geometry tasks rather than interface complexity.
- Consistency for ease of navigation: UI elements follow a consistent structure and interaction pattern across different modules (gameplay, drawing, AI), making it easier for users to learn once and apply everywhere.
- Feedback-rich interaction: Every action, from drawing a shape to answering a question, produces immediate and clear visual or auditory feedback, reinforcing correct understanding and enabling error correction.

C.1.1 Color Psychology and Visual Hierarchy

Given that the app is aimed at high school students studying a mentally taxing subject, the emotional effect of color was a major consideration. Research-informed design choices were applied as follows:

- **Green** was selected as the primary theme color, associated with calmness, focus, and reduced anxiety - supporting sustained learning in subjects like geometry, which many students find intimidating.
- **Yellow** was used for highlights and call-to-actions, evoking energy, creativity, and attentiveness without being overwhelming.
- All text and icons were displayed with high-contrast color pairs (black/white backgrounds with grayscale foreground elements) to ensure legibility across different devices and lighting conditions.

APPENDIX C: CONCEPT AND THEME

This palette was also iterated upon. In early versions of the UI, the team used more saturated hues, which tested poorly on mobile screens due to glare and visual fatigue. The final version introduced a “tone-down” adjustment to create a softer, more readable appearance across all screens.

C.1.2 Iterative Prototyping and Version Testing

The UI went through multiple design cycles. Two major layout prototypes were created and tested:

- Version 1 used bright color tones and more decorative elements but it not suitable for the objective of application

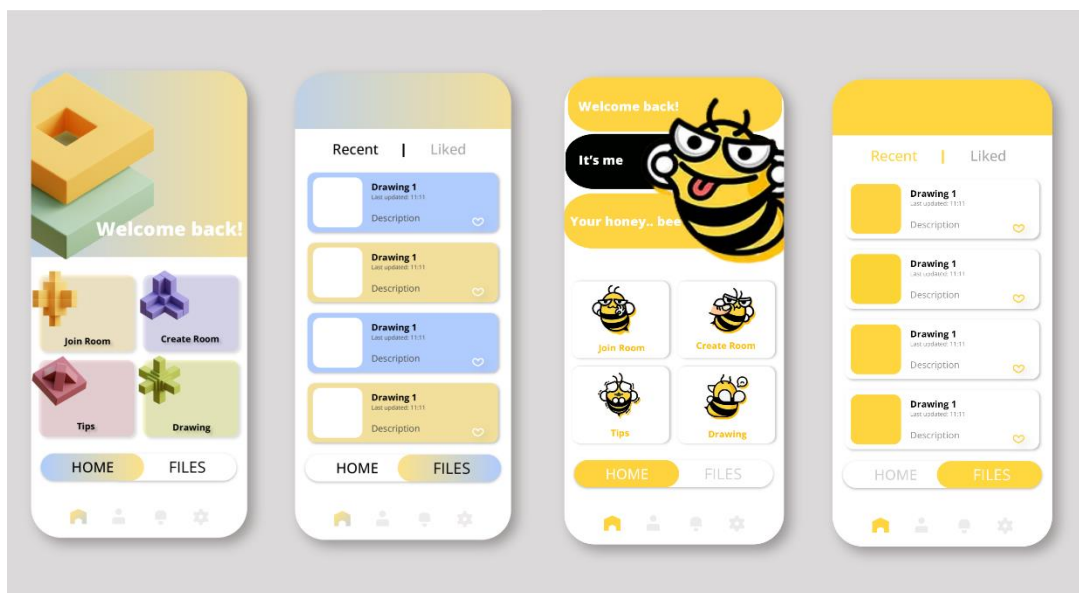


Figure C.1 Version 1 of UI

- Version 2 had aesthetic appeal, feedback revealed it caused eye fatigue over prolonged usage and lacked visual balance.

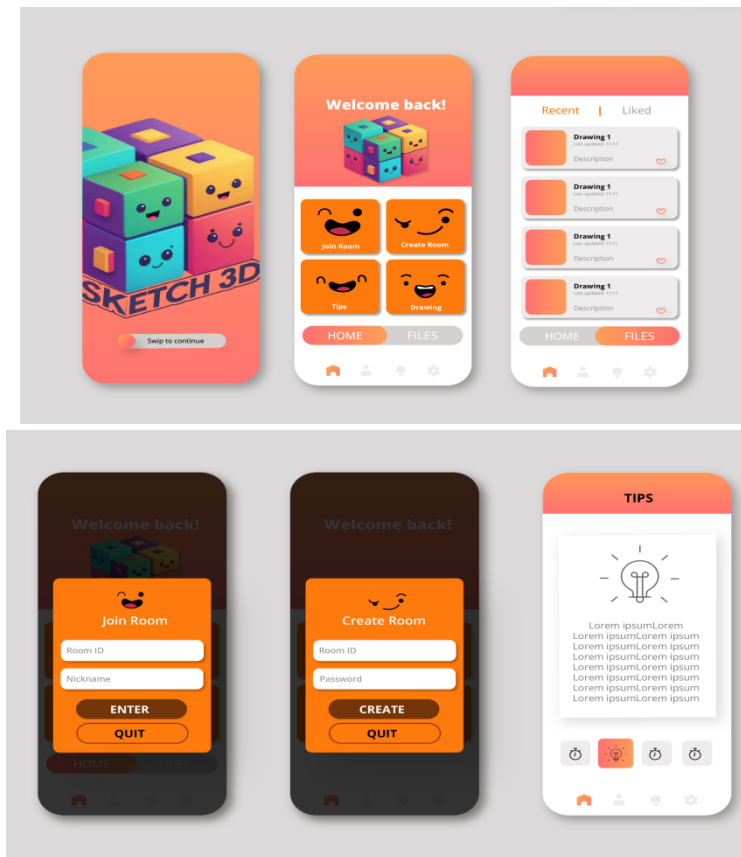


Figure C.2 Version 2 of UI

- Version 3 seems like the closest to the final version with matching objectives and once again, it caused eye fatigue over prolonged usage.

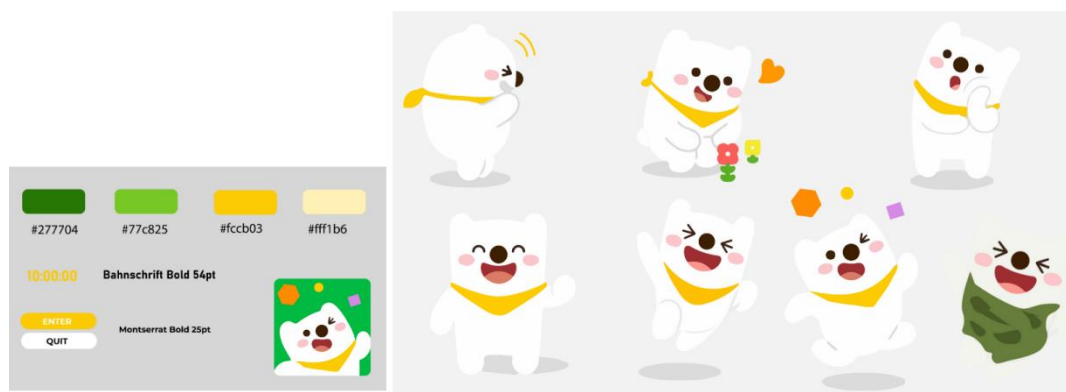


Figure C.3 Version 3 of UI: Mascot and Color themes

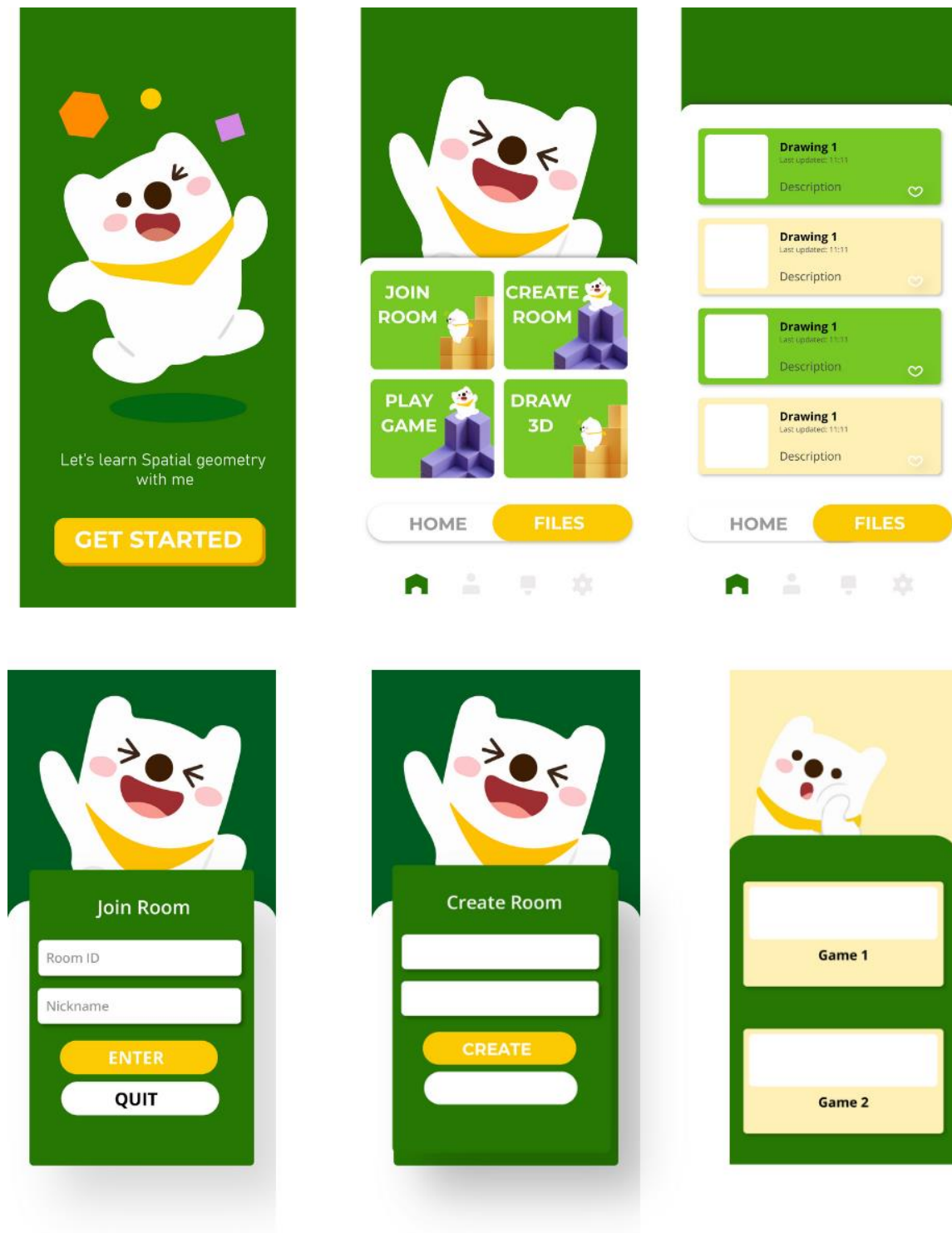


Figure C.4 Version 3 of UI

- The final version is basically the 'down-tone' of version 3 - simplified spacing and reduced color saturation. This version performed better in usability testing and was selected as the baseline for refinement.

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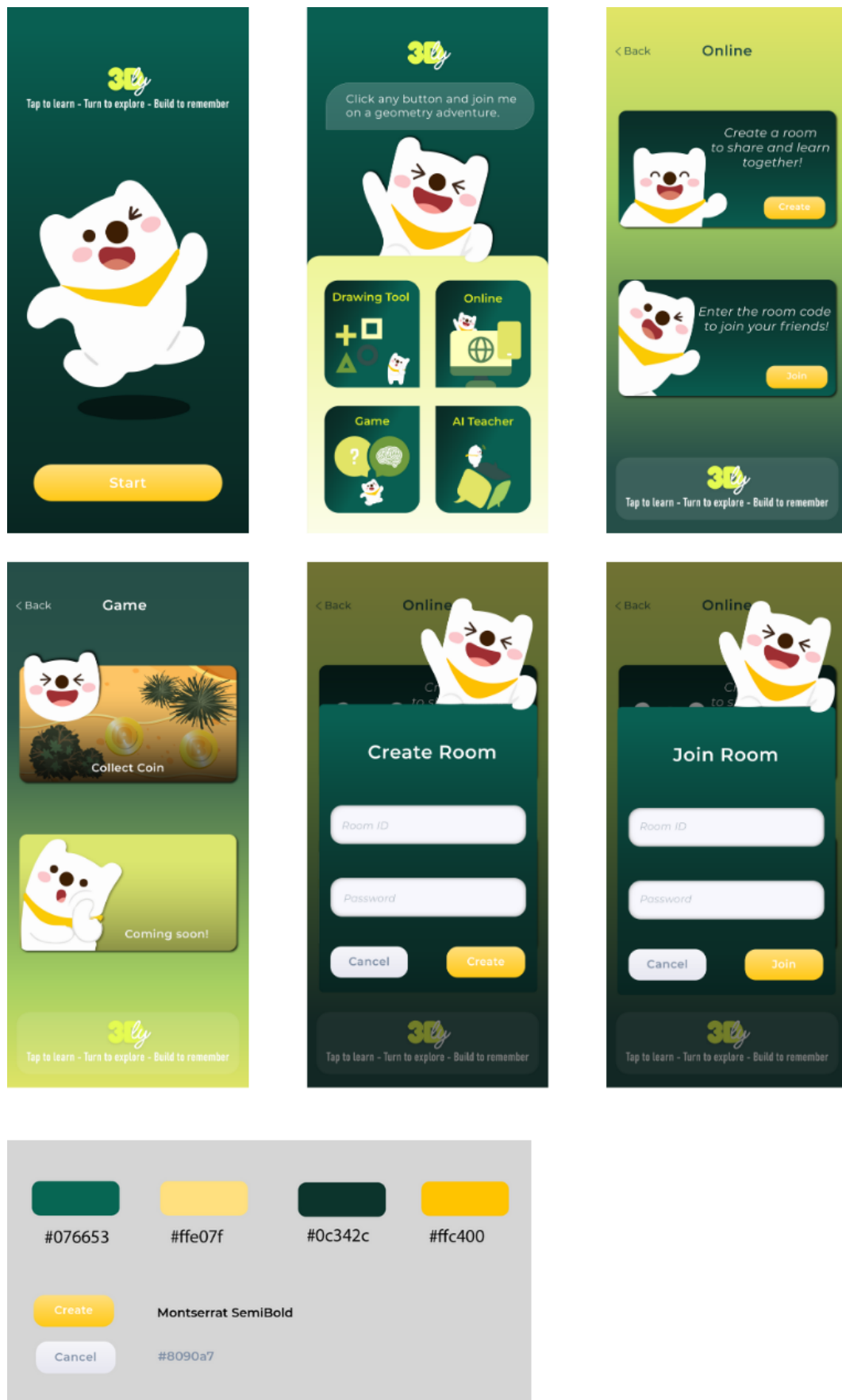


Figure C.5 Final version of UI

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Each iteration involved internal walkthroughs, expert reviews (within the team), and heuristic evaluations. Functional clarity, visual harmony, and modular reusability were key criteria in selecting the final version.

C.1.3 Survey-Based Validation

To ground design decisions in real-world needs, the team conducted our own survey involving **92 high school students** from multiple provinces across the country. The survey confirmed that:

- Most students felt traditional geometry teaching methods were either ineffective or only moderately helpful.
- Students expressed a preference for mobile-first, visually interactive tools with minimal setup.
- Engagement was significantly higher when visual feedback and interactive tasks were used.

These findings directly informed interface and feature placement, including reducing text-heavy content, embedding quick-tap buttons for geometry construction, and designing flashcard systems that favored image and formula visualization.

C.1.4 Mascot-Driven Visual Identity

To increase user engagement and brand memorability, the team introduced an in-app mascot - a friendly white bear, designed to represent approachability and emotional support. Although the mascot does not yet have a defined name, it serves as a guide in onboarding screens, tutorials, and game modules.

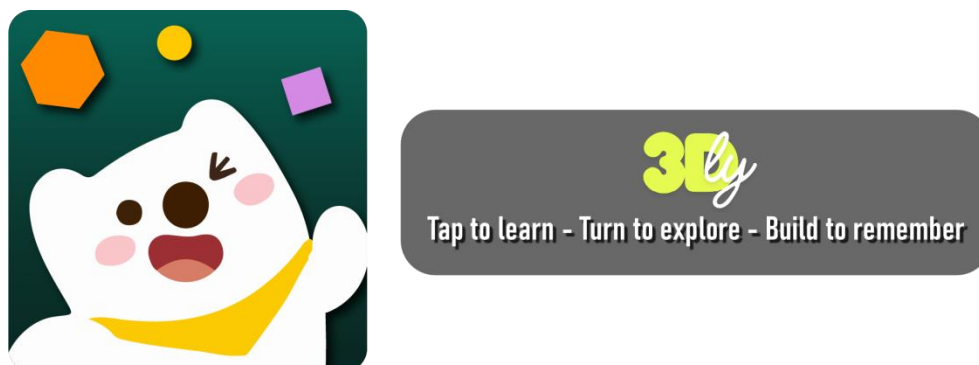


Figure C.6 App identified logo

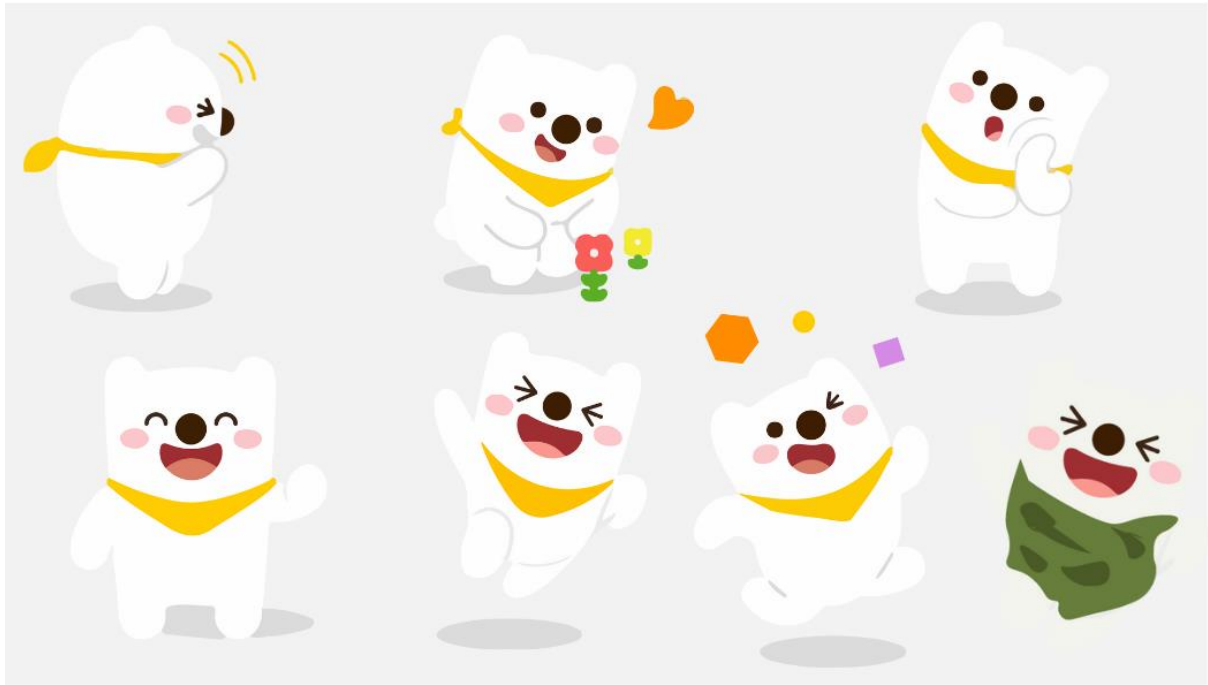


Figure C.7 Mascot detail

The inclusion of the mascot was not purely decorative. In line with UX literature on visual branding, mascots create emotional attachment, enhance recall, and reduce perceived formality in educational interfaces. This aligns well with the app's aim to demystify spatial geometry and make the subject feel more accessible.

C.2. FUNCTIONAL OVERVIEW & NAVIGATION DESIGN

C.2.1 Functional Modules

The application is divided into four primary functional modules, each targeting a different mode of learning:

1. 3D Geometry Drawing Tool

This allows students to construct spatial geometric shapes by placing and connecting points in a virtual 3D environment. The interface supports basic shape creation, measurements, and movement, mimicking paper-based construction but with more visual control.

2. Game-Based Learning (Collect Coin)

A gamified environment where students control a character to collect coins while answering geometry-related questions. This module combines reflex-based navigation with active recall, reinforcing knowledge through playful interaction.

3. Online Collaboration Mode

Enables users to host or join real-time shared drawing sessions. A Room ID system allows one user (the host) to draw while others (guests) view or optionally interact, supporting teacher-led or peer-based remote learning.

D.2.2 Navigation Structure

The app's navigation is based on a *hub-and-spoke model* with the *Home Screen* as the central hub. Each functional module is treated as a separate spoke, navigated via clearly labeled buttons.

This model was chosen for its simplicity and low cognitive cost, especially for student users who may not be digitally fluent. Every screen has a prominent back button, reducing the fear of “getting lost” during use.

Navigation within modules is linear and goal-directed:

- The game module uses a progressive level structure.
- The drawing tool includes modal overlays for Room Info or AI chat.

C.2.3 Screen Navigation Map

Below is a screens list table:

No.	Screen Name	Description
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APPENDIX C: CONCEPT AND THEME

1	Start Screen	Logo screen shown on app launch and button ‘Start’
2	Home Screen	Central hub with buttons to access game, drawing tool, online mode
3	Game Menu	List of games, currently is Collect Coin and a card for Coming soon game
4	Game Menu (Collect Coin)	Menu for starting the game, selecting level, accessing tutorial
5	Game Tutorial	Instructions on how to play Collect Coin game
6	Level Selection	Choose unlocked levels to play
7	Gameplay – Collect Coin	Main game interface: character movement, obstacle avoidance, coin pickup
8	Question Popup	Appears when picking up a coin; contains geometry questions and answers
9	Game Win Screen	Display score and options to next level, retry or return to menu
10	Game Over Screen	Display loss reason and options to retry or return to menu
11	Pause Menu	Temporarily stop the game; options to resume, retry, or exit
12	Drawing Tool Main Screen	Interface for creating and manipulating 3D geometric shapes
13	Room Info Popup	Shows room ID, password, and visibility toggle (host only)
14	Join Room Popup	Enter room ID and password to join a new collaborative drawing session
15	Create Room Popup	Enter room ID and password to create a new collaborative drawing session

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16	AI Assistant Chat	Chat interface with integrated GPT assistant for help and explanation
17	Settings	Language, sound, vibration, and app customization options

Table C.1 Screen Navigation Map

All modules support backward navigation, and modal popups (e.g., room info, questions) use dimmed overlays to retain spatial hierarchy.

C.2.4 Component Reusability and UI Standards

To maintain consistency and reduce development overhead, the following UI components were standardized and reused across modules:

- Top Title Bar: Displays current module, back button, and quick access to settings.
- Floating Action Buttons: Used for toggling room settings, or triggering AI chat.
- Modal Overlays: Unified styling for popup windows (e.g., Room Info, Question Prompt).
- Feedback Messages: Custom toast layer for actions like success, error, or retry tips.

All components follow mobile-first principles: touch targets are minimum 48x48dp, labels use clear Montserrat family fonts, and animations are short but informative.

C.3. SCREEN SPECIFICATIONS

C.3.1 Screen Naming Convention

Each screen in the application is assigned an identifier in the format **Screen xx**, where:

- **xx** is a two-digit number indicating the screen's position in the navigation hierarchy.
- Screens are grouped logically by module (e.g., game, drawing tool), and numbered sequentially for clarity.

This naming convention ensures consistency across documents, source code references, and team communication during UI implementation.

C.3.2 Format for Screen Description

Each screen is specified using the following structure:

- Screen ID
- Screen Name
- Purpose: Describes the screen's function or user goal.
- Navigation:
 - Navigation In: From which screen(s) the user can arrive here.
 - Navigation Out: Where the user can go from this screen.

This structure provides both technical clarity and usability context, allowing UI designers, developers, and testers to work from a shared reference.

C.3.3 Site Map (Screen Navigation Map)

The following diagram shows the navigation structure of the application. It reflects how the user moves between screens across different modules: gameplay, drawing tools, and online collaboration.

Each module originates from the Home Screen, with all transitions following simple, intuitive flows to minimize cognitive load.

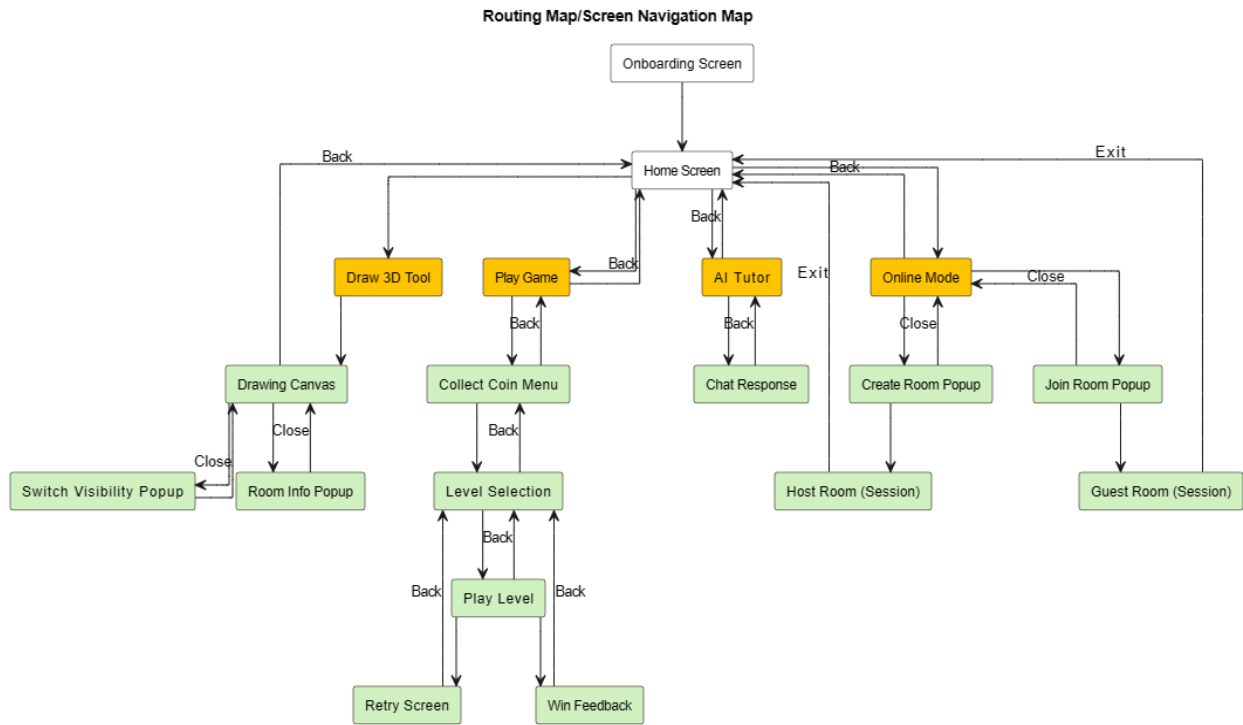


Figure C.8 Screen Navigation Map

C.4. Screen Specifications

C.4.1 Screen 01 – Start Screen



Figure C.9 Start screen

Screen Name: Start Screen

Purpose: Welcome screen displaying the logo, mascot, tagline, and a start button to enter the app.

Navigation:

- **In:** App Launch
- **Out:** Screen 02 – Home Screen

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No	Required (Input / Validation)	UI Component	Source	Note
1	N/A	Image – App logo	Static	Positioned at the top center “Tap to learn – Turn to explore – Build to remember”
2	N/A	Image – Mascot	Local (Asset)	White bear mascot, playful pose, centered on screen
3	Yes	Button – Start	Static	Yellow gradient button; navigates to Home Screen

Table C.2 Start screen Specifications

C.4.2 Screen 02 – Home Screen

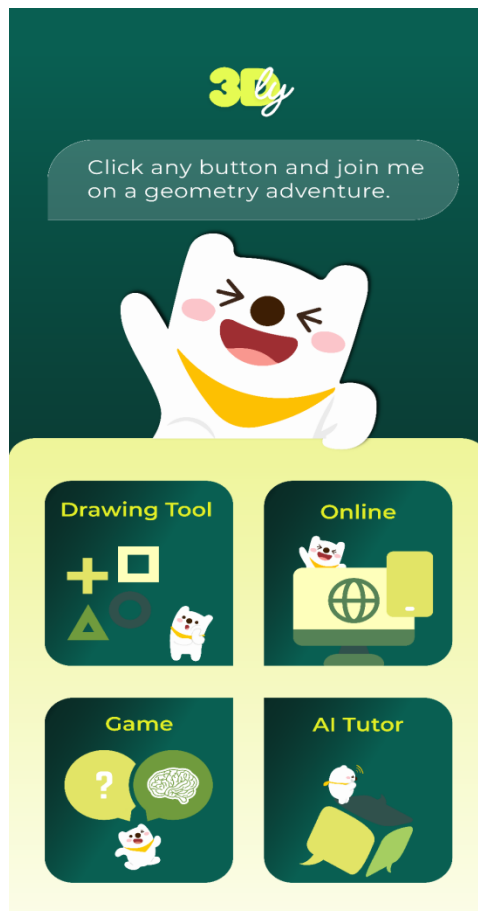


Figure C.10 Home screen

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Screen Name: Home Screen

Purpose: Main navigation hub that allows users to access different learning modules.

Navigation:

- **In:** Screen 01 – Start Screen
- **Out:** Screen 03, 12

No	Required	UI Component	Source	Note
1	N/A	Image – App logo	Static	Top of screen
2	N/A	Image – Instruction	Static	“Click any button and join me on a geometry adventure.”
3	N/A	Image – Mascot	Local (Asset)	Bear leaning over panel
4	Yes	Button – Drawing Tool	Static	Navigates to drawing module
5	Yes	Button – Online Mode	Static	Navigates to online mode
6	Yes	Button – Game	Static	Navigates to Game Menu (Collect Coin)

Table C.3 Home screen Specifications

C.4.3 Screen 03 – Game Menu

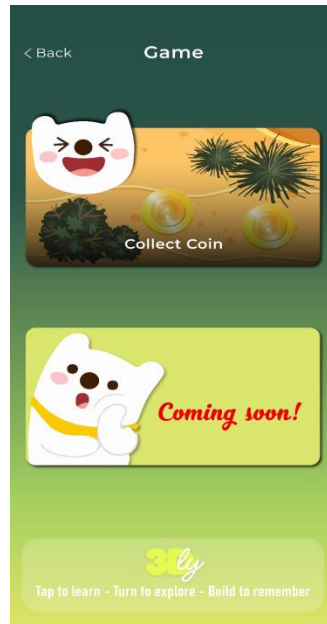


Figure C.11 Game Menu

Screen Name: Game Menu

Purpose: Allows the user to select a game. Currently, only “Collect Coin” is active.

Navigation:

- **In:** Screen 02 – Home Screen
- **Out:** Screen 04 – Game Menu (Collect Coin)

No	Required	UI Component	Source	Note
1	N/A	Text – Header	Static	Title: “Game”, with back icon
2	Yes	Card – Collect Coin	Static	Active card; shows coin + obstacle visuals
3	N/A	Card – Coming Soon	Static	Inactive card; placeholder for future game
4	N/A	Text – App Slogan	Static	Displayed at bottom: app slogan line

Table C.4 Game Menu Specifications

C.4.4 Screen 04 – Game Menu (Collect Coin)

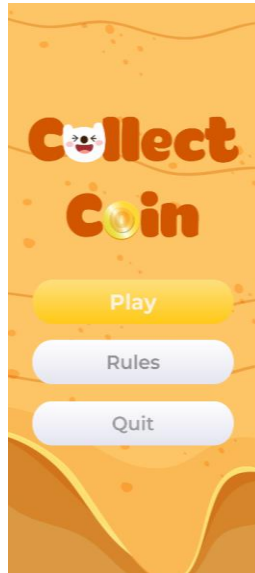


Figure C.12 Game Menu

Screen Name: Game Menu – Collect Coin

Purpose: Allows the user to start the game, view tutorials, or exit back to the game menu.

Navigation:

- **In:** Screen 03 – Game Menu
- **Out:** Screen 06, 05, or back to 03

No	Required	UI Component	Source	Note
1	N/A	Text – Header	Static	“Collect Coin” title with mascot face icon
2	Yes	Button – Play	Static	Navigates to Level Selection (Screen 06)
3	Yes	Button – Rules	Static	Navigates to Game Tutorial (Screen 05)
4	Yes	Button – Quit	Static	Returns to Game Menu (Screen 03)
5	N/A	Background Image	Static	Yellow tone background with desert visual

Table C.5 Game Menu Specifications

C.4.5 Screen 05 – Game Tutorial



Figure C.13 Game Tutorial screen

Screen Name: Game Tutorial

Purpose: Provides players with detailed instructions on how to play the Collect Coin game, including movement, objectives, scoring, and how questions appear.

Navigation:

- In: Screen 04 – Game Menu (Collect Coin)
- Out: Screen 04 – Game Menu (Collect Coin)

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No	Required	UI Component	Source	Note
1	N/A	Text – Header Title	Static	“How to Play” or similar instructional title
2	N/A	Text – Instruction 1	Static	“Control the mascot to collect coins on the map.”
3	N/A	Text – Instruction 2	Static	“Avoid hitting bushes, worms, or obstacles to keep your stars.”
4	N/A	Text – Instruction 3	Static	“Answer questions correctly to earn coins. Wrong answers cost stars.”
5	N/A	Text – Stars Rule	Static	“Each level has 3 stars. Lose all = Game Over.”
6	N/A	Text – Completion	Static	“Complete a level by collecting all coins.”
7	Yes	Button – Back	Static	Return to Game Menu (Screen 04)

Table C.6 Game Tutorial screen Specifications

C.4.6 Screen 06 – Level Selection

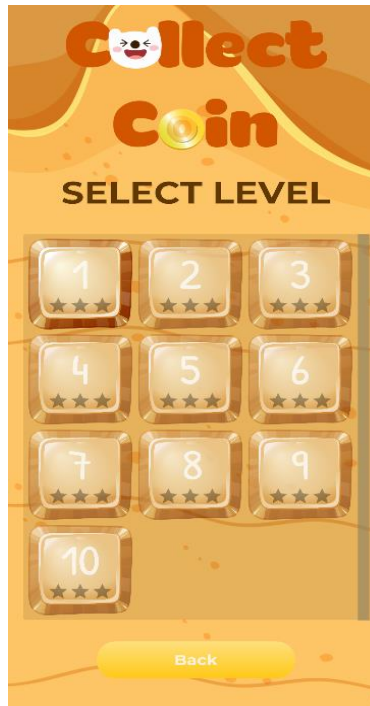


Figure C.14 Level Selection Screen

Screen Name: Level Selection

Purpose: Allows the player to choose a game level before starting gameplay.

Navigation:

- **In:** Screen 04 – Game Menu (Collect Coin)
- **Out:** Screen 07 – Gameplay

No	Required	UI Component	Source	Note
1	N/A	Text – Title	Static	“Select Level” displayed under game logo
2	Yes	Button – Level 1-10	Generated (by loop)	Each level is a button with stars based on score
3	Yes	Button – Back	Static	Navigates back to Game Menu (Screen 04)

Table C.7 Level Selection screen Specifications

C.4.7 Screen 07 – Gameplay – Collect Coin

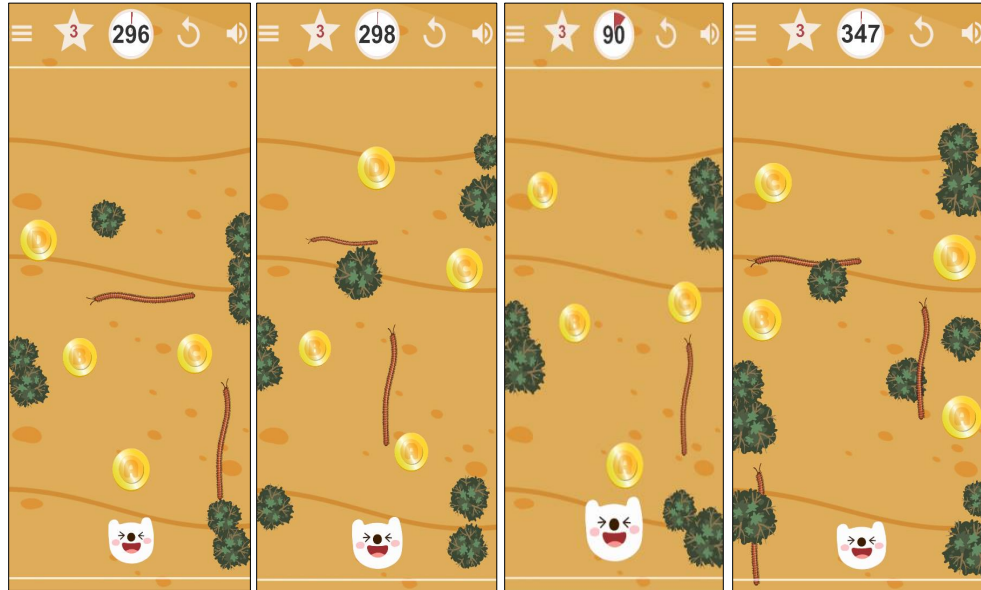


Figure C.15 Gameplay – Collect Coin screen

Screen Name: Gameplay – Collect Coin

Purpose: Main game screen where the player moves the mascot to collect coins and avoid obstacles.

Navigation:

- **In:** Screen 06 – Level Selection
- **Out:** Screen 08 (Question Popup), Screen 09 (Win), Screen 10 (Game Over),
Pause Menu

No	Required	UI Component	Source	Note
1	N/A	Icon – Star Count	Dynamic (from logic)	Number of lives remaining
2	N/A	Timer – Countdown	Dynamic (from logic)	Total time left for the level
3	Yes	Icon – Pause	Static	Opens Pause Menu
4	N/A	Button – Restart	Static	Restarts current level

APPENDIX C: CONCEPT AND THEME

5	N/A	Icon – Sound	Static	Mute/unmute toggle
6	Yes	Coins (A, B, C...)	Generated (by logic)	Collected by moving mascot toward them
7	No	Obstacles – Bushes	Static	Stationary obstacles
8	No	Obstacles – Worms	Animated	Move vertically; must be avoided
9	Yes	Player Mascot	Dynamic	Controlled by swipe/touch

Table C.8 Gameplay – Collect Coin screen Specifications

C.4.8 Screen 08 – Question Popup



Figure C.16 Question Popup screen

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Screen Name: Question Popup

Purpose: Displayed when the mascot picks up a coin, requiring the player to answer a multiple-choice geometry question.

Navigation:

- **In:** Screen 07 – Gameplay
- **Out:** Back to Gameplay (correct or wrong → affects stars/score)

No	Required	UI Component	Source	Note
1	N/A	Text – Question ID	Dynamic (from logic)	E.g., “Question 1”
2	Yes	Text – Question	Dynamic (from data)	Geometry-related multiple choice
3	Yes	Button – Answer A	Dynamic (from data)	Selectable
4	Yes	Button – Answer B	Dynamic (from data)	Selectable
5	Yes	Button – Answer C	Dynamic (from data)	Selectable
6	Yes	Button – Answer D	Dynamic (from data)	Selectable

Table C.9 Question Popup screen specification

C.4.9 Screen 09 – Game Win Screen

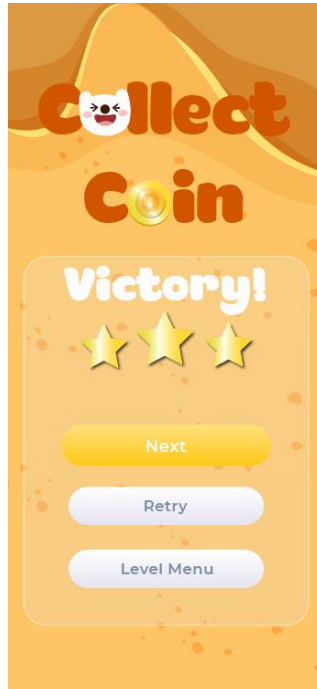


Figure C.17 Game Win Screen

Screen Name: Game Win Screen

Purpose: Displayed when the player completes the level successfully.

Navigation:

- **In:** Screen 07 – Gameplay
- **Out:** Screen 06 – Level Selection, or next level

No	Required	UI Component	Source	Note
1	N/A	Text – “Victory!”	Static	Celebratory message
2	Yes	Button – Next	Static	Moves to next unlocked level
3	Yes	Button – Retry	Static	Replay current level
4	Yes	Button – Level Menu	Static	Return to Level Selection (Screen 06)

Table C.10 Game Win Screen specification

C.4.10 Screen 10 – Game Over Screen

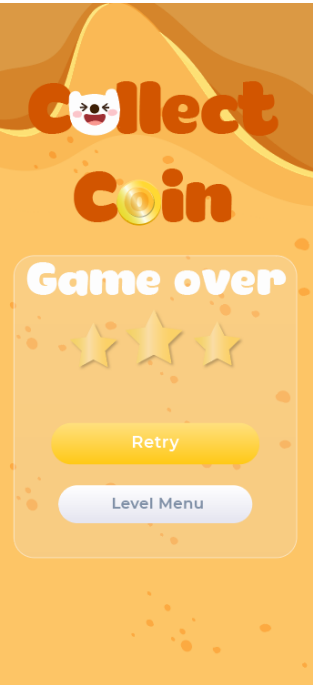


Figure C.18 Game Over Screen

Screen Name: Game Over Screen

Purpose: Displayed when the player fails the level (e.g., hits obstacles, time out, runs out of stars).

Navigation:

- **In:** Screen 07 – Gameplay
- **Out:** Screen 06 – Level Selection, or retry

No	Required	UI Component	Source	Note
1	N/A	Text – “Game Over”	Static	Message shown on failure
2	Yes	Button – Retry	Static	Retry current level
3	Yes	Button – Level Menu	Static	Return to Level Selection (Screen 06)

Table C.11 Game Over Screen specification

C.4.11 Screen 11 – Drawing Tool Main Screen

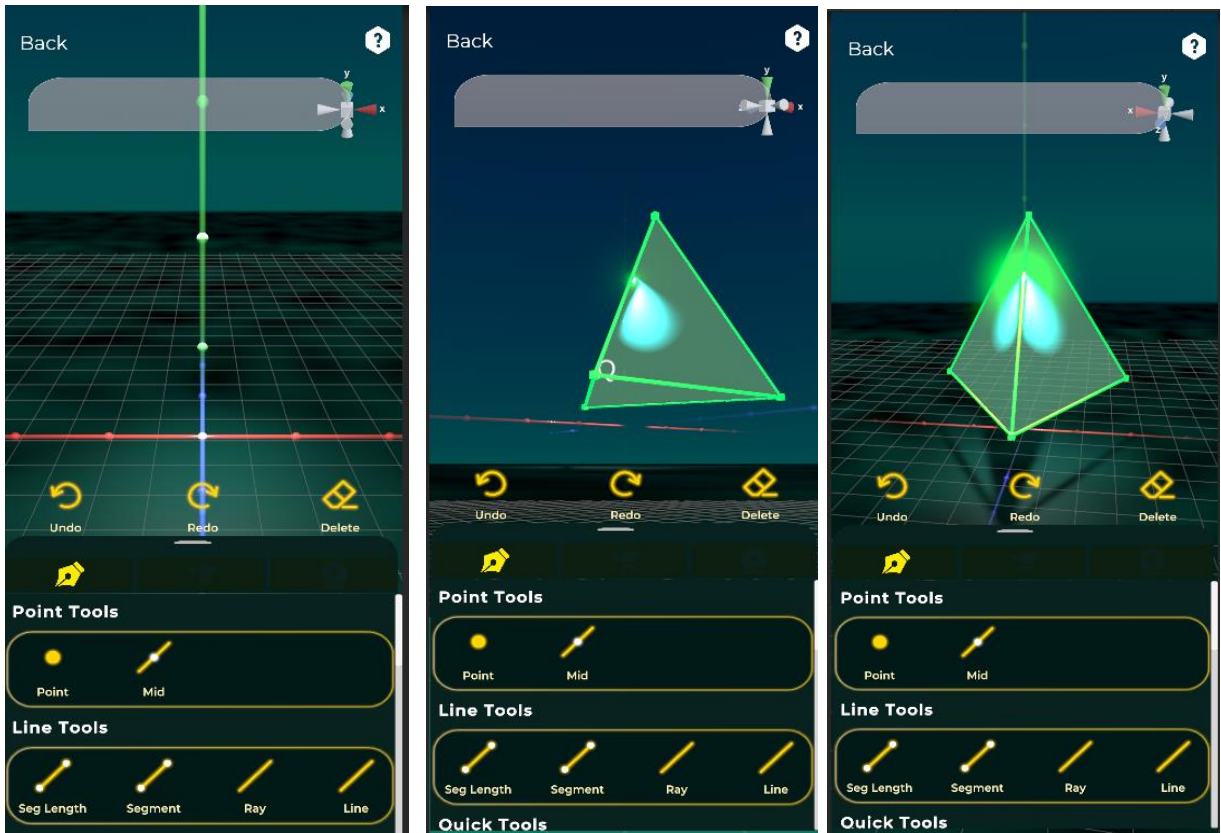


Figure C.19 Drawing Canvas

Screen Name: Drawing Canvas

Purpose: Main screen for constructing, editing, and interacting with 3D geometric models in a coordinate space. The user can select from multiple categories of tools (Basic, Spawn, Edit), get room info, chat with AI helper, undo/redo actions, and return to home.

Navigation:

- **In:** Screen 02 – Home Screen
- **Out:**
 - Screen 13 – Room Info Popup (if in online mode)
 - Screen 16 – AI Assistant (help icon)
 - Back to Screen 02 – Home

No	Required	UI Component	Source	Note
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APPENDIX C: CONCEPT AND THEME

1	Yes	Button – Back	Static	Return to Home Screen (Screen 02)
2	Yes	Icon – Rotate Left	Static	Rotate 3D canvas view counter-clockwise
3	Yes	Icon – Rotate Right	Static	Rotate 3D canvas view clockwise
4	Yes	Icon – Help (?)	Static	Opens AI Tutor or guidance popup
5	N/A	Coordinate Grid 3D	Dynamic (runtime)	Full 3D grid space with X, Y, Z axes and geometry elements
7	Yes	Button – Undo	Static	Undo last action
8	Yes	Button – Redo	Static	Redo undon action
9	Yes	Panel – Tool Menu	Static	Expandable bottom sheet with all tool categories
10	N/A	Section – Basic Tools	Static (UI layout)	Group: Move, New Line, Cube, Pyramid
11	N/A	Section – Edits	Static (UI layout)	Group: Duplicate tools from Basic (Move, Cube...) for post-spawn edit
12	Yes	Tool – Move	Static	Used to drag/move points or shapes
13	Yes	Tool – New Line	Static	Draws a new 3D line between selected points
14	Yes	Tool – Cube	Static	Spawns cube directly in canvas

APPENDIX C: CONCEPT AND THEME

15	Yes	Tool – Pyramid	Static	Spawns a square-based pyramid
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Table C.12 Drawing Canvas specification

C.4.12 Screen 12 – Room Info Popup

The image shows a 'Room Info' popup screen with a dark green background. At the top, the title 'Room Info' is displayed in white. Below the title are two white, rounded rectangular input fields. The first field contains the placeholder text 'Room ID' in a light gray font. The second field contains the placeholder text 'Password' in a light gray font. At the bottom of the screen, there are two rounded rectangular buttons. The left button is light gray with the text 'Cancel' in dark gray. The right button is yellow with the text 'Copy' in dark gray.

Figure C.20 Room Info Popup

Screen Name: Room Info Popup

Purpose: Provides current room ID, visibility status, and copy/share functionality in multiplayer sessions.

Navigation:

- **In:** Screen 12 – Drawing Canvas
- **Out:** Back to Drawing Canvas

No	Required	UI Component	Source	Note
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APPENDIX C: CONCEPT AND THEME

1	N/A	Text – Room ID	Dynamic (from logic)	Unique identifier of the current room
2	N/A	Text – Room Password	Dynamic (from logic)	If room is password protected
3	Yes	Button – Copy ID	Static	Copy Room ID to clipboard
4	Yes	Button – Close	Static	Close popup and return to Drawing Canvas

Table C.13 Room Info Popup screen specification

C.4.13 Screen 13 – Join Room Popup

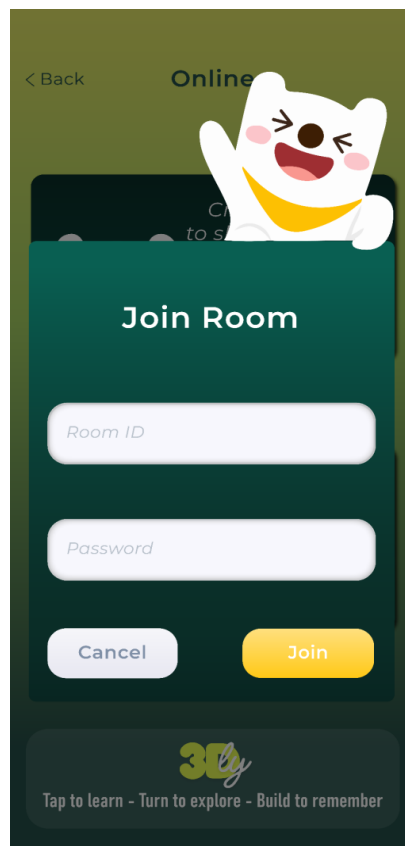


Figure C.21 Join Room Popup

Screen Name: Join Room Popup

APPENDIX C: CONCEPT AND THEME

Purpose: Allows a guest user to enter an existing public room by providing the Room ID and a nickname.

Navigation:

- **In:** Screen 02 – Home → Online Mode → Join
- **Out:**
 - Screen 12 – Drawing Canvas (as guest)
 - Back to Online Menu (via Quit)

No	Required	UI Component	Source	Note
1	Yes	Text Field – Room ID	Input (validated)	Room ID to join
2	Yes	Text Field – Nickname	Input (validated)	Guest’s display name in session
3	Yes	Button – Join	Static	Attempt to join room, goes to Drawing Canvas (guest)
4	Yes	Button – Quit	Static	Close popup and return to Online Menu

Table C.14 Join Room Popup screen specification

C.4.14 Screen 14 – Create Room Popup

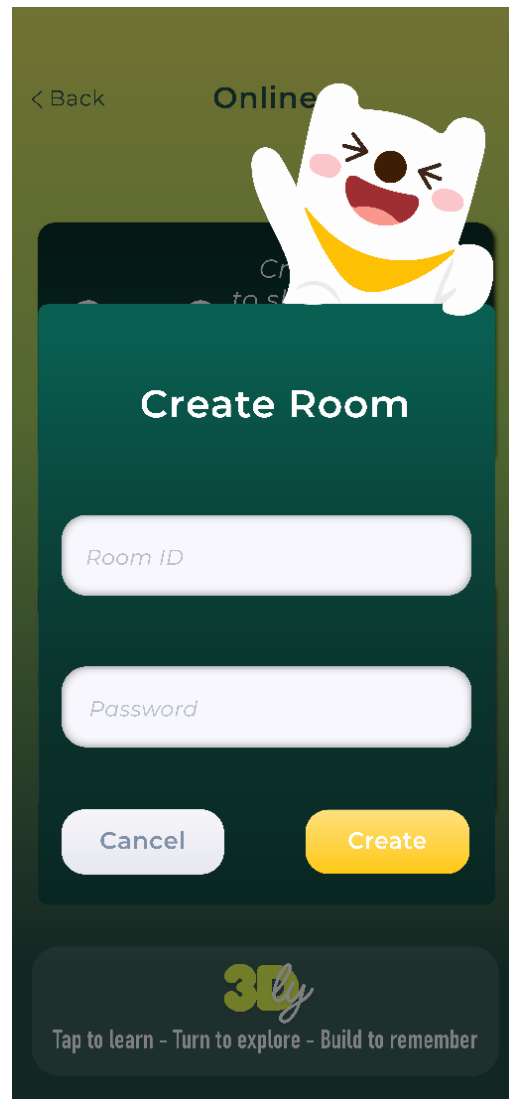


Figure C.22 Create Room Popup

Screen Name: Create Room Popup

Purpose: Allows a host to create a new public/private room by setting an ID and optional password.

Navigation:

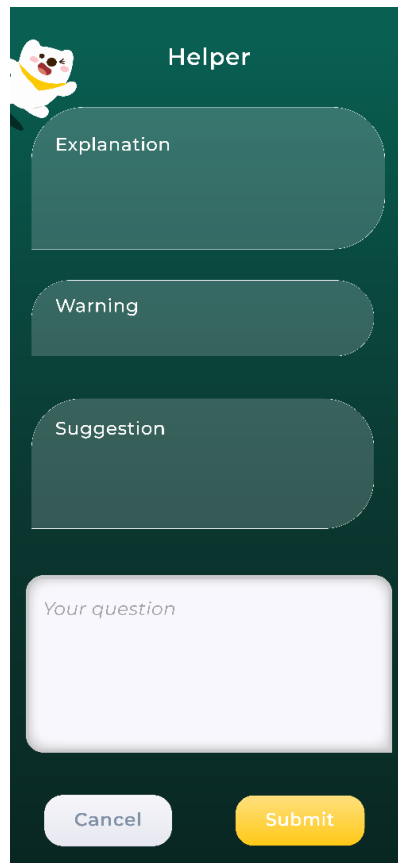
- **In:** Screen 02 – Home → Online Mode → Create
- **Out:**
 - Screen 12 – Drawing Canvas (as host)
 - Back to Online Menu (via Quit)

APPENDIX C: CONCEPT AND THEME

No	Required	UI Component	Source	Note
1	Yes	Text Field – Room ID	Input (validated)	Must be unique; identifies the room
2	No	Text Field – Password	Optional Input	Optional for private sessions
3	Yes	Button – Create	Static	Creates and enters room as host
4	Yes	Button – Quit	Static	Close popup and return to Online Menu

Table C.15 Create Room Popup screen specification

C.4.15 Screen 15 – AI Assistant Chat



The image shows a mobile app screen titled "Helper" with a small white cat icon in the top left corner. The screen has a dark green background. It features four rounded rectangular input fields stacked vertically, each with a label: "Explanation", "Warning", "Suggestion", and "Your question". The "Your question" field is the largest and has a light gray background. At the bottom of the screen, there are two buttons: a light gray "Cancel" button and a yellow "Submit" button.

APPENDIX C: CONCEPT AND THEME

Figure C.23 AI Assistant Panel

Screen Name: AI Assistant Panel

Purpose: Allows users to ask natural-language questions about geometry during drawing sessions. Powered by GPT.

Navigation:

- **In:** Screen 12 – Drawing Canvas (via Help icon)
- **Out:** Back to Drawing Canvas

No	Required	UI Component	Source	Note
1	Yes	Text Field – User Input	Input	Enter natural language question
2	Yes	Button – Send	Static	Sends message to GPT
3	N/A	Chat Bubble – User	Dynamic (from input)	Shows user’s message
4	N/A	Chat Bubble – Response	Dynamic (from API)	Response from AI based on geometry problem/question
5	Yes	Button – Close	Static	Exit chat and return to Drawing Canvas

Table C.16 AI Assistant Panel specification

APPENDIX D. GAME DESIGN DOCUMENT – COLLECT COIN

D.1 Game Overview

- Title: Collect Coin
- Genre: Educational Puzzle / Spatial Reasoning
- Target Platform: Android, Windows (Unity cross-platform)
- Target Audience: High school students (Grades 10–12), mathematics educators
- Purpose: Reinforce spatial geometry knowledge through interactive gameplay. By collecting coins and solving embedded 3D geometry challenges, players consolidate key concepts in a low-pressure, high-feedback environment.

D.2 Learning Objectives

- Reinforce 3D transformation and slicing concepts aligned with the national curriculum.
- Assess spatial reasoning and geometric comprehension indirectly via gameplay.
- Provide immediate, low-stakes feedback to support cognitive development.
- Increase student motivation through meaningful gamification.
- Collect data for identifying patterns in student difficulties for future analysis.

D.3 Core Gameplay Loop

1. **Start Level:** Player enters a level populated with mascot, coins, and obstacles.
2. **Swipe Movement:** Slide the mascot (a bear) across the board; mascot continues in a straight line until obstructed.
3. **Coin Encounter:** When a coin is touched, movement halts and a multiple-choice geometry question appears.
4. **Answer Question:** User selects an answer. If correct, the coin is collected. If incorrect, a star is deducted.
5. **Progression:** Player resumes moving to collect remaining coins.
6. **Completion Criteria:** Level is completed when all coins are collected. Level fails when all 3 stars are lost, or no coins are reachable.

D.4 Game Mechanics

Mechanic	Description
Movement	Directional swipe control. Movement continues until hitting an obstacle.
Collision Logic	Trees and animals act as fixed obstacles. Prevent straight paths.
Star System	3 stars per level. Each incorrect answer removes one star. 0 stars = level failure.
Retry Feature	Retry is allowed without penalty. Encourages learning via iteration.
Feedback	Visual and auditory cues for correct/incorrect answers. Immediate clarification if incorrect.
Forgiving Design	Answering wrong does not immediately end the game. Encourages exploration.

Table D.1 Game Mechanics

D.5 Question System

- Question Type: Geometry-related, multiple-choice, image-based. Often involves slicing, sectioning, and transformation.
- Trigger: Coin collision.
- Visual Context: Diagrams or 3D previews accompany questions.
- Tagging Criteria: Each question is tagged with:
 - Concept (e.g., prism cross-section)
 - Difficulty
 - Error trends (from survey data)

D.6 Level Design

Level	Difficulty Type	Description
1–5	Basic	– Easy to answer questions

		– Low density of obstacles, easy to moving and collecting coins
6–10	Advanced	– Hard to answer questions, multi-directional planning – High density of obstacles and complex routing

Table D.2 Level Design

- **Design Philosophy:** Gradual increase in spatial planning and question difficulty.
- **Progression Mechanics:**
 - Time available increases in later levels to support reasoning.
 - Coin placement encourages multi-step strategies.
 - Maps designed to restrict easy access, encouraging geometric thinking.

D.7 Assessment and Analytics Integration

- **Loggable Data:**
 - Accuracy per question and per concept.
 - Retry frequency.
 - Level completion time.
 - Navigation patterns (e.g., inefficient routes).
- **Use Case:**
 - Teachers use logs to identify conceptual weaknesses.
 - Offline logs can be extracted manually or automatically in classroom builds.
 - Future AI tutor integration may utilize these logs to suggest practice material or visualization tasks.

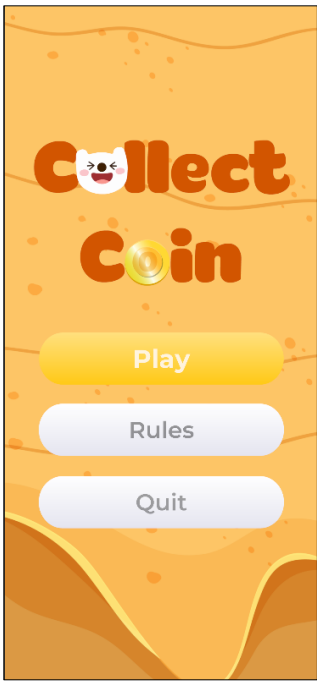
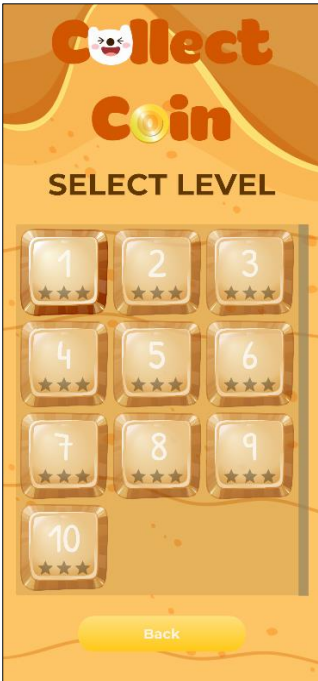
D.8 Pedagogical and UX Rationale

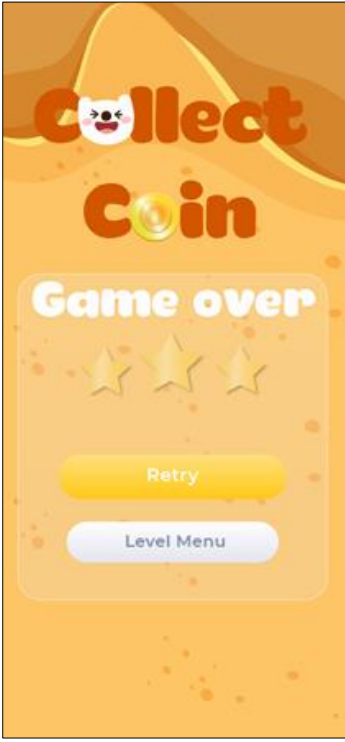
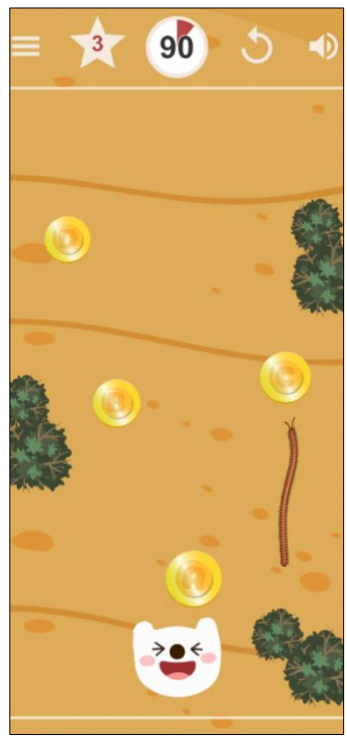
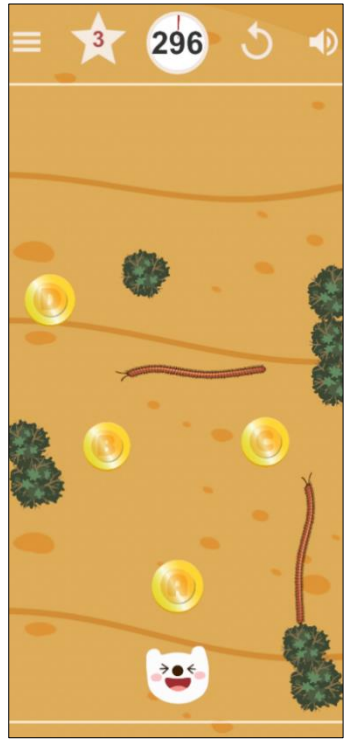
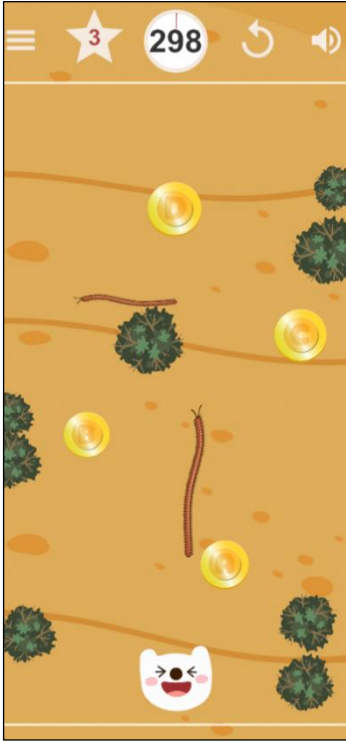
- **Van Hiele Alignment:** Levels reinforce spatial reasoning levels 0–2.
- **Engagement Model:** Combines fun (gameplay) with function (diagnostic questions).

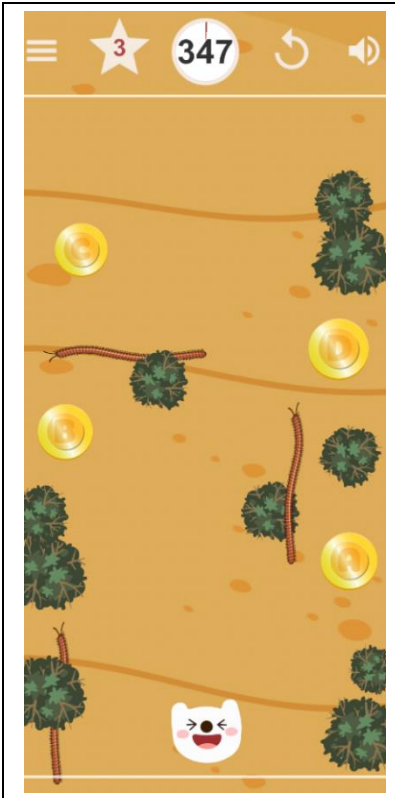
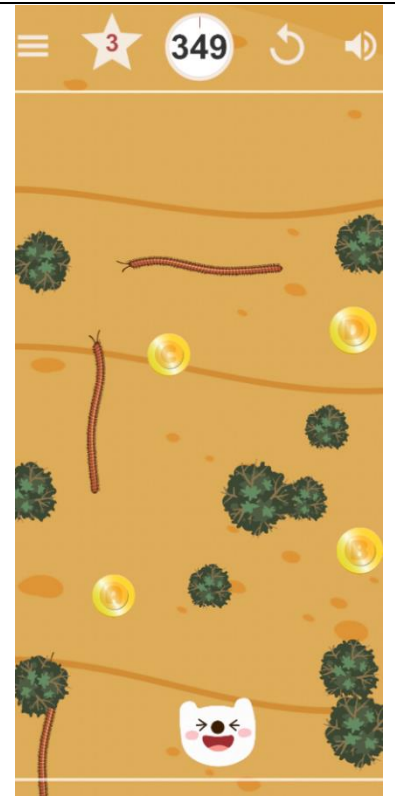
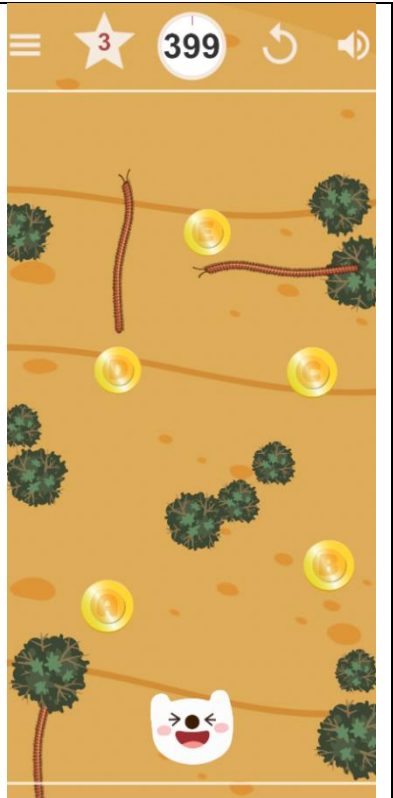
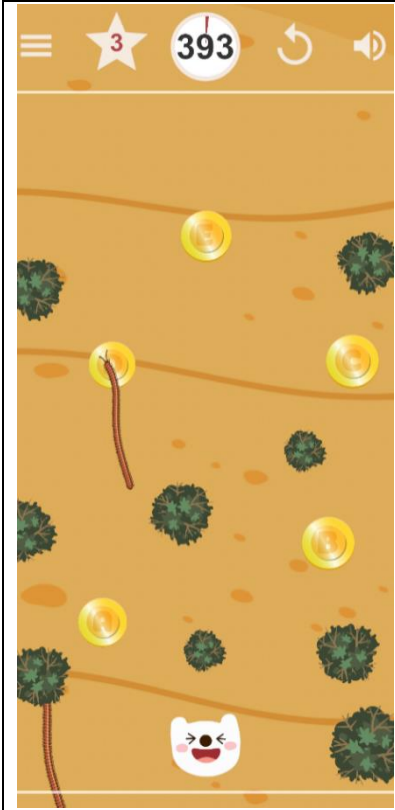
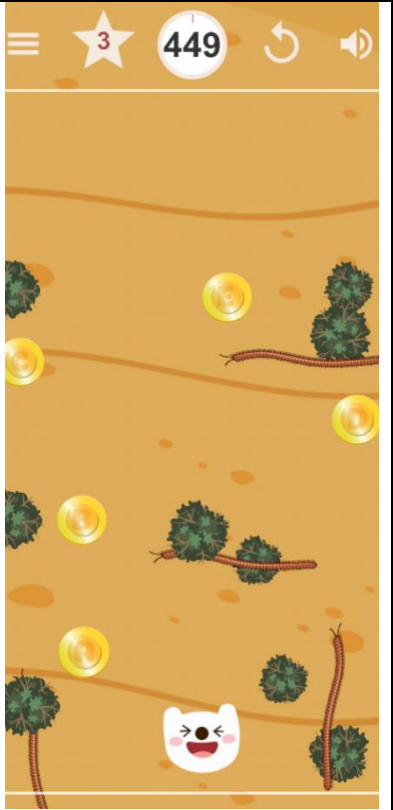
APPENDIX D: GAME DESIGN DOCUMENT – COLLECT COIN

- **UI Principles:** Calm green/yellow palette to reduce cognitive fatigue. Font and layout adapted for mobile-first design.
- **Feedback Loops:** Visual confirmation, retry suggestions, and explanatory feedback help form positive learning reinforcement.
- **Motivational Flow:** Consistent challenge without anxiety. Questions act as gates, not punishments.

D.9 Screenshots

		
Menu screen	Select level	Rules

		
Pause pop-up	Game over pop-up	Win game pop-up
		
Level 1	Level 2	Level 3

		
Level 4	Level 5	Level 6
		
Level 7	Level 8	Level 9

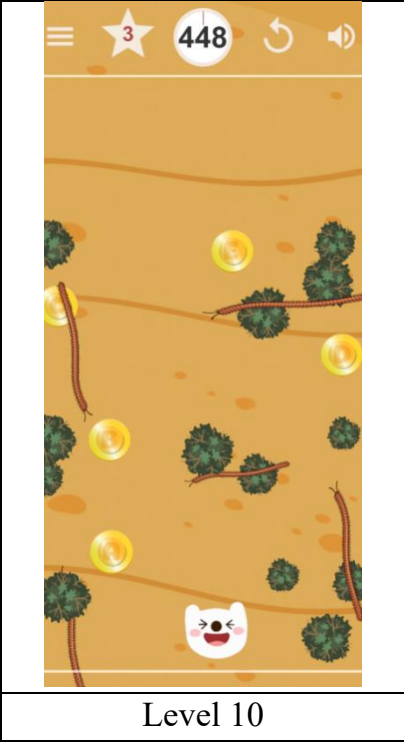


Table D.3 Game screenshots

D.10 Summary

Collect Coin transforms the assessment of geometric understanding into an interactive and motivational experience. By aligning game design with cognitive research and curriculum goals, it supports both formative assessment and deep engagement. The data it produces enhances system adaptivity and informs future pedagogical improvements. It exemplifies how game mechanics can be harnessed to strengthen STEM education while maintaining student enjoyment.

WORK DISTRIBUTION

Student ID	Team Member	Responsibilities	Completion
21110028	Lam Nguyen Huy Hoang	- Identified four system pillars (3D Drawing Tool, AI Tutor, Educational Game, Online Collaboration) - Designed abstract Shape base class with polymorphic data - Applied dependency injection for shape settings and pivot logic - Developed shape composition system- Implemented design patterns via ShapeNetworkSync - Created unified draw and constraint handler - Built snapping, labeling, undo/redo features - Designed JSON schema and ChatGPTClient for AI Geometry - Built scoring algorithm and question tracking - Developed Controller and slicing physics - Integrated Whisper API and GPT logic - Refactored recursion bug - Built LabelSetting - Created AIInputPanel and linked to AIInputController - Developed QuestionLoader and feedback animation	100%
21110001	Diep Hoai An	- Conducted literature review on cognitive demands - Analyzed cognitive barriers - Applied Cognitive Load Theory and Van Hiele Model - Designed and analyzed large-scale student survey - Synthesized 5 educational frameworks - Built learning progression roadmap - Proposed system-thinking model	100%

		- Conducted morphological analysis of tools (GeoGebra, etc.) - Mapped assessment into MBTI-like system - Designed metadata system and dynamic profiling - Connected geometry question bank - Created tag-aware randomization - Built onboarding UI, icons, and menus - Defined and designed app identifiers - Created dedicated test scenes - Illustrated UI and ported to Unity	
21110045	Nguyen Dang Khoa	- Designed internal shape settings UI - Built TTS/Whisper clients and voice handler - Integrated SALSA LipSync for model - Finalized voice-to-animation pipeline - Created gameplay log tracker - Built foundation of Network Transmitter - Implemented network session logic and Network Sync foundation - Designed lobby and cross-scene communication - Developed session authentication - Designed public/private session navigation	80%